



**Semiconductor Physics
Laboratory, Inc.**

Prielle Kornélia u. 2. 1117 Budapest, Hungary

Tel.: +36 1 505 4690 Fax: +36 1 505 4695

www.semilab.com

WT-2000

User Manual



**Measurement System for the
Semiconductor Industry**

The Semilab Model WT-2000 is covered under various international patents, including U.S. patents 5,237,266; 5,406,214; 5,580,828; and 6,653,850.

Copyright 2007 by Semilab Semiconductor Physics Laboratory, Inc. All rights reserved.

Trademarks

Semilab and the Semilab logo are trademarks of Semilab Semiconductor Physics Laboratory, Inc. WT-2000 is a trademark of Semilab Semiconductor Physics Laboratory, Inc.

Note: The photo on the front page is illustration only. The actual instrument can be different.

Table of Contents

Table of Contents	iii
List of Tables	x
List of Figures	xi
1. Introduction	1-1
1.1 <i>Measurement Capabilities</i>	1-1
1.1.1 μ -PCD	1-1
1.1.2 Charge-PCD	1-2
1.1.3 SPV	1-2
1.1.4 Fe Concentration Mapping	1-2
1.1.5 Oxide Characterization with V-Q	1-2
1.1.6 JPV Measurements	1-2
1.1.7 Other Capabilities	1-3
1.2 <i>User Manual</i>	1-3
1.3 <i>Contacting Semilab</i>	1-4
2. Equipment Overview	2-1
2.1 <i>System Configuration</i>	2-1
2.2 <i>System Software</i>	2-2
2.3 <i>System Dimensions</i>	2-2
2.4 <i>Facilities Requirements</i>	2-7
2.4.1 Environmental Requirements	2-7
2.4.2 Power Specifications	2-8
2.4.3 Vacuum Requirements	2-8
3. Safety Precautions	3-1
3.1 <i>Safety Symbols</i>	3-1
3.2 <i>Safety Interlocks</i>	3-2
3.2.1 Flash Interlock	3-3
3.3 <i>Emergency Motor Stop</i>	3-4
3.4 <i>Mechanical Safety</i>	3-4
3.5 <i>Laser Safety</i>	3-5
3.5.1 Lasers in the WT-2000	3-5
3.5.2 μ -PCD Unit	3-6
3.5.3 Aux laser	3-1
3.5.4 SPV Laser	3-8
3.5.5 VQ Unit	3-9
3.6 <i>Automatic Cassette Loader Safety</i>	3-9

4.	Installation	4-1
4.1	<i>Unpacking the System</i>	4-1
4.1.1	Unpacking the WT-2000 System Main Unit	4-1
4.1.2	Unpacking Other System Components	4-3
4.2	<i>Cable Installation</i>	4-3
4.2.1	Line Voltage Selection	4-5
4.2.2	System Power and Interconnections	4-5
4.3	<i>Stage Connections</i>	4-10
4.4	<i>Flash lamp interconnections</i>	4-11
4.4.1	Flash lamp	4-12
4.4.2	Flash power unit	4-12
4.4.3	Flash lamp interface	4-13
4.4.4	Flash power cable	4-13
4.4.5	Flash trigger cable	4-14
4.4.6	Digital I/O cable	4-14
4.5	<i>Installation of Charge PCD Option</i>	4-1
4.5.1	Installation of the Console Stage	4-16
4.6	<i>Installation of Indexer</i>	4-17
4.7	<i>Start-up and Initialization</i>	4-19
5.	Theory of Operation	5-1
5.1	<i>μ-PCD Measurement</i>	5-1
5.1.1	Laser Excitation	5-1
5.1.2	Detection	5-1
5.1.3	Evaluation	5-4
5.1.4	Bulk and Surface Recombination	5-5
5.1.5	Hardware Configuration for Lifetime Measurement	5-7
5.1.6	Automatic Parameter Setting in the μ -PCD Measurement	5-7
5.1.7	Effect of Bias Light on μ -PCD Measurements	5-8
5.2	<i>QSS-μPCD Measurement</i>	5-9
5.2.1	Background Information: Photoconductance and excess carrier concentration	5-10
5.2.2	Steady-state condition	5-10
5.2.3	Transient conditions	5-11
5.2.4	QSS- μ PCD procedure and measured parameters	5-11
5.2.5	Microwave measurements of photoconductance decay	5-12
5.2.6	Nonexponential PCD decay	5-13
5.2.7	Determination of emitter saturation current	5-14
5.2.8	Bulk and Surface contribution to τ_{eff}	5-16
5.2.9	Evaluation of the upper limit of surface recombination or lower limit of τ_b	5-17
5.2.10	Surface limited lifetime in as-cut and etched PV wafers	5-18
5.2.11	Effect of traps and elimination of this effect by steady illumination	5-18
5.3	<i>Charge-PCD Measurement</i>	5-19
5.3.1	Oxidized Wafers	5-19
5.3.2	Bare Wafers	5-19
5.4	<i>Surface Photovoltage Measurements</i>	5-19
5.4.1	Automatic Parameter Setting in the SPV Measurement	5-23
5.5	<i>Iron Contamination Analysis by the μ-PCD and SPV Techniques</i>	5-24
5.5.1	Fe-B Pair Dissociation and Association in a Single Point	5-25
5.6	<i>Measurement guide</i>	5-26
5.7	<i>V-Q Measurements</i>	5-1
5.7.1	Surface Potential	5-28
5.7.2	Q_{eff}	5-28

5.7.3	V-Q (T_{ox} , V_{fb} , D_{it} , Q_{total})	5-29
5.7.4	Q_m	5-30
5.7.5	E_{tunnel}	5-31
5.7.6	Plasma Damage Monitoring	5-31
5.8	<i>Junction Photovoltage Measurement</i>	5-31
5.8.1	Edge effect in the JPV measurement method	5-33
6.	Quick Start Guide	6-1
6.1	<i>Performing a Manual Measurement</i>	6-1
6.1.1	Check System Status	6-2
6.1.2	Load the Sample	6-2
6.1.3	Perform Wafer Finder Procedure	6-3
6.1.4	Set Signal Parameters	6-3
6.1.5	Set Measurement Parameters	6-4
6.1.6	Set Measurement Type and Start Measurement	6-4
6.1.7	Save or Print the Data	6-4
6.2	<i>Performing Automatic Measurements</i>	6-5
6.3	<i>Automatic Measurements with Recipes</i>	6-6
6.4	<i>Starting the WT-2000</i>	6-8
6.5	<i>Shutting Down the WT-2000</i>	6-9
7.	Using Wintau32 Software	7-1
7.1	<i>The Main Window</i>	7-1
7.1.1	The Menu Bar	7-2
7.1.2	The Toolbars	7-3
7.1.3	The Status Bar	7-3
7.1.4	The Recorder Value Bar	7-3
7.2	<i>Using the Menus</i>	7-3
7.3	<i>The Toolbar</i>	7-4
7.4	<i>The Measurement Toolbar</i>	7-5
7.5	<i>Buttons area</i>	7-7
7.6	<i>Data Types</i>	7-8
7.7	<i>Software Modes</i>	7-8
7.7.1	Data Evaluation Mode	7-9
7.7.2	Measurement Mode	7-9
7.8	<i>Setting User Access Levels</i>	7-9
7.8.1	Access Level Restrictions	7-9
7.8.2	Defining Users	7-9
7.8.3	Log In and Log Out	7-11
7.8.4	User Management	7-12
7.8.5	Customizing the User Input Fields	7-12
7.9	<i>Getting Help</i>	7-13
8.	Wintau32 Software Menus	8-1
8.1	<i>File Menu</i>	8-2
8.2	<i>Edit Menu</i>	8-3
8.3	<i>View Menu</i>	8-4
8.4	<i>Measure Menu</i>	8-5

8.5	<i>Macro Menu</i>	8-8
8.6	<i>Process Menu</i>	8-9
8.7	<i>Arithmetic Menu</i>	8-10
8.7.1	Arithmetic Requiring a Constant Value:	8-11
8.7.2	Arithmetic without a Constant:	8-11
8.7.3	Corrections, Correlations, Calculations:	8-11
8.7.4	LBIC calculations	8-13
8.7.5	Arithmetic with another Map:	8-14
8.8	<i>Print Menu</i>	8-14
8.9	<i>Access Menu</i>	8-14
8.10	<i>Window Menu</i>	8-15
8.11	<i>Help Menu</i>	8-16
9.	Displaying Data	9-1
9.1	<i>Opening Files</i>	9-1
9.1.1	Changing the File Display Mode	9-2
9.1.2	Selecting Files to Open	9-3
9.1.3	Working with the File List	9-3
9.2	<i>The Map Window</i>	9-5
9.3	<i>Display Options</i>	9-6
9.3.1	Zoom	9-6
9.3.2	Changing the Displayed Map Colors	9-7
9.3.3	View Options	9-7
9.4	<i>The Measurement Information Window</i>	9-12
9.5	<i>Using Masks and Erasing Data Points</i>	9-14
9.6	<i>Copying Maps and Adding Comments</i>	9-16
9.6.1	Using the Right Mouse Button on the Map	9-16
9.6.2	Editing Comments	9-17
9.7	<i>Data Distributions Windows</i>	9-18
9.7.1	Histogram	9-18
9.7.2	Cumulative Distribution	9-19
9.7.3	Line Scan	9-20
9.8	<i>Saving Data</i>	9-20
9.9	<i>Link Files</i>	9-20
9.9.1	Linking Files	9-21
9.9.2	Displaying Linked Files	9-21
9.10	<i>Printing Data</i>	9-21
9.10.1	Print Setup	9-22
9.10.2	Printing Displayed Maps	9-23
9.10.3	Printing from Files	9-24
10.	Measurement Setup Options	10-1
10.1	<i>Selecting the Measure Mode</i>	10-2
10.2	<i>Initializing the System</i>	10-2
10.3	<i>Setting the Sample Size</i>	10-2
10.4	<i>Loading and Unloading a Sample</i>	10-3
10.5	<i>Wafer Finder</i>	10-4
10.6	<i>Resuming Map Measurements</i>	10-5

10.7	Corona	10-5
10.8	Recipes	10-6
10.9	Hardware Info	10-7
10.10	Hardware Test	10-7
10.10.1	ATX00 Tab	10-8
10.10.2	Finder Tab	10-9
10.10.3	Interlock Tab	10-9
10.10.4	Interlock2 Tab	10-11
10.10.5	Indexer Tab	10-12
10.11	Measure Remote Functions for WSOS2 Software	10-12
10.12	Measure Options Tabs	10-13
10.12.1	Info Options	10-14
10.12.2	Height Options	10-14
10.12.3	Loading Options	10-17
10.12.4	Single Options	10-19
10.12.5	Auto Options	10-22
10.12.6	u-PCD Options	10-24
10.12.7	QSS u-PCD Options	10-25
10.12.8	SPV Options	10-25
10.12.9	Fe Options	10-27
10.12.10	Charge Options	10-29
10.12.11	Eddy Options	10-30
10.12.12	V-Q Options	10-31
10.12.13	Cassette Options	10-35
10.12.14	Interface Options	10-38
10.12.15	Wave Options	10-39
10.13	Recorder Windows and Evaluation Windows	10-39
10.13.1	Transient Recorder (μ -PCD and C-PCD Modes)	10-40
10.13.2	SPV Recorder (SPV Mode)	10-54
10.13.3	V-Q Recorder (V-Q Mode)	10-1
10.13.4	JPV Recorder	10-63
11.	Macros	11-1
11.1	Programming Environment	11-2
11.1.1	The Macro Menu	11-2
11.1.2	The Macro Editor Window	11-2
11.1.3	Macro Editor Toolbar	11-3
11.1.4	The Macro Syntax Window	11-4
11.1.5	The Macro Watch Window	11-4
11.2	Lexical Grammar of Macro Programs	11-5
11.2.1	Comments	11-5
11.2.2	Tokens	11-6
11.2.3	Declarations	11-9
11.3	Statements	11-12
11.3.1	Labeled Statements	11-12
11.3.2	Goto Statement	11-12
11.3.3	If Statements	11-12
11.3.4	While Statement	11-12
11.3.5	For Statement	11-13
11.3.6	Break Statement	11-13
11.3.7	Continue Statement	11-13
11.4	Writing Macro Programs	11-13
11.4.1	Call Function	11-13
11.4.2	Components	11-14

12.	System Operation	12-1
12.1	<i>Preparation for Use</i>	12-2
12.2	<i>Loading the Sample</i>	12-3
12.2.1	Manual Loading	12-3
12.2.2	Automatic Wafer Loading	12-5
12.2.3	Loading a sample on the console	12-7
12.3	<i>Wafer Finder</i>	12-8
12.4	<i>Setting the Signal Parameters</i>	12-9
12.4.1	Autosetting	12-9
12.4.2	Signal Parameters for μ -PCD and C-PCD Measurements	12-10
12.4.3	Signal Parameters for SPV Measurements	12-11
12.4.4	Signal Parameters for V-Q Measurements	12-11
12.4.5	Signal Parameters for JPV Measurements	12-12
12.5	<i>Setting the Measurement Parameters</i>	12-12
12.6	<i>Making a Measurement</i>	12-14
12.6.1	Single Point Measurements	12-14
12.6.2	X-Point Measurements	12-14
12.6.3	Map Measurements	12-16
12.6.4	Fe Concentration Measurement	12-18
12.6.5	Making C-PCD Measurements	12-18
12.6.6	Using Linear Corona Charge	12-19
12.6.7	Mobile Charge Measurement	12-19
12.7	<i>Saving Data</i>	12-20
12.7.1	Saving SPV Maps as Linked Files	12-20
12.8	<i>Making Multiple Measurements</i>	12-20
12.9	<i>Recipes</i>	12-21
12.9.1	Creating a Recipe	12-21
12.9.2	Loading a Recipe	12-24
12.10	<i>Automatic Measurements</i>	12-25
12.10.1	Systems with Manual Wafer Loading	12-25
12.10.2	Systems with Automatic Wafer Loading	12-26
12.11	<i>Automatic Measurements Using Recipes</i>	12-30
12.11.1	Systems with Manual Wafer Loading	12-30
12.11.2	Systems with Automatic Wafer Loading	12-31
13.	Calibration	13-1
13.1	<i>Calibration of the μ-PCD Bias Light Option</i>	13-1
13.2	<i>SPV Calibration</i>	13-2
13.3	<i>Corona Calibration</i>	13-2
13.4	<i>V-Q Calibration</i>	13-3
13.5	<i>JPV Head Calibration</i>	13-4
13.5.1	Calibrating the LED intensity	13-4
13.5.2	Calibrating the JPV voltage by the metal reference plate	13-4
13.5.3	Calibrating to the external quantum efficiency of the samples	13-6
14.	Maintenance	14-1
14.1	<i>Cleaning the System</i>	14-1
14.2	<i>Lubrication</i>	14-2
14.3	<i>Replacement of the flash bulb</i>	14-3

14.3.1	Steps of replacing the flash bulb	14-4
14.4	<i>Replacing the μ-PCD bias lamp</i>	14-4
14.4.1	Steps of replacing the bias halogen bulb	14-4
14.5	<i>Replacing the pins of 4pp</i>	14-5
14.5.1	Steps of replacing the 4 point probe	14-5
15.	Appendix	15-1
15.1	<i>Installing Wintau32 Software</i>	15-1
15.1.1	Requirements	15-1
15.1.2	Installing the Software from the CD-ROM	15-1
15.1.3	Windows Networking Configuration	15-2
15.1.4	Interface Settings in Wintau32	15-4
15.1.5	Software Upgrades	15-4
15.1.6	Registration	15-5
16.	Index	16-1

List of Tables

Table 1-1	Conventions for Software Interface	1-4
Table 2-1	Measurements of Main Unit Manual Load Systems	2-2
Table 2-2	Measurements of Main Unit Systems with 13-Wafer Indexer	2-2
Table 2-3	Measurements of Main Unit Systems with 25-Wafer Indexer	2-3
Table 2-4	Measurements of Computers (All Systems)	2-3
Table 3-1	Safety Symbols	3-1
Table 3-2	Standard μ -PCD Laser Source Details	3-7
Table 3-3	μ -PCD AUX Laser Source Details	3-8
Table 3-4	SPV Laser Source Details	3-9
Table 3-5	VQ Bias Laser Source Details	3-9
Table 4-1	Summary of Back Panel Connections	4-5
Table 4-2	Summary of Flash Connections	4-11
Table 4-3	Moving Parts in the Main Unit	4-21
Table 8-1	Organization of File Menus	8-1
Table 9-1	Data Review Options	9-1
Table 10-1	Typical Head Heights	10-15
Table 11-1	Organization of Macro Topics	11-1
Table 12-1	Organization of System Operation Procedures	12-1
Table 12-2	Mapping Capability for V-Q Measurements	12-17
Table 14-1	Periodic Maintenance	14-1
Table 14-2	Guaranteed usage of flash bulbs	14-3
Table 15-1	Computer Requirements for Wintau32 Software	15-1

List of Figures

Figure 2-1: WT-2000 system with optional flash unit and 25-wafer indexer	2-4
Figure 2-2: WT-2000 system with optional hot chuck, optional flash unit, and optional 25-wafer indexer	2-5
Figure 2-3: WT-2000 system with manual load	2-6
Figure 2-4: WT-2000 system with manual load and optional hot chuck	2-7
Figure 3-1: Interlock override button	3-2
Figure 3-2: Interlock override button warning label	3-2
Figure 3-3: μ -PCD laser trigger LED	3-3
Figure 3-4: Laser warning label on inner door	3-3
Figure 3-5: Emergency Motor Stop switch	3-4
Figure 3-6: Warning label on the inside bottom and outside bottom cover of the WT-2000.	3-4
Figure 3-7: Warning label on the front panel	3-5
Figure 3-8: Safety labels on the measuring head	3-5
Figure 3-9: Location and beam divergences of lasers in WT-2000	3-6
Figure 3-10: Laser apertures	3-6
Figure 3-11: Laser source module in the main unit	3-7
Figure 3-12: Laser radiation warning label on the housing of the laser source module	3-7
Figure 3-13: Laser radiation warning label on the housing of the AUX laser source module	3-8
Figure 3-14: Laser warning label on doors	3-9
Figure 3-15: Do not leave anything under the indexer plate.	3-10
Figure 3-16: Location of warning label on cassette loader platform	3-10
Figure 3-17: Warning label on cassette loader platform	3-10
Figure 4-1: Unpacking the WT-2000 main unit	4-2
Figure 4-2: Back panels of the two computers (Measuring on top, User on bottom)	4-3
Figure 4-3: The back panel of the WT-2000 system	4-4
Figure 4-4: UTP Interconnect cable	4-6
Figure 4-5: μ -PCD cable	4-6
Figure 4-6: AUX cable	4-6
Figure 4-7: Motor A and Motor B cable	4-7
Figure 4-8: Mouse/Keyboard Y cable	4-7
Figure 4-9: Cables connected to KVM switch (KVM switch model may be different)	4-8
Figure 4-10: Digital I/O cable	4-8
Figure 4-11: UTP cable for LAN connection	4-8
Figure 4-12: Ground cable	4-9

Figure 4-13: Mains cable	4-9
Figure 4-14: Optional vacuum pump	4-10
Figure 4-15: Vacuum tubing with two different fittings	4-10
Figure 4-16: Stage interconnections	4-11
Figure 4-17: The flash lamp mounted with the reflector onto the top of the WT-2000	4-12
Figure 4-18: The flash power unit	4-12
Figure 4-19: Flash interface	4-13
Figure 4-20: Flash power cable	4-13
Figure 4-21: The flash trigger cable	4-14
Figure 4-22: The Digital I/O cable	4-14
Figure 4-23: Main unit with cover removed, showing pin corona charger (3)	4-15
Figure 4-24: Installation of upper corona wire	4-15
Figure 4-25: Upper corona wire after installation	4-16
Figure 4-26: Stage connections	4-16
Figure 4-27: Console stage	4-17
Figure 4-28: Indexer	4-17
Figure 4-29: Indexer mounting points on main unit	4-18
Figure 4-30: Indexer mounting screws	4-18
Figure 4-31: Mounting the indexer	4-18
Figure 4-32: Indexer cable	4-19
Figure 4-33: Connection of the indexer cable to the indexer	4-19
Figure 4-34: WT-2000 with the main unit open	4-20
Figure 4-35: Moving parts in the main unit	4-20
Figure 4-36: Do not leave anything under the indexer plate.	4-21
Figure 4-37: Power switches	4-22
Figure 4-38: The <i>Main Window</i>	4-23
Figure 4-39: The <i>Status Bar</i>	4-23
Figure 4-40: <i>Down</i> status	4-23
Figure 4-41: Message displayed when warm-up period is not complete	4-24
Figure 4-42: <i>Initialize</i>	4-24
Figure 4-43: The <i>Height</i> tab of the <i>Measurement Options</i> window	4-25
Figure 4-44: List of available measurement modes	4-26
Figure 4-45: <i>Transient Recorder</i> window	4-26
Figure 5-1: Laser excitation	5-1
Figure 5-2: Detection by microwave	5-2
Figure 5-3: Resonance curves of the microwave probe with a sample underneath. (MWDC – the DC signal in the function of microwave frequency) When the sample is in equilibrium, the curve is the red line. When the sample is injected, excess carriers are generated by the illumination, the conductivity and reflectance is changed, the curve is the blue line. The vertical green line shows the frequency used during the measurement, retuned so that the change of the signal is maximal.	5-2

Figure 5-4: Transient, measured on a sample. As the carriers return to equilibrium exponentially, the signal changes accordingly. The signal is descending, according to Figure 5-5, b) case.	5-3
Figure 5-5: The MWDC curves in equilibrium and after injection. If the frequency is set to position a), the signal grows during recombination to the original state, if the frequency is set to b) position, the signal descends.	5-4
Figure 5-6: Transient with increasing signal, according to Figure 5-5, a) case.	5-4
Figure 5-7: Block diagram of the sequence of processes during the μ -PCD measurement	5-5
Figure 5-8: Theoretical correlation curves of the measured lifetime versus bulk lifetime	5-6
Figure 5-9: Simplified diagram of the WT-2000 system	5-7
Figure 5-10: Structure of the integrated μ -PCD measuring head	5-7
Figure 5-11: Steady-state condition created by illumination	5-10
Figure 5-12: Photoconductance decay after short duration laser pulse	5-11
Figure 5-13: Illustration of QSS- μ PCD. Note that it contains steady-state condition in Figure 5-11 and laser induced transient condition (encircled) as in Figure 5-12.	5-12
Figure 5-14: Illustration of the QSS- μ PCD technique of laser pulse induced photoconductance decay, $\Delta\sigma(t)$ is measured using a reflected microwave signal.	5-13
Figure 5-15: Example of nonexponential u-PCD transient	5-14
Figure 5-16: Injection level dependence of QSS- μ PCD decay lifetime for passivated p^+ or n emitter.	5-15
Figure 5-17: Inverse of τ_{eff} vs. Δn illustrating the determination of J_0 .	5-15
Figure 5-18: The wafer map of J_0 measured with QSS- μ PCD	5-16
Figure 5-19: Error in using simplified analytical expression for overall surface contribution to τ_{eff} .	5-17
Figure 5-20: Band diagram showing recombination centers and minority carrier traps. Recombination centers capture both electrons and holes, annihilating them. The traps capture and subsequently emit only one type of carrier.	5-18
Figure 5-21: Setup for SPV measurement	5-20
Figure 5-22: Penetration depth of the laser in silicon as a function of the wavelength	5-21
Figure 5-23: SPV signal as a function of light intensity	5-21
Figure 5-24: Light intensity divided by the SPV signal, as a function of the light penetration depth	5-22
Figure 5-25: SPV amplitude maps measured with 670 nm and 980 nm wavelength lasers	5-23
Figure 5-26: Diffusion length and relative surface recombination velocity maps calculated from the previous maps	5-23
Figure 5-27: Fe-B dissociation and association over time	5-25
Figure 5-28: Kelvin probe measurement setup	5-28
Figure 5-29: Evaluation of V-Q curve	5-29
Figure 5-30: Total charge in the oxide	5-30
Figure 5-31: Schematic of JPV measurement method	5-33
Figure 6-1: Steps to perform a manual measurement	6-2
Figure 6-2: Steps for automatic measurement (for manual load and automatic wafer load)	6-5
Figure 6-3: Steps for automatic measurement with recipe (for manual load and automatic wafer load)	6-7
Figure 6-4: The <i>Main Window</i>	6-8
Figure 6-5: The <i>Measurement Options</i> window	6-9
Figure 7-1: The <i>Wintau32 Main Window</i>	7-2

Figure 7-2: The <i>Menu Bar</i>	7-2
Figure 7-3: The <i>Wintau32 Toolbar</i> and <i>Measurement Toolbar</i>	7-3
Figure 7-4: <i>Status Bar</i>	7-3
Figure 7-5: <i>Recorder Value Bar</i>	7-3
Figure 7-6: The menu bar	7-3
Figure 7-7: The <i>Toolbar</i>	7-4
Figure 7-8: The <i>Measurement Toolbar</i>	7-5
Figure 7-9: <i>User Administration</i> window	7-10
Figure 7-10: Creating a new user	7-10
Figure 7-11: The Wizard <i>Confirm</i> window	7-11
Figure 7-12: Users created by the Wizard	7-11
Figure 7-13: Right mouse click on <i>Input List</i> window	7-12
Figure 7-14: Changing the input field	7-13
Figure 8-1: The <i>Wintau32 Menu Bar</i>	8-1
Figure 8-2: The <i>File</i> menu	8-2
Figure 8-3: The <i>Edit</i> menu	8-3
Figure 8-4: The <i>View</i> menu	8-4
Figure 8-5: The <i>Measure</i> menu	8-6
Figure 8-6: The <i>Macro</i> menu	8-8
Figure 8-7: The <i>Process</i> menu	8-9
Figure 8-8: The <i>Arithmetic</i> menu	8-10
Figure 8-9: The <i>Print</i> menu	8-14
Figure 8-10: The <i>Access</i> menu	8-15
Figure 8-11: The <i>Window</i> menu	8-15
Figure 8-12: The <i>Help</i> menu	8-16
Figure 9-1: <i>Open File</i> dialog window	9-2
Figure 9-2: The menu that appears after right mouse click on file list	9-4
Figure 9-3: <i>Configure Columns</i> dialog window	9-5
Figure 9-4: <i>Main Window</i> with a map and the <i>Measurement Information</i> window	9-6
Figure 9-5: Using the Zoom feature	9-7
Figure 9-6: <i>General</i> tab of <i>View Options</i> window	9-8
Figure 9-7: <i>Color Scale</i> tab of <i>View Options</i> window	9-10
Figure 9-8: <i>Auto Scaling</i> options	9-11
Figure 9-9: <i>Floating</i> tab of <i>View Options</i> window	9-11
Figure 9-10: <i>Measurement Information</i> window (μ -PCD mode)	9-12
Figure 9-11: The menu that appears after right mouse click on <i>Measurement Information</i> window	9-13
Figure 9-12: <i>Measurement Settings</i> information window (μ -PCD)	9-14
Figure 9-13: Masks on the map	9-15
Figure 9-14: <i>Masks</i> tab on the <i>Measurement Information</i> window	9-15
Figure 9-15: Right mouse button pop-up menu on the map	9-16

Figure 9-16: Pop-up menu on the <i>Comment</i> window	9-17
Figure 9-17: <i>Histogram</i> tab in <i>Distribution</i> window	9-18
Figure 9-18: Menu displayed by right mouse click on the histogram area	9-19
Figure 9-19: Cumulative distribution	9-19
Figure 9-20: Line Scan	9-20
Figure 9-21: The <i>Linked Data Files</i> window	9-21
Figure 9-22: The <i>Link Loaded Map</i> window	9-21
Figure 9-23: Print Setup – Normal Size	9-22
Figure 9-24: Print Setup – Mini Size	9-23
Figure 10-1: <i>Measure</i> menu options	10-1
Figure 10-2: Drop-down list of measurement methods	10-2
Figure 10-3: <i>Wafer Size</i> options	10-3
Figure 10-4: <i>Wafer Finder</i> options	10-4
Figure 10-5: <i>Corona</i> menu options	10-5
Figure 10-6: <i>Linear Charge Front-side</i> settings	10-5
Figure 10-7: <i>Recipe</i> menu options	10-6
Figure 10-8: The <i>Hardware Info</i> window	10-7
Figure 10-9: <i>ATX00</i> tab	10-8
Figure 10-10: <i>Finder</i> tab	10-9
Figure 10-11: <i>Interlock</i> tab	10-9
Figure 10-12: <i>Interlock2</i> tab	10-11
Figure 10-13: <i>Indexer</i> tab	10-12
Figure 10-14: <i>Measure Remote</i> functions	10-12
Figure 10-15: The <i>Options</i> window has several tabs	10-13
Figure 10-16: The <i>Info</i> tab with the Input List active	10-14
Figure 10-17: The <i>Height</i> tab	10-15
Figure 10-18: <i>Loading</i> options	10-17
Figure 10-19: Settings for Optical Finder	10-18
Figure 10-20: <i>Single</i> tab options (shown in all modes)	10-19
Figure 10-21: <i>Prescan Points Setup</i> window	10-20
Figure 10-22: <i>Single</i> tab options for μ -PCD with Bias Light	10-21
Figure 10-23: <i>Auto</i> tab	10-22
Figure 10-24: <i>u-PCD</i> tab	10-24
Figure 10-25: QSS <i>u-PCD</i> options	10-25
Figure 10-26: <i>SPV</i> tab options	10-26
Figure 10-27: <i>Diffusion Length Correction</i> window	10-27
Figure 10-28: <i>Fe</i> tab options for μ -PCD and SPV	10-28
Figure 10-29: <i>Charge</i> tab options	10-29
Figure 10-30: <i>Eddy</i> tab options	10-30

Figure 10-31: V-Q tab options (V-Q measurement method shown)	10-31
Figure 10-32: <i>Automatic Settings</i>	10-32
Figure 10-33: Manual settings (typical values for p-type wafers are shown)	10-33
Figure 10-34: Options for Q_m measurement method	10-34
Figure 10-35: <i>Cassette</i> tab	10-36
Figure 10-36: <i>Check Cassette Offset</i> window	10-37
Figure 10-37: <i>Cassette Offset Setting</i> window	10-37
Figure 10-38: <i>Interface</i> tab	10-38
Figure 10-39: <i>Wave</i> tab	10-39
Figure 10-40: <i>Transient Recorder</i> window	10-40
Figure 10-41: Right-click in the area shown by the red box to display additional options	10-44
Figure 10-42: <i>Autosetting Options</i>	10-45
Figure 10-43: <i>Bias Light</i> options	10-46
Figure 10-44: Menu in the Transient recorder	10-46
Figure 10-45: <i>Microwave DC</i> curve	10-47
Figure 10-46: <i>Laser Control Monitor</i>	10-48
Figure 10-47: <i>Lifetime – Laser Power</i>	10-48
Figure 10-48: <i>Lifetime – Time</i>	10-49
Figure 10-49: <i>Amplitude – Head Height Setup</i> window	10-49
Figure 10-50: <i>Transient Evaluation</i> window	10-50
Figure 10-51: Right-click menu in <i>Transient Evaluation</i> window	10-51
Figure 10-52: Qss u-PCD recorder	10-52
Figure 10-53: Info window of a Qss map	10-53
Figure 10-54: Qss-uPCD evaluation	10-53
Figure 10-55: Functions of Qss evaluation	10-54
Figure 10-56: The reciprocal of the lifetime in the function of the injection level. J_0 is calculated from the fitted line.	10-54
Figure 10-57: <i>SPV Recorder</i>	10-55
Figure 10-58: Pop-up menu for <i>Bias Light</i> options	10-57
Figure 10-59: Right-click menu in <i>SPV Recorder</i>	10-57
Figure 10-60: The <i>Functions</i> menu in the <i>SPV Recorder</i>	10-58
Figure 10-61: <i>Amplitude – Flux</i>	10-58
Figure 10-62: <i>Diffusion Length – Amplitude</i>	10-59
Figure 10-63: <i>Regression – Amplitude</i>	10-59
Figure 10-64: <i>Error Function – Amplitude</i>	10-60
Figure 10-65: Calibration Curves	10-60
Figure 10-66: <i>V-Q Recorder</i> window	10-61
Figure 10-67: Functions in the <i>V-Q Recorder</i> window	10-62
Figure 10-68: <i>Corona Test</i> window	10-62
Figure 10-69: <i>V-Q Evaluation</i> window (V-Q measurement shown)	10-63

Figure 10-70: <i>JPV Recorder</i> window	10-64
Figure 10-71: Functions in the <i>JPV Recorder</i> window	10-65
Figure 10-72: <i>SHR Setup</i> window for <i>pn-type noise limit</i>	10-66
Figure 11-1: The <i>Macro</i> menu	11-2
Figure 11-2: The <i>Macro Editor</i> window	11-3
Figure 11-3: Toolbar in <i>Macro Editor</i> window	11-3
Figure 11-4: The <i>Macro Syntax</i> window	11-4
Figure 11-5: The <i>Macro Watch</i> window	11-5
Figure 11-6: Right-click menu in <i>Macro Watch</i> window	11-5
Figure 11-7: <i>Reserved Words</i> tab of <i>Macro Syntax</i> window	11-7
Figure 11-8: Built-in Wintau32 macro language identifiers in <i>Objects</i> tab	11-8
Figure 11-9: Built in objects in <i>Objects</i> tab	11-9
Figure 11-10: Created object shown in <i>Object</i> tab of <i>Macro Syntax</i> window	11-10
Figure 11-11: Created object shown when Hide Inherited is not selected	11-11
Figure 11-12: Example of inverted color scale	11-15
Figure 11-13: Example of change in Eddy averaging mode	11-16
Figure 11-14: Functions in a <i>Recorder</i> window (μ -PCD shown)	11-17
Figure 11-15: Example of previously saved curve	11-18
Figure 11-16: Example of change in raster	11-20
Figure 12-1: The <i>Height</i> tab of the <i>Measurement Options</i> window	12-2
Figure 12-2: Selecting the measurement method	12-3
Figure 12-3: Sub Measurement Mode drop-down list in the <i>Buttons area</i> (μ -PCD method shown)	12-3
Figure 12-4: Selecting the sample size	12-4
Figure 12-5: <i>Load Sample</i> window for manual loading	12-4
Figure 12-6: Selecting the wafer size	12-5
Figure 12-7: Mounting the cassette	12-6
Figure 12-8: Select the wafer for automatic loading.	12-7
Figure 12-9: Typical appearance of the <i>Main Window</i> after the Wafer Finder procedure (μ -PCD mode shown)	12-8
Figure 12-10: <i>Transient Recorder</i> window	12-10
Figure 12-11: <i>SPV Recorder</i> window	12-11
Figure 12-12: <i>Input List</i> tab for μ -PCD measurement	12-12
Figure 12-13: Selecting the number of points to measure	12-15
Figure 12-14: A 9-point measurement pattern	12-15
Figure 12-15: After x-point measurement, the measured values are displayed for each point.	12-16
Figure 12-16: Measurement timer	12-17
Figure 12-17: Map statistics in the <i>Input List</i>	12-18
Figure 12-18: Linear charge control window	12-19
Figure 12-19: The <i>Loading</i> tab of the <i>Options</i> window	12-21
Figure 12-20: The <i>Select Samples</i> window	12-23

Figure 12-21: Saving a recipe	12-24
Figure 12-22: Reloading previously saved recipes	12-24
Figure 12-23: Drop-down menu next to Auto Measure button	12-26
Figure 12-24: Drop-down menu next to Auto Measure button for systems with automatic wafer loading capability	12-27
Figure 12-25: <i>Select Samples</i> window	12-28
Figure 12-26: <i>Auto</i> tab	12-29
Figure 12-27: Drop-down menu next to Measure Recipe(s) button	12-30
Figure 12-28: Recipe selection window with two recipes selected using <Ctrl> and left mouse button	12-31
Figure 12-29: Drop-down menu next to Measure Recipe(s) button for systems with automatic wafer loading capability	12-31
Figure 12-30: Recipe selection window with two recipes selected using <Ctrl> and left mouse button	12-32
Figure 12-31: Selecting samples and setting file names for a recipe	12-33
Figure 13-1: Typical calibration curve for Bias Light Intensity vs. Bias Power	13-1
Figure 13-2: Corona calibration menu item	13-2
Figure 13-3: Successful calibration	13-2
Figure 13-4: Calibration curve of the corona charger	13-3
Figure 13-5: The <i>Calibrate</i> option in the <i>V-Q Recorder</i> window	13-3
Figure 13-6: Right click on the intensity slider bar to check on calibrate intensity	13-4
Figure 13-7: Right click on the intensity slider bar to set Average Setup	13-5
Figure 13-8: JPV Averaging Setup table. Point measurement averaging in the first column.	13-5
Figure 13-9: Calibrating the JPV to EQE	13-7
Figure 14-1: Parts of the WT-2000 that need lubrication.	14-3
Figure 14-2: Replacing the bias halogen bulb	14-5
Figure 14-3: Measuring head of WT-2000 with a 4pp. Remove front cover by the screws at the arrows.	14-6
Figure 14-4: Remove the plug of the 4 point probe (A) and unscrew the two screws of the probe holder (B)	14-6
Figure 15-1: UTP connection notification	15-2
Figure 15-2: <i>Network Connections</i> window	15-2
Figure 15-3: <i>WT Properties</i> window	15-3
Figure 15-4: <i>Internet Protocol Properties</i> window	15-3
Figure 15-5: <i>Interface</i> tab on <i>Measurement Options</i> window	15-4
Figure 15-6: Registration window	15-5

Information furnished by Semilab is believed to be accurate and reliable. However, no responsibility is assumed by Semilab for its use, nor for any infringements of patents or other rights of third parties which may result from its use. No license is granted by implication or otherwise under any patent or patent rights of Semilab. The information contained herein is subject to change without notice.

The system is designed to be safe in case of normal operation. However, Semilab bears no responsibility for consequences of improper use, or any action not described here in the manual or by a Semilab personnel. Do not use the equipment or any part of it for purposes they are not designed for, e.g. do not stand on the top of them, do not lean against them, do not eat or drink on them, do not place coffee cups on them etc.

1. Introduction

Welcome to the community of WT-2000 owners.

The WT-2000 system is a platform that can be equipped with a variety of measuring options. This allows characterization of silicon wafers in diverse ways. It is a useful tool for incoming wafer inspection, quality control and process monitoring in the wafer and IC manufacturing processes.

The WT-2000 can be equipped with the following measurement options:

- μ -PCD and C-PCD for determination of minority carrier lifetime
- SPV for measuring diffusion length
- Iron (Fe) concentration mapping
- V-Q for oxide and interface characterization
- JPV for determination of sheet resistance

Not all measurement options can be included in one WT-2000 system. Please consult Semilab to learn which measurement options are possible for your system.

1.1 Measurement Capabilities

1.1.1 μ -PCD

The microwave induced photoconductive decay method is the most common way of measuring minority carrier lifetime in semiconductors. This method excels due to its reliability, good reproducibility and the short measurement time that permits of making lifetime maps with high resolution.

Bulk lifetime can be measured by passivating the surface of the semiconductor. One of the most common passivation methods is the patented chemical passivation of Semilab.

✓ **NOTE: Use of this method is only legal if a chemical passivation license is purchased from Semilab.**

Together with the passivation license we also give a passivation kit and a manual that describes the chemical passivation method in details.

1.1.2 Charge-PCD

Charge-PCD (C-PCD) is an improvement of the μ -PCD technique that allows measuring bulk lifetime even on non-passivated, bare wafers. A continuously deposited charge passivates the surface, even if there is no good quality thermal oxide on it. However, the sample must have at least a thin native oxide layer.

1.1.3 SPV

The Surface Photovoltage (SPV) technique is an effective method for monitoring and characterization of heavy metal contaminations and crystal growth originated defects in bulk silicon wafers. The system provides fast, non-contact, non-destructive whole wafer mapping of bare and oxidized crystalline silicon.

SPV measurements can be performed on polished wafers with resistivity greater than 0.01 Ω -cm. The sample must be characteristic p or n type. If the surface of the sample is oxidized, the sample must be in depletion.

1.1.4 Fe Concentration Mapping

Fe (iron) concentration measurement can be performed in both μ -PCD and SPV measurement modes. Both methods provide rapid, non-destructive measurements yielding high-resolution maps. In the case of p-type silicon doped by boron (B), iron contamination can be separated from other types of contamination and other sources that reduce the lifetime and diffusion length.

1.1.5 Oxide Characterization with V-Q

The Voltage-Charge (V-Q) option offers a non-contact, preparation free, high-speed oxide characterization measurement technique. The system provides fast non-contact, non-destructive whole wafer mapping (or contour map) of samples with oxide on top.

All necessary data for determining the main oxide physical parameters is acquired. The system includes the Surface potential measuring method, Plasma damage monitoring (PDM), mobile charge measurement, effective charge, tunnel field and V-Q evaluation.

1.1.6 JPV Measurements

JPV (Junction Photovoltage) is a useful characterization tool for semiconductor manufacturing. This sheet resistance measurement option as an alternative method to the four-point probe gives the ability to measure near surface layer sheet resistance maps (homogeneity characterization) on semiconductor material having p-n or n-p structure (diffused or implanted layers).

1.1.7 Other Capabilities

Semilab manufactures a variety of products with specific configurations for different applications. The WT-2000 also has the following measurement capabilities, which are not typically used in the semiconductor industry:

- LBIC for diffusion length and internal quantum efficiency evaluation on solar cells
- Eddy Current for measurement of resistivity
- P/N type determination

This manual may occasionally reference these capabilities, which may not be present in your system.

1.2 User Manual

Semilab provides the following manuals with the WT-2000:

User Manual

User Manual covers the operation of the WT-2000. The information is grouped into several sections:

Chapters **1** through **4** provide information about the system configuration and specifications, as well as safety precautions. Installation instructions are also provide.

Chapter **5** describes theory of operation for various options available in the system.

Chapter **6** is a Quick Start Guide, giving a general overview of the measurement procedure, and Chapter **7** gives an introduction to the Wintau32 software interface.

Detailed information about the software interface is provided in Chapters **8** through **11**, and detailed measurement instructions are provided in Chapter **12**.

Chapters **13** to **14** include calibration and maintenance information.

The Appendix (Chapter **15**) provides procedures for software upgrades as well as supplemental information about specialized measurement techniques possible with the WT-2000.

Diagnostics and Troubleshooting Manual

Diagnostics and Troubleshooting Manual is for defining the source of possible failures of the WT-2000 and solving problems.

Conventions

This manual includes three kinds of statements that require special attention by the user, which are enclosed within a border for emphasis.

✓ NOTE: <i>Note statements highlight or clarify information.</i>

! CAUTION: *Caution statements identify conditions or practices that could result in damage to the system or to wafers being processed.*

⚠ WARNING: *Warning statements identify conditions or practices that could endanger personnel, resulting in injury or loss of life.*

This manual uses several conventions in the instructions for the software interface, listed in Table 1-1.

Table 1-1
Conventions for Software Interface

Operator Interface	Convention
Software menu listings	<i>bold italic text</i>
Buttons and areas of the window or screen that are selected by clicking the mouse	bold text
Names of windows and screens	<i>italic text</i>
Names of areas within windows and screens	<i>italic text</i>
Keyboard entries	<text in angle brackets>

1.3 Contacting Semilab

Only personnel trained and certified in maintenance of the WT-2000 are allowed to service the tool. If you need help with the WT-2000, please contact Semilab for assistance. Semilab has world-wide service and support centers. Contact one of the locations listed below:

Semilab Headquarters

Office: Semilab Semiconductor Physics Laboratory Co. Ltd.
Address: Prielle Kornélia u. 2. H-1117 Budapest, Hungary
Tel.: +36 1 505 4690
Fax: +36 1 505 4695
General information: semilab@semilab.com
Sales inquiries: sales.support@semilab.com
Technical inquiries: technical.support@semilab.com

Germany: +49 3375 902632 | thomas.jaehrling@semilab.com
France: +33 675 067040 | nicolas.laurent@semilab.com

China

Office: Semilab China
Address: 3rd Floor, B2 Building, BETWIN, No. 889, Shangcheng Road, Pudong New Area, Shanghai, China
Tel: +86 21 58362889
Fax: +86 21 58362887
Email: info.china@semilab.com
Web: www.semilab.com.cn

Japan

Office: Semilab Japan K.K.
Address: 6th Floor, YS Building, 2-15-10, Shin-Yokohama,
Kohoku-ku, Yokohama, Kanagawa 222-0033 JAPAN
Tel: +81 45 620 7960
Fax: +81 45 476 2146
Email: info.japan@semilab.com
Web: www.semilab-j.jp

Korea

Office: Semilab Korea Co. Ltd.
Address: #996-3 Youngtong-dong, Youngtong-Gu, Suwon-Si,
Kyunggi-do, S. Korea 443-851, Daewoo Worldmark Officetel 102-508.
Tel: +82 31 273 2010
Fax: -
Email: info.korea@semilab.com

Singapore

Office: Semilab (S.E.A.) Pte Ltd.
Address: 45 Cantonment Road, Singapore 089748, Singapore
Tel: +65 82000117
Fax: -
Email: info.singapore@semilab.com

USA

Office: Semilab USA LLC.
Address: 47 Manning Road, Billerica, MA, USA 01821
Tel: +1 508 647 8445
Fax: +1 508 647 8419
Email: info.usa@semilab.com

If your problem appears to be the result of a fault with the equipment, please take notes, as accurately as possible, of the conditions leading to the problem. Consult Diagnostics and Troubleshooting Manual for tests you can perform that appear relevant to your problem. Document test results and error codes using the **Print Screen** function as described in the **Note** below.

✓ **NOTE:** Use the **Print Screen** function to capture the Wintau32 screens showing test results and error codes. When the appropriate screen appears, press the <Print Screen> button on your keyboard. Press <Ctrl-Esc> to display the Windows Start Menu. Select **Programs | Accessories | Paint** from the Start Menu to start the Paint program. Select **Edit | Paste** to paste the bitmap of the screen into the Paint program. Select **File | Print** to print the image or **File | Save** to save it to a file.

2. *Equipment Overview*

The WT-2000 system is a computer-controlled system which is suitable for recording, storing and displaying several kinds of maps (depending on the measurement modes; see section 1.1) of silicon wafers of different sizes. Standard wafers can be loaded manually as well as automatically from a cassette.

2.1 System Configuration

Each system contains the same basic units:

- WT-2000 main unit with scanning capability
- Sample stage (maximum wafer size: 200 mm or 300 mm)
- Two computers, including a Measuring computer (System 1, DOS) and a User computer (System2, Windows)
- Peripherals including keyboard, mouse, LCD monitor, and color printer
- Vacuum pump

Several measurement heads are available. The measurement capabilities of your system are defined by the measurement heads installed:

- μ -PCD head for lifetime and Fe concentration measurements
- SPV head for diffusion length and Fe concentration measurements
- V-Q measurement (Kelvin probe) for PDM, Q_{eff} , surface potential, V-Q, E_{tunnel} , and Q_m measurements
- JPV measurement head providing Junction Photovoltage measurements of sheet resistance

Additionally, the system may contain any of the following optional units:

- Console stage for use with the Charge-PCD option
- Indexer for automatic wafer handling from a wafer cassette
- Flash unit for whole wafer, providing high intensity illumination (for Fe measurement)
- Bias Light unit for high intensity continuous illumination (used for μ -PCD or SPV)

2.2 System Software

The WT-2000 main unit is controlled by the WSOS2 software, which runs on the DOS (Measuring) computer. The Wintau32 software with the user-friendly interface runs on the Windows computer (User).

2.3 System Dimensions

The tables below list the dimensions and weight of the WT-2000 main unit (without the computers). Table 2-4 lists the dimensions of the computer (PC, monitor, and keyboard).

Table 2-1
Measurements of Main Unit
Manual Load Systems

	WT-2000	With Optional Hot Chuck
Height (with open door)	960 mm	960 mm
Height with optional light tower	1000 mm	1000 mm
Width	650 mm	1150 mm
Depth	730 mm	750 mm
Weight	50 kg	90 kg
Weight with optional flash unit	51 kg	90 kg

Table 2-2
Measurements of Main Unit
Systems with 13-Wafer Indexer

	WT-2000	With Optional Hot Chuck
Height (with open door)	960 mm	960 mm
Height with optional light tower	1000 mm	1000 mm
Width	980 mm	1480 mm
Depth	730 mm	750 mm
Weight	55 kg	90 kg
Weight with optional flash unit	56 kg	90 kg

Table 2-3
Measurements of Main Unit
Systems with 25-Wafer Indexer

	WT-2000	With Optional Hot Chuck
Height (with open door)	960 mm	960 mm
Height with optional light tower	1000 mm	1000 mm
Indexer distance below system level	140 mm	140 mm
Width	980 mm	1480 mm
Depth	730 mm	750 mm
Weight	55 kg	90 kg
Weight with optional flash unit	56 kg	90 kg

Table 2-4
Measurements of Computers (All Systems)

Typical Dimensions	
Height	650 mm
Width	480 mm
Depth	700 mm

The following figures show overall views of the whole system indicating its dimensions, for use in determining the footprint and room needed for proper operation.

Servicing of the WT-2000 is performed from all sides, depending on the nature of the service. In planning the facilities, please allow 750 mm clear space on each side of the WT-2000 system.

Figure 2-1 shows the dimensions of a WT-2000 system with optional flash unit and optional 25-wafer indexer. Note that the WT-2000 with optional 13-wafer indexer has the same dimensions except that it does not require the indicated 140 mm distance below the base of the system.

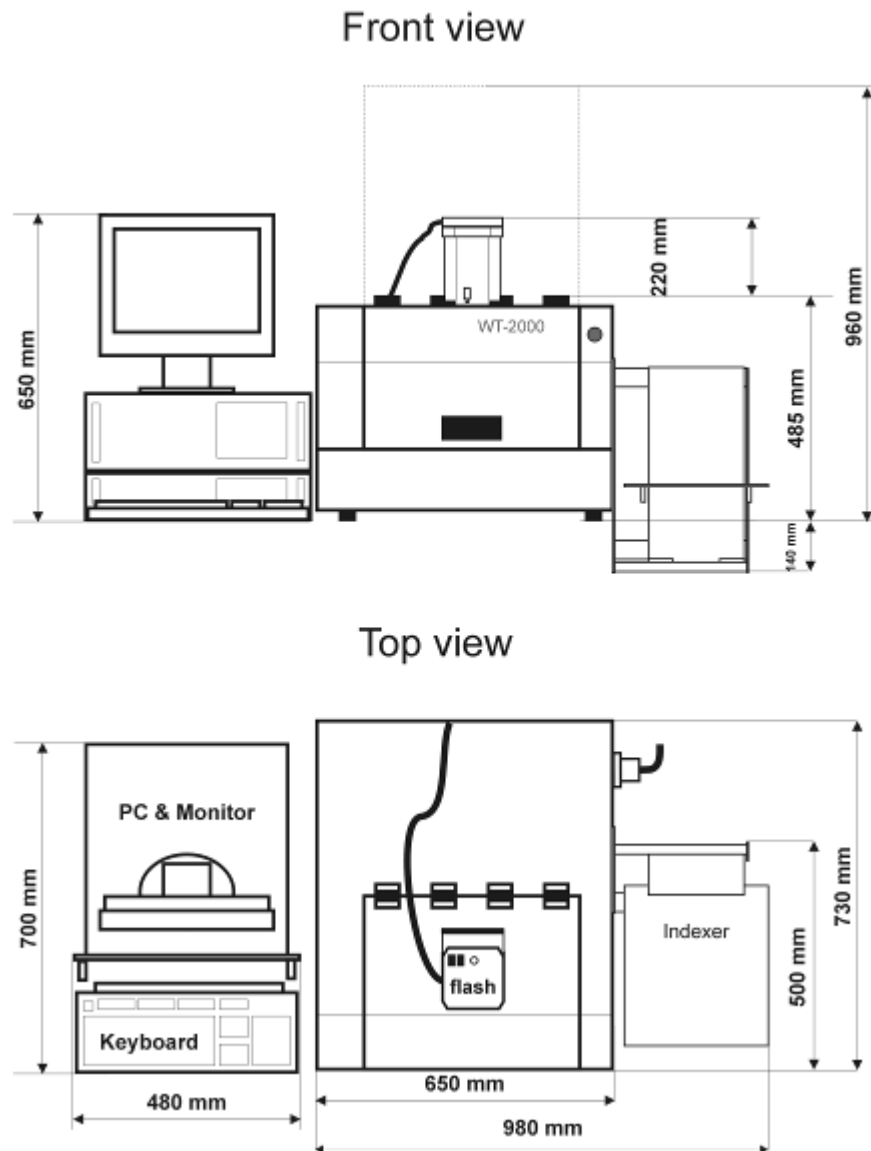


Figure 2-1: WT-2000 system with optional flash unit and 25-wafer indexer

Figure 2-2 shows the dimensions of a “long” WT-2000 system with optional hot chuck, optional flash unit, and optional 25-wafer indexer. The WT-2000 with optional 13-wafer indexer has the same dimensions except that it does not require the indicated 140 mm distance below the base of the system.

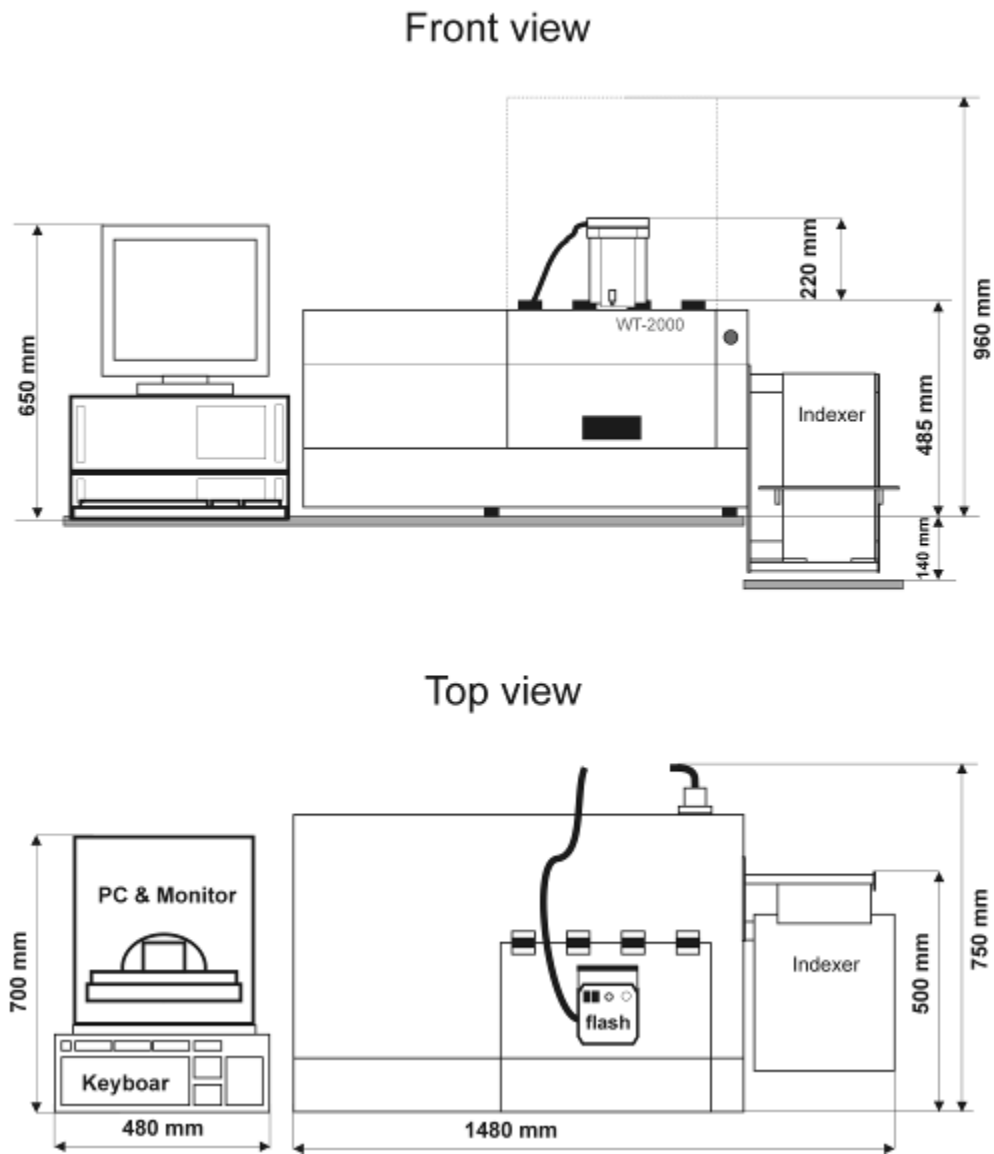


Figure 2-2: WT-2000 system with optional hot chuck, optional flash unit, and optional 25-wafer indexer

Figure 2-3 shows the dimensions of a manual-load WT-2000 system with optional flash unit. Figure 2-4 shows dimensions of a “long” manual load WT-2000 system with optional hot chuck, as well as the optional flash unit.

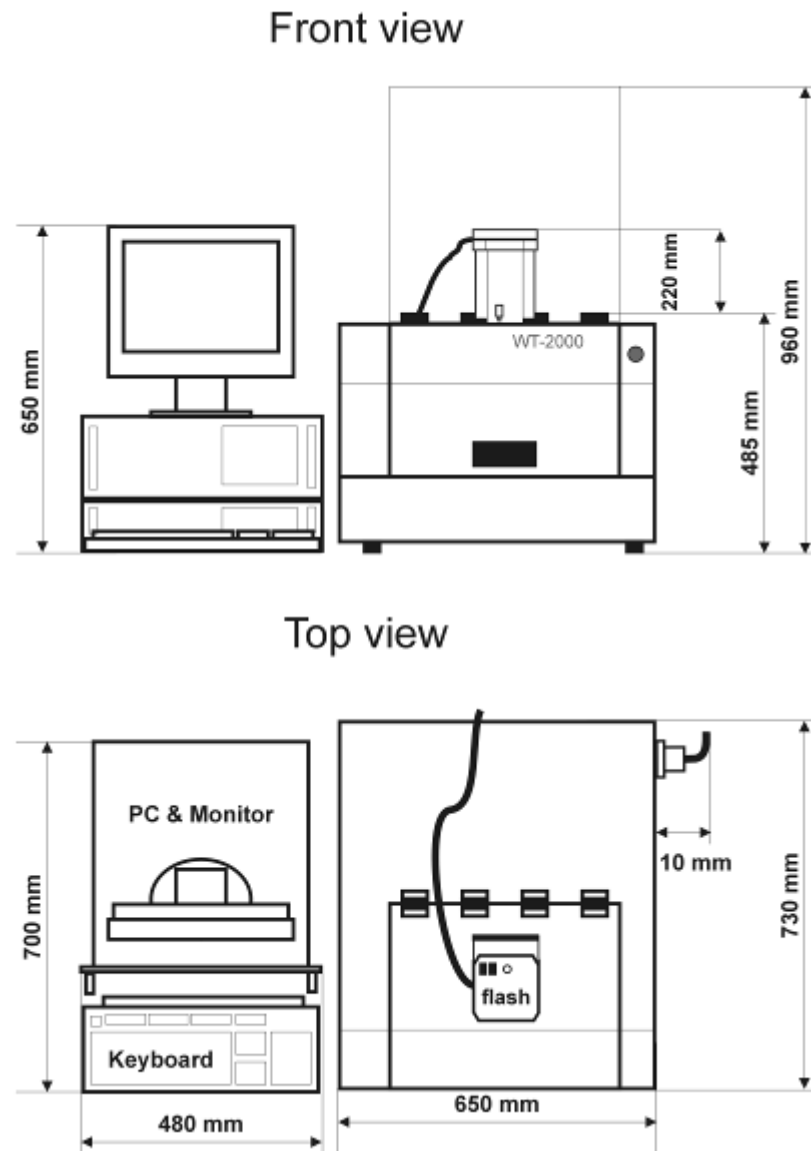


Figure 2-3: WT-2000 system with manual load

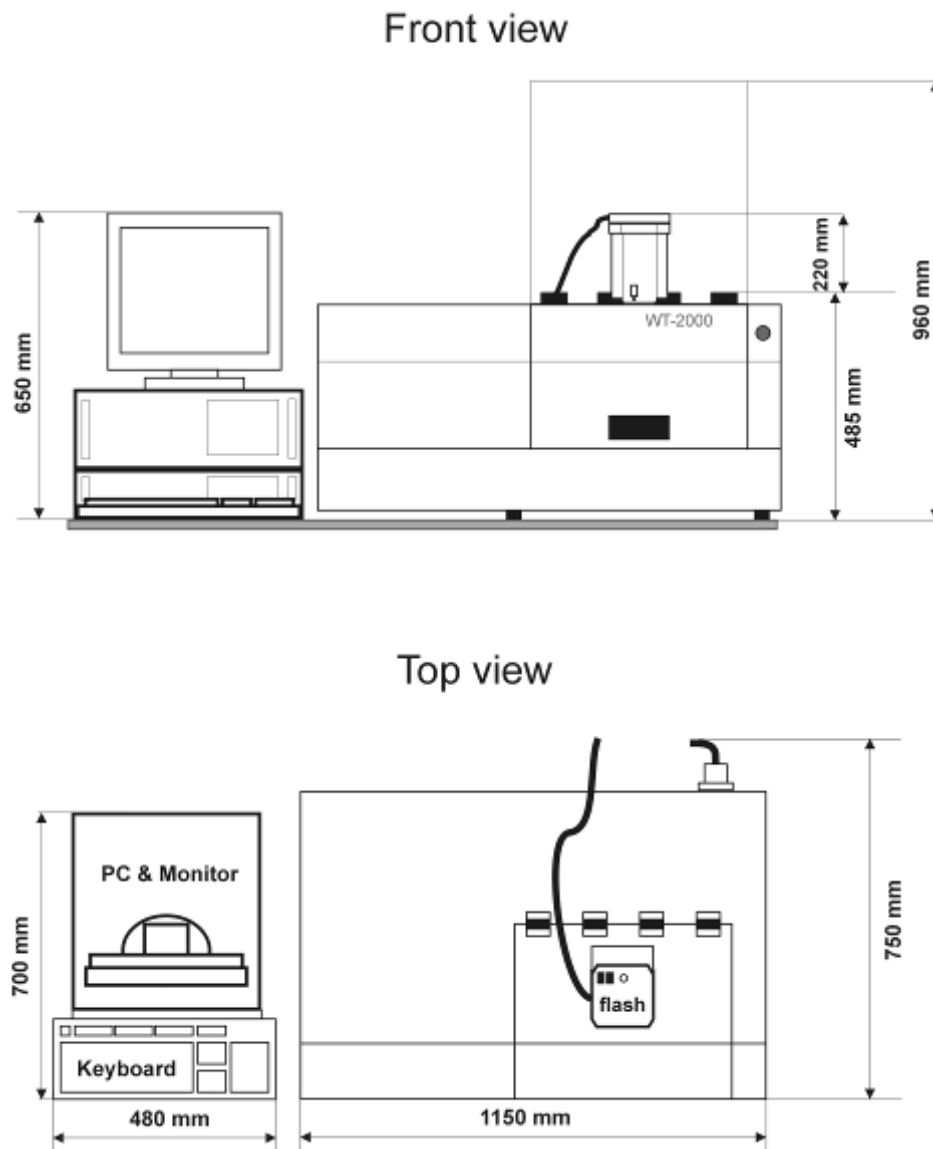


Figure 2-4: WT-2000 system with manual load and optional hot chuck

2.4 Facilities Requirements

2.4.1 Environmental Requirements

The WT-2000 has to be kept and operated in a building, properly protected from rain, snow, wind, and the natural environment.

Ambient temperature	Between 15°C and 35°C nominal, varying <1°C per hour
Relative humidity	Between 20 % and 60 %
Clean room	Class 10,000 or better
Vibration sensitivity	VC-A

2.4.2 Power Specifications

Specifications in the table below concern the main unit only.

Input voltage	115 Vac or 230 Vac
Line type	Single phase, 2 wire + ground
Line frequency	50/60 Hz
Line connection	1.8 m long, 3-wire
Power consumption	Maximum 600 VA
Power line connection	1.8 m long, 3-wire 3 x 0.75 mm ² power cable
Class of hazardous electrical shock	Class I, protective conductor with green/yellow color
EMC classification	Group 1*, Class A** according to EN55011
Overvoltage group	II, according to IEC 664
Number of connections:	Systems without flash unit: 4 • (2 PC, 1 monitor and 1 vacuum pump) Systems with optional flash unit: 5 • (2 PC, 1 monitor, 1 vacuum pump and 1 flash)

* *Group 1 ISM equipment*: group 1 contains all ISM equipment in which there is intentionally generated and/or used conductively coupled radio-frequency energy which is necessary for the internal functioning of the equipment itself.

** Class A equipment is equipment suitable for use in all establishments other than domestic and those directly connected to a low voltage power supply network which supplies buildings used for domestic purposes.

The machine behavior in case of power loss is designed to cause no harm on samples and no damage to the equipment. Unloading of the loaded wafer is possible after recovering. All unsaved data will be lost, and additional data loss is possible because of unexpected Windows shutdown.

2.4.3 Vacuum Requirements

If your system does not use the vacuum pump recommended by Semilab, the WT-2000 system has the following vacuum requirements.

Vacuum air pressure	From 0.2 bar (minimum) to 0.5 bar (maximum)
Vacuum line tube	6 mm or 1/4" outer diameter (Swagelok)
Air flow power	6 norm. l/min.

3. Safety Precautions

The WT-2000 is designed to be completely safe when operated and maintained by properly trained personnel using procedures described by Semilab. Semilab assumes no responsibility or liability for the operating procedures or safety procedures adopted by the user. The user is responsible for operating the WT-2000 system in a safe manner.

In this chapter, Semilab provides safety information regarding the WT-2000 system design and its intended use. Read this chapter thoroughly before operating or servicing the WT-2000 system.

Table 3-1 Organization of Safety Precautions

Section name	Section
Safety Symbols	3.1
Safety Interlocks	3.2
Emergency Motor Stop	3.3
Mechanical Safety	3.4
Laser Safety	3.5

The measuring head of WT-2000 system may contain μ -PCD, LBIC, SPV, JPV, and V-Q (Kelvin) voltage-measuring units depending on the actual configuration.

3.1 Safety Symbols

The following symbols in Table 3-2 appear within this manual and on safety labels at various locations on the WT-2000 system. These symbols are used to alert the user to potential safety hazards.

**Table 3-2
Safety Symbols**

	Ground (earth) terminal
	Electric shock hazard

	Laser radiation hazard
	Pinch Point caution.

3.2 Safety Interlocks

The access panel (door) is equipped with an interlock. The door may be opened during normal operation e.g. sample loading or unloading or even for maintenance purposes.

If the door is opened during operation of the Class 1M (μ -PCD) laser product the software gives the notice “*Laser interlocked! Close the cover!*” then disables the triggering of the Class 1M laser diode(s) and suspends the measurement. For maintenance or service purposes, this interlock can be overridden by the red button shown in Figure 3-1 below.

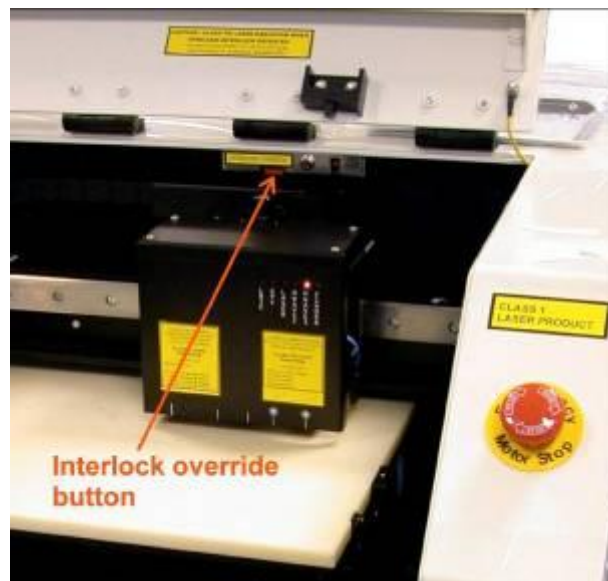


Figure 3-1: Interlock override button

The interlock override remains active until the next time the cover is closed. When the interlock is overridden, a red LED flashes continuously next to the override button, warning the user that laser triggering is enabled. The interlock override button warning label can be seen in Figure 3-2.



Figure 3-2: Interlock override button warning label

When the door is open and the interlock is overridden, the laser triggering can be checked by looking at the indicator on the front of the head unit (Figure 3-3). The μ -PCD laser emits Class 1M radiation directed down perpendicular to the sample or stage only when the **LASER** LED lights.

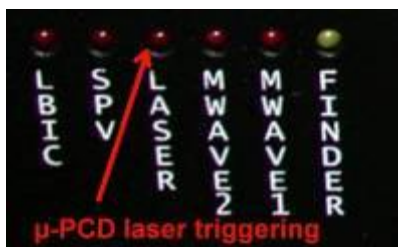


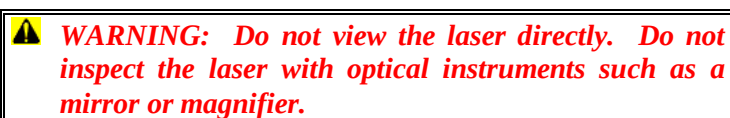
Figure 3-3: μ -PCD laser trigger LED

There is a warning label (Figure 3-4) notifying the user about possible Class 1M invisible laser radiation placed on the inner side of the door.



Figure 3-4: Laser warning label on inner door

During maintenance or inspection of the μ -PCD Class 1M laser product, do not use a mirror to inspect it especially from a distance closer than 10 cm, as it may be hazardous. Also do not use any other optical tool, such as a magnifier.

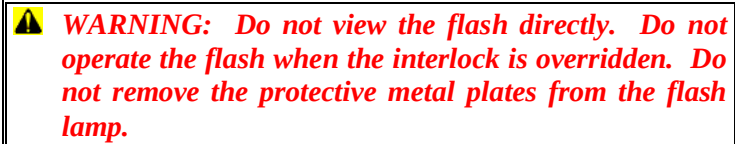


3.2.1 Flash Interlock

WT-2000 systems with optional Fe contamination evaluation have a flash lamp which produces very high intensity light. If this light flashed in an operator's eyes, it could cause momentary loss of vision. For this reason, safety measures prevent the operator from being exposed to the light.

The flash lamp is interlocked to prevent it from operating when the door is open. The interlock is driven by the same hardware as the microwave and the laser interlock, so if the microwave and laser interlock is overridden, the flash interlock is also overridden.

The flash lamp is mounted on the top of the door. When the door is open, the reflector of the flash is directed at the operator. To protect the operator from seeing the flash, a set of metal plates hides the flash lamp when the door is open. These plates move away from the lamp when the door is closed, to allow the flash to reach the sample being measured.



3.3 Emergency Motor Stop

There is an Emergency Motor Stop switch on the front side of the main unit (Figure 3-5).



Figure 3-5: Emergency Motor Stop switch

Pressing the red knob turns off the power to all the motors of the system. The Emergency Motor Stop switch can be deactivated by turning the knob in a clockwise direction and releasing it.

3.4 Mechanical Safety

The moving parts can be stopped by hand force. If there is something in the way of a moving part that blocks the movement, the part stops, and the software gives an error message. After such an event, initialization is needed.

Because of this, no serious injury is possible to be caused by the moving parts of the WT-2000. However, a warning label is placed on the inside bottom and outside front cover of the WT-2000 to prevent people from putting their limbs in the way, as seen on Figure 3-6



Figure 3-6: Warning label on the inside bottom and outside bottom cover of the WT-2000.

3.5 Laser Safety

3.5.1 Lasers in the WT-2000

Systems with μ -PCD, SPV, VQ or LBIC units contain lasers that can emit visible and invisible laser radiation that may be harmful to human eyes. In these cases a warning label is mounted on the front panel (Figure 3-7) and measuring head (Figure 3-8) of the WT-2000 system.



Figure 3-7: Warning label on the front panel

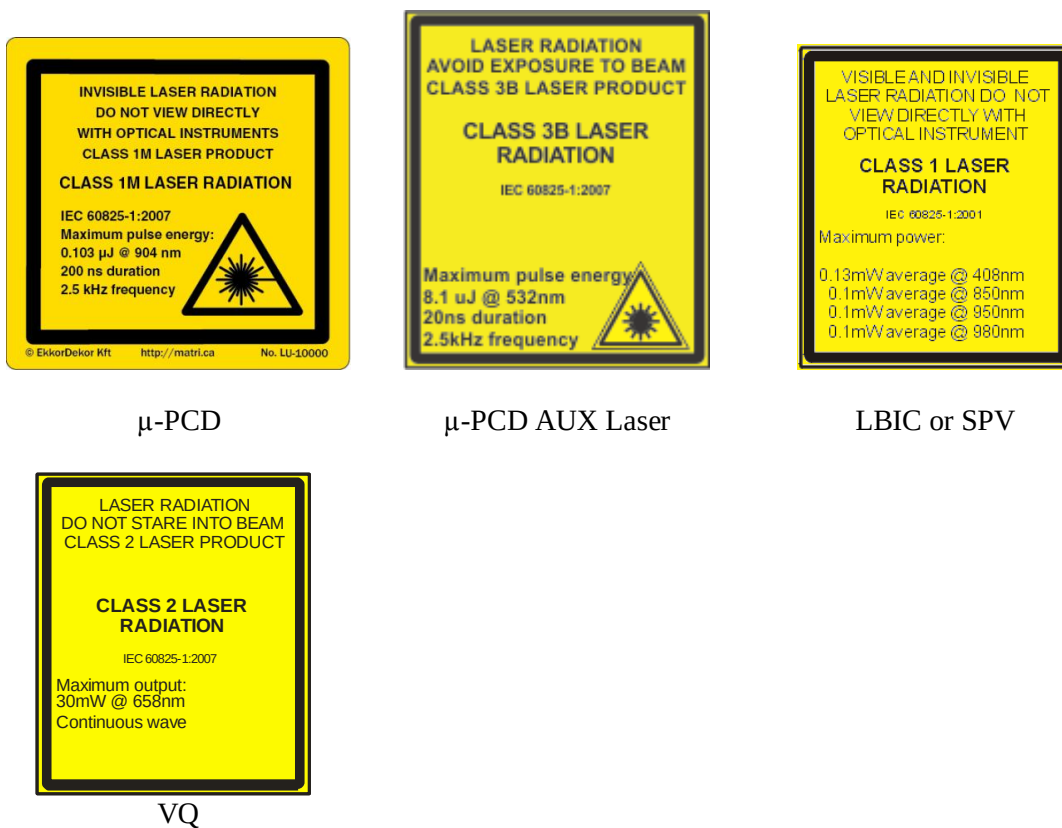


Figure 3-8: Safety labels on the measuring head

The location of the μ -PCD and LBIC heads and the beam divergences are shown in Figure 3-9. The location of the SPV laser sources is almost the same as the LBIC lasers, but the intensity is much lower, and the beam divergence is much higher than in case of LBIC, so the SPV lasers are safer than the LBIC lasers.

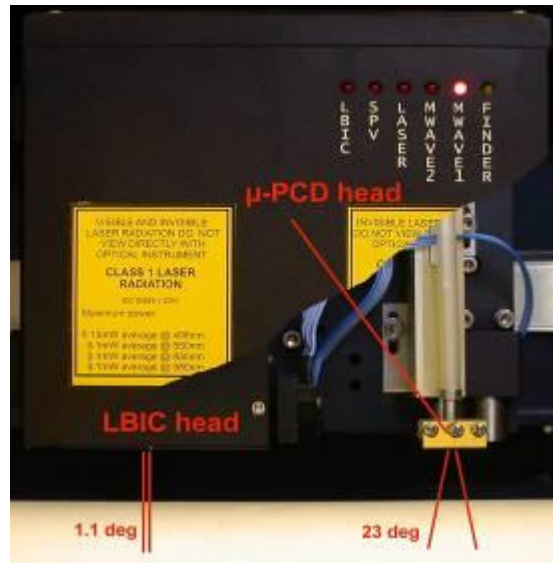


Figure 3-9: Location and beam divergences of lasers in WT-2000

The apertures are shown in Figure 3-10.

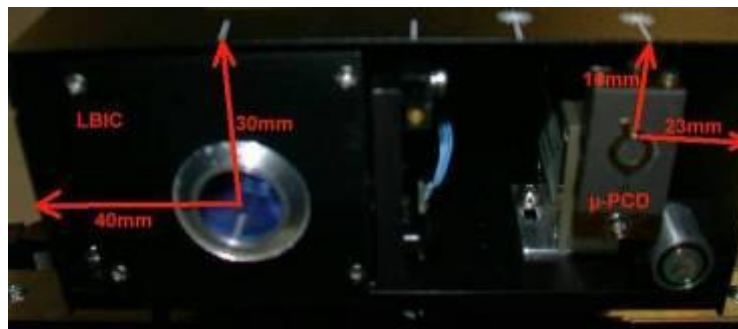


Figure 3-10: Laser apertures

⚠ WARNING: Do not view the laser directly. Do not inspect the laser with optical instruments such as a mirror or magnifier.

3.5.2 μ-PCD Unit

The laser source of the μ-PCD unit is not built into the measuring head but in the rear compartment of the main unit. The light from the embedded laser source is directed to the measuring head via a fiber optic cable. In all the cases the invisible laser radiation is emitted down perpendicular to the wafer placed below the measuring head. During normal operation there is no risk of exposure to laser radiation.

The laser source of the standard infrared μ-PCD measuring unit is **Class 1M** according to standard EN 60825-1:2007. Laser source details are shown in Table 3-3.

Table 3-3
Standard μ -PCD Laser Source Details

Type	EG&G PGAU4S12 InGaAs laser diode
Wavelength	905 $\pm$ 10 nm
Pulse width	200 ns
Maximum repetition frequency	2.5 kHz
Laser beam divergence	0.4 radian
Maximum output	0.103 μ J per pulse

The embedded μ -PCD laser source is built into the rear compartment of the WT-2000 main unit as shown in Figure 3-11 below. The laser source has its own protective housing with a warning label as shown in Figure 3-12.

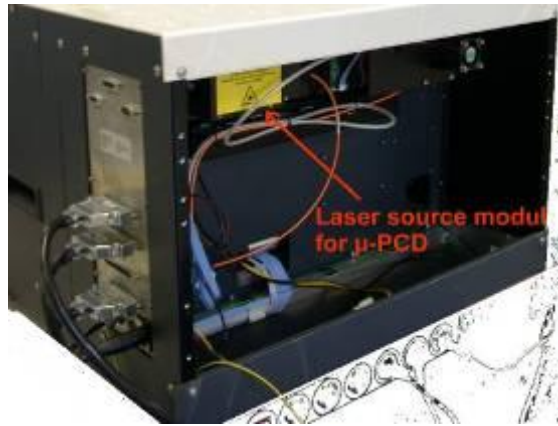


Figure 3-11: Laser source module in the main unit

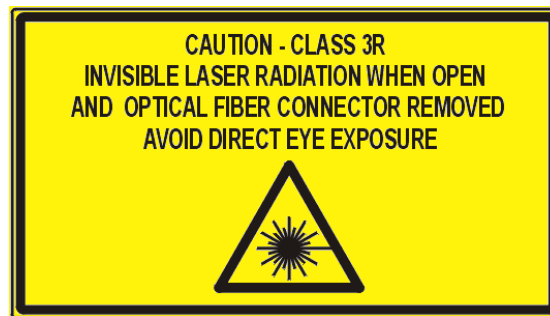


Figure 3-12: Laser radiation warning label on the housing of the laser source module

3.5.3 Aux laser

System with Aux laser option contains a Class 4 laser source. The wavelength is 532 nm, the power of the laser source is 500 mW during normal operation. Laser source details are shown in Table 3-4.

The embedded AUX laser source is built into the laser box of the WT-2000 main unit and it has its own protective housing with a warning label as shown in Figure 3-13. The light from the

embedded laser source is directed to the measuring head via a fiber optic cable. The laser source has its own protective housing with a warning label. The laser is interlocked when the door is open, and the interlock cannot be overridden.

⚠ WARNING: Class 4 laser is very harmful. Do not remove the revetment of the laser source!

⚠ WARNING: Removal of panels that allow exposure to the Class 4 laser when the WT-2000 is switched on is strictly prohibited.



Figure 3-13: Laser radiation warning label on the housing of the AUX laser source module

**Table 3-4
μ-PCD AUX Laser Source Details**

Type	Explorer 532 - 2Y OEM version
Wavelength	532 nm
Pulse width	20 ns
Maximum repetition frequency	2.5 kHz
Laser beam divergence	0.4 radian
Maximum output	8.1 μJ per pulse

3.5.4 SPV Laser

There are four or eight c.w. laser sources built in the SPV measuring head. Their light might be both visible and invisible for human eyes.

The SPV head laser is **Class 1**. Laser source details are listed in Table 3-5. Available wavelengths are: 1013nm, 980nm, 950nm, 850nm, optional: 900nm, 880nm, 780nm, 660nm. Standard wavelengths for a 4 laser setup are 850, 950, 980, and 1013nm. The actual wavelengths may vary by ± 10 nm.

Table 3-5
SPV Laser Source Details

Pulse repetition frequency	412 Hz
Duration of a single pulse	1.25 ms (50% duty factor)
Maximum output	0.1 mW average, 0.2 mW during pulse
Laser beam divergence	2-3 rad

3.5.5 VQ Unit

There is one bias laser source built in the VQ head. The VQ bias unit is a **Class 2 laser product** according to standard EN 60825-1:2007. Laser source details are shown in Table 3-6.

Table 3-6
VQ Bias Laser Source Details

Type	HL6501MG AlGaInP laser diode
Wavelength	658 nm (645 nm - 665 nm)
Usage	Continuous
Laser beam divergence	0.2 radian
Maximum output	30 mW
Beam diameter	5 mm

If the lid of the WT-2000 is open and the interlock is overridden the WT-2000 becomes a Class 2M product as indicated by a warning label on the lid of the WT-2000 (Figure 3-14).



Figure 3-14: Laser warning label on doors

3.6 Automatic Cassette Loader Safety

The optional cassette loader platform moves up and down during system initialization as well as during the automatic loading procedure.

! CAUTION: Do not leave anything under the indexer plate.



Figure 3-15: Do not leave anything under the indexer plate.

⚠ WARNING: Do not place your hand between the cassette loader platform and the main part of the WT-2000 system during operation.

The loading arm removes wafers from the cassette. A warning label is placed on the cassette loader platform (Figure 3-16 and Figure 3-17) to warn against removing the cassette while the arm is moving.

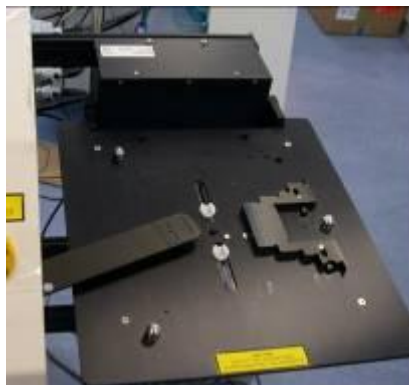


Figure 3-16: Location of warning label on cassette loader platform

CAUTION
WATCH THE LOADING ARM
WHEN REMOVING THE CASSETTE

Figure 3-17: Warning label on cassette loader platform

4. Installation

Table 4-1 Organization of Installation chapter

Section name	Section
Installation	4.1
Cable Installation	4.2
Stage Connections	4.3
Flash lamp interconnections	4.4
Hiba! A hivatkozási forrás nem található.	Hiba! A hivatkozási forrás nem található.

4.1 Unpacking the System

Before installing the system, unpack and inspect each component as described in the following sections.

4.1.1 Unpacking the WT-2000 System Main Unit

Unpack and remove the WT-2000, the industrial computer and other units (depending on system specification) from the crates. Check the crate contents according to the packing list. Inspect the system if there is any damage described below. Be sure that the exterior of the instrument was not damaged. Any deformation on the exterior of the instrument can influence the operation of it. Please report the damage to the shipper immediately.

Open the lid of the main unit and remove red plastic fastening pieces in the following order, referencing the numbered images in Figure 4-1:

- [1.] First, remove the plastic arm fastener (numbered 1 on image).
- [2.] Next, remove the ones restricting stage movements (numbered 2 on image).
- [3.] The next step is to make the head move freely by removing its fastener (numbered 3 on image).
- [4.] When done, move the head to the opposite end position and remove the fastener of the bridge (numbered 4 on image).
- [5.] Finally remove the plastic fasteners of the stage (numbered 5 on image).

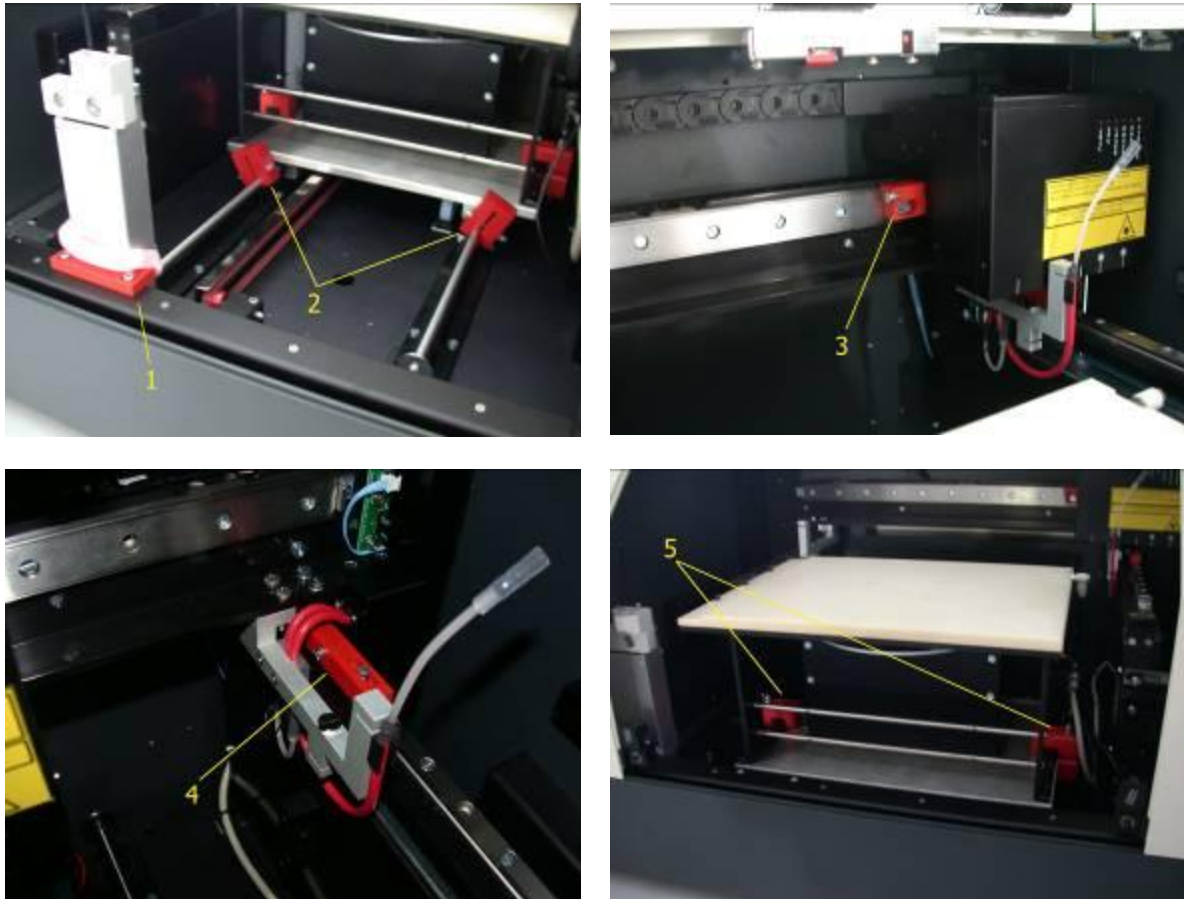


Figure 4-1: Unpacking the WT-2000 main unit

Check the movement of the measuring head, the bridge, and the stage.

! CAUTION: Do not touch the bottom of the measuring heads when checking the movement.

The measuring head, the bridge, and the stage should move without any restraint over their full movement range.

DC motors ensure the movement of the measuring head, bridge and stage under normal operation, so they can slightly restrict the movement of the measuring head and the stage while trying to move them. In spite of this, the above mentioned movement check should be performed.

Slide the stage to the rear end position. Be sure there is no fracture on the stage.

! CAUTION: If any part of the instrument got loose during shipment, it can cause an inaccuracy in the result of the measurement. Be sure to tighten any loose part before use.

4.1.2 Unpacking Other System Components

Unpack all other components for the system. Refer to each component's manual for specific unpacking instructions. Place all computer peripheral devices (computers, monitor, mouse, keyboard and printer) in their desired locations.

Connection information for system components is provided in section 4.2.2.

4.2 Cable Installation

Both end connectors of the cables used in WT-2000 system have identical connectors, and they may be connected in either direction. The only exception is the KVM switch and its related components. Cable installation directions in this section refer to specific sockets on the back panels of the two computers and on the back panel of the WT-2000 system. The numbered sockets on each back panel are shown in Figure 4-2 and Figure 4-3. In Figure 4-2, the upper computer is Measuring (DOS, System 1), and the lower computer is User (Windows, System 2).

✓ **NOTE:** *The computers shipped with the WT-2000 may be different from the computers shown in Figure 4-2. Use Figure 4-2 only as a reference for the connector types on the back panels.*



Figure 4-2: Back panels of the two computers (Measuring on top, User on bottom)



Figure 4-3: The back panel of the WT-2000 system

Table 4-2 provides a short description of the cables and their connections to the back panels. Additional detail is provided in the following sections.

Table 4-2
Summary of Back Panel Connections

First Connection	Second Connection	Cable to Use
1	LAN	UTP for LAN
2	2	UTP interconnect
3	KVM switch	KVM's Monitor Out
4	KVM switch	Y cable
5	12	Motor A
6	13	Motor B
7	15	AUX
8	14	μ-PCD
9	Mains power	Mains cable
10	19	Ground cable
11	22	Power cable
20	Vacuum pump	Air cable
21	Vacuum pump	Air cable (only for systems with C-PCD)
16	Indexer	Indexer cable
17	External device (eg. flash)	I/O cable

4.2.1 Line Voltage Selection

Check the Measuring and User computers to see if a red voltage selector switch is present. Typically, the voltage selector switch is a red switch near the mains power connection on the back panel. If the switch is present, set the switch position to correspond to the available power line voltage (115 V ac or 230 V ac).

! CAUTION: Do not connect the system to mains power until you have checked the position of the line voltage selector switch.

4.2.2 System Power and Interconnections

⚠ WARNING: All electrical connections of the system should be performed with the power switched off and the line cable unplugged!

Both end-connectors of the cables used in WT-2000 system have identical profile and it is allowed to plug them in any direction. The only exception is the KVM switch and its related components. The following list gives a short description of the cables and their interconnection. For socket numbering, please refer to Figure 4-2 and Figure 4-3.

UTP Interconnect Cable

The UTP (Unshielded Twisted Pair) Interconnect cable is used to establish the TCP/IP interconnection between the User (Windows, System 2) and Measuring (DOS, System 1) computers. The two end-connectors should be plugged into the sockets numbered 2.



Figure 4-4: UTP Interconnect cable

μ -PCD Cable

This cable makes the connection between the WT-2000 lifetime measuring system and the transient recorder in the Measuring PC. One end of this cable should be plugged into the socket numbered 8 on the Measuring computer and the other into the socket numbered 14 on the WT-2000 main unit.



Figure 4-5: μ -PCD cable

AUX Cable

This cable is for power and interlock connection between WT-2000 main unit and the Measuring computer. One end of this should be connected to socket number 7 on the Measuring computer, while the other end is connected to number 15 on the main unit.



Figure 4-6: AUX cable

Motor A and B Cables

Motor cables connect the motor driver card of the measuring computer to the individual movement units (stage, bridge, head etc.). The Motor A and Motor B cables are identical and

can be changed freely. One end of Motor A cable should be plugged to connector number.5 on the Measuring PC and the other to connector number 12 on the main unit. The Motor B cable should interconnect connector number.6 on the PC and number 13 on the main unit.



Figure 4-7: Motor A and Motor B cable

Indexer Cable (Optional)

This cable is used with the automatic wafer loading option. This cable makes the connection between the back panel of the WT-2000 and the automatic wafer loader. Connect one end to connector number 16 on the back panel of the main unit. The other end will be connected to the indexer, after the indexer is installed.

Mouse/Keyboard Y Cables

Two Mouse/Keyboard Y cables (Figure 4-8) are provided to connect both mouse and keyboard to the single PS/2 connector of both the Measuring PC and the User PC. A Y cable should be connected to the keyboard (purple) and mouse (green) connector of both output branches of the KVM switch (Figure 4-9). One of the Y cables should be then connected to PS/2 connector of the User computer, and the other to the Measuring computer.



Figure 4-8: Mouse/Keyboard Y cable

KVM Switch and Related Cables

Connect the monitor cables to the connectors numbered 3 on the computers and to the KVM switch as shown in Figure 4-9. (Actual KVM model may be different from that shown in the figure.) Connect the peripherals (keyboard, mouse and monitor) to the appropriate connections of the KVM switch. For further information, please refer to the Mouse/keyboard Y cable description above and the manual for the KVM switch.



Figure 4-9: Cables connected to KVM switch (KVM switch model may be different)

Digital I/O cable (optional)

The optional digital I/O cable connects the main unit and external devices (for example, to the flash lamp interface for systems with the flash option). Connect the digital I/O cable to connector number 17 on the WT-2000 back panel and then to the external device.



Figure 4-10: Digital I/O cable

UTP Cable for LAN Connection

Another UTP cable is also delivered with the system for connection to a Local Area Network. It should be plugged to connector number 1 on the User computer, and the other end plugged into the LAN wall connector or hub.



Figure 4-11: UTP cable for LAN connection

Ground Cable

This cable connects the housing (ground) of the WT-2000 main unit and that of the dual computer. Make the connection between connectors number 11 and number 19 using this cable.



Figure 4-12: Ground cable

Mains Cable

Two mains power cables are delivered for the dual computer. Connect them to the sockets numbered 9.



Figure 4-13: Mains cable

In addition to these cables, all peripheral devices (monitor, printer etc.) have their own mains cables. Connect all these as directed in the user manuals for the devices.

Vacuum tubes

If house vacuum line is used, connect the line to connector number 20 (see Figure 4-3) of the main unit. If using the optional vacuum pump, connect the vacuum tubing between connector number 20 (VACUUM) and the inlet of the vacuum pump (number 2 in Figure 4-14). The supplied vacuum tubing has two different fittings, shown in Figure 4-15. Fitting number 1 must be connected to the pump, and fitting number 2 must be connected to the main unit.



Figure 4-14: Optional vacuum pump



Figure 4-15: Vacuum tubing with two different fittings

If air flow is needed (e.g. for C-PCD measurement), use a second vacuum pump. Connect vacuum tubing between the exhaust outlet on the pump (number 1 in Figure 4-14) and connector number 21 (AIR) on the main unit (see Figure 4-3).

4.3 Stage Connections

When finished making all interconnections described in section 4.2.2, make the connections to the sample stage.

There are three connectors on the right side of the stage see Figure 4-16:

- Vacuum connection, number 2 on Figure 4-16
- Stage ground connection, number 1 on Figure 4-16
- Wafer lift control connection (only with automatic wafer loading option), number 3 on Figure 4-16

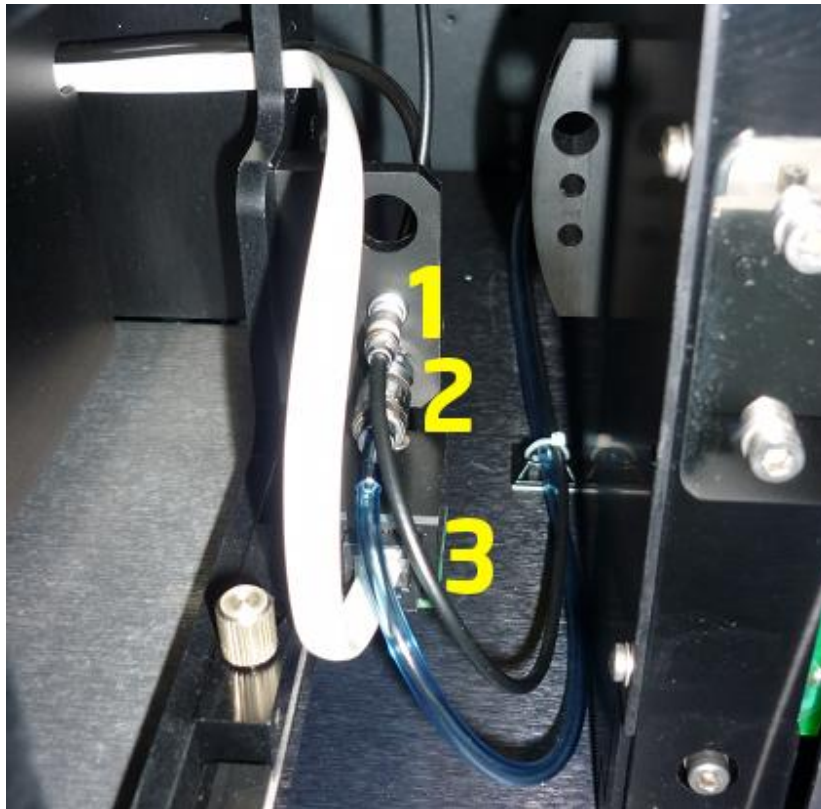


Figure 4-16: Stage interconnections

4.4 Flash lamp interconnections

If Fe measurement option is purchased with the system, a flash lamp unit is also installed with the system. The flash lamp is supported by a power unit and an interface unit, and sometimes a transformer. The interconnections between these parts and the system are described below.

The following table summarizes the parts and connections of the flash system.

**Table 4-3
Summary of Flash Connections**

First Connection	Second Connection	Cable to Use
Flash lamp 1	Flash power supply 1	Flash power cable
Flash power supply 2	Flash interface 5	PC power cable
Flash power supply 3	Flash interface 2	Flash trigger cable
Flash interface 1	Back panel 17	Digital I/O cable
Flash interface 3	Mains or transformer	Mains cable
Flash interface 4	DOS PC 10	Grounding cable

4.4.1 Flash lamp

The flash lamp is mounted on the top of the WT-2000 for iron contamination measurement. Its only socket delivers the power to the lamp by the flash power cable.



Figure 4-17: The flash lamp mounted with the reflector onto the top of the WT-2000

4.4.2 Flash power unit

The flash power unit supplies the flash lamp with the large amount of charge that is needed for flashing. It has the following sockets:

- 1 – power out: to the flash lamp
- 2 – power in: from the flash interface
- 3 – trigger in: from the flash interface



Figure 4-18: The flash power unit

4.4.3 Flash lamp interface

The flash lamp interface ensures the communication between the WT-2000 and the flash lamp, gives the trigger to the flash power unit, and controls the power that charges the power unit. It has the following sockets:

- 1 – Digital I/O: to the back panel, socket 17
- 2 – trigger out: to the flash power unit
- 3 – power in: from the mains or the transformer (110 V or 230 V)
- 4 – grounding: to the DOS PC, socket 10
- 5 – power out: to the flash power unit

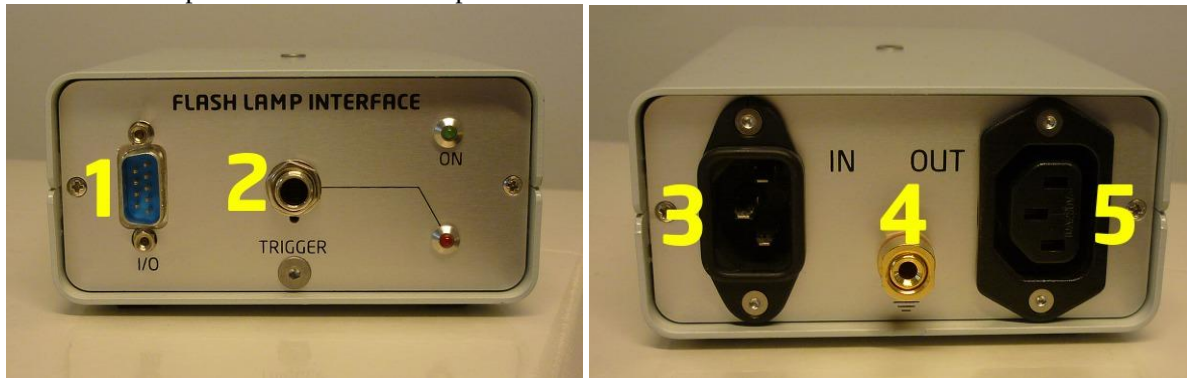


Figure 4-19: Flash interface

- ✓ **NOTE:** If a 230 V to 110 V transformer is shipped with the system, always use it before the power in (3) socket of the flash interface. In these cases, the flash power unit can accept only 110 V. 230 V causes failures.

✓ If no transformer is shipped with the system, the interface can be connected to the 230 V mains without problem.

4.4.4 Flash power cable

The flash power cable delivers the current pulse from the flash power unit to the flash lamp when fleshing. It has to be plugged into socket 1 on the lamp, and socket 1 on the power unit.



Figure 4-20: Flash power cable

4.4.5 Flash trigger cable

The flash trigger cable delivers the trigger signal from the flash interface, socket 2, to the flash power unit, socket 3.



Figure 4-21: The flash trigger cable

4.4.6 Digital I/O cable

The digital I/O cable ensures communication between the WT-2000 and the flash interface. It has to be plugged into the back panel, socket 17, and the flash interface, socket 1.



Figure 4-22: The Digital I/O cable

4.5 Installation of Charge PCD Option

The WT-2000 with the Charge-PCD option uses a pin corona charger, an upper corona wire, and a lower corona wire. A special console stage permits charging the bottom of the sample when the lower corona wire is used.

The pin corona charger is factory installed (shown as number 3 in Figure 4-23). Note that the system in Figure 4-23 also contains an SPV head (number 1) as well as the μ -PCD head (number 2).



Figure 4-23: Main unit with cover removed, showing pin corona charger (3)

The corona wires must be installed by the user. Note that the lower corona wire can only be used with the console stage (see section 4.5.1).

To install the corona wire, plug the corona unit to the appropriate slot and connect the red wire. Installation of the upper corona wire is shown in Figure 4-24 and Figure 4-25. The lower corona wire is installed in a similar manner.



Figure 4-24: Installation of upper corona wire

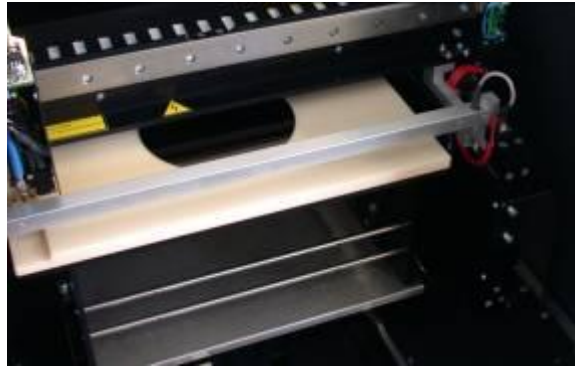


Figure 4-25: Upper corona wire after installation

4.5.1 Installation of the Console Stage

A standard stage is factory installed in the WT-2000. Measurements using the C-PCD option require the console stage. To change the standard stage to the console stage, follow the steps below.

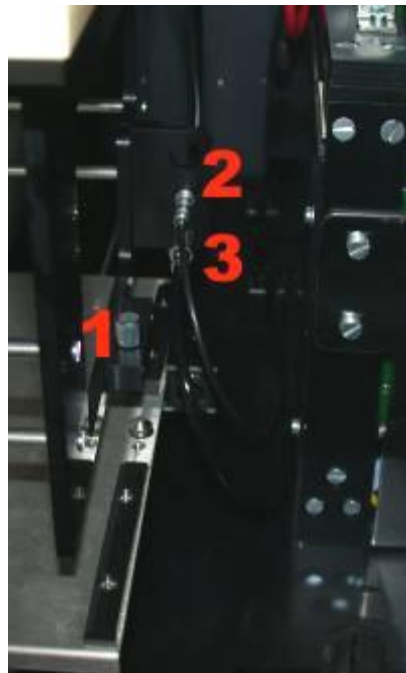


Figure 4-26: Stage connections

- [1.] Pull out the ground wire (item 2 in Figure 4-26)
- [2.] Unplug the vacuum tube (item 3 in Figure 4-26).
- [3.] Release the mounting screws (item 1) by hand. There are two screws, one on each side of the stage.
- [4.] If the stage is for automatic wafer loading, there is an additional connector plugged in for the wafer lift control. Detach this connector before removing the stage.
- [5.] Remove the stage.

- [6.] Place console stage in the main unit.
- [7.] Connect the ground cable (item 2 in Figure 4-27).
- [8.] Attach the console stage using the mounting screws (item 1 in Figure 4-27). There are two screws, one on each side of the stage.
- [9.] Several wafer holders are available for different wafer diameters. The wafer holder can be changed to a different diameter by removing the screws (item 3) and replacing the wafer holder.

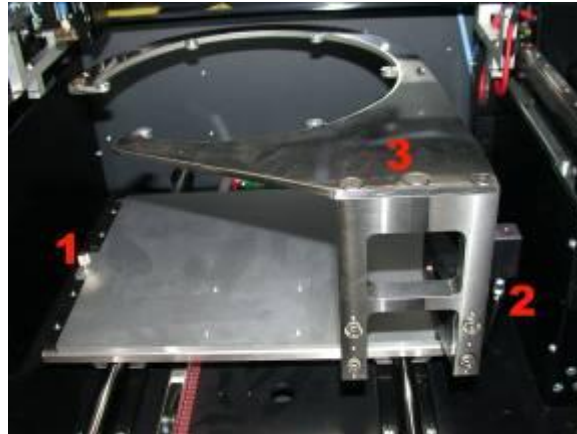


Figure 4-27: Console stage

Use the **Measure | Wafer Size** menu item (section 10.3) to indicate that a console stage is installed.

If the wafer holder must be changed, use the **Measure | Change Console** menu item.

4.6 Installation of Indexer

The optional indexer unit is shown in Figure 4-28.



Figure 4-28: Indexer

The indexer is attached to the main unit at three mounting points on the main unit, shown as item 1 in Figure 4-29.



Figure 4-29: Indexer mounting points on main unit

Using the mounting screws provided (Figure 4-30), attach the indexer to the main unit as shown in Figure 4-31. Tighten the screws using a 5mm hexagonal key.



Figure 4-30: Indexer mounting screws



Figure 4-31: Mounting the indexer

Connect the indexer cable (Figure 4-32) to the indexer as shown in Figure 4-33.



Figure 4-32: Indexer cable



Figure 4-33: Connection of the indexer cable to the indexer

After installing the indexer, check the parallelism of the stage, indexer and arm with a spirit-level tool and make fine adjustments if needed.

4.7 Start-up and Initialization

After the hardware connections are made properly, the WT-2000 measuring system is ready for use. For the first use, follow the sequence below:

- [1.] Open the lid of the main unit (Figure 4-34).

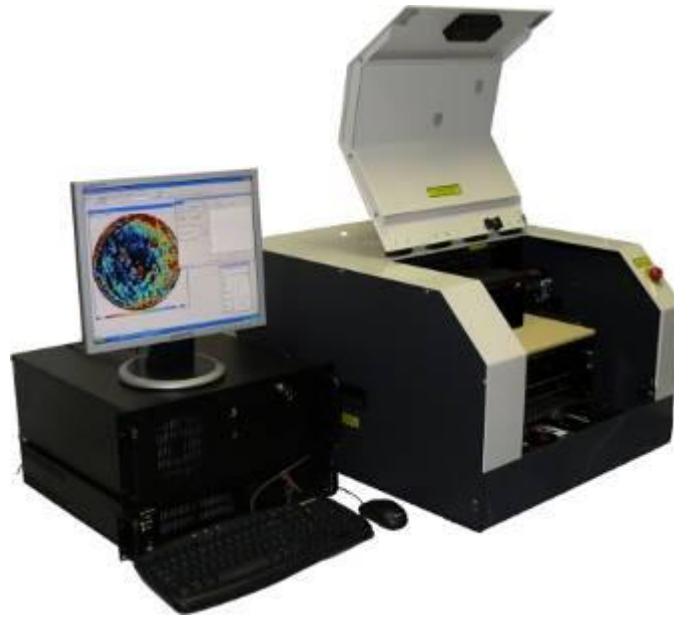


Figure 4-34: WT-2000 with the main unit open

- [2.] Verify that no red plastic fastening pieces are left in the main unit (see section 4.1.1).
- [3.] Verify that the measuring head, the stage and the bridge move freely as shown in Figure 4-35 and Table 4-4.

! CAUTION: Do not touch the bottom of the measuring heads when checking the movement.

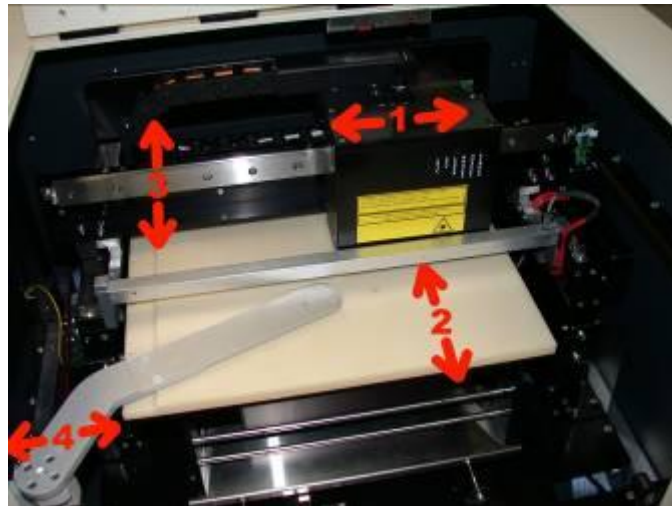


Figure 4-35: Moving parts in the main unit

Table 4-4
Moving Parts in the Main Unit

Image Number	Name	Movement Direction
1	Head	Left to right
2	Stage	Front to back
3	Bridge	Front to back
4	Arm	Left to right

- [4.] Verify that the indexer is not blocked in its movement path. (For example, do not leave a sample holder under the indexer plate.)



Figure 4-36: Do not leave anything under the indexer plate.

- [5.] Turn on the power for the User computer and then for the Measuring computer, by pressing the power switch on each computer (Figure 4-37).



Figure 4-37: Power switches

- [6.] Switch to the Windows-operation system, if necessary, and double-click the Wintau32 icon .

✓ **NOTE:** *Switching between the User (Windows) and the Measuring (DOS) computer can be done by double-pressing the <Ctrl> key on the keyboard, or by double-pressing <Scroll Lock> and pressing the up/down arrow. For details, please refer to the owner's manual of the shipped KVM switch.*

✓ **NOTE:** *The full path of Wintau32 software (by default) is: C:\Wintau\Wintau32.exe*

- [7.] After the Wintau32 software is loaded, the *Main Window* is displayed (see Figure 4-38).

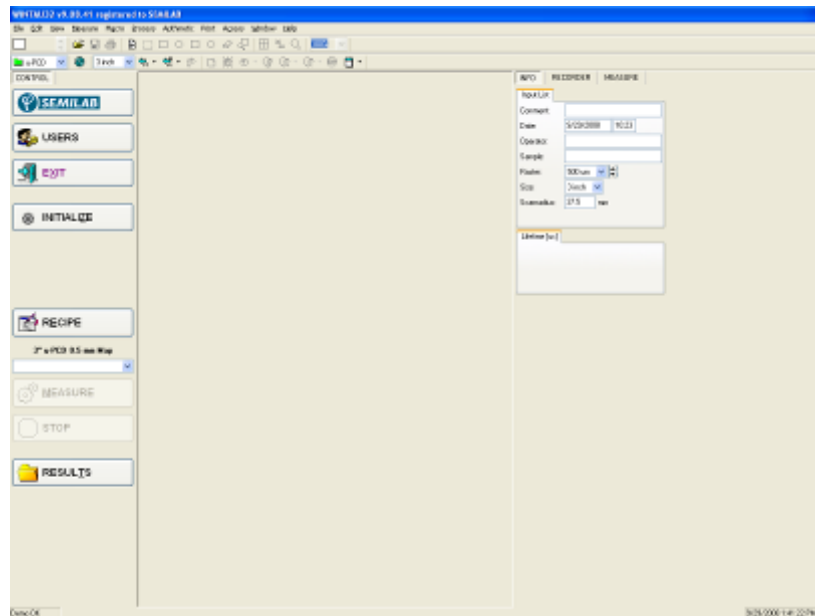


Figure 4-38: The Main Window

- [8.] When in Measurement Mode, the software checks the connection between WSOS2 and Wintau32 every 15 seconds and indicates the status of the connection in the bottom-left of the *Status Bar* at the bottom of the Main Window (see Figure 4-39). Verify that the status is *OK* and the version of the WSOS2 (or CMD) is also displayed.

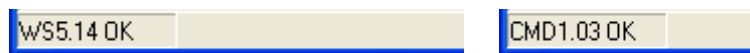


Figure 4-39: The Status Bar

- [8.1.] If the connection is not available (e.g. the DOS side is switched off or not connected to the computer on which the Wintau software is running) the status is *Down* (Figure 4-40). In this case, recheck all connections and verify that the DOS computer is on. See Diagnostics and Troubleshooting Manual for more information.



Figure 4-40: Down status

✓ **NOTE:** It is possible to use the Wintau software without the measurement capabilities. The Measurement Mode can be set in the Register Wintau window (*Help | Register* menu item) by checking the Use measurement capabilities if available box if the registration is valid. (See section 15.1.6.)

- [9.] Wait 10 minutes for the measurement electronics to warm up. The Wintau32 software measures this warm-up time and will provide a warning message if the system is used before the warm-up period has completed. (See Figure 4-41.)

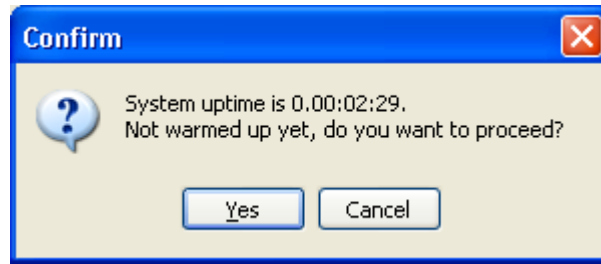
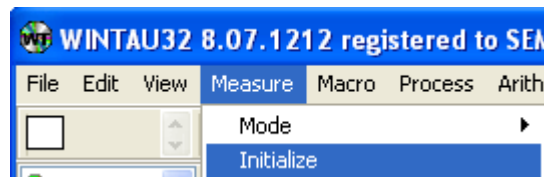


Figure 4-41: Message displayed when warm-up period is not complete

- [10.] Initialize the system by selecting the **Measure | Initialize** menu item or the **INITIALIZE** button (Figure 4-42).



Or

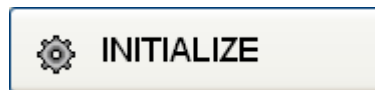


Figure 4-42: Initialize

- [11.] After clicking the **Initialize** menu item, the *Height* tab of the *Measurement Options* window may appear (Figure 4-43). This window displays the head heights for your WT-2000 system. Click **OK** to close the window.
- [11.1.] The head heights shown in the Height tab are customized for your WT-2000 system. It is recommended to record the head height values so that they are easily available if they need to be checked at a later date.

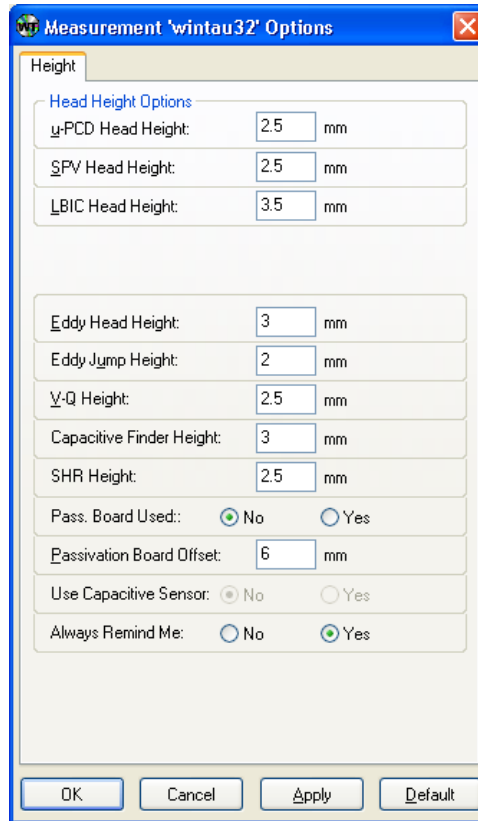


Figure 4-43: The *Height* tab of the *Measurement Options* window

! ***CAUTION:** The head heights shown in the *Height* tab are customized for your WT-2000 system. Do not change these heights. An improper head height setting can either cause a damage of the sample or the head (too low position) or decreased measuring sensitivity (too high position).*

- [12.] During the initialization procedure, the head, the stage and the bridge (and the arm if built in) make a series of movements. This procedure initializes and checks the motors and mechanics.

! ***WARNING:** Do not place your hand (or anything) inside the main unit during Initialization.*

- [13.] After the initialization procedure has finished, click the **Load Wafer**  icon and then click **Cancel**. This action releases the motor restraints. Now the head, the bridge and the stage can be easily moved by hand.
- [14.] Next, check every measurement mode on the system. The measurement modes are shown in the drop-down list in the *Measurement Toolbar* of the *Main Window* (see Figure 4-44).

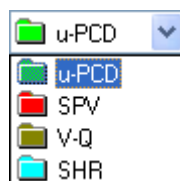


Figure 4-44: List of available measurement modes

✓ **NOTE:** *The software screens on your WT-2000 system may look different from the figures shown in this manual, depending on the options installed in the system.*

- [15.] If the WT-2000 has μ -PCD capability, check this mode first. Click the drop-down menu (Figure 4-44) and select **u-PCD**.

! **CAUTION:** *Be sure that the measurement mode is selected before placing a sample on the stage. The measurement mode determines the correct head height.*

- [16.] When testing a measurement mode, always use a sample known to be measurable by the selected mode. For μ -PCD measurement, use the test sample provided by Semilab.
- [17.] Move the head above the sample.
- [18.] Click the  icon. The *Transient Recorder* window appears (Figure 4-45).

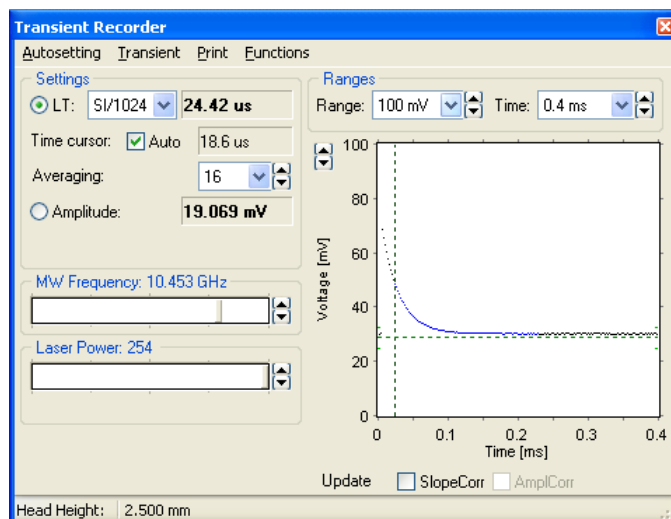


Figure 4-45: Transient Recorder window

- [19.] Select **Autosetting** in the menu bar of this window.
- [19.1.] When there is proper operation of the signal excitation and detection units, a smooth decay curve should be positioned in the *Scope* window as shown in Figure 4-45.

- [19.2.] If there is no signal (flat line) or the noise is larger than half of the whole range, see Diagnostics and Troubleshooting Manual to determine if a hardware problem exists.
- [20.] Continue testing each measurement mode installed on the system.
 - [20.1.] For each measurement mode, use a sample known to be measurable by the selected mode.
 - [20.2.] Referring to the operation procedures in section 12.1, perform the Compensate procedure for the JPV measurement mode if it is installed in your system. The Compensate procedure must be performed before a sample is loaded onto the stage.
 - [20.3.] Referring to the operation procedures in section **12.4.1**, perform the Autosetting procedure for the SPV mode if it is installed in your system.
- [21.] If no problems are observed during the Autosetting procedures, refer to the calibration procedures in section **13.5** and perform any necessary calibration for the JPV measurement mode if it is installed in your system.
- [22.] Referring to section 10.7, calibrate the linear corona charger if it is installed in your system.
- [23.] The WT-2000 system is now ready for use.

5. Theory of Operation

5.1 μ -PCD Measurement

The microwave-induced photoconductive decay (μ -PCD) method is a popular way of measuring minority carrier lifetime in semiconductors. The method excels due to its reliability, good reproducibility, and short measurement time that permit making high-resolution lifetime maps.

As the name implies, the technique involves the optical excitation and the signal detection by the microwave system.

5.1.1 Laser Excitation

The pulse of an infrared semiconductor laser generates free electron-hole pairs under the illuminated area on the sample (Figure 5-1). Since the penetration depth of the 904 nm wavelength light is about 30 μm in silicon, the free carriers are generated close to the front surface of the sample.

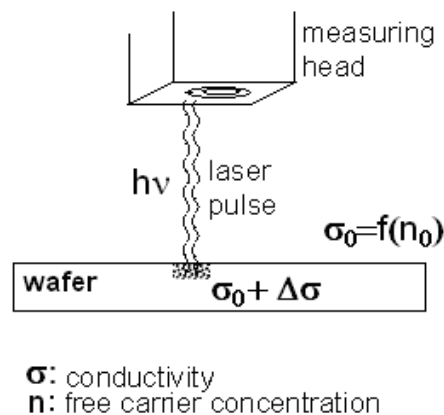


Figure 5-1: Laser excitation

5.1.2 Detection

Since the free electrons and holes recombine, their concentration and so the conductivity of the sample decreases after the excitation. The decaying conductivity can be monitored by detecting the microwave reflectivity, because the reflected microwave power depends on the conductivity of the sample (Figure 5-2). It is measured as a function of time.

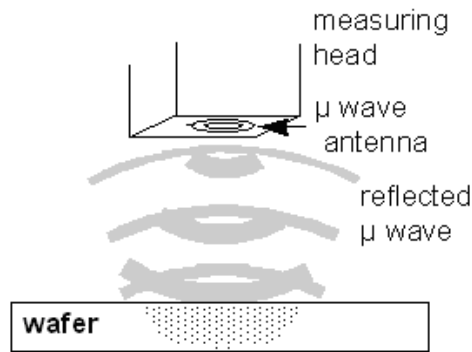


Figure 5-2: Detection by microwave

The ring-shaped microwave antenna is used near to its resonance frequency, in the range of 10.0 GHz – 10.5 GHz. The sample and the antenna above it, compose a resonator, that's resonance curve can be seen on Figure 5-3, red line. As the conductivity of the sample increases due to the excess carriers generated by the illumination, the resonance curve shifts and gets deformed (Figure 5-3, blue line). As the excess carriers recombine, the conductivity of the sample returns to equilibrium, and the resonance curve gets back to its original state.

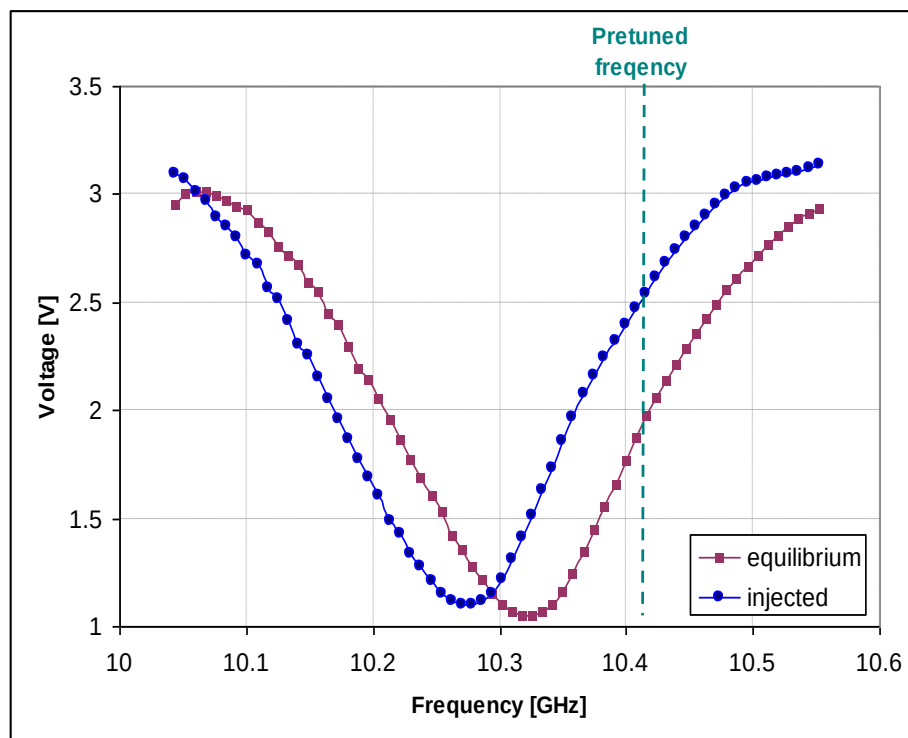


Figure 5-3: Resonance curves of the microwave probe with a sample underneath. (MWDC – the DC signal in the function of microwave frequency) When the sample is in equilibrium, the curve is the red line. When the sample is injected, excess carriers are generated by the illumination, the conductivity and reflectance is changed, the curve is the blue line. The vertical green line shows the frequency used during the measurement, retuned so that the change of the signal is maximal.

During measurement, the microwave frequency is set to a predefined frequency (vertical, green, broken line on Figure 5-3). As the curve gets back to the original state, the measured signal of

the antenna changes. Because the recombination velocity is proportional to the number of excess carriers, the recombination is exponential in time, so the change of the signal is exponential too. (Figure 5-4).

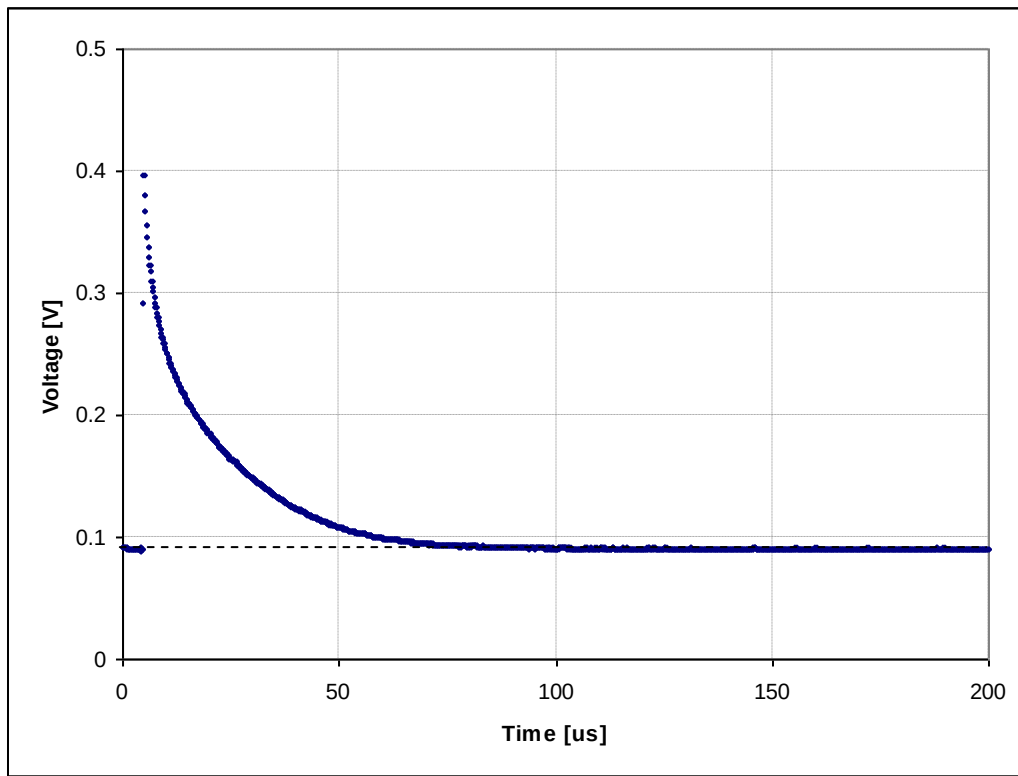


Figure 5-4: Transient, measured on a sample. As the carriers return to equilibrium exponentially, the signal changes accordingly. The signal is descending, according to Figure 5-5, b) case.

If the signal is higher right after illumination than in the equilibrium, the signal is descending during recombination, as it is indicated on Figure 5-5, b) case. The transient on Figure 5-4 belongs to this case.

In some cases, the frequency is tuned to a position, where the signal of the injected sample is below the signal of the sample in equilibrium (Left part of the graphs on Figure 5-3). In this case, the transient is reversed, the signal is growing by time during the recombination.

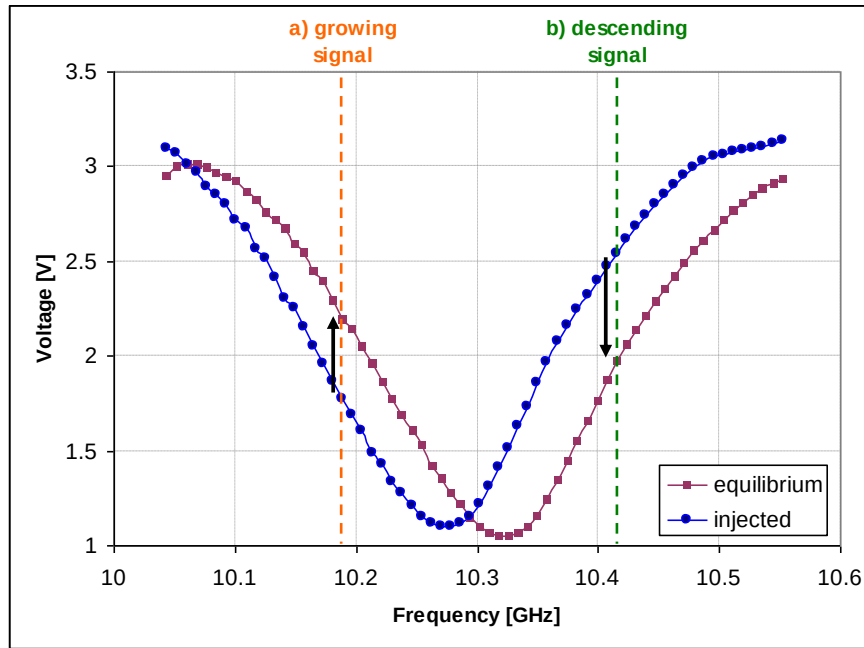


Figure 5-5: The MWDC curves in equilibrium and after injection. If the frequency is set to position a), the signal grows during recombination to the original state, if the frequency is set to b) position, the signal descends.

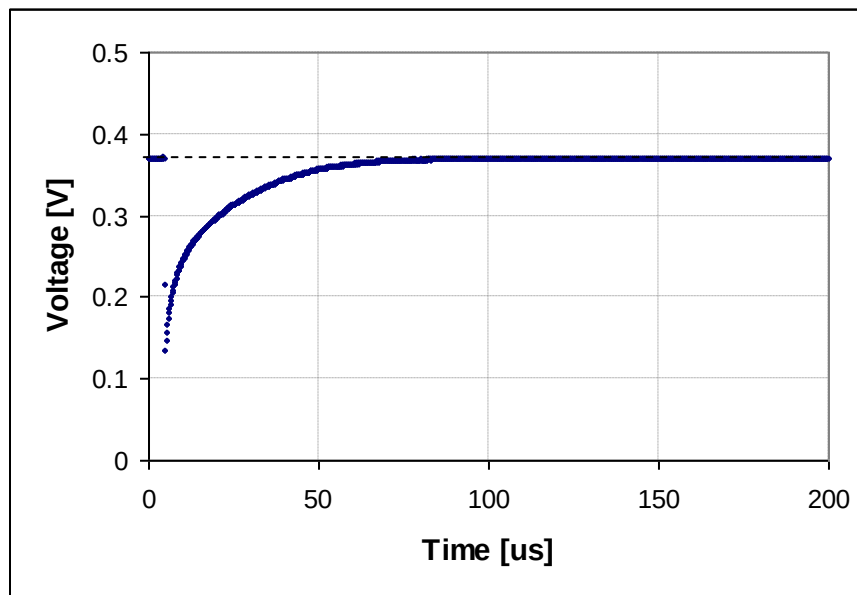


Figure 5-6: Transient with increasing signal, according to Figure 5-5, a) case.

5.1.3 Evaluation

The measured microwave reflectivity decay is fitted with an exponential curve and the obtained time constant is recorded as the measured, effective lifetime in the given position of the sample.

5.1.4 Bulk and Surface Recombination

The redistribution (homogenization) of the free carriers takes place during and immediately after the excitation. After this relatively fast event there are two main parallel processes until the thermal equilibrium is reached (Figure 5-7):

- bulk recombination
- carrier diffusion to the sample surface and then surface recombination

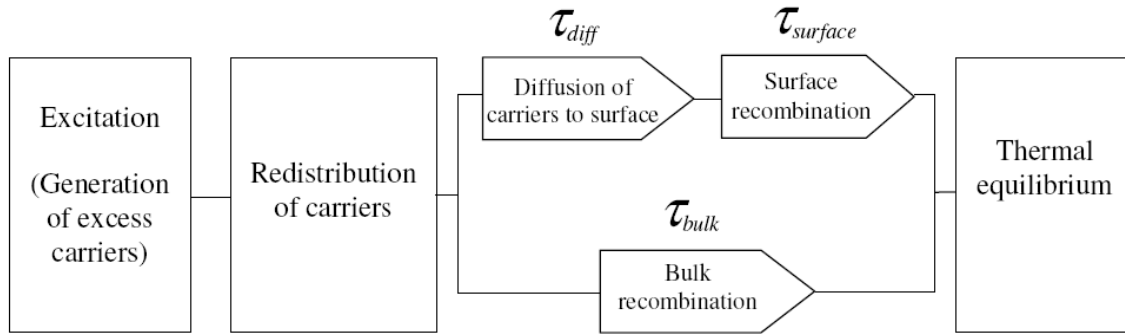


Figure 5-7: Block diagram of the sequence of processes during the μ-PCD measurement

The measured lifetime is characteristic of the whole recombination process as it is described with the following equations:

$$\frac{1}{\tau_{meas}} = \frac{1}{\tau_{bulk}} + \frac{1}{\tau_{diff} + \tau_{surf}} \quad (5.1)$$

$$\tau_{diff} = \frac{d^2}{\pi^2 \cdot D_{n,p}} \quad (5.2)$$

$$\tau_{surf} = \frac{d}{2 \cdot S} \quad (5.3)$$

Where:

- τ_{bulk} is the bulk recombination lifetime that is characteristic of material quality.
- τ_{diff} is the characteristic time necessary for carrier diffusion to one of the surfaces of the wafer from place of excitation. It is determined by the wafer thickness and the diffusion constant of minority carriers.
- τ_{surf} is the characteristic time of surface recombination, determined by surface recombination velocity. It is short for polished bare wafers ($S \approx 50000 \text{ cms}^{-1}$) and long for passivated wafers ($S < 10 \text{ cms}^{-1}$).
- τ_{meas} is the measured lifetime.
- d is wafer thickness.

- D_n and D_p are diffusion constants of the minority carriers, for electrons and holes.
- S is the surface recombination velocity.

As described above in the equations, the sample surface may have a large influence on the lifetime because of the surface recombination process. Figure 5-8 shows theoretical data of the basic equation calculated for three different surface recombination velocities.

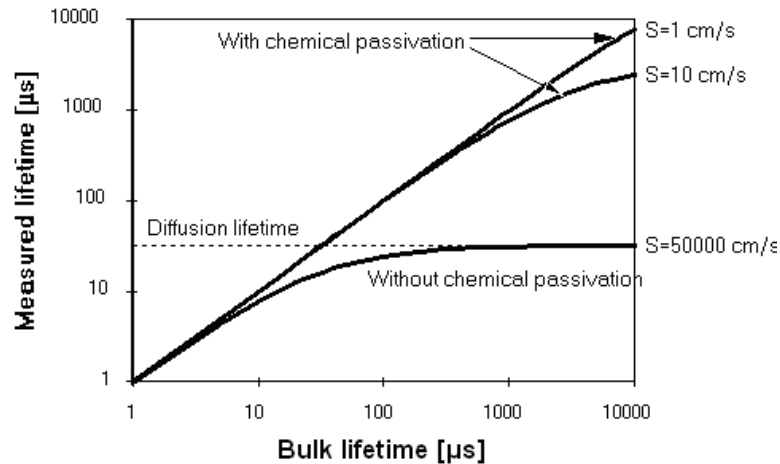


Figure 5-8: Theoretical correlation curves of the measured lifetime versus bulk lifetime

The two steps of the surface recombination process, the free carrier diffusion to the surface and the recombination at the surface are described with two independent parameters (τ_{diff} and τ_{surf}). If the surface recombination velocity is very high, τ_{surf} is very small (typically on bare surfaces), the carrier diffusion is the limiting process i.e. τ_{diff} describes the whole surface recombination process. The longer the bulk lifetime is, the more the diffusion lifetime determines the measured lifetime (Figure 5-8).

Since the aim of the measurement is to determine the contamination dependent bulk lifetime, the surface recombination process should be suppressed i.e. the sample surface should be passivated. There are two commonly used ways of surface passivation: thermal oxidation and chemical passivation. Depending on the efficiency of the passivation, bulk lifetimes much longer than the diffusion lifetime can be measured. Although the measured lifetime depends on both the bulk and the surface recombination processes, in special cases it refers to either the bulk or the diffusion lifetime.

In cases of high contamination level, if the measured lifetime is much shorter than the diffusion lifetime, then the bulk lifetime is measured. The bulk lifetime is also measured if the surface is passivated.

If the contamination level is low, the bulk lifetime is much longer than the diffusion lifetime. But if the surface recombination velocity is high (bare sample), then the measured lifetime is the diffusion lifetime.

In this case, the lifetime map is very uniform, since the diffusion lifetime depends only on the sample thickness and the diffusion constant of the minority carriers, two parameters that typically do not change within a sample.

5.1.5 Hardware Configuration for Lifetime Measurement

The WT-2000 is fully computer-controlled. The measured transients are recorded by the transient recorder card. The mapping above the sample surface is controlled by the motion controller (Figure 5-9).

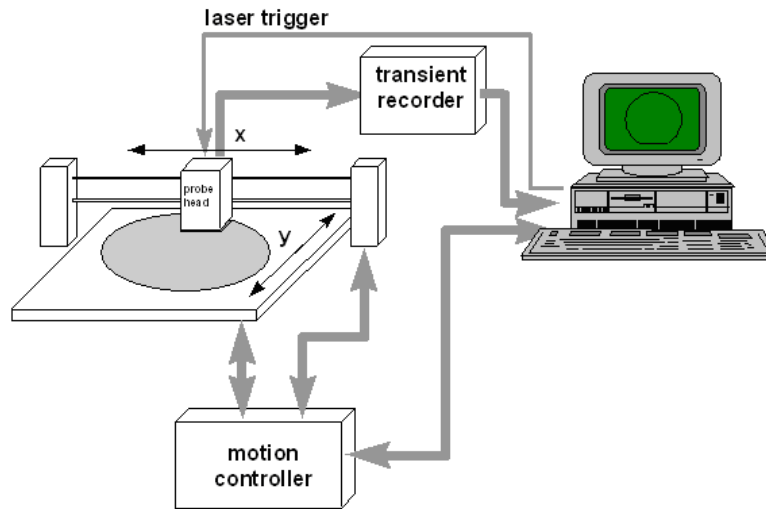


Figure 5-9: Simplified diagram of the WT-2000 system

The whole microwave system is integrated in the measuring head of the WT-2000. The infrared light source and sensor (optical finder) makes it possible to determine the sample position and size before the measurement.

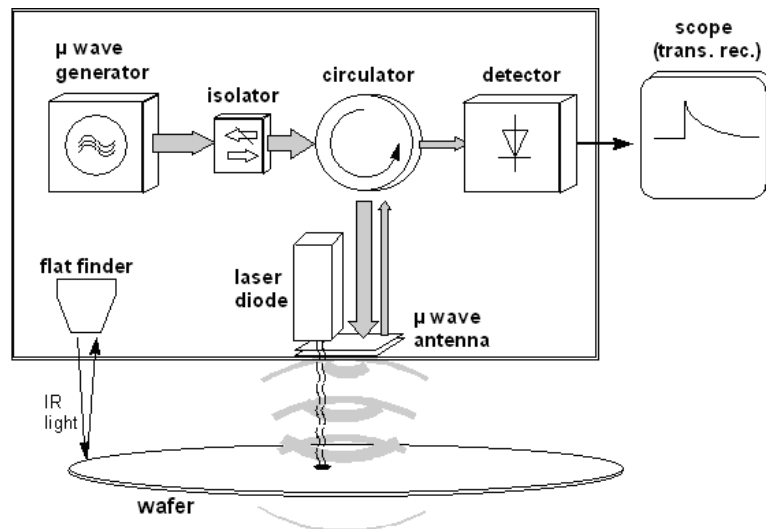


Figure 5-10: Structure of the integrated μ -PCD measuring head

5.1.6 Automatic Parameter Setting in the μ -PCD Measurement

The automatic setting—Autosetting—of the μ -PCD measuring parameters should find an optimal setting for the measurement. The best possible “wide-band” evaluation is necessary, that is in a concrete time window the lifetime should be evaluated even if it changes orders of magnitude, supposing that the voltage range and the microwave tuning are set correctly. Because of this “wide-band” behavior which also results in gathering noise from a broad

frequency range, the evaluation of the individual transients and hence the Autosetting procedure requires a signal to noise ratio much higher than one ($S/N \gg 1$).

The Autosetting procedure works in serial mode, i.e. the individual parameters are set in turn as described below.

- The parameters are set to their initial value.
- Next the voltage range is decreased until the amplitude of the transient (or its recorded first part) is optimal for the A/D conversion, and at the same time the extreme values of the transient are also far from the scope borders (AutoRange).
- Due to the laser pulse, there is a step at the beginning of the transient that should be maximized by changing the microwave frequency in large steps with the necessary increase of the voltage range (pre-tuning).
- With this not optimal but adequate tuning, an approximately optimal time window can be found which results in valid lifetime data within the time window. Especially in case of strongly distorted transients, it can occur that the resulting time window is longer than the optimal one. After the time window setting the number of averaging is corrected to an approximate optimum value (AutoTimeWindow).
- After setting the time window the fine tuning is performed by searching for the maximum of the correlation function that characterizes the “energy” of the transient weighted by an exponential decay of optimal lifetime value. This way the tuning is independent of both the distortion and the fact that the evaluation is reasonable or not. In general this process gives better microwave frequency than the pre-tuning procedure (tuning).
- After the tuning procedure the time window is corrected if necessary. If the time window correction consists of more than one step, then the tuning process is repeated followed by a new time window control.
- The first step before mapping is the activation of the PreScan procedure. The sample is measured on some points, and an average lifetime and its standard deviation can be estimated. If the average value as well as the standard deviation is too high or too low, then the time window increases or decreases to its next possible value.

5.1.7 Effect of Bias Light on μ -PCD Measurements

The bias light is a controlled intensity continuous white light illumination for μ -PCD measurement. The optical fiber of μ -PCD system equipped with bias light is a concentric structure. There is a single fiber in the center for the pulsed excitation laser light and there are several fibers around this center one for continuous illumination during measurement.

The main idea of using bias light is to generate a large number of excess carriers. These carriers are continuously generated and recombine due to the surface states. By this technique the surface states are continuously “occupied,” thus the excess carriers generated by the excitation pulse has a much smaller probability to recombine at surface states. As a conclusion this method works as passivation of the surface.

It is especially useful for enhancing nitride passivation layer effectiveness. It also works on oxidized wafers. The passivation effect of bias light on bare silicon is small.

5.2 QSS- μ PCD Measurement

Microwave detected photoconductance decay, μ PCD, is one of the most common lifetime measurement techniques used in silicon photovoltaics. The industry proven Semilab μ PCD technique determines the effective excess carrier decay lifetime, τ_{eff} directly as the time constant of exponential signal decay. It requires no calibration, no wafer parameters and has advantage of high resolution mapping.

In 2010 Semilab introduced a new version of the technique termed, QSS- μ PCD that gives enhanced accuracy, new capabilities and broader applications. QSS- μ PCD uses two superimposed light beams: 1- the laser beam that is pulsed in order to measure the photoconductance decay and 2- the steady light beam with intensity that can be scanned up to 2suns. This second beam brings several benefits, the most important of which is the ability to measure and control the value of injection level, Δn .

In QSS- μ PCD the steady-state condition is virtually preserved during the photonductance decay as long as the time dependent concentration of carriers injected by the laser, $\delta n_{laser}(t)$ is much lower than the $\Delta n_{generated}$ **Δn generated** by the steady-state light beam.

Issues important for PV and corresponding improvements realized in QSS- μ PCD are listed below.

Issue	Answer	Explanation
QSS-μPCD		
1. Can the injection level, Δn is determined?	Yes	Δn is obtained as $\Delta n = G \cdot \tau_{eff}$, where G is the steady generation rate determined from illumination intensity.
2. Can the injection level be scanned?	Yes	Scanning is done by scanning steady illumination intensity.
3. Can the measurement be done at 1sun?	Yes	This condition is achieved by selecting steady illumination intensity 1sun.
4. Can the implied V_{oc} at 1sun be evaluated?	Yes	Knowledge of Δn at 1sun allows the calculation of the implied V_{oc} .
5. Can the emitter saturation current, J_0 be determined?	Yes	Simultaneous measurement of τ_{eff} and Δn allow the determination of J_0 including first ever noncontact J_0 mapping.
6. Are the measurements more accurate and repeatable?	Yes	For 3 reasons: 1- steady illumination makes the PCD transient more exponential; 2- the effect of trapping is eliminated; 3- the effect of space charge modulation is eliminated.

5.2.1 Background Information: Photoconductance and excess carrier concentration

Photoconductance, PC, is the change of electrical conductance caused by illumination with photon energy, $h\nu$ larger than the energy gap $E_g=1.12$ eV. Photogenerated electrons and holes are generated are free to move in the conduction and valence band, respectively. They are referred to as “excess charge carriers” or simply excess carriers. The term “excess” means that they are an addition to carriers present at thermal equilibrium in the dark.

The photoconductivity is $\Delta\sigma=q(\Delta n\mu_n+\Delta p\mu_p)$ where Δn and Δp are the concentration of excess electrons and holes, respectively; and μ_n and μ_p are the corresponding mobilities. When trapping is negligible, $\Delta n = \Delta p$. In such a case, the photoconductivity becomes

$$\Delta\sigma = q\Delta n(\mu_n+\mu_p)$$

For excess carrier lifetime measurement, Δn is of importance. If mobility changes caused by photo-injection of excess carriers are small, the photoconductance can be used in order to obtain steady state Δn value or to follow a time decay of Δn in transient measurement. The steady-state condition and the transient condition are two distinct cases in lifetime measurement. The QSS- μ PCD technique combines both of them.

5.2.2 Steady-state condition

This condition is illustrated in Figure 5-11. Illumination is turned on at $t = 0$. This starts the photo-generation of excess carriers at a rate, G per cm^3s , proportional to illumination intensity.

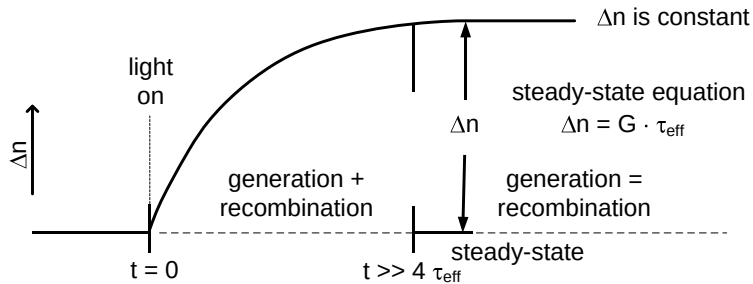


Figure 5-11: Steady-state condition created by illumination

With increase of Δn the excess carrier recombination rate $R = \Delta n / \tau_{eff}$ also increases. The steady-state is established when $R = G$ and Δn reaches the saturation value given by:

$$\Delta n = G \cdot \tau_{eff}$$

In the QSS- μ PCD technique, the equation above provides a means for obtaining the injection level, Δn using τ_{eff} and the corresponding precalibrated values of G . In other techniques, such as the quasi-steady-state photoconductance QSSPC, or photoluminescence QSSPL, the equation is used to obtain the effective lifetime as $\tau_{eff} = \Delta n / G$.

The term effective lifetime is used because it contains a contribution not only from recombination in the bulk, but also from the surface recombination.

The value of Δn at steady-state is also referred to as the injection level.

5.2.3 Transient conditions

The transient condition that is used in the μPCD technique is illustrated in Figure 5-12. A short duration laser pulse generates excess carriers increasing the wafer conductance under the illuminated area. After illumination, the excess carrier concentration decays with time due to electron-hole recombination. This causes a corresponding decay of the conductance. For small mobility changes, photoconductance transient reflects the transient of Δn .

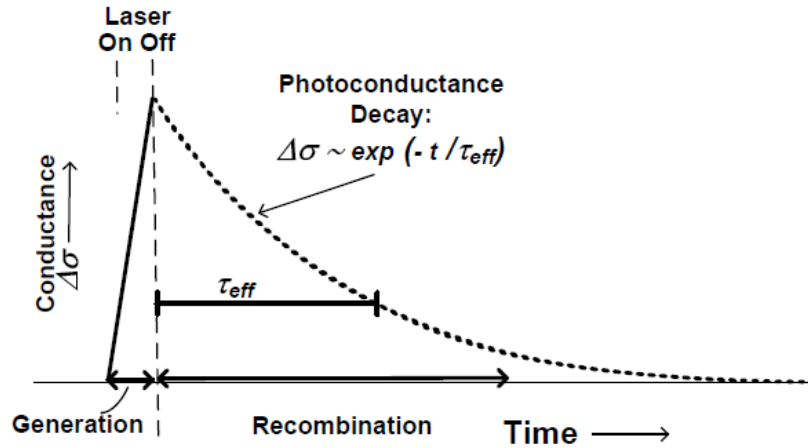


Figure 5-12: Photoconductance decay after short duration laser pulse

The continuity equation with surface recombination boundary condition predicts, in an ideal case, an exponential Δn decay:

$$\Delta n = \text{const} \cdot \exp(-t / \tau_{\text{eff}})$$

with a time constant equal the effective lifetime, τ_{eff} .

To determine the time constant, the transient can be measured in arbitrary units. Therefore, μPCD does not require any calibration or wafer parameters. It is however, very important that it operates in a linear range where the microwave reflected signal is proportional to the photoconductance and thus to Δn .

4-14

5.2.4 QSS- μPCD procedure and measured parameters

In QSS- μPCD technique a decay lifetime measurement is done using a transient condition imposed on the steady-state condition. First the steady light is turned on to achieve the steady-state condition as illustrated in Figure 5-11. Then the short laser pulse generates additional free carriers δn_{Laser} and carrier decay time constant is measured with μPCD as illustrated in Figure 5-11. Two conditions are met: 1- the steady-state illumination and the corresponding generation rate G is varied slowly compared to the lifetime and 2- the laser induced δn_{Laser} is small compared to steady-state Δn . This assures that the decay proceeds under condition that is very close to that in the steady-state. Therefore, it is referred to as quasi-steady-state, “QSS”, photoconductance decay.

The scanning sequence in QSS- μPCD is shown in Figure 5-13.

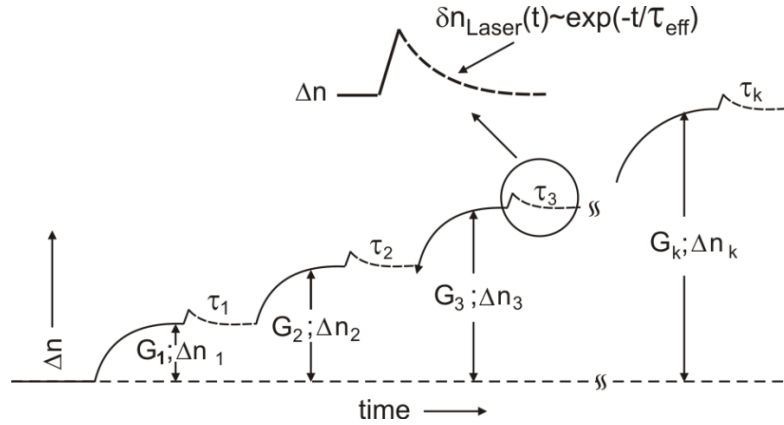


Figure 5-13: Illustration of QSS-μPCD. Note that it contains steady-state condition in Figure 5-11 and laser induced transient condition (encircled) as in Figure 5-12.

The steady-state illumination intensities are varied in steps $I_1, I_2 \dots I_k$. Corresponding decay lifetimes are measured $\tau_1, \tau_2, \dots \tau_k$. The plot τ vs. I is the intensity scan.

To determine the injection levels $\Delta n_1, \Delta n_2 \dots \Delta n_k$, the values of generation rates $G_1, G_2 \dots G_k$ are needed. They are assumed uniform across the wafer thickness and are calculated following the procedure used in silicon PV

$$G[\text{in cm}^{-3} \text{ s}^{-1}] = 2.38 \times 10^{17} I[\text{suns}]/d[\text{cm}]$$

Where d is the wafer thickness and I is the solar equivalent QSS light intensity. G_k and corresponding τ_k enable to calculate Δn_k from the equation above.

The QSS-μPCD gives the following data set: $I[\text{suns}]$; $\tau_{\text{eff}}[\mu\text{s}]$; $\Delta n[\text{cm}^{-3}]$. Three important measured characteristics are: (a) τ_{eff} vs. I ; (b) τ_{eff} vs. Δn and (c) Δn vs. I . Their applications are as follows:

- (a) – τ_{eff} at the desired intensity, typically 1sun.
- (b) – determination the emitter saturation current J_0 from τ_{eff} vs. Δn
- (c) – estimation of the implied open circuit voltage for desired I from Δn vs. I .

$$\text{Implied } V_{\text{oc}}[\text{V}] = kt/q \ln[\Delta n(N_A + \Delta n)/n_i^2]$$

Where n_i is the intrinsic carrier concentration in silicon, calculated according to revised PV silicon values (e.g. $n_i = 1.02 \times 10^{10} \text{ cm}^{-3}$ at 300 K)

5.2.5 Microwave measurements of photoconductance decay

Semilab QSS-μPCD uses the industry proven reflected microwave technique for monitoring the laser induced photoconductance decay similar to that previously used in μ-PCD.

Arrangement with steady light beam and PCD probing laser pulses is shown in **Figure 5-14**

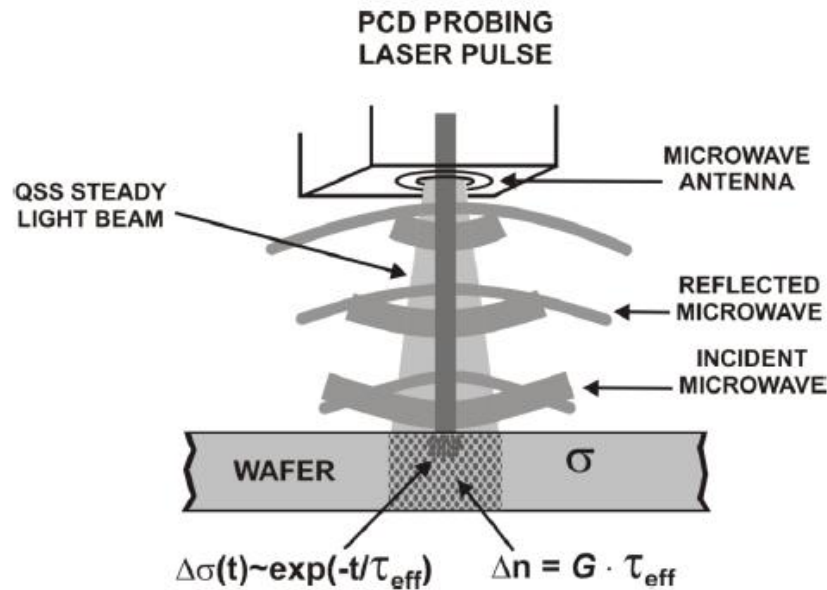


Figure 5-14: **Illustration of the QSS-μPCD technique of laser pulse induced photoconductance decay, $\Delta\sigma(t)$ is measured using a reflected microwave signal.**

The ring-shaped microwave antenna is used near the resonance frequency 10.0GHz-10.5GHz. The wafer and antenna form a resonator. As the electrical conductance of the wafer increases, the resonance curve shifts. As the photoconductance decays after the laser is turned off, the resonance curve returns to its original state.

The measured microwave dc-voltage signal is proportional to wafer conductivity change; therefore time decay of this signal provides a measure of the photoconductance decay. By fitting signal decay to exponential time dependence, photoconductance decay constant is obtained.

5.2.6 Nonexponential PCD decay

Reliable determination of τ_{eff} by μ -PCD requires an exponential decay. This poses no problem for short lifetime wafers such as PV as-cut wafers and etched wafers without passivation dielectrics. However, for PV wafers with passivation and after diffusion and passivation, the transients measured without steady illumination are very often nonexponential. Such a transient is illustrated in Fig. 4-11 together with the exponential fitting curve that is used to determine τ_{ef} . It is seen that the fitting is not good and thus a determination of τ_{ef} is not reliable.

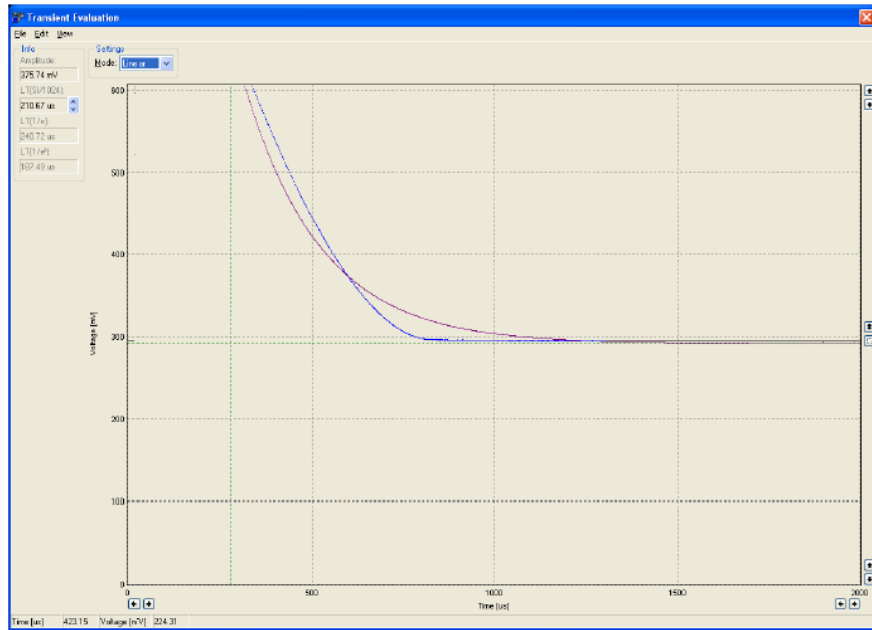


Figure 5-15: Example of nonexponential u-PCD transient

In QSS- μ PCD, even low intensity (0.1sun) steady illumination significantly improves transient exponentiality. Above 0.3suns QSS- μ PCD transients are typically very well exponential. Difficulty with nonexponential transient arises due to the strong recombination dependence on the injection level. In passivated PV-silicon this is due to surface recombination. The steady-state illumination used in QSS- μ PCD reduces dependence on the injection level, improving exponentiality of the transient and, in consequence, the accuracy of the technique.

5.2.7 Determination of emitter saturation current

Simultaneous measurement of τ_{eff} and Δn facilitates emitter passivation measurement illustrated in Figure 5-16 and Figure 5-17. The measurements were done on a n-base CZ silicon test wafer with a two-sided symmetrical p^+ emitter passivated with high quality SiO_2/SiN dielectric stack.

The descending τ_{eff} range that is used for emitter monitoring is within the exponential transient range.

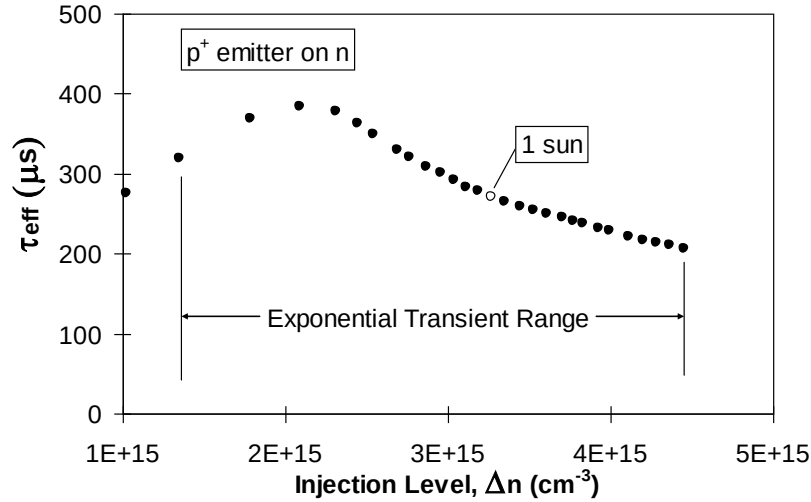


Figure 5-16: Injection level dependence of QSS-μPCD decay lifetime for passivated p⁺ or n emitter.

The plot of $1/\tau_{eff}$ vs. Δn , in that range, is linear with $R^2 = 0.998$ as shown in Figure 5-17. This enables a determination of the emitter saturation current J_0 from the slope using the Kane and Swanson method.

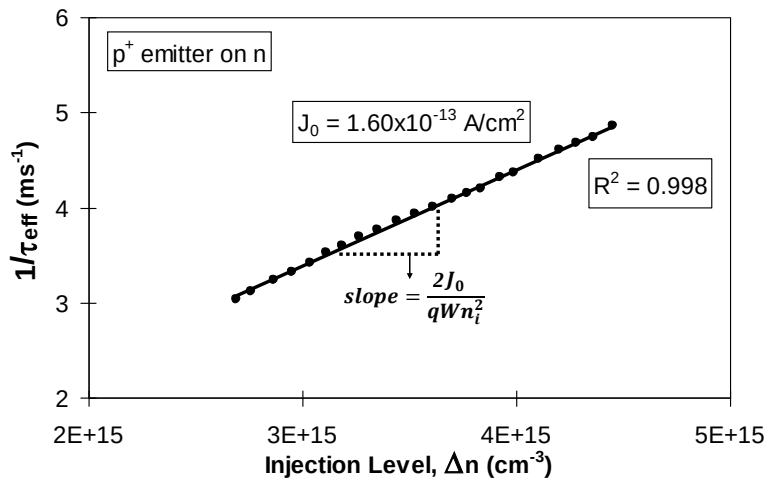


Figure 5-17: Inverse of τ_{eff} vs. Δn illustrating the determination of J_0 .

J_0 measurement capability available in QSS-μPCD is also used for the whole wafer J_0 mapping. An example map is shown in Figure 5-18. In mapping J_0 is obtained at each site from the slope of $1/\tau_{eff}$ using two τ_{eff} maps measured at two light intensities.

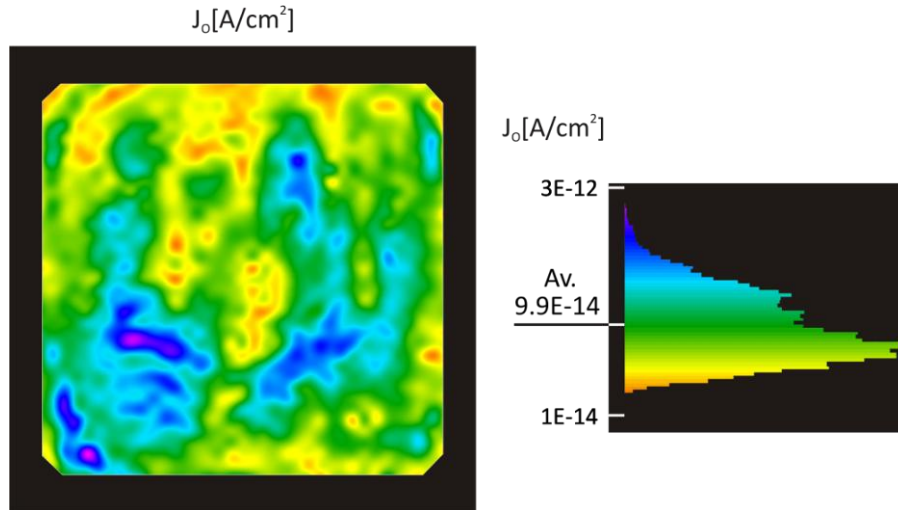


Figure 5-18: The wafer map of J_0 measured with QSS- μ PCD

5.2.8 Bulk and Surface contribution to τ_{eff}

The excess carrier decay involves recombination of photogenerated excess carriers in the bulk and at the wafer surfaces. In the absence of trapping, the dominant mode decays exponentially with a time constant τ_{eff} given by the equation

$$\frac{1}{\tau_{eff}} = \frac{1}{\tau_b} + D\beta^2 \quad 4-17$$

where τ_b is the bulk lifetime, $D\beta^2$ is the surface term, β is the root of equation

$$\tan\left(\frac{\beta d}{2}\right) = \frac{S}{\beta D} \quad 4-18$$

where d is the wafer thickness, S is the surface recombination velocity, (assumed the same for both wafer surfaces) and D is the minority carrier diffusivity.

For PV wafers two limiting cases of low and high surface recombination velocity are of special interest.

$$D\beta^2 (S \rightarrow 0) = \frac{2S}{d} = \frac{1}{\tau_{Surf.Re c.}} \quad 4-19$$

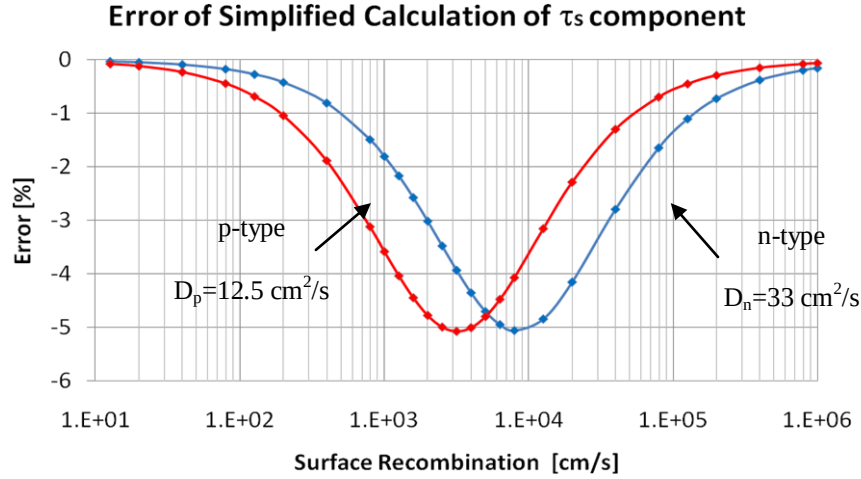
$$D\beta^2 (S \rightarrow \infty) = \frac{\pi^2 D}{d^2} = \frac{1}{\tau_{diff.}} \quad 4-20$$

For intermediate S values, an approximate analytical expression for $D\beta^2$ is

$$D\beta^2 = \frac{1}{\tau_{Diff.} + \tau_{Surf.Re c.}} \quad 4-21$$

This expression slightly underestimates the true $D\beta^2$ value. The maximum underestimation is 5%, and it occurs when $\tau_{diff} = \tau_{Surf.Rec.}$. (See Figure 5-19). The error of approximate expression diminishes toward low S and high S values. In these two practically important cases of $S < 500$ cm/s and $S > 5000$ cm/s, the error reduces to 2.5% or less.

Maximum underestimation is only 5%. It occurs only when $\frac{d^2}{\pi^2 D} = \frac{d}{2S}$.



sample thickness $d = 200\mu\text{m}$ maximum error at $S = \frac{\pi^2 D}{d^2}$

Figure 5-19: Error in using simplified analytical expression for overall surface contribution to τ_{eff} .

High S approximation applies to as cut wafers, wafers after saw damage etch and wafers after texturing. If in high S wafers the bulk lifetime is quite long, (τ_b , typically $20\mu\text{s}$ or higher), then $\tau_{eff} \rightarrow \tau_{diff}$ providing a means to obtain the minority carrier diffusivity and the minority carrier mobility values from μPCD measurement:

$$\frac{1}{\tau_{eff}} \rightarrow \frac{1}{\tau_{diff}} = \frac{\pi^2}{d^2} \cdot D_{n,p} = \frac{\pi^2}{d^2} \left(\frac{kT}{q} \cdot \mu_{n,p} \right) \quad 4-22$$

Low surface recombination corresponds to wafers after passivation. In this case:

$$\frac{1}{\tau_{eff}} = \frac{1}{\tau_b} + \frac{2S}{d} \quad 4-23$$

When two surfaces are different, $2S$ shall be replaced with $S_f + S_b$. (S_f corresponds to front surface and S_b to back surface).

5.2.9 Evaluation of the upper limit of surface recombination or lower limit of τ_b

The equation above can be used to evaluate the upper limit of the surface recombination in passivated wafers. Neglecting $1/\tau_b$, one gets

$$S_{\max} = \frac{d}{2\tau_{\text{eff}}}$$

4-24

In QSS- μ PCD τ_{eff} is measured typically at a steady-state light intensity of 1Sun.

Example: p-Si 200 μ m thick passivated on both sides with Si₃N₄ shows increasing of τ_{eff} vs. intensity that saturates above 0.4suns. At 1Sun τ_{eff} is 500 μ s. Using this value in eqn. 4-24 one gets $S_{\max} = 20$ cm/s. This means that the surface recombination is 20 cm/s or less.

The same data can be used to evaluate lower limit of the bulk lifetime $\tau_{b\min}$. By neglecting surface recombination term one gets $\tau_{b\min} = \tau_{\text{eff}}$. For the example given above $\tau_{b\min} = 500\mu\text{s}$. This means that τ_b is 500 μ s or more.

5.2.10 Surface limited lifetime in as-cut and etched PV wafers

For p-Si wafers with doping of $1.5 \times 10^{16} \text{ cm}^{-3}$ in as cut form, after saw damage etching or after texturing etch, τ_{eff} shall be smaller or equal to diffusion limit τ_{diff} given by eqn. 4-20. For 200 μ m wafers this limit is about 1.7 μ s, while for a 140 μ m wafer it is about 1.1 μ s. Values of τ_{eff} below these limits indicate a very short bulk lifetime i.e. very poor quality, heavy

contaminated silicon. Such bulk lifetime is $1/\tau_b = 1/\tau_{\text{eff}} - 1/\tau_{\text{diff}}$.

5.2.11 Effect of traps and elimination of this effect by steady illumination

When minority carrier traps are present, as shown in Figure 5-20, some excess electrons become trapped and Δn no longer equals Δp . This may significantly change the photoconductance value and the photoconductance decay process. Traps illustrated in Figure 5-20 are at the energy levels that capture only one type of free carrier, hold them temporarily and then release them back to the band from which the carriers were captured. For high trap concentrations, the transient decay of photoconductance is not a simple exponential and it may contain a long time constant tail caused by slow release of the carriers from traps. The effect can be reduced using additional steady illumination that continually creates excess carriers and keeps traps filled during the PCD measurement. This was the original use of steady-state illumination in PCD, termed "bias light". In QSS- μ PCD, this role is played by the QSS illumination.

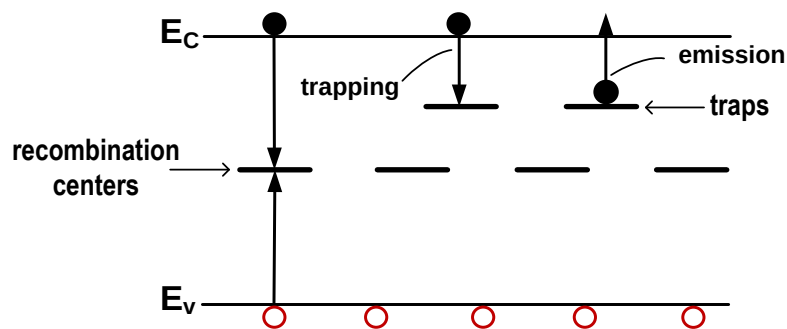


Figure 5-20: Band diagram showing recombination centers and minority carrier traps. Recombination centers capture both electrons and holes, annihilating them. The traps capture and subsequently emit only one type of carrier.

5.3 Charge-PCD Measurement

The Charge-PCD system involves the standard μ -PCD technique as well as the ability to perform continuous corona charge deposition on the sample. The C-PCD system can be used for measuring both bare and oxidized samples.

The Charge-PCD (C-PCD) measurement is based on the μ -PCD measurement. The laser excitation, the microwave detection and the Autosetting procedure are the same in both techniques. The main difference is that during the C-PCD measurement there is a *continuous* corona charging performed on one or both sides of the wafer. This way the bulk lifetime can be measured even on non-passivated, bare wafers because the continuously deposited charge passivates the surfaces even if there is no good quality thermal oxide on them.

To achieve good surface passivation efficiency by charging, there should be at least a thin native oxide layer on the surfaces of the sample. The deposited charge creates an electric field at the SiO_2 -Si interface, which attracts the opposite sign free carriers and repels the identical sign carriers. This way the electric field separates the electrons and holes decreasing the probability of the recombination.

Depositing negative charge on a p-type silicon sample attracts the majority carriers (the positive holes) while it repels the minority carriers (the negative electrons). This creates majority carrier accumulation at the surface (very high hole concentration). The concentration of the minority carriers is practically negligible at the surface, therefore the probability of surface recombination is very low. As a result, the surface passivation is very effective.

Depositing positive charge on p-type silicon sample attracts the minority carriers, the negative electrons while repelling the majority carriers, the positive holes. This creates majority carrier depletion at the surface (very low hole concentration). In this case, there is also minority and majority carrier separation at the surface just like in case of accumulation, and the surface passivation is also very effective. However the passivation efficiency of the accumulation is better than that of depletion, so for the Charge-PCD measurement the accumulation condition is recommended (negative charge).

5.3.1 Oxidized Wafers

If the wafer is oxidized, the deposited charge remains on the top of the oxide because the oxide is an insulator layer. Therefore, there is no need to continuously charge the wafer during the measurement; it is enough to charge it before the lifetime measurement (linear corona charge deposition + standard μ -PCD technique).

5.3.2 Bare Wafers

Linear corona charge deposition is not as useful for bare wafers. Very thin oxide layers or the native oxide (on a bare wafer) cannot hold the deposited charge for a long time due to the charge tunneling effect. Therefore, continuous charge deposition is necessary to keep the surface charged during the measurement (Charge-PCD technique).

5.4 Surface Photovoltage Measurements

The Surface Photovoltage (SPV) measurement provides information about both bulk and surface parameters of a semiconductor material in a non-destructive way, without touching the surface of the sample. The diffusion length of the minority carriers a bulk property, the relative

surface recombination velocity refers to the surface and the total surface potential (band bending) which depends on the surface charge and the doping concentration characterizes both the bulk and the surface.

SPV measurements have several advantages over μ -PCD lifetime measurements:

- The sample surface does not need to be passivated to get the bulk lifetime.
- Much shorter lifetimes i.e. diffusion lengths ($L_D = \sqrt{D\tau}$) can be measured.
- SPV signal can be obtained on very low resistivity material ($<0.1 \Omega\text{-cm}$), below the limits of the μ -PCD technique.

The drawbacks of the SPV measurement are:

- The slower measurement speed (not only one but several wavelengths are used).
- Surface preparation is needed in most cases to create a surface potential barrier.
- The thickness of the oxide layer on oxidized samples should be known.

The other essential difference between the two techniques is the injection level which influences the lifetime value itself. The SPV technique uses light sources orders of magnitude less intensity than the ones used in the μ -PCD technique.

The primary condition of the SPV measurement is a potential barrier, i.e. a space charge region (scr), at the surface. The sample is illuminated with light which has higher energy than the forbidden gap of the measured semiconductor. (For silicon, $E_{gap} > 1.12 \text{ eV}$, corresponding to wavelengths less than 1100 nm.) Some of the generated carriers diffusing to the surface reduce the surface potential barrier compared to the grounded backside of the sample. Since the illumination is chopped with a given frequency (50 % duty factor), the periodic change of surface potential can be measured capacitively by a probe above the sample surface (see **Figure 5-21**). The measured voltage is the SPV signal.

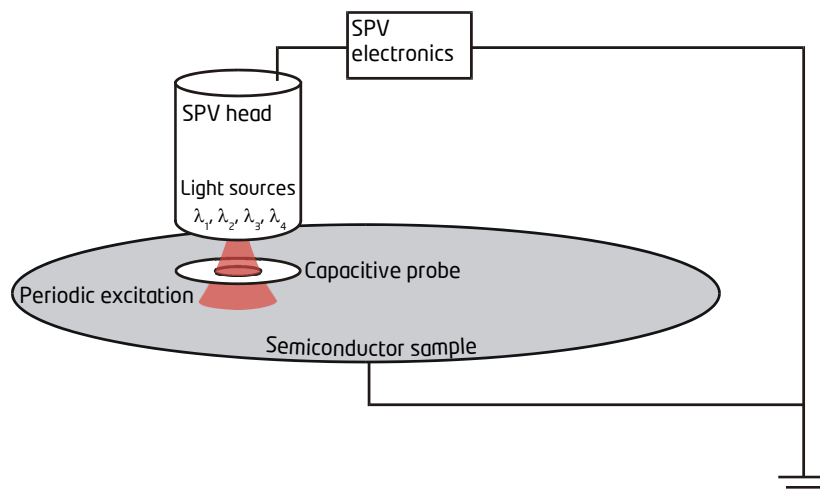


Figure 5-21: Setup for SPV measurement

The sign of the signal depends on the conduction type of the material (p or n). Increasing the light intensity, the SPV signal increases until it reaches the saturation, the so-called flat-band condition. The maximum possible SPV signal is determined by the band bending. This

saturated value is characteristic of the sample. The longer the wavelength of the excitation light, the higher the light intensity that is necessary to reach saturation. This is because longer wavelengths of light penetrate more deeply (Figure 5-22 shows the penetration depth versus wavelength curve for silicon). With the deeper penetration, fewer generated charge carriers can reach the surface by diffusion because of the bulk recombination. Therefore to get the same SPV signal with different wavelength lights, a higher light intensity (i.e. photon flux) is necessary for longer wavelengths of light (see Figure 5-23).

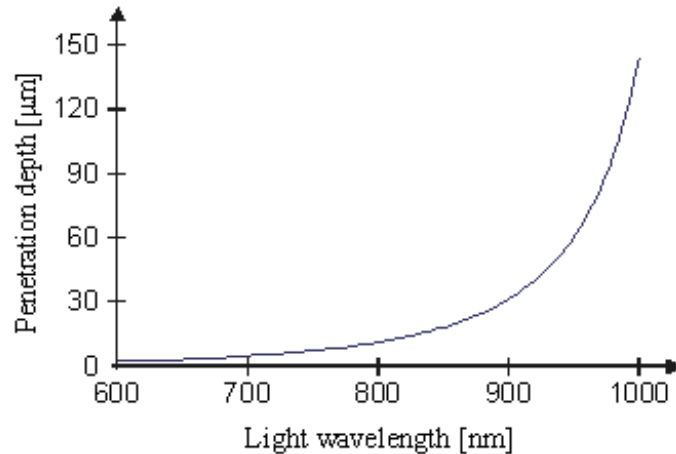


Figure 5-22: Penetration depth of the laser in silicon as a function of the wavelength

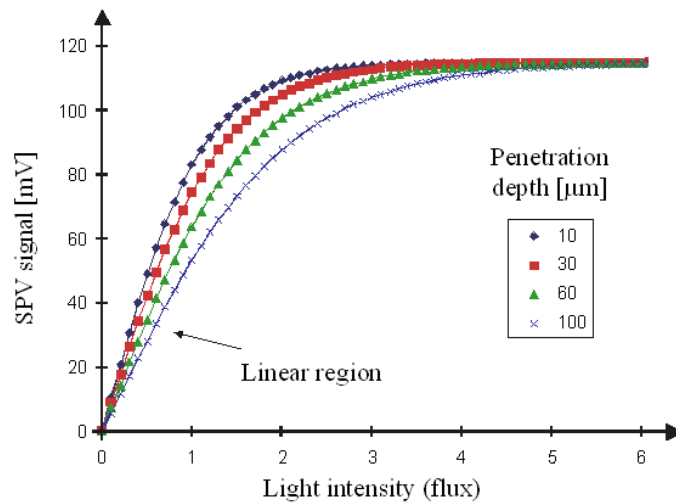


Figure 5-23: SPV signal as a function of light intensity

The saturated SPV signal directly gives the band bending. The standard SPV measurement is performed in the initial, linear region of the SPV signal vs. light intensity curve to obtain the diffusion length. In the linear region, the ratio of the light intensity (photon flux) over the SPV signal is constant for each wavelength. The diffusion length can be calculated from these quotients measured with applying different wavelength lights. It is necessary to perform SPV measurements with at least two different wavelengths to get the diffusion length.

Figure 5-24 shows a graph of light intensity divided by the SPV signal as a function of the light penetration depth, for two different surface recombination velocities. The data points are fitted to straight lines, which are extrapolated to their intersection with the X axis. The intersection with the X axis gives the diffusion length, while the slope of the line is related to the surface

recombination velocity. In Figure 5-24, the upper line has a higher surface recombination velocity S than the lower line.

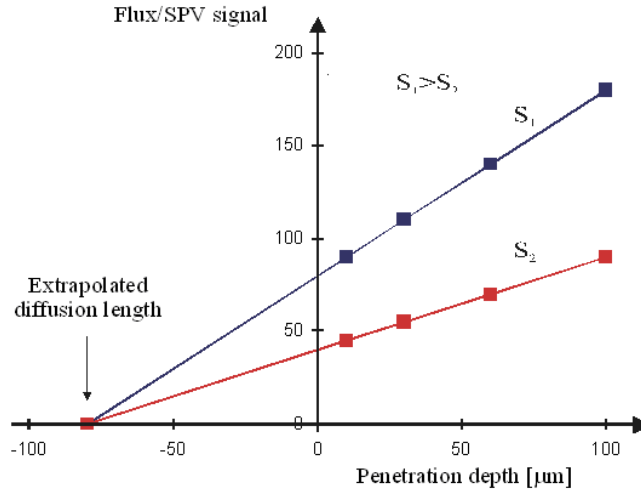


Figure 5-24: Light intensity divided by the SPV signal, as a function of the light penetration depth

The surface recombination decreases the SPV signal, because not all of the charge carriers which reach the surface can take part in decreasing the surface potential barrier, some recombine by the surface recombination. Therefore the higher the surface recombination velocity is, the smaller the amplitude of the SPV signal is, independent of the light penetration depth. A smaller SPV signal means a higher ratio of light intensity to SPV signal, i.e. larger slope of the fitted straight line. Since there is a slope without any surface recombination, because of the bulk recombination, the absolute value of the surface recombination velocity cannot be obtained only using the measured SPV data.

Although this straight line fitting is the most frequently used way of evaluation, the extrapolated diffusion length value is true only if the following four conditions are fulfilled:

$$T - w > 4L_D \quad (5.4)$$

$$w \ll L_D \quad (5.5)$$

$$w \ll \frac{1}{\alpha} \quad (5.6)$$

$$T - w \gg \frac{1}{\alpha} \quad (5.7)$$

where T is the thickness of the sample, w is the thickness of the space charge layer, L_D is the diffusion length and $1/\alpha$ is the penetration depth of the light (one over the absorption coefficient). In this case the general solution of the diffusion equation can be simplified to the following equation (5.8) which describes a straight line whose intersection on the X axis is $-L_D$.

$$\frac{\Phi}{V} = C \cdot \left(L_D + \frac{1}{\alpha} \right) \quad (5.8)$$

where Φ is the light flux, V is the detected SPV signal, and C is defined as a constant.

Figure 5-25 shows two SPV signal (amplitude) maps measured on an oxidized silicon wafer. The applied wavelengths of the excitation lights were 670 nm and 980 nm, having 4 μm and 100 μm penetration depths (see Figure 5-22). Figure 5-26 shows two maps of diffusion length and relative surface velocity, calculated from the two maps of SPV amplitude.

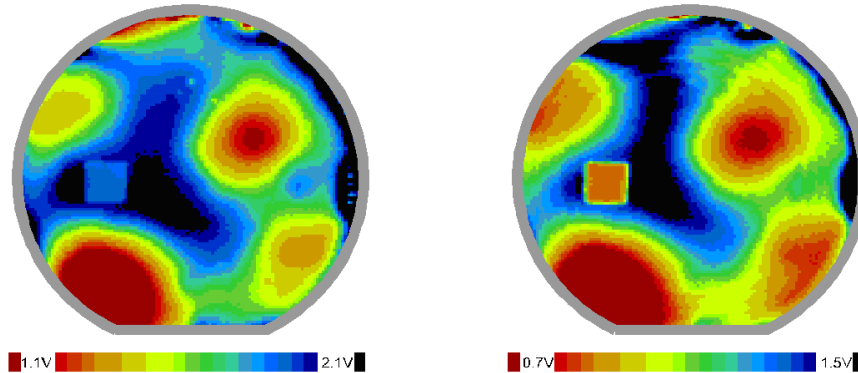


Figure 5-25: SPV amplitude maps measured with 670 nm and 980 nm wavelength lasers

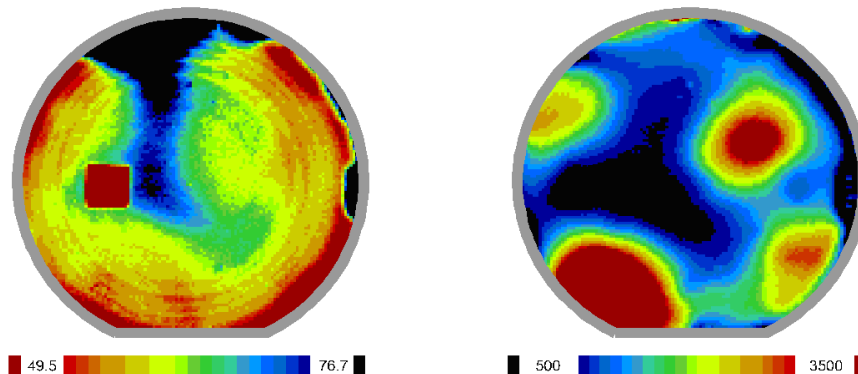


Figure 5-26: Diffusion length and relative surface recombination velocity maps calculated from the previous maps

The SPV amplitude maps contain both bulk and surface related information. These are nearly entirely separated in the calculated diffusion length and surface recombination velocity maps. The typically bulk related ring structure and the small square can be seen only on the diffusion length map. (The small square has different diffusion length because of an intensive illumination treatment for other investigations.) However, the surface related larger spots appear basically on the surface recombination maps. The SPV map measured with the deeper penetration depth light contains bulk related information, therefore it is more similar to the calculated diffusion length map than the SPV map measured with the less deeply penetrating light.

5.4.1 Automatic Parameter Setting in the SPV Measurement

In the case of zero electrical noise, the optimal working point would be an SPV amplitude much smaller than the thermal voltage (approximately 26 mV at room temperature), otherwise non-linear behavior would appear and thus distorted diffusion length would be evaluated. However when decreasing the SPV amplitude the signal to noise ratio also decreases, hence the reliability

of the evaluated diffusion length values falls to unacceptable level. It means that we have to find a compromise between the linearity and the signal to noise ratio requirements.

The linearity of the SPV amplitude vs. light flux curves may differ considerably in different samples due to the differences of the surface conditions. There are also wafers showing non-linear behavior, while there are samples with surprisingly linear characteristics even at amplitudes larger than 100 mV.

Taking into account the considerations above, we constructed a complex error function. This is the relative error of the evaluated diffusion length divided by the fourth power of the slope of the SPV amplitude vs. light flux characteristics. The Autosetting procedure sets the amplitude at the minimum value of the above described error function. In the case of non-linear characteristics, it sets low amplitude when increasing the light flux because of the rapidly falling slope. However in the case of linear characteristics, it sets high amplitudes due to the increasing signal to noise ratio at higher light flux.

The steps of the automatic parameter setting are as follows:

- It measures the SPV amplitude vs. light flux characteristics at all wavelengths.
- From the SPV amplitude vs. light flux characteristics, the monotone behavior is tested for all wavelengths. If the SPV amplitudes begin to decrease at a given flux value, the software will neglect the data above this flux level.
- From the characteristics, the highest amplitude values will be extracted. If these amplitude values have the same polarity they will be used in the calculations. If the curves recorded at different wavelengths have different polarity the Autosetting procedure breaks.
- After these steps, the program calculates the error function described above and searches for its minimum as a function of the SPV voltage. The light flux for each wavelength is set to obtain approximately the SPV voltage, which belongs to the minimum error.
- Using the amplitude, the program calculates the noise and sets the necessary averaging.

5.5 Iron Contamination Analysis by the μ -PCD and SPV Techniques

Both the μ -PCD and the SPV techniques measure general recombination parameters (lifetime and diffusion length) influenced by all the recombination centers, contaminants and defects. However, the iron in p-type material can be identified because of its unique behavior.

The most often detected contaminant in silicon is iron. In p-type silicon iron is present as an iron-boron (Fe-B) pair. This pair is a very effective recombination center. The Fe-B pair dissociates to interstitial iron (Fe_i) and boron after heat treatment (200°C for 10 minutes with fast quenching after the treatment) or high intensity illumination. This unique behavior offers a possibility for its identification because the recombination efficiency of the Fe-B pair and that of the Fe_i is different. At high injection level, which is used in the μ -PCD technique, the Fe-B pair causes lower lifetime than the Fe_i , so during the dissociation process the lifetime increases. This can be recorded by in situ lifetime measurement (see Figure 5-27). Once iron is separated from boron an association process takes place even at room temperature which results in decreasing lifetime. Its time constant can be some hours or days depending on the boron concentration (i.e. the resistivity) of the wafer under test.

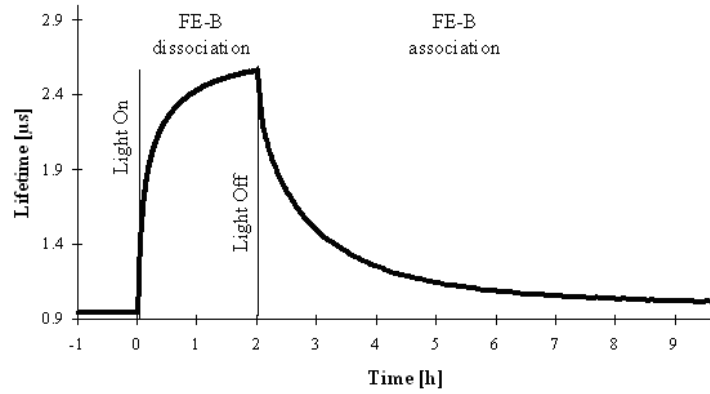


Figure 5-27: Fe-B dissociation and association over time

Figure 5-27 shows a typical sequence. The sample was illuminated with fast repeated laser pulses to break the Fe-B pairs. After switching on the laser, the lifetime increases very quickly at the beginning then more slowly to a saturation value. After switching off the illumination, the interstitial iron again forms a pair with boron, and the lifetime decreases with a given time constant to the original value. The Fe concentration can be calculated from the lifetime measured before and after the dissociation with the following formula:

$$N_{Fe} = C_{\mu-PCD} \cdot \left(\frac{1}{\tau_{before}} - \frac{1}{\tau_{after}} \right) \quad (5.9)$$

where $C_{\mu-PCD} = 3.4E13 \text{ } \mu\text{s}/\text{cm}^3$.

The above equation (5.9) is true in the case of both passivated and bare surfaces.

5.5.1 Fe-B Pair Dissociation and Association in a Single Point

On a passivated surface, the measured lifetime is actually the bulk lifetime, since the surface recombination does not play an important role. On a bare surface, the measured lifetime is usually much lower than the bulk lifetime, but equation (5.9) is true since the difference between the measured and the bulk lifetime does not change due to the dissociation process, so the second term of the following equation (5.10) disappears when substituting the bulk lifetime into the above formula (5.9).

$$\frac{1}{\tau_{meas}} = \frac{1}{\tau_{bulk}} + \frac{1}{\tau_{diff} + \tau_{surf}} \quad (5.10)$$

Therefore:

$$N_{Fe} = C_{\mu-PCD} \cdot \left(\frac{1}{\tau_{meas_bef.}} - \frac{1}{\tau_{meas_aft.}} \right) = C_{\mu-PCD} \cdot \left(\frac{1}{\tau_{bulk_bef.}} - \frac{1}{\tau_{bulk_aft.}} \right) \quad (5.11)$$

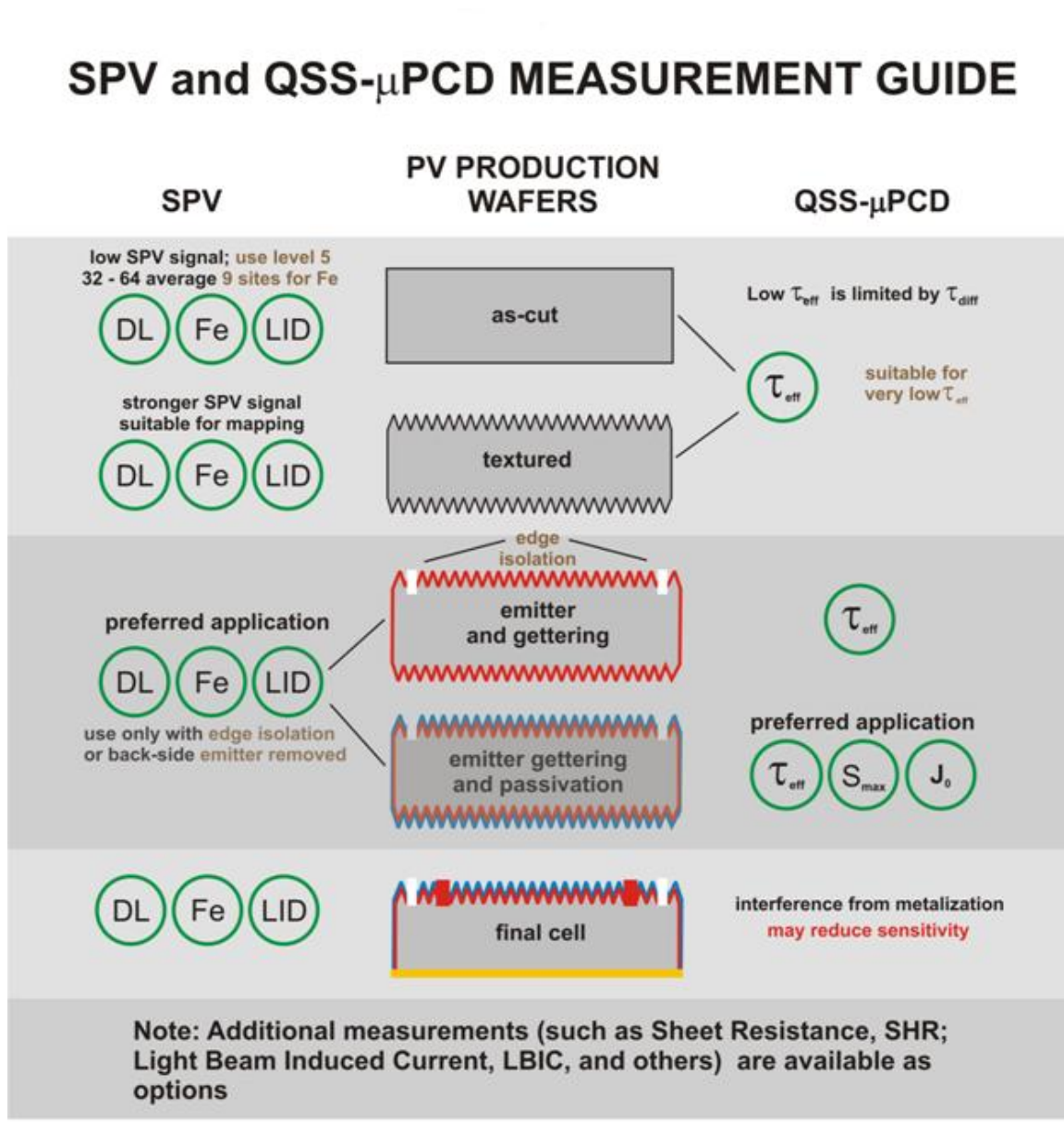
This way the iron concentration on bare wafers can also be measured, although the accuracy of the measurement is less than in case of the oxidized samples due to the small change of the measured lifetime.

The SPV measurement uses injection level orders of magnitude less than the μ -PCD technique. The recombination efficiency of the Fe-B pair is much lower while that of the iron interstitial is much higher at the low, SPV injection level than at the high, μ -PCD injection level; so the diffusion length (lifetime) decreases due to the dissociation. The equation to calculate the iron concentration from the SPV diffusion length measurements:

$$N_{Fe} = C_{SPV} \cdot \left(\frac{1}{L_{after}^2} - \frac{1}{L_{before}^2} \right), \quad (5.12)$$

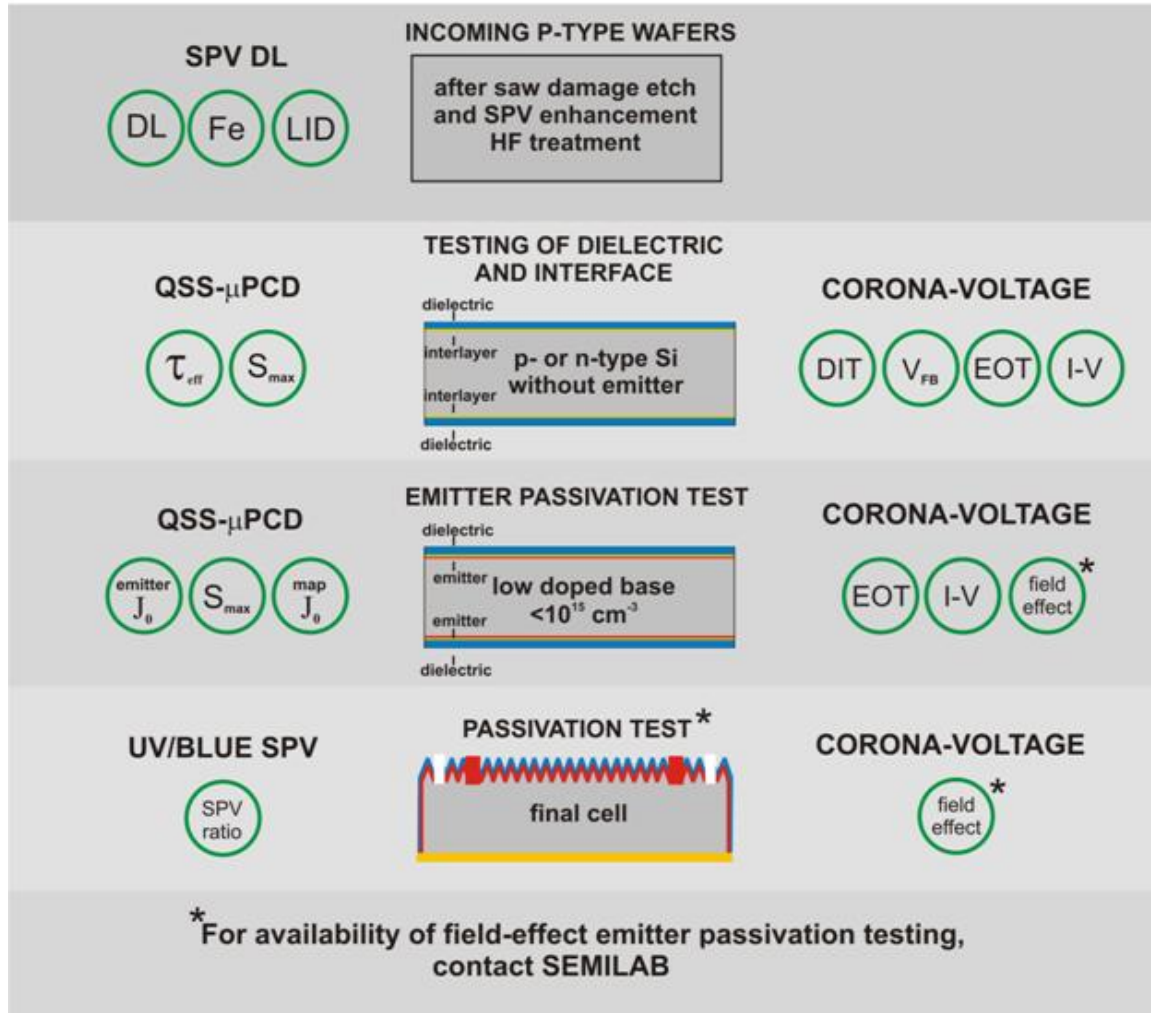
where $C_{SPV} = 1E16 \mu m^2/cm^3$.

5.6 SPV and QSS- μ PCD Measurement Guide



RECOMMENDED MEASUREMENTS

USING MONITOR WAFERS AND TEST STRUCTURES



5.7 V-Q Measurements

In the V-Q measurement method, a homogeneous charge is deposited on the entire surface by a corona wire. After the charging process, a Kelvin probe performs non-contact measurements. This versatile method can be used to determine a variety of parameters:

- Surface Potential, V_s
- Effective charge, Q_{eff}
- Oxide thickness, T_{ox}
- Flat band voltage, V_{fb}
- Density of interface states, D_{it}

- Total charge, Q_{tot}
- Mobile charge, Q_m
- Tunnel electric field, E_{tunnel}
- Plasma damage monitoring, PDM

5.7.1 Surface Potential

Surface potential (V_s) measurement is actually a capacitance measurement, where one plate of the capacitor is the sample while the other one is a vibrating plate placed in the probe head. This arrangement is known as a Kelvin probe.

The mean distance between the probe and the sample, the amplitude of vibration, and the frequency of the vibration are kept constant during the measurement. If there is a potential difference between the Kelvin probe and the sample surface, an AC current flows through the capacitance, which originates from the moving charge due to the existing electric field and the vibrating capacitance. By changing the DC voltage on the vibrating plate, a condition with no moving charge can be reached. In this case, the DC voltage on the vibrating plate directly gives the surface potential of the sample.

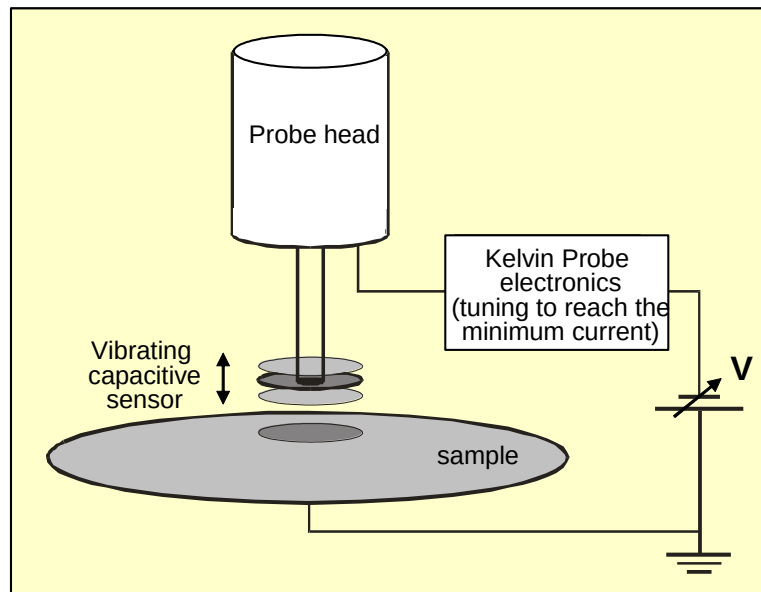


Figure 5-28: Kelvin probe measurement setup

5.7.2 Q_{eff}

During the measurement the sample is illuminated with high intensity light. The effective charge Q_{eff} gives the amount of charge caused by the difference in the band bending between the “dark” and “bright” Kelvin probe measurements. The effective charge is calculated with the following formula:

$$Q_{\text{eff}} = \sqrt{1.3\text{E}7 \cdot N_d \cdot \Delta U} \quad (5.13)$$

where N_d is the dopant concentration and ΔU is the difference between the dark and bright voltages.

5.7.3 V-Q (T_{ox} , V_{fb} , D_{it} , Q_{total})

The V-Q measurement method allows non-contact determination of the oxide thickness (T_{ox}), the flat-band voltage (V_{fb}), the density of interface states (D_{it}) and the total charge (Q_{tot}).

The measurement is based on charge deposition on the top of the oxide layer and a subsequent measurement of surface potential. This procedure is repeated with a predefined charge step and until a final charge value is reached. As a result there will be a series of surface potential value-charge amount value data pairs (V-Q curve). The C-V curve can be then easily be derived from the charge Q and the voltage U according to the equation $C=dQ/dU$.

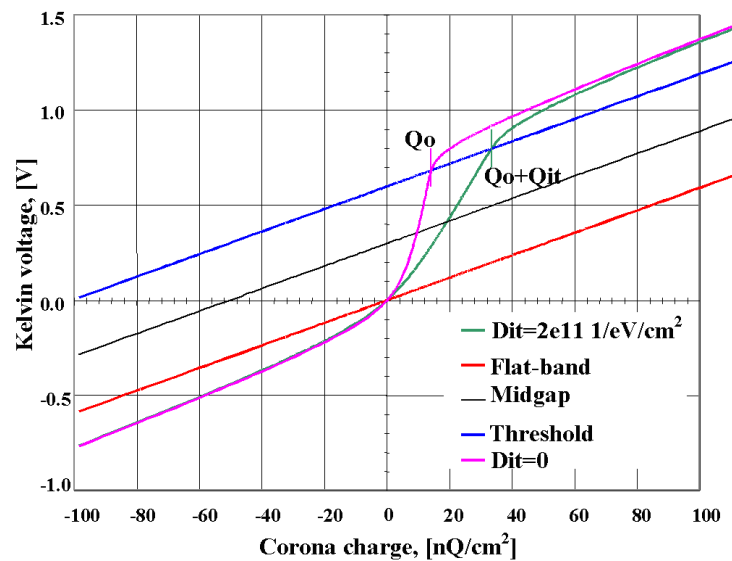


Figure 5-29: Evaluation of V-Q curve

Both ends of the V-Q curve approximate a straight line with the same slope. This slope is caused by charging the oxide layer itself (and not the interface states) i.e. the surface potential change is predominantly caused by the voltage change across the oxide layer. The oxide thickness, T_{ox} , can be calculated from this slope: oxide capacitance $C_{ox} = dQ/dU$, and $T_{ox} = \epsilon/C_{ox}$.

Fitting both ends of the curve results in parallel lines. The line parallel to these and placed at half distance between the fitted ones will be the midgap line. Determining the two lines separated by the Fermi level both upwards and downwards results in the threshold and the flatband lines (see Figure 5-29). The flatband voltage (V_{fb}) and threshold positions are simply the intersection of the appropriate line and the measured V - Q curve.

The interface state density qualifies the silicon-oxide interface. For the interface state density calculation a theoretical V-Q curve is calculated assuming $D_{it}=0$. After that D_{it} can be calculated from Q_0 and Q_0+Q_{it} (see Figure 5-29).

The image above (Figure 5-29) shows an ideal V-Q curve by means of the flat band condition occurs at 0 charge. In real measurements the flat band condition occurs at a given charge state, which means that there is a built-in charge caused by mobile ions, trapped charge in the oxide, fixed charge in the interface (contaminants) and trapped charge in the interface. (See Figure

5-30.) Because of this the total charge Q_{tot} is the sum of the above mentioned and the deposited charge on the surface.

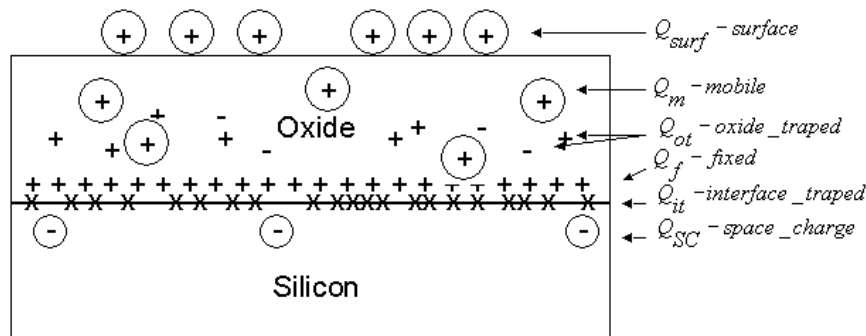


Figure 5-30: Total charge in the oxide

5.7.3.1 Dark and Bright Measurements

With the V-Q option, a homogeneous charge is deposited on the entire surface by a corona wire. The charge is varied in steps. After each charging step, a dark and a bright Kelvin probe measurement are performed. The charge and measurement steps are repeated to produce a dark and bright V-Q curve. The bright curve is generally used only for specific calculations:

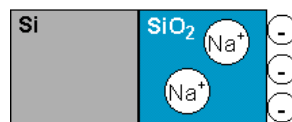
- T_{ox} is calculated from the slope of the bright V-Q curve in the accumulation region.
- V_{fb} is calculated from the surface potential curve by interpolation (intersection of dark and bright V-Q curves).

5.7.4 Q_m

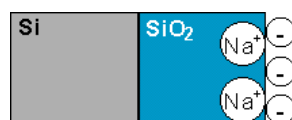
During the measurement of Q_m , mobile charge is moved across the oxide by using a heat cycle. An electric field is applied using corona charge deposition. The change in surface potential that occurs as a result of the heat cycle is directly proportional to the amount of mobile charge in the oxide.

The measurement steps are the following:

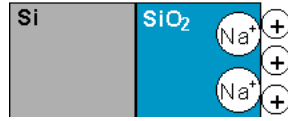
- To set the initial state: apply negative corona charge,



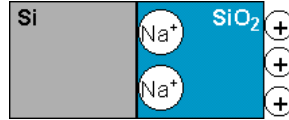
- and temperature stress (typically 4 minutes at 180°C).



- Apply positive corona charge:



- Measure the surface potential V_{ox1} to create a map of the wafer.
- Apply temperature stress (typically 4 minutes at 180°C).



- Measure the surface potential V_{ox2} to create a map of a wafer.

At each point of the map, mobile charge Q_m is calculated using the formula below:

$$Q_m = \epsilon \epsilon_{ox} \text{Const} \frac{\Delta V_{ox}}{T_{ox}} \quad (5.14)$$

5.7.5 E_{tunnel}

During tunnel electric field E_{tunnel} measurement, the wafer is charged in steps until the surface voltage saturates. This saturation is caused by the tunneling across the oxide layer induced by the high electric field of the deposited charge. The tunnel field can be calculated from the saturation voltage divided by the oxide thickness.

The tunnel electric field can be measured using E+, E- and E+/E- measurements. In the E+ measurement case, the wafer surface is charged positively, in the case of E- negatively, while E+/E- both positively and negatively.

5.7.6 Plasma Damage Monitoring

The PDM measurement method is used with a monitor wafer to check the functionality of a plasma etching tool.

If a sample with different oxide thickness than 1000 Å is used, the measured values are corrected to a 1000 Å case. This correction standardizes the measurement to allow easy comparison from sample to sample.

5.8 Junction Photovoltage Measurement

The measurement of the lateral spread of the junction photovoltage (JPV) makes it possible to get information on the sheet resistance of implanted or diffused layers of p-n junctions. Let we illuminate the wafer on a round spot with a given radius (R_0) and a chopping frequency f . In the simplest case we suppose a homogeneous illumination with a density of J . Similarly let we suppose that all of the generated electron-hole pairs will appear as photocurrent in the p-n junction, and only the reflection on the silicon surface decreases this photocurrent.

The generated electron hole pairs partly charges the depleted region capacitance, Cd , resulting in the appearance of a photovoltage, U , and at the same time according to the inhomogeneous photovoltage distribution this charge will begin to spread laterally with a current density of, I . Of course there will appear an ohmic leakage current as well across the depleted region, which can be characterized by its resistance Rd . At the edge of the wafer we may have an additional leakage to the ground which may be described by a resistance value of Re : $U = IRe$. In the absence of negligible leakage at the edge the spreading current will be zero there. In the simplest case we have a round shaped wafer, and we illuminate it just in its center. If the wafer is homogeneous the photovoltage and spreading current values will depend on the radius only.

According to the processes described above the change of the photovoltage and the spreading current as a function of the distance from the illumination center, r can be written by two differential equations as follows:

$$\left(\frac{\partial U}{\partial r}\right) = -I \left(\frac{R_s}{2r\pi}\right)$$

$$\left(\frac{\partial I}{\partial r}\right) = 2r\pi \left(J - \left(\frac{1}{R_d} + (i\omega)C_d\right)U \right),$$

where R_s is the spreading resistance value of the top layer while $\omega=2\pi f$. Outside the illuminated area ($r > R_0$), the value of J is zero.

It can be shown that if the sheet resistance of the substrate is not negligible we will measure the same value. In this case we will be able to measure a similar photovoltage at the bottom, with an amplitude inversely proportional to the sheet resistance of the substrate. If the top layer and the substrate have the same sheet resistance value the photovoltage values will be the same but with opposite polarity.

The solution of these equation above are Bessel functions with complex variable. It seemed to be easier to handle these equations with numerical methods where we described the photovoltage and the spreading current as:

$$U = U_\gamma + iU_i$$

$$I = I_\gamma + iI_i$$

In the calculation the step in the radius was always determined by the actual characteristic length of the spreading L_s , i.e. at high frequencies the step should be much finer than at low frequency values.

$$L_s = \sqrt{\frac{\left(\frac{2R_d}{R_s}\right)}{\sqrt{1 + (R_s C_d \omega)^2} + 1}}$$

At small enough frequency values:

$$L_s(0) = \sqrt{\frac{R_d}{R_s}}$$

At high enough frequency values:

$$S_s(\omega) = \sqrt{\frac{2}{R_s C_d \omega}}$$

In the illuminated area we used the Taylor series approximation for the photovoltage and the spreading current.

The photovoltage was sensed by two rings, an internal and an external one, for schematic see Figure 5-31. It proved to be necessary to calculate the real average across the width of the rings, instead of calculating the photovoltage at the mean radius of this curves. (At high frequencies there is a considerable change in the photovoltage values even in the mm-range.)

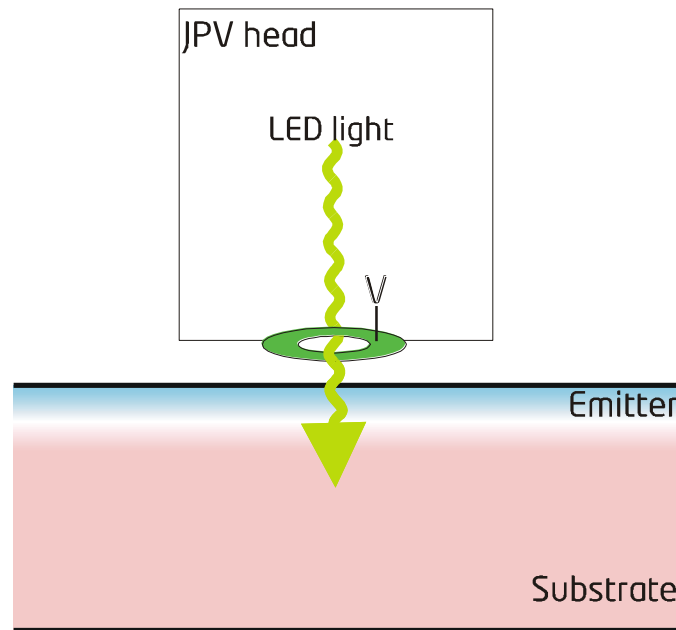


Figure 5-31: Schematic of JPV measurement method

5.8.1 Edge effect in the JPV measurement method

In JPV measurement method we use the surface photovoltage to determine the sheet resistivity, R_s . However, measuring close to the edge of the wafer, a significant deviation can develop from real values of the photovoltage. There are two effects which have to be considered, namely the finite size of measuring head and the finite size of wafer. The main effect is caused by the second one.

Effect of the finite wafer size: The phenomena can be easily understood by the following way. The laterally spreading current cannot go out from the wafer, at the edge it reflects back and causes an additional signal to the measured value. Its characteristic length is the same as mentioned above (L_s) and can reach a notable distance. When the measuring point is away from the middle of the round wafer, the equations with the required boundary conditions becomes difficult to solve both numerically and analytically. A solution can be given by using the same method as in the conventional electromagnetic theories, when it is supposed there is a mirror illumination of the original one in such a way that the boundary conditions are satisfied. The measured values of the photovoltage arise from both the original and mirror illumination. This method needs solutions of the equations, which are Bessel-functions with complex argument as

it was mentioned before. These functions have some difficulty to handle, however taking some considerations they can be expressed in usable forms. Making this one can get a theoretical description of the effect originating from the finite wafer size. This solution depends on the physical parameters R_s , R_d and C_d and with a fit procedure via these parameters finite wafer size effect can be well eliminated.

Effect of the finite size of the measuring head: The junction photovoltage is measured by a capacitive ring with a size about 8 mm diameter. The illumination is at the center of the ring, so close enough to edge of the wafer only a part of the sensor measures the signal from the sample. To compensate this, a geometric factor can be applied to the measured data.

6. Quick Start Guide

The WT-2000 system is a versatile measuring system with many different capabilities. The user-friendly software permits customizing nearly every aspect of the measurement for different sample types. Each measurement, however, consists of a basic sequence of steps that is followed. Once the sequence is mastered, a user will find it simple to measure many types of samples.

This chapter provides a brief overview of the steps required to perform a measurement. Detailed information on each step is provided in Chapter 12.

Measurements can be performed with several levels of operator intervention. A fully manual measurement requires the operator to perform all setup before the measurement, and to manually save and print the results. The setup can be automated using a recipe to store a set of measurement conditions. Additional operator actions—such as saving and printing the results—can be performed without operator intervention using the “automatic measurement” capability.

This Quick Start Guide outlines the steps required to perform the following types of measurements:

- Fully manual measurements (section 6.1)
- Automatic measurements with manual setup (section 6.2)
- Automatic measurements with recipes (section 6.3)

Additionally instructions are provided for turning on the power to the WT-2000 system (section 6.4) as well as instructions for shutting off the power (section 6.5).

6.1 Performing a Manual Measurement

This section describes the method to perform a fully manual measurement in which the operator sets all parameters and manually performs all procedures.

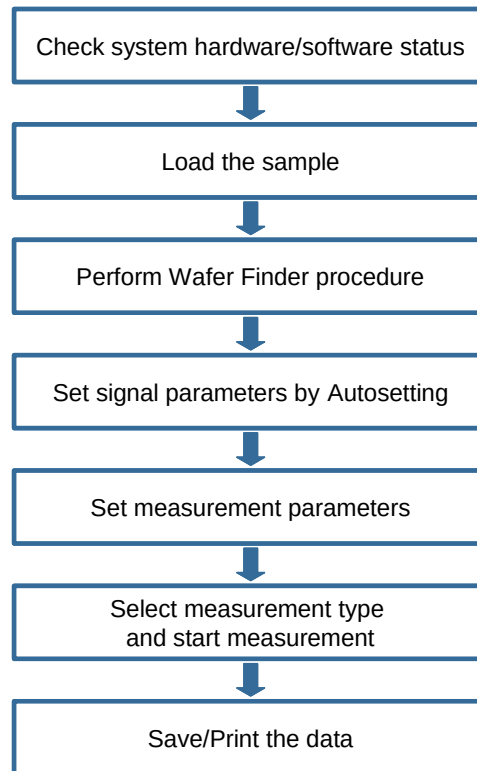


Figure 6-1: Steps to perform a manual measurement

6.1.1 Check System Status

Before starting the measurement, check to see that the hardware and software are ready. For complete details, see section 12.1.

- [1.] Initialize the system if required, using the **Measure | Initialize** menu item.
- [2.] Select the measurement mode using **Measure | Mode** or the drop-down list in the *Measurement Toolbar*.

6.1.2 Load the Sample

To load the sample manually, follow the steps below. For complete details, see section 12.2.1.

- [1.] Check the *Wafer Shape* in the *Measurement Options* window.
- [2.] Set the sample size using **Measure | Wafer Size** or the drop-down list in the *Measurement Toolbar*.
- [3.] Click the **Load Sample** button . Open the lid, then place the sample on the stage.

! CAUTION: Please, place the wafer in the middle of the stage as accurately as possible! If a hot-chuck is installed in the WT-2000, placing the wafer improperly can make the wafer stuck in the hot-chuck. Semilab refuses responsibility for damage of the wafer or the WT-2000 that comes from inaccurate positioning of the wafer during manual loading.

- [4.] Click **OK**, then close the lid.

To load the sample using the optional indexer, follow the steps below. For complete details, see section 12.2.2.

- [1.] Check to see that the lid is closed.
- [2.] Set the sample size using **Measure | Wafer Size** or the drop-down list in the *Measurement Toolbar*.
- [3.] Install the cassette on the indexer.
- [4.] Click the **Load from Cassette** button .
- [5.] In the *Load Sample* window, select the wafer to be loaded.
- [6.] Loading will be performed automatically.

6.1.3 Perform Wafer Finder Procedure

After the sample is placed on the stage, complete the Wafer Finder procedure so that the system detects the placement of the sample on the stage. A complete description of this procedure is in section 12.3.

- [1.] Perform the procedure by selecting the **Measure | Wafer Finder** menu item.

6.1.4 Set Signal Parameters

Signal parameters are those settings which optimize the measurement signal. Most measurement methods use the Autosetting procedure for this purpose. Signal parameter setting is discussed in detail in section 12.4.

For μ -PCD and C-PCD:

- [1.] Perform the Autosetting procedure by clicking the  button in the *Measurement Toolbar*.

For SPV:

- [1.] Enter the oxide thickness value in the *Input List* tab.
- [2.] Perform the Autosetting procedure by clicking the  button in the *Measurement Toolbar*.

For V-Q:

No autosetting is required.

For JPV:

No autosetting is required, but the calibration file must be loaded.

- [1.] In the *SHR Recorder* window, use the **Functions | Load Calib. File** menu item to load the correct calibration file.

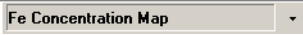
6.1.5 Set Measurement Parameters

Measurement parameters describe the sample being measured, including the date of the measurement, operator name, size of the sample, and similar identifying information. See section 12.5 for more information.

- [1.] Enter the measurement parameters in the *Input List* tab of the *Measurement Information* window.

6.1.6 Set Measurement Type and Start Measurement

The user must select the measurement pattern and measurement type (if applicable) before starting the measurement. Brief descriptions of these options are given below. See section 12.6 for more information.

- [1.] If linear corona charge is to be used before the μ -PCD or SPV measurement, deposit the charge using the **Measure | Corona** menu before proceeding further.
- [2.] For an x-point measurement:
 - [2.1.] Select the number of points using the drop-down menu next to the **X-Point** button  on the *Measurement Toolbar*.
 - [2.2.] Perform the measurement by clicking the **X-Point** button.
- [3.] For a map:
 - [3.1.] Start the measurement by pressing the **Map/Resume** button  on the *Measurement Toolbar*.
- [4.] For Fe concentration measurements:
 - [4.1.] Enable the Fe concentration method by using the *Fe* tab in the *Measurement Options* window, or by selecting the **Fe** sub-measurement mode from the drop-down list  in the *Measurement Toolbar*.
 - [4.2.] Select the number of points from the drop-down arrow to the right of the **Auto Measure** button  on the *Measurement Toolbar*.
 - [4.3.] Start the measurement by clicking the **Auto Measure** button .

Additional procedures for measuring mobile charge (using V-Q) are provided in section 12.6.

6.1.7 Save or Print the Data

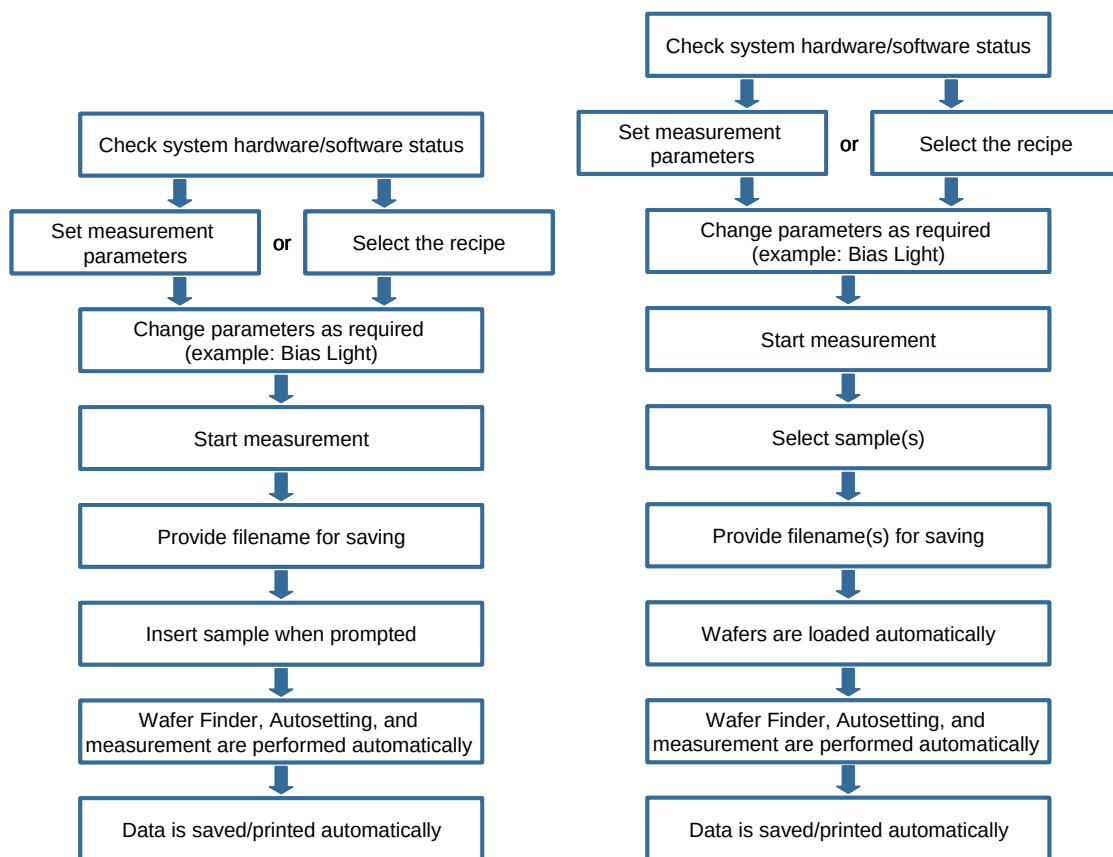
- [1.] Use **File | Save** to save the active data (see section 9.8 for more information).
- [2.] Save linked files if desired (see section 9.9 for more information).

- [3.] Print data as described in section 9.10.

6.2 Performing Automatic Measurements

Automatic measurements allow pre-selection of certain operator actions—such as saving and printing results—which are then performed automatically, without further operator intervention, as part of the measurement.

Automatic measurements may be performed with manual parameter setting or with recipes that use a previously saved set of parameters. (Recipe creation is described in section 12.9.1.) The WT-2000 system has a **Measure Recipes** feature which automatically begins the measurement after specifying the recipe. However, sometimes it is necessary to modify the signal parameters before beginning the measurement. This procedure is shown in Figure 6-2. See section 12.10 for more information.



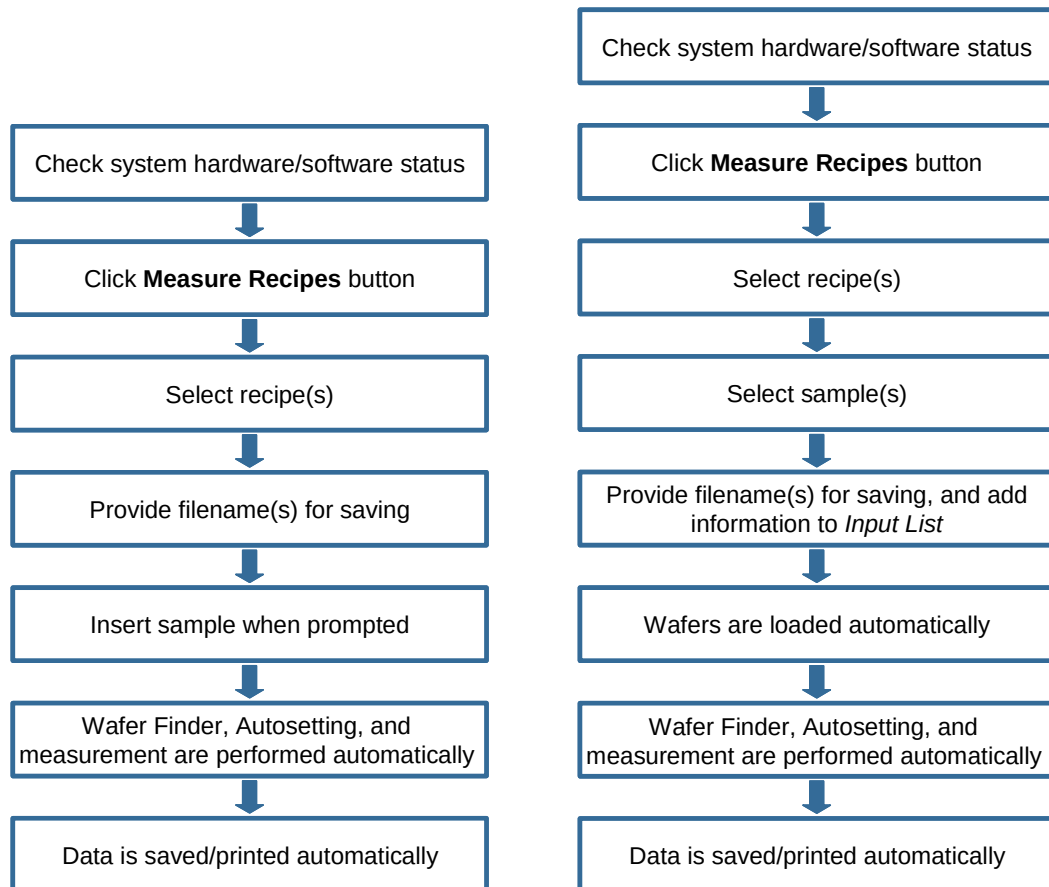
**Figure 6-2: Steps for automatic measurement
(for manual load and automatic wafer load)**

- [1.] Check the system status as described in section 6.1.1.
- [2.] Set the measurement parameters in the *Measurement Options* window. See section 12.10 for details. Alternately, load a recipe to use a previously-saved set of parameters.

- [2.1.] If recipes are stored in the Wintau32 default location, choose a recipe from the **Recipe List** drop-down menu  in the *Buttons area*.
- [2.2.] To select a recipe from a different location, use the **Measure | Recipe** menu.
- [3.] If a special parameter must be set (such as bias light intensity), set it before continuing.
- [4.] Start the measurement by clicking on the **Auto Measure**  button.
- [5.] For systems with manual loading:
 - [5.1.] When prompted, provide filename(s) for saving the data.
 - [5.2.] Insert the sample when prompted.
- [6.] For systems with automatic loading:
 - [6.1.] When prompted, select wafer(s) for measurement.
 - [6.2.] Provide filename(s) for saving the data.
 - [6.3.] Verify *Auto* settings.
 - [6.4.] Wafers will be automatically loaded.
- [7.] The Wafer Finder and Autosetting procedures will be performed automatically if selected in the *Measurement Options* window.
- [8.] Data is saved and printed automatically if selected in the *Measurement Options* window.

6.3 Automatic Measurements with Recipes

If signal parameter modification is not required, the fully automatic **Measure Recipes** feature begins the measurement after the recipe is specified. This feature also allows multiple recipes to be used on each sample. For systems with automatic wafer loading, it is possible to specify different recipes for different wafers in the cassette. The procedure is shown in Figure 6-3 and described below. For more information, see section 12.11.



**Figure 6-3: Steps for automatic measurement with recipe
(for manual load and automatic wafer load)**

- [1.] Check the system status as described in section 6.1.1.
- [2.] Start the measurement by clicking on the **Measure Recipes**  button.
- [3.] Select the recipe(s) to be used for the measurement.
- [4.] For systems with manual loading:
 - [4.1.] When prompted, provide filename(s) for saving the data.
 - [4.2.] Insert the sample when prompted.
- [5.] For systems with automatic loading:
 - [5.1.] When prompted, select wafer(s) for measurement.
 - [5.2.] Provide filename(s) for saving the data.
 - [5.3.] Enter information in the *Input List*.
 - [5.4.] Wafers will be automatically loaded.
- [6.] The Wafer Finder and Autosetting procedures will be performed automatically if selected in the *Measurement Options* window.

- [7.] Data is saved and printed automatically if selected in the *Measurement Options* window.

6.4 Starting the WT-2000

The procedure below assumes that the system is sitting idle with all power off. If the system has been recently used, perform only those checks that are appropriate.

- [1.] Switch on the main power of the computers (see section 4.7).
 [2.] Switch on the vacuum pump.
 [3.] Verify that there is no sample remaining on the stage.

! CAUTION: Any sample left on the stage can cause damage either in the system or in the sample itself.

- [4.] Close the WT-2000 top cover.
 [5.] Start the Wintau32 Operating Software by double clicking the appropriate icon .

✓ **NOTE:** The operating software can also be started by running the Wintau32.exe file from the command prompt with additional startup options.

- [6.] After the Wintau32 software is loaded, the *Main Window* is displayed (Figure 6-4).

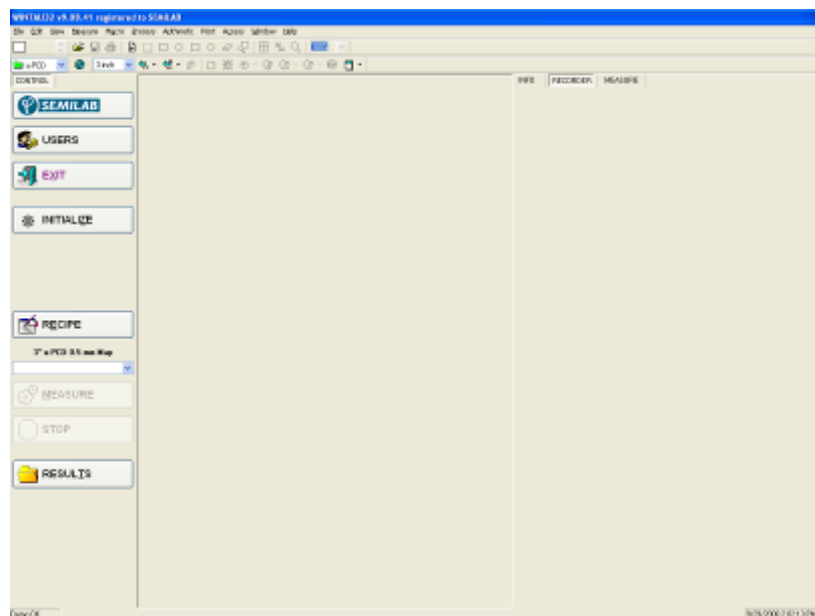


Figure 6-4: The Main Window

✓ **NOTE:** If the software is used for the first time, check that the Use measurement capabilities check box is checked in the *Help | Register* menu. See section 15.1.6.

- [7.] Wait 10 minutes to allow the measurement electronics to warm up.
- [8.] Activate the **Initialize** function in the **Measure** menu. During the initialization procedure the stage, the bridge and the measuring head make a series of movements. This procedure initializes and checks the motors and mechanics.

✓ **NOTE:** *Initialization is necessary only once after startup of the system. Even if you exit the Wintau software and start it again, no initialization is needed again if the WSOS2 system (on the computer with DOS operating system) was not switched off.*

- [9.] Depending on the setup of your software, the *Measurement Options* window may appear (Figure 6-5). This window displays the head heights for your WT-2000 system. Click **OK** to close the window.

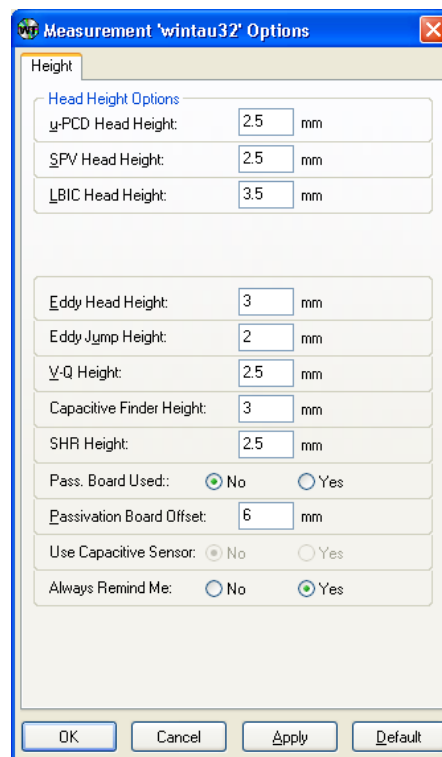


Figure 6-5: The *Measurement Options* window

6.5 Shutting Down the WT-2000

- [1.] Before shutting down the WT-2000 system, remove any sample on the stage.
- [2.] Shut down the Wintau32 software by clicking the **Exit** button  or by using the **File | Exit** menu item.
- [3.] Shut down the Windows operating system by using the Windows **Shut Down** feature.

! CAUTION: To avoid any unexpected situation always quit the software and the operating system (shut down Windows) before switching off the computer.

- [4.] Switch off the vacuum pump.
- [5.] If the User computer does not shut down completely after Windows has ended, turn it off by pressing the power switch.
- [6.] Press the power switch on the Measuring computer to turn it off (see section 4.7).

7. Using Wintau32 Software

This chapter provides a short introduction to the user interface of the Wintau32 software. Sections 7.1 to 7.4 describe the elements of the *Main Window*. The chapter also discusses the following topics:

- Data Types (section 7.6)
- Software Modes (section 7.7)
- Setting User Access Levels (section 7.8)
- Getting Help (section 7.9)

Table 7-1 Organization of Wintau32 Software chapter

Section name	Section
The <i>Main Window</i>	7.1
Using the Menus	7.2
The Toolbar	7.3
The Measurement Toolbar	7.4
Buttons area	7.5
Data Types	7.6
Software Modes	7.7
Setting User Access Levels	7.8
Getting Help	7.9

A new interface of the software is made to make using the Wintau32 more convenient. The change is valid for versions 9.00 and higher. All functions can be reached from both the old and the new interfaces. This manual describes the new interface.

7.1 The Main Window

When you start the Wintau32 software, the *Main Window* appears (Figure 7-1).

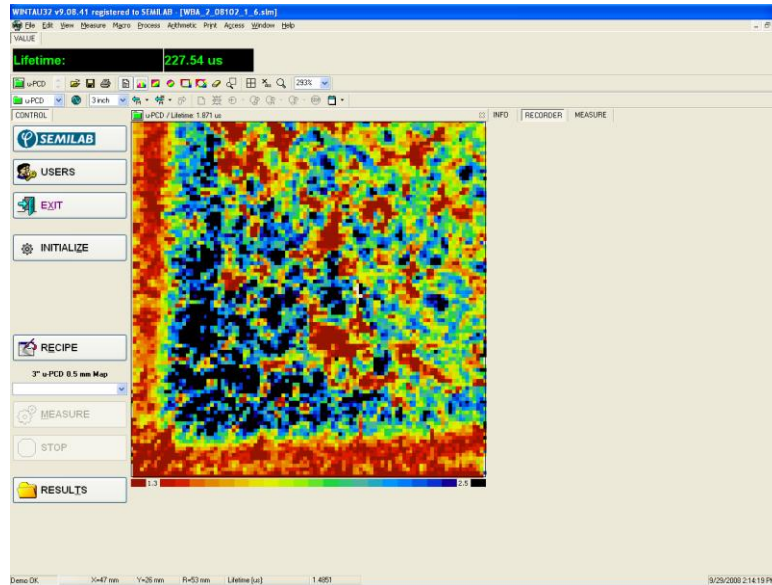


Figure 7-1: The Wintau32 Main Window

The window has six parts:

- *Menu Bar* at the top of the window just below the window title
- *Recorder Value Bar* below the *Menu Bar*
- *Toolbars* below the *Menu Bar*
- *Fast Access Button* area at the left side of the screen
- *Status Bar* at the bottom of the main window
- The working area between the *Toolbar* and the *Status Bar*.

✓ **NOTE:** *The software screens may look different from the figures, depending on the options installed in your WT-2000 system.*

7.1.1 The Menu Bar

The *Menu Bar* at the top of the *Main Window* allows access to various functions for file management, data review, and measurement.

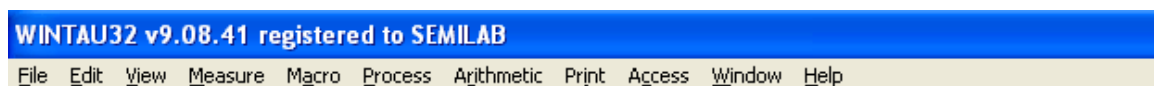


Figure 7-2: The Menu Bar

The software version is shown in the *Menu Bar*, as well as the name of the licensed user.

The different menu items are described in section 7.2.

7.1.2 The Toolbars

There are two toolbars in the *Main Window*: the Wintau32 operational *Toolbar* and the *Measurement Toolbar*.



Figure 7-3: The Wintau32 Toolbar and Measurement Toolbar

The *Toolbar* has icons for performing useful functions for file management and map display. The *Measurement Toolbar* has icons that correspond to many items in the **Measure** menu.

Toolbars can be shown or hidden by the **View | Toolbar** menu item. Clicking on a *Toolbar* with the right mouse button and selecting **Hide** hides the *Toolbar* also.

The *Toolbar* icons are described in sections 7.3 and 7.4.

7.1.3 The Status Bar

The *Status Bar* displays the process indicator during processes such as file loading to show the percentage that is already done of the running process. Additionally, it shows the location of the selected point of the active map (see Figure 7-4).



Figure 7-4: Status Bar

7.1.4 The Recorder Value Bar

The *Recorder Value Bar* displays the measurement results at the location specified in the *Status Bar*. (See Figure 7-5.)



Figure 7-5: Recorder Value Bar

7.2 Using the Menus

All the options in the *Main Window* menu are listed in this section. Each menu item has several options below it.

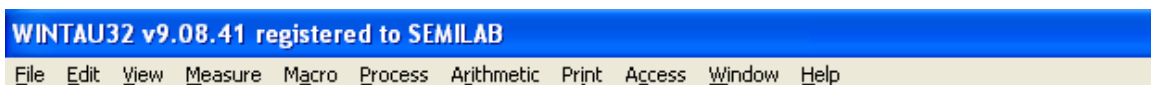


Figure 7-6: The menu bar

<i>File</i>	The <i>File</i> menus are used to load, save or display data files in many formats.
<i>Edit</i>	<i>Edit</i> menu options allow operations on displayed data. .
<i>View</i>	<i>View</i> options change the appearance of the displayed data.
<i>Measure</i>	The <i>Measure</i> menu contains many items that control the measurement to be performed.
<i>Macro</i>	The macro language has been implemented in Wintau32 to give users a wider range of control over software capabilities.
<i>Process</i>	The <i>Process</i> menu is used to change map images. Image processing allows subtle data variations to be apparent. .
<i>Arithmetic</i>	Under the <i>Arithmetic</i> menu you can do several kinds of calculations on a map.
<i>Print</i>	Use the <i>Print</i> menu items to print the active data files.
<i>Access</i>	The <i>Access</i> menu items are used to manage users and their access levels.
<i>Window</i>	The <i>Window</i> menu listings control the appearance and placement of pop-up (child) windows.
<i>Help</i>	The options in the <i>Help</i> menu are used to display the online help and to register the software.

7.3 The Toolbar

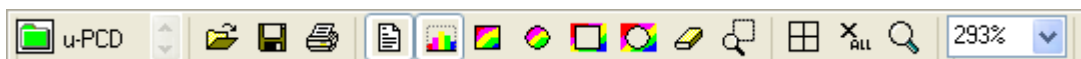


Figure 7-7: The Toolbar

The *Toolbar* has buttons for performing useful functions for file management and map display. If the cursor is moved over a button, a tool-tip label will display the name of the button.



When displaying a linked data file in the *Main Window*, switches the display between the different linked images.



Opens a previously-saved file containing measurement data.



Saves the active measurement data.



Prints the active measurement data.



Causes the *Measurement Information* window to appear or disappear. See section 9.4.

 Displays or hides the *Distribution* window. See section 9.7.

 Inclusive Rectangle Mask. See section 9.5 for a description of masks.

 Inclusive Circle Mask. See section 9.5 for a description of masks.

 Exclusive Rectangle Mask. See section 9.5 for a description of masks.

 Exclusive Circle Mask. See section 9.5 for a description of masks.

 The Erase button allows erasing points from the selected map by marking them with a left mouse button click. See section 9.5 for an explanation of this button.

 This is the Instant Zoom. The mouse cursor changes its shape to a magnifier glass, and when you click on the map a rectangular area appears with the magnified part of the map. Keeping the mouse button pressed, you can scan over the map area.

 If you have more maps loaded, pressing this button causes the software to resize all windows and arranges them in a grid (“tiled”).

 If you have more map windows loaded then you can close all of them by pressing this button.

 After pressing this button you can zoom in by clicking on the map with the left mouse button and zoom out using the right mouse button while the mouse cursor is on the map area.

 You can use the predefined zoom scales by using the drop-down list or by typing the desired value in the box. The zooming factor can be set from 25% to 999%.

7.4 The Measurement Toolbar

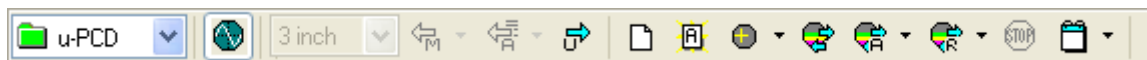
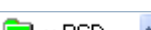
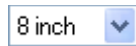


Figure 7-8: The *Measurement* Toolbar

The drop-down lists and buttons of the *Measurement Toolbar* correspond to the appropriate menu items in the *Measure* menu except the Stop button.

If the cursor is moved over a button, a pop-up label will display the name of the button.

	Main Measuring Mode (μ -PCD, LBIC, EDDY, etc.)
	Recipe List (displaying all recipes stored in the default directory)
	Recorder (displays or hides the <i>Recorder</i> window for the selected measurement method)



Wafer Size



Load Sample (including automatic initialization). The drop-down arrow next to the button  allows selection of the wafer shape and the installed stage.



Load from Cassette (including automatic initialization). This button appears only for systems with automatic sample loading.



Unload Sample



Clicking New will clear all results from the previous measurement without confirmation message.



Performs the Autosetting procedure before the measurement.



X-Point Measurement (the drop-down arrow next to the button allows choosing the number of points)



Map/Resume



Auto Measure. Performs automatic measurements (section 12.9). This button has multiple functions, depending on the measurement mode. The drop-down arrow next to this button  permits other measurement patterns.

- *Repetitive Mode* causes the sample to be measured repeatedly until cancelled.
- *Selection* is available on systems with automatic loading capability. When this option is selected, the *Select Samples* window will appear to allow the user to select the wafers to be measured.
- *Single Sample* is available on systems with automatic loading capability. When this option is selected, the user will be limited to selecting only one wafer in the *Select Samples* window.



Measure Recipe(s). Use this button to select and run previously-saved recipes. The drop-down arrow next to this button  sets the measurement conditions.

- *Repetitive Mode* allows the user to select another recipe immediately after the current one is completed.
- If *Recipe Confirmation* is selected, the *Measurement Options* window will appear before starting the measurement, to allow the user to review (and possibly change) the recipe settings.

- If *Keep Original Recipe(s)* is selected, the recipe will run with any changes made during Recipe Confirmation. However, the original recipe will not be changed.



Stops the measurement (same as pressing the <Esc> key). The measurement can be continued by pressing the Resume button or activating the *Measure | Resume* menu item.



Opens the *Measurement Options* window. Measurement Options can be changed only by engineers.

The *Measurement Toolbar* can be shown or hidden by the *View | Measurement Toolbar* menu item. Clicking on the *Measurement Toolbar* with the right mouse button and selecting *Hide* also hides the *Measurement Toolbar*.

7.5 Buttons area

This area contains the most frequently used functions during operation of the WT-2000. It includes the following buttons:



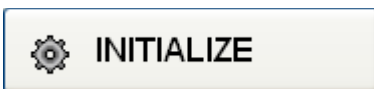
Calls the *About Wintau32* window.



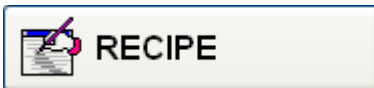
- *USERS* calls the User Administration window, if no user is defined yet.
- *LOG-IN* is possible if one or more users are defined, and no one is logged in.
- *LOG-OFF* makes the user log off if the user is logged in.



Closes the Wintau32 software



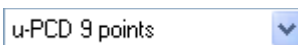
Runs system initialization.



Calls the *Select Recipe* window. Recipes can be selected for each measurement mode from the c:\wintau32 folder. If no recipe is selected, the software offers to create a new recipe.

3" Fe Concentration 0.5 mm Map

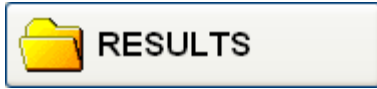
Sub Measurement Mode (the drop-down arrow allows selection from the different measurement options that are available for each mode)



Recipe List displays all recipes stored in the default directory



Stops the measurement (same as pressing the <Esc> key). The measurement can be continued by pressing the Resume button or activating the *Measure | Resume* menu item.



Semilab map files can be opened by this button

7.6 Data Types

Wintau has a file format that supports several kinds of data. It may contain μ -PCD, SPV or other types of measurements. This information is displayed in the *Toolbar*, and in all the lists (such as *Open File Window*) that display several data files. There is a little icon and the type of the map (μ -PCD, SPV etc.) on the left side of the *Toolbar*. The color of the icon is different depending on the origin of the map. The data type can have the following values:

	μ -PCD	Lifetime
	C-PCD	Charge-PCD lifetime map
	TDL	Diffusion Length calculated from Lifetime
	TDL	Diffusion Length calculated from SPV maps
	FEB	Fe-B concentration calculated from Lifetime
	FEB	Fe-B concentration calculated from SPV diffusion length maps
	LNK	Linked file (explained in section 9.9)
	SRV	Surface Recombination Velocity
	SPV	Surface PhotoVoltage
	LBIC	Light Beam Induced Current map
	EDDY	Resistivity map
	V-Q	V-Q map
	SHR-JPV	Sheet resistance and implant dose monitoring

7.7 Software Modes

The software is operated in two different modes, Data Evaluation Mode and Measurement Mode.

7.7.1 Data Evaluation Mode

When reviewing previous measurements, the Wintau32 software is used in Data Evaluation Mode. In this operation mode, the software can load and save maps, evaluate data and print the maps, but performing measurements is not possible.

7.7.2 Measurement Mode

The Measurement Mode of the Wintau software has additional features for performing measurements, setting measuring parameters etc. To be in Measurement Mode, the appropriate hardware must be installed: the WT-2000 main unit and a dual computer setup.

✓ **NOTE:** *The Measurement Mode can be set in the Register Wintau window (Help | Register menu item) by checking the Use measurement capabilities if available box if the registration is valid.*

The **Measure** menu and the *Measurement Toolbar* are only available in Measurement Mode. Additionally, certain menu items are not available unless the software is in Measurement Mode.

7.8 Setting User Access Levels

Three different access levels are defined in Wintau32 software: Engineer level, Operator level and User level.

7.8.1 Access Level Restrictions

Engineer level access gives full functionality of Wintau32. Additionally, Engineer-level users have the ability to customize user input fields (see section 7.8.5).

Operator-level users can make measurements and review data. Operator level has the following restrictions:

- Cannot modify any of the measurement options
- Has no access to **Macro**, **Process** and **Arithmetic** menus
- **Hardware Test**, **System Diagnostics** and **Remote** options are unavailable
- **User Administration** is unavailable
- Cannot register software

User-level users can review data. User level access has restrictions besides the Operator level restrictions:

- No measurement capability.

7.8.2 Defining Users

When the WT-2000 is first installed, the software starts with Engineer access level privileges and no defined users.

! CAUTION: An Engineer level access user should be defined first, because defining other access level user and closing user administration window will make the User Administration menu item unavailable. In such a case there will be no user to have the access to the User Administration menu and there will be no possibility to use software at Engineer level.

To define a user, select the **User Administration** menu item. Clicking on it the *User Administration* window appears (Figure 7-9).

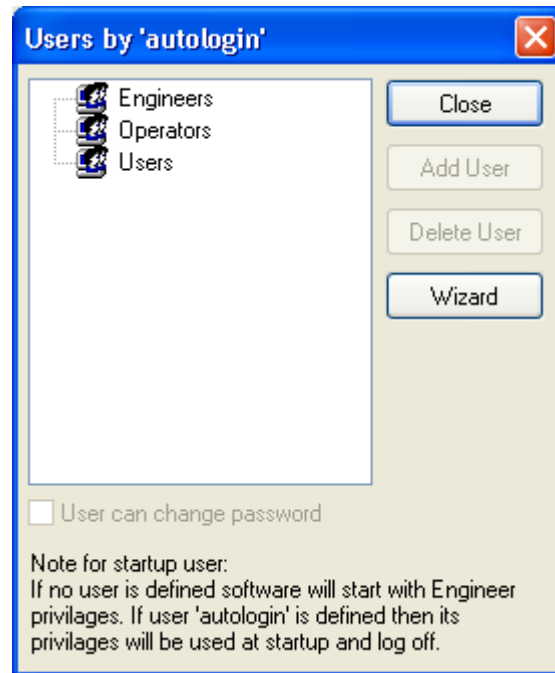


Figure 7-9: User Administration window

Select the desired access level by clicking on its name, then click the **Add User** button. A window appears (Figure 7-10) requiring the user to define a user name and a password.

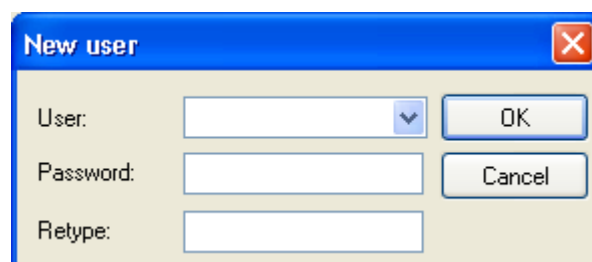


Figure 7-10: Creating a new user

Enter the *User* and *Password* if desired (and *Retype* the password), then click **OK**. If there is no password defined, then login only requires entering the user name.

✓ NOTE: If a different level of access other than Engineer is defined first, a warning message appears. "No Engineer user is defined yet!"

7.8.2.1 Using the Wizard

When the WT-2000 is first installed, a Wizard can automatically create two users: an Engineer-level user (*engineer*) and an Operator-level user (with autologin). In the *User Administration* window, click the **Wizard** button. A pop-up window (Figure 7-11) warns that all existing users will be deleted.

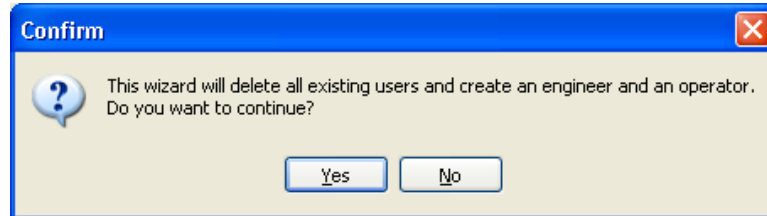


Figure 7-11: The Wizard Confirm window

After clicking **Yes**, the *New User* window appears with the username *engineer*. Enter the password for this user. The Operator-level user will be created with no password. The list of users created by the Wizard is shown in Figure 7-12.

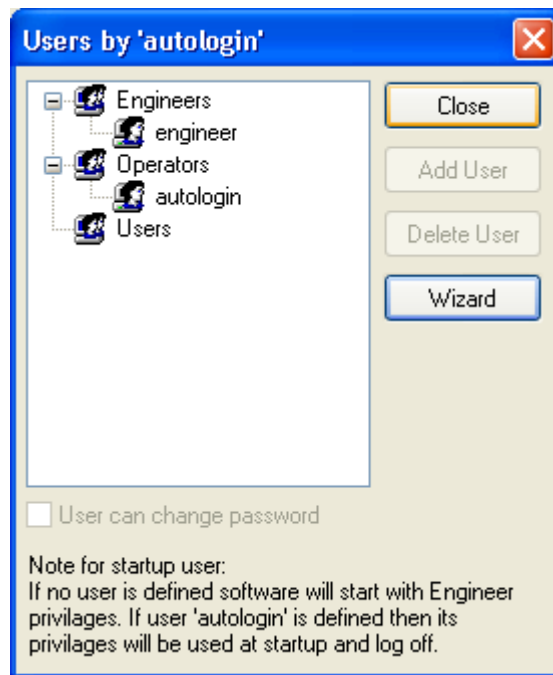


Figure 7-12: Users created by the Wizard

7.8.3 Log In and Log Out

After defining users, Wintau32 will start with a login box forcing the user entering a user name and a password. In case of login failure, the software cannot start.

It is possible to define a default access level for Wintau32. Creating a user with the username *autologin* (such as shown in Figure 7-12) will cause Wintau32 to start with the access level defined for the *autologin* user. Under these conditions, Wintau32 will not ask for a password.

Once the Wintau32 software has started, use the **Log In as Different User** menu item to change to a different access level.

Logging out is possible in **Log Off** menu.

7.8.4 User Management

Users can be deleted with the **Delete User** button in the *User Administration* window. Select the user to be deleted, then click **Delete User**.

The password of the current user can be changed in the **Change Password** menu.

7.8.5 Customizing the User Input Fields

Users with Engineer-level access can add or remove *User Input* fields in the *INFO* window by right clicking on the tab name in the bottom part of this window (where measurement results are displayed). See Figure 7-13.

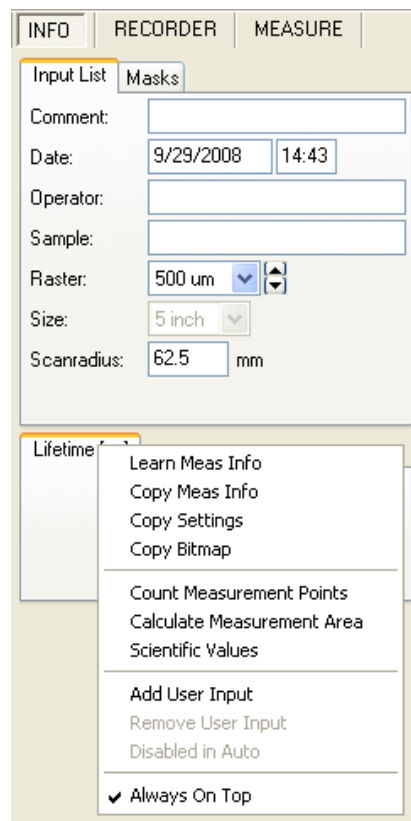


Figure 7-13: Right mouse click on *Input List* window

After selecting the **Add User Input** menu item, you are prompted to enter the name of the user-defined parameter. Four input fields can be added in this way.

By right-clicking on the name of an input field, it can be set to read-only. The name of the field can also be changed.

The set input fields are used at all the measurements and saved with map files until they are removed by a user with Engineer-level access.

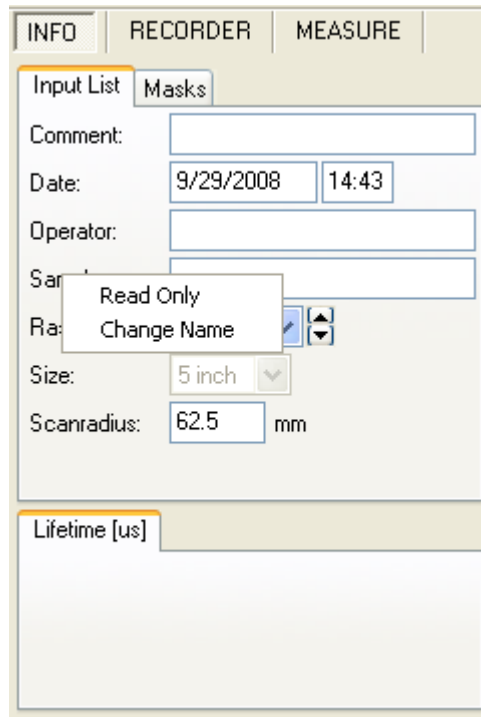


Figure 7-14: Changing the input field

7.9 Getting Help

To view the online help, select **Help | Help**. The *Wintau Help* will appear. This file contains information about the menu options in the Wintau32 software.

This manual is also available in a searchable PDF file with hyperlinked references.

If you are unable to find the answer to your questions in the online help or the manual, Semilab is happy to answer questions by telephone or e-mail. Please see the contact information in section 1.3.

8. Wintau32 Software Menus

This chapter includes a listing of all menus in the Wintau32 *Menu Bar* (see Figure 8-1). Detailed information on using the menus can be found in later chapters, as indicated by cross-references for the different menu commands.

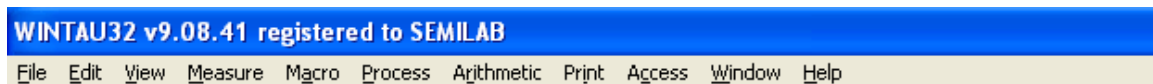


Figure 8-1: The Wintau32 Menu Bar

The chapter is organized according to the order in the *Menu Bar*.

**Table 8-1
Organization of File Menus**

Menu Title	Section
<i>File</i> Menu	8.1
<i>Edit</i> Menu	8.2
<i>View</i> Menu	8.3
<i>Measure</i> Menu	8.4
<i>Macro</i> Menu	8.5
<i>Process</i> Menu	8.6
<i>Arithmetic</i> Menu	8.7
<i>Print</i> Menu	8.8
<i>Access</i> Menu	8.9
<i>Window</i> Menu	8.10
<i>Help</i> Menu	8.11

8.1 File Menu

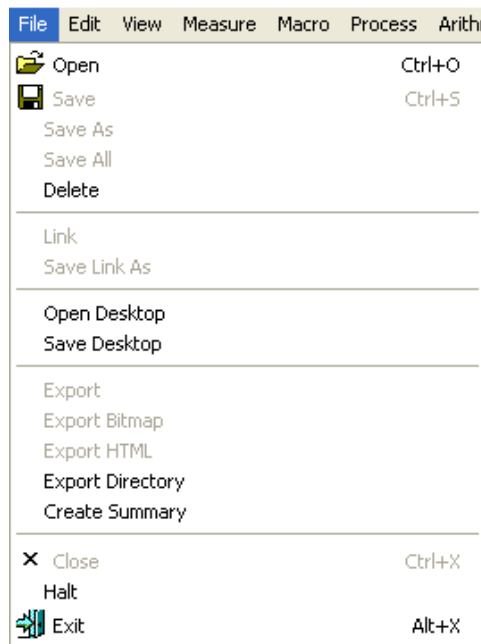


Figure 8-2: The File menu

Open	Opening data files (see section 9.1)
Save	Saving data files (see section 9.8).
Save As	Saving data files with different path and name.
Save All	Saving all open data files.
Delete	If you no longer need to store some data files on your hard disk, you can select the files here that you want to delete. Before deletion you will be prompted to confirm that you really want them to be deleted.
Link	Linking selected data files together (see section 9.9).
Save Link As	This function has similar functionality as Save As , but it is used when a linked file should be saved or saved with a different name. In this case, the main file and the linked ones will also be saved. The names of linked files are generated from the filename of main file (*.lnk).
Save Desktop	The actual desktop can be saved to a file with *.dsk extension. The desktop file stores the actually loaded data files and their position and size. A previously saved desktop can be reloaded via the Load Desktop menu item.
Load Desktop	Loading previously saved desktop.
Export	This function is for exporting the data files in ASCII format (text). It is useful if you want to process your data by some other application, and this application is capable of reading ASCII files. The exported file will be saved in the same directory as the original data file, the filename will

	be the same as the original one, but the extension is changed to .txt .
Export Directory	Exporting directory to ASCII form.
Export Bitmap	This function exports map images in bitmap format. The bitmap file will be saved in the same directory as the original data file, the filename will be the same as the original, but the extension is changed to .bmp .
Export HTML	This function allows exporting the selected map to the Web based .html file. When selected, the .jpg images of the map and distributions are created together with the .html file in the same directory as the map file. (If you want to display the .html file on a different computer, make sure that the .jpg files are also in the same directory as the .html file.)
Close	You can close the active window with this menu command or by pressing the right-top button of the active window.
File Halt	You can shut down Wintau32 software, when there is a serious error or crash. Using this option all the unsaved data and settings will be lost!
Exit	You can exit Wintau with this menu command or by pressing the right-top button of the <i>Main Window</i> or pressing the  button on the <i>Toolbar</i> .

8.2 Edit Menu

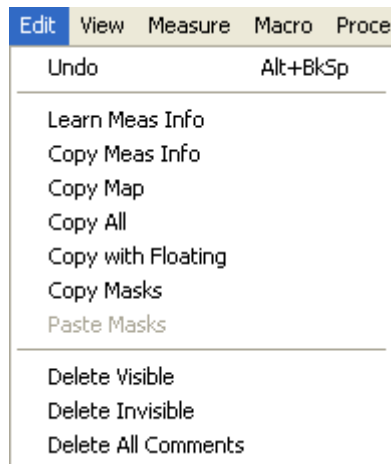


Figure 8-3: The Edit menu

Undo	It undoes the last change that was performed on the data file. However be aware that not every modification can be undone.
Duplicate	Opens a new window, and copies the contents (data) of the active window. This function does not load the data file from hard-disk. This is useful when you want to make a comparison after some processing to the original map.

Copy Meas Info	This function copies all the measurement information to the clipboard as a text.
Copy Map	This function copies the map itself (without the wafer image) as a bitmap to the clipboard.
Copy All	This function copies the map with the color scale as a bitmap to the clipboard.
Copy with Floating	It copies the map with the color scale and the possible Comments and Data Distributions as a bitmap to the clipboard.
Copy/Paste Masks	If you have masks placed on a map, and you want to use the same mask info to another map, use Edit Copy Masks , then Edit Paste Masks .
Delete Visible	This function deletes all currently visible data points from the map. It is useful if you want to delete a whole region of data. There should be at least one mask on the map to have this menu item available.
Delete Invisible	Delete all currently invisible (masked) data points from the map. It can be useful if you want to delete a whole region of data.
Delete All Comments	This function removes all the Comment windows from the map.

8.3 View Menu

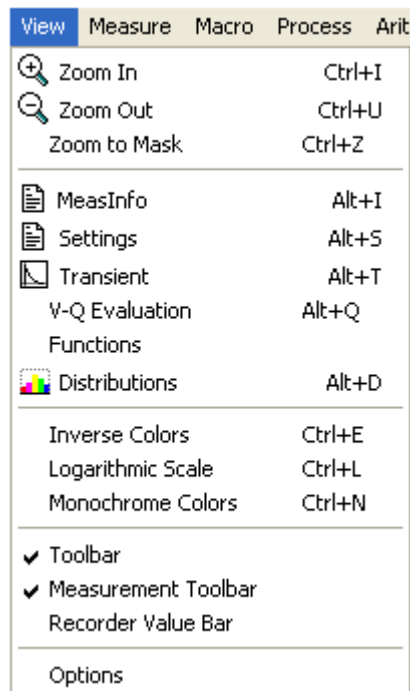


Figure 8-4: The View menu

Zoom In	Decreases the displayed size of the map (see section 9.3.1).
Zoom Out	Increases the displayed size of the map (see section 9.3.1).

<i>Zoom to Mask</i>	This option is useful when one or more masks are applied on the map. In such a case this option enlarges the active map so that the visible parts of the masked map fill the active map window.
<i>Meas Info</i>	Displays <i>Measurement Information</i> window (see section 9.4).
<i>Settings</i>	Displays the <i>Measurement Settings</i> window (see section 9.4).
<i>Transient</i>	Displays the <i>Transient Evaluation</i> window.
<i>V-Q Evaluation</i>	Opens the <i>V-Q Evaluation</i> window used in semiconductor analysis.
<i>Functions</i>	This option is used for opening any type of Function data file created in a <i>Recorder</i> window.
<i>Distributions</i>	Displays the <i>Distributions</i> window (see section 9.7).
<i>Inverse Colors</i>	Changes to Inverse Colors or back to regular (see section 9.3.2).
<i>Log. Scale</i>	Changes to Logarithmic Scale or back (see section 9.3.2)
<i>Monochrome Colors</i>	Changes to Monochrome Colors or back (see section 9.3.2).
<i>Toolbar</i>	Displays and hides the <i>Toolbar</i> .
<i>Measurement Toolbar</i>	Displays and hides <i>Measurement Toolbar</i> (see section 7.4). This menu item is only available in Measurement Mode.
<i>Rec. Value Bar</i>	Shows or hides measured results at the bottom of the Wintau32 <i>Main Window</i> . The display size can be changed by right-clicking the <i>Recorder Value Bar</i> and selecting the <i>Size</i> option. This <i>Recorder Value Bar</i> can be hidden also if the <i>Hide</i> right-click option is selected.
<i>Options</i>	The <i>View Options</i> window appears (see section 9.3.3). This window is used to set the default settings for the map display.

8.4 Measure Menu

The ***Measure*** menu (Figure 8-5) contains measurement-related commands. Items in this menu are only available when the system is in Measurement Mode (see section 7.7.2).



Figure 8-5: The *Measure* menu

- Mode** You can set measurement method (μ -PCD, SPV, etc., depending on your system configuration). See section **Hiba! A hivatkozási forrás nem található..**
- Initialize** It checks the movements of the measuring head and the sample stage (X-Y movements) and searches for the end positions for the head and the stage movements. It sets the indexer to zero position. No sample loading and mapping can be performed until the Initialization procedure is complete.
- Recorder** Clicking this menu item the *Recorder* window of the currently set measurement method appears. This window is the basic control panel of the measurements, all the measurement related parameters can be modified here.
- Temperature Controller** Displays the *Temperature Controller* window, used with the optional cryostat.
- Heat Treatment** Displays pop-up windows for entering the heating and cooling times for mobile charge measurements using the V-Q measurement method.
- Wafer Size** This menu item is used to select the size and shape of the sample to be measured, as well as the type of stage to be used.
- Load Sample** This menu item is used for manual loading of a sample before measurement.

<i>Load from Cassette</i>	This menu item is used for automatic loading of a wafer from the indexer.
<i>Unload Sample</i>	If the sample was loaded manually, this menu item allows you to remove the sample from the stage manually. If the optional indexer was used, this command puts the wafer back into the same slot of the cassette from where it was loaded.
<i>Unload from Chuck</i>	When this menu item is selected, the indexer arm removes a wafer from the optional hot chuck before it has cooled.
<i>Change Table</i>	If a table is used for the measurement stage, use the <i>Change Table</i> option to change the table. If the console stage is installed, this menu item will be listed as <i>Change Console</i> instead.
<i>Indexer Home</i>	Moves the indexer to re-establish its home position. In the home position, the plate has moved completely up and the arm has moved to the left end position.
<i>Wafer Finder</i>	Before map measurements use <i>Wafer Finder</i> to specify the exact contour of the previously loaded sample.
<i>New</i>	Selecting this option a new map measurement can be started on the currently loaded and found wafer. If there is a finished or interrupted map on the desktop, clicking <i>New</i> will clear the whole measurement without confirmation message. Before starting the measurement itself (<i>Measure</i> <i>Map</i>) the measurement parameters can be modified.
<i>Autosetting</i>	Before any measurement the correct parameters should be set for the given sample and measurement method. All the parameters can be set manually, but it is strongly recommended to use the automatic parameter setting procedure (Autosetting) to get reproducible and user-independent results.
<i>x-Points</i>	If a full wafer map is not desired, a selected number of points can be measured instead. The number of the measured points can be specified in the <i>Measurement Options</i> window, <i>Single</i> tab, and this specified number of points will appear as “x” in the menu item name. See section 12.6.2 for additional information about x-point measurements.
<i>Map</i>	Starts the map measurement with the previously set parameters.
<i>Resume</i>	In case the measurement was aborted, it can be continued by selecting this option.
<i>Auto</i>	This option allows the user to perform an automated x-point or map measurement. Operations normally performed by the operator, such as saving or printing the file, are done automatically without user intervention. See section 12.10.
<i>Flash</i>	Flashes the sample. A pop-up window is used to specify the number of flashes.

Corona	This option is used to set the linear corona charge settings.
Recipe	Measurement recipes are used for repeatedly performing measurements with the same settings. See section 12.9 for more information about creating and using recipes.
Hardware Info	There is status information about the hardware in the <i>Hardware Info</i> window.
Hardware Test	The <i>Hardware Test</i> window makes it possible to check the digital input and output communication between the computer and the optical sensors, switches, vacuum valves and moving parts such as the measuring head and the sample stage etc. See Diagnostics and Troubleshooting Manual. for more information about this window.
System Diagnostics	Runs the system diagnostics check. See Diagnostics and Troubleshooting Manual.
Remote	The software which directly controls the WT-2000 main unit (WSOS2) runs on the DOS computer. The Remote commands affect the WSOS2 software. See section 15.1.5.2 for more information.
Options	Displays windows for setting measurement options.

8.5 Macro Menu

The Macro language has been implemented in Wintau32 to give users a wider range of control over software capabilities. It is a special program language similar to BASIC or DOS batch programming. Macros are explained in Chapter 11.

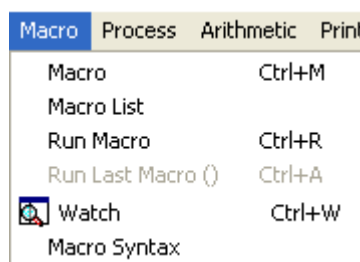


Figure 8-6: The Macro menu

Macro	Displays the available macros in the <i>Macro Editor</i> window (see section 11.1.2).
Macro List	Displays the available macros.
Run Macro	Runs a macro from the open dialog box.
Run Last Macro	Runs the last executed macro.
Watch	Displays the <i>Macro Watch</i> window (see section 11.1.5).
Macro Syntax	Displays the <i>Macro Syntax</i> window (see section 11.1.4).

8.6 Process Menu

The **Process** menu is used to process map images. It is only available for an engineer level user. Image processing allows subtle data variations to be apparent. The various processing options are described below, along with examples of their usage.

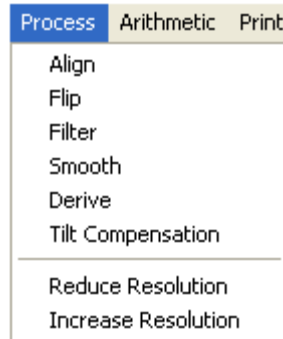


Figure 8-7: The Process menu

Align

This function rotates (aligns) the wafer image to the orientation determined by the angle, which can be set in the upcoming window. This is an irreversible transformation of the map data; however, it does not affect the statistics of the measurement.

Flip

Flipping the sample horizontally or vertically acts to create a mirror image of the original display. It is useful when you measure both sides of a sample, and want to make a comparison from the same viewpoint.

Filter

If for some reason (for example at the edge of the sample) you have extreme high or low values, that impairs your statistics, and you can use this function to replace the false values with their interpolated value calculated from the neighboring values.

Smooth

This function makes the map smoother i.e. substitutes every data point by the average value of the given data point and the four neighboring points. It reduces the short-range (in the range of the resolution) noise of the map. This is an irreversible transformation. It decreases the deviation of the map data.

Derive

This function is useful for detecting short-range, subtle non-uniformities in a map, typical of slip lines, striation rings, and uneven annealing of implanted wafers.

The derive procedure consists of the following steps:

1. The map is smoothed n times
2. The smoothed map is divided by the original map point by point
3. The result is smoothed m times.

The numbers n and m can be set in a pop-up window after calling the function, as *before division* and *after division*. Increasing these numbers eliminates the shortest range deviations that are in the order of the resolution. This can remove the noise while accentuating the deviation.

Derive function is irreversible, and gives approximately 1 as result.

Tilt Compensation

Tilt compensation of the map. This process eliminates tilt such as may occur due to artifacts from any measured map. Tilt can be caused, for example by a corona wire that is not set parallel with the stage, in the case of a Kelvin measurement.

Reduce Resolution

You can reduce the resolution of the map. In other words, a map with 1 mm raster size can be transformed to a map with 2 mm raster size. This way the number of points in the data structure will be one quarter of the original. This is useful if you want to reduce the size of the data file (this will reduce it roughly to a quarter of the original size).

Increase Resolution

You can increase the resolution of the map. In other words, a map with a 1 mm raster size can be transformed to a map with 0.5 mm raster size. The values for the new positions are calculated by interpolation from the original values. The number of points in the data structure will be four times the original. This is useful if you want to measure quickly, but still you want the map image as nice as possible.

8.7 Arithmetic Menu

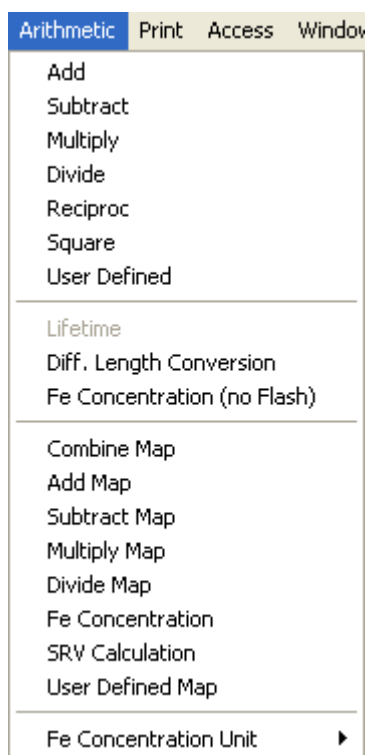


Figure 8-8: The Arithmetic menu

The Arithmetic menu is only available for an engineer level user. When a map is displayed, the **Arithmetic** menu options are available.

Under **Arithmetic** menu you can do several kinds of calculations on a map. All the calculations are performed point by point on the whole map.

The options that appear in the **Arithmetic** menu will depend on the configuration of your system. Although some options are only for specific measurement modes, you can use other **Arithmetic** options for maps made using any measurement method.

8.7.1 Arithmetic Requiring a Constant Value:

If the arithmetic operation requires the use of a constant, the value of the constant can be entered in the upcoming window.

Add	This function adds the constant to every data value.
Subtract	This function subtracts the constant from every data value.
Multiply	This function multiplies every data value with the constant.
Divide	This function divides every data value with the constant.

8.7.2 Arithmetic without a Constant:

Reciproc	It calculates the reciprocal value of every data value.
Square	It calculates the square value of every data value.
User Defined	Every data value can be modified by a macro.

8.7.3 Corrections, Correlations, Calculations:

Lifetime	Converts SPV diffusion length to lifetime
Diff.L.Conversion	Converts μ -PCD lifetime to diffusion length
Diff.Length Calc.	Diffusion length is calculated from SPV amplitude maps. Clicking on this menu item a window appears with the calculation parameters. • <i>Si Type</i> can be changed with the related radio buttons. • Oxide thickness can be set (same value appears in the <i>SPV Recorder</i> window). • Surface constant is an arbitrary constant. Use the default value (0.5) for calculation. • Calculate Regression Map. SPV diffusion length is calculated by linear fitting the SPV signals of the different wavelength lasers. If this box is checked the regression of the linear fitting is recorded as a map. • Checkboxes for different wavelengths. Only the checked ones are used for diffusion length calculation.
Correct Diff.Length/Correlation	The appearing window gives the possibility to use thickness correction and apply correlation function for the SPV measurement. • Calculate Regression Map: SPV diffusion length is

calculated by linear fitting the SPV signals of the different wavelength lasers. If this box is checked the regression of the linear fitting is recorded as a map.

- **Thickness Correction:** The wafer thickness should be entered in μm units and the backside Surface Recombination Velocity (SRV) in cm/s units. If SRV is not known, use default value for calculation (**10000** cm/s). This function corrects the measured SPV diffusion length depending on the wafer thickness.
- **Diff.Length Correlation:** This function is a correlation function between different SPV measuring methods. Using default values corrects measured values to correlate well with the other commonly used SPV method.

Fe Concentration (no Flash)

This option is used to calculate Fe concentration on μ -PCD maps made without using the flash. It assumes that Fe is the only contamination present, and uses a fixed constant to calculate Fe based on the measured lifetime.

Fe Concentration

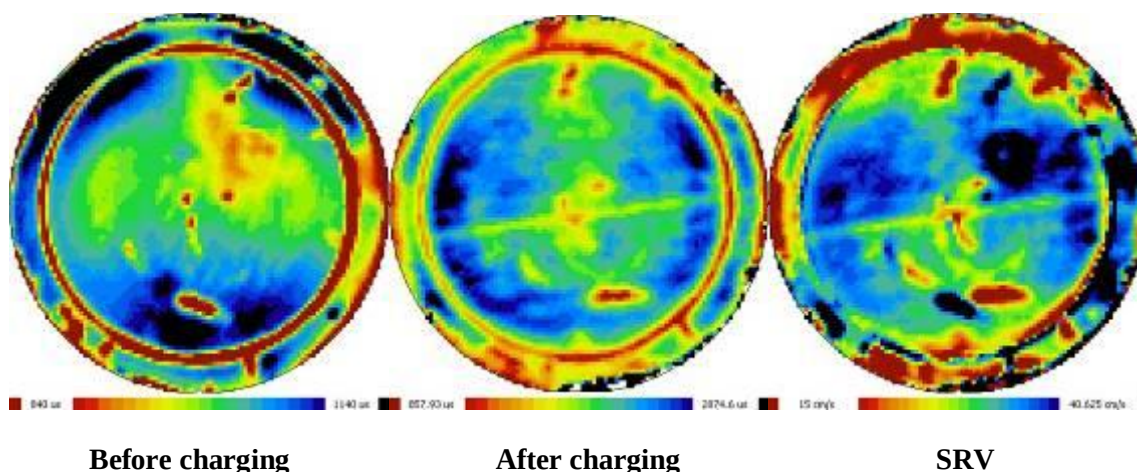
It converts the lifetime or diffusion length value from two maps into Fe Concentration value.

SRV Calculation

Calculates surface recombination velocity (SRV) on a μ -PCD map. If the corona charge passivates the surface such that lifetimes are different before and after the charge, this option calculates the SRV value before the charge was applied.

It needs two μ -PCD maps of the same wafer, one before charge, one after charge. Open both maps, and activate the map before charging. After calling SRV Calculation, select the map after charging. The SRV map replaces the active map. Give the wafer thickness. The calculation gives value only if the 'after' map has higher lifetime than the 'before' map.

The map after charging is regarded as real bulk lifetime map, the surface is supposed to be perfectly passivated. The backside of the wafer is regarded to be perfectly passivated, both before and after charging.



Fe Concentration Unit	Allows changing Fe concentration to different units: 1/cm ³ , 1E10/cm ³ (0.2 ppt), or ppt (5E10/cm ³).
Resistivity Calculation	Only available if 4pp option is installed in the system. It calculates Resistivity from the 4pp results. This function needs the wafer thickness that has to be entered in the pop-up window. If the sample is a block, enter 0 .

8.7.4 LBIC calculations

Lifetime	Converts LBIC diffusion length to lifetime
Diff.L.Conversion	Converts lifetime to diffusion length
Diff.Length Calculation	Diffusion length is calculated from LBIC I_{SC} and reflectance maps. Clicking on this menu item a window appears with the calculation parameters. • Remove points: the points measured over the metal lines can be removed from the map. Metal lines give higher reflection and lower current values. By defining the range where reflection and current results are expected, the values outside of these ranges are regarded as invalid on the diffusion length map. • Calculate Regression Map: LBIC diffusion length is calculated by linear fitting the Gflux/LBIC (=1/IQE) signals of the different wavelength lasers. If this box is checked the regression of the linear fitting is recorded as a map. • Textured: Use this option in case of monocrystalline solar cells that are textured with pyramids. The pyramids change the effective penetration depth of the lasers that is taken in account when this option is checked. • Checkboxes for different wavelengths. Only the checked ones are used for diffusion length calculation. Use only wavelengths that has a penetration depth that is much more than the junction depth and not more than the half of the wafer thickness. For ordinary solar cells, the proper wavelengths are the following: 750 nm < λ < 1000 nm
LBIC Calculation and LBIC0 Calculation	Compensates the short circuit current maps with the reflectance maps by the function $y=x/(1-R)$ point by point, and reverts this compensation. This is a historical feature kept for compatibility reasons.
External Quantum Efficiency	Calculates external quantum efficiency from the short circuit current maps. The current maps are replaced by the EQE maps, so it is advised to save the link under a different name.
Internal Quantum Efficiency	Calculates internal quantum efficiency from the short circuit current maps and the reflectance. The current maps are replaced by the IQE maps, so it is advised to save the link under a different name.

8.7.5 Arithmetic with another Map:

The other map can be selected in the upcoming window. All the calculations are performed point by point on the whole map.

Combine Map

Two complementary maps can be combined. If the sample is very inhomogeneous, it is possible to measure the sample two times with two different settings. After this, two complementary maps of different regions can be combined into one map.

Add Map

It adds the other map data to the active map data.

Subtract Map

It subtracts the other map data from the active map data.

Multiply Map

It multiplies the active map data by the other map data.

Divide Map

It divides the active map data by the other map data.

User Defined Map

Every data can be modified by a macro.

8.8 Print Menu

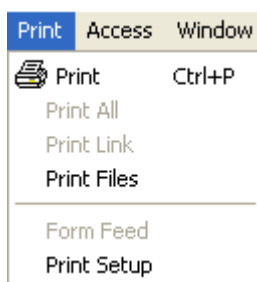


Figure 8-9: The *Print* menu

Print

It prints the active data file (map).

Print All

It prints all loaded data files.

Print Link

It prints linked data files (see section 9.9).

Print Files

It prints data files selected from the file list.

Form Feed

It finishes the printing procedure.

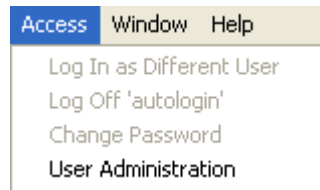
Print Setup

You can set the printer parameters in the *Print Setup* window.

You can find detailed information about printing in section 9.10.

8.9 Access Menu

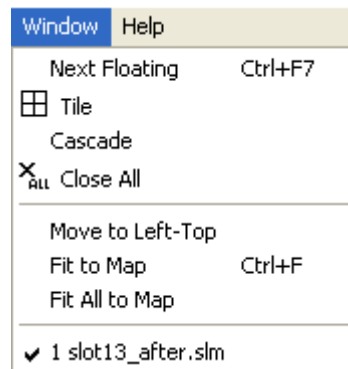
The **Access** menu items are used to manage users and their access levels. User access is explained in section 7.8.

**Figure 8-10: The Access menu**

Log In as Different User	Logs off the current user and allows immediate login as a different user.
Log Off	Logs off the current user.
Change Password	Allows changing the password of the current user.
User Administration	Displays the <i>User Administration</i> window, for adding and deleting users (see section 7.8.2).

8.10 Window Menu

The **Window** menu listings control the appearance and placement of pop-up (child) windows.

**Figure 8-11: The Window menu**

Next Floating	It activates next floating window like <i>Measurement Information</i> or <i>Distribution Window</i> if they are visible.
Tile	It arranges the child windows (maps) as much as possible in a so-called “Tile” styled grid, selecting the optimal size and position for them.
Cascade	It arranges the child windows in a so-called “Cascade” style, selecting the default size and position for them.
Close All	It closes all child windows.
Move to Left-Top	Moves the active map window to left-top position.
Fit to Map	It sets the window size according to the map size to its optimum value.
Fit All to Map	It sets all the child windows according to the map size to their optimum

value.

Additionally, the *Window* menu lists all open windows. The active window is selected with a check mark. To change the active window, click on the name of the window.

8.11 *Help* Menu

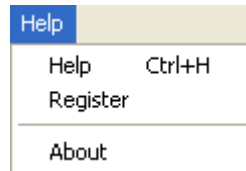


Figure 8-12: The *Help* menu

Help

It displays the on-line help. See section 7.9.

Register

For registering Wintau refer to section **15.1.6**.

About

It displays some basic information about the Wintau software and the computer.

9. *Displaying Data*

This chapter demonstrates methods for improving data display for both recently measured samples and previously saved data. It also includes information on opening, saving, and printing data files. Table 9-1 shows the organization of this chapter.

Table 9-1
Data Review Options

Data Review Topic	Section
Opening Files	9.1
The Map Window	9.2
Display Options	9.3
The <i>Measurement Information</i> Window	9.4
Using Masks and Erasing Data Points	9.5
Copying Maps and Adding Comments	9.6
Data Distributions Windows	9.7
Saving Data	9.8
Link Files	9.9
Printing Data	9.10

9.1 Opening Files

You can open one or several previously saved data files for evaluation. This function can be activated by choosing the **File | Open** menu item or by pressing the  *Toolbar* button.

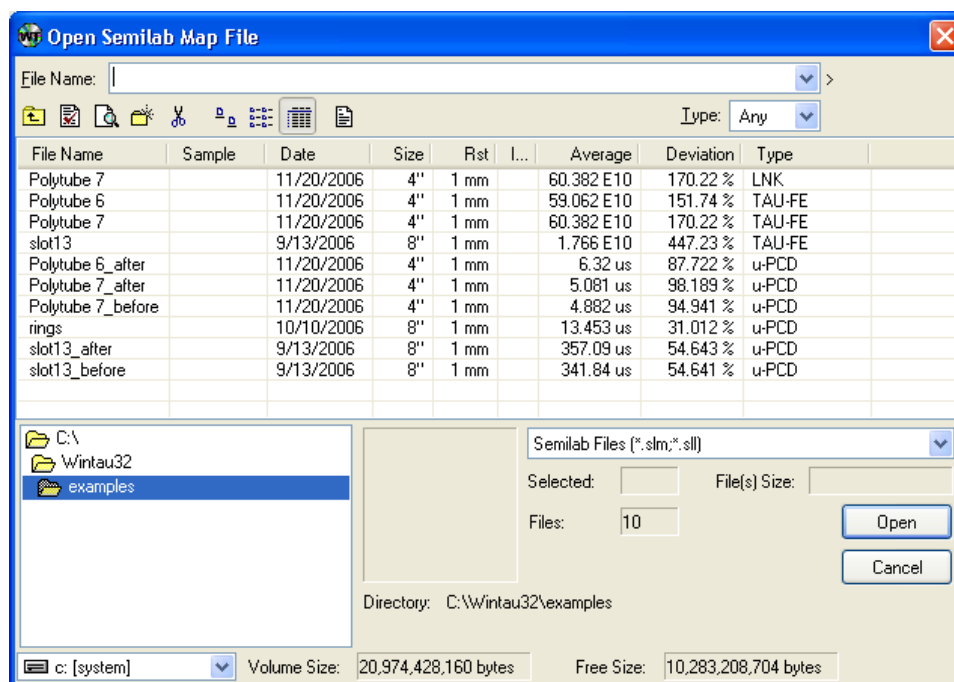


Figure 9-1: Open File dialog window

The *Open File* window (Figure 9-1) contains a listing of the WT-2000 files in the active directory. Change the directory by navigating in the directory tree at the bottom left of the window. To navigate to a different drive, use the drop-down list beneath the directory tree.

9.1.1 Changing the File Display Mode

The main part of the *Open File* window is a list with several columns that display the data files in the current directory. In the columns the main property of the data files can be seen such as *File Name* and *Sample* name, etc. This is the default display mode. You can use the following buttons to change the listing mode:



Shows the so-called previews of the files. The same preview is shown below as you change the active file in the list box.



Only the file names are shown. It is useful when you have many files in one directory; additional information of the active file is shown at the bottom of the window.



This is the default display mode, when all the important information of the files is shown in columns.

Additional features in this open dialog:



Refreshes the file list. This features creates a new file listing (**filelist.res**) based on the contents of the current directory, adding new files and removing files no longer in the directory.



Find File. When you have many data files on your hard disk sometimes it is difficult to find specific files. This function helps you find data files that have some common characteristics, such as size or sample name. There are two types of

characteristic: text and list style.

Comment is an example of a text style characteristic. For this kind you can use wild cards: * (matches any unspecified long text) and ? (matches any character) to match text in the files; any other character will be compared. Comparisons are not case-sensitive. If you do not want to match *Comment* then a * would match any *Comment*.

The *Size* is an example of a list style characteristic. It means that you can select a specific *Size* value to match. Selecting **ANY** would match any *Size*.



You can create a new directory under the current directory.



You can delete the selected files. If you no longer need to store some data files on your hard disk, here you can select those files to delete. Before they are deleted you will be prompted to confirm that you really want to delete them.

9.1.2 Selecting Files to Open

To open a file you have several options:

- Type in the file name, and click **OK** (or press the <Enter> key)
- Click on the file name, and click **OK** (or press the <Enter> key)
- Double-click the filename
- Click on the file name drop-down list and select the file name from the previously opened files, then click **OK** (or press the <Enter> key)

To open several files, hold down the <Ctrl> key while clicking on the files. Alternately, click on the first file you want to open, and while holding down the <Shift> key click the last file. All the files between the first and the last click will be selected.

You can rename the active file by clicking once again on the file, typing a new name, and then pressing the <Enter> key.

9.1.3 Working with the File List

If you click the right mouse button on the file list area, a menu appears.

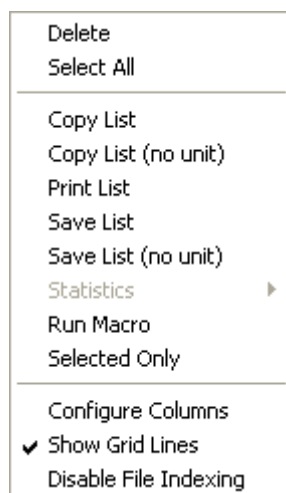


Figure 9-2: The menu that appears after right mouse click on file list

Delete	A file can be removed.
Select All	Select all files of the list.
Copy List	Copies file names and all the related information to the Clipboard.
Copy List (no unit)	The same as the Copy List with the difference that the copied values do not contain the physical units (μm , ohm, etc.).
Print List	Prints out the filenames and related information.
Save List	Creates a résumé file with *.res extension. This file stores the file list, with data file properties, in tab-delimited format.
Save List (no unit)	The same as the Save List except that the saved values do not contain the physical units (μm , ohm, etc.)
Statistics	When this is clicked above a column header with the right mouse button, statistics can be obtained from the values of column.
Run Macro	Runs a macro. See macro description in User Manual Vol. 1.
Selected Only	If you choose Selected Only , you can use the options above for the selected files. (To select items use the left mouse button, or for multiple selections press <Ctrl> and left mouse button.)
Configure Columns	Displays the <i>Configure Columns</i> window (Figure 9-3), used to hide or make columns visible.

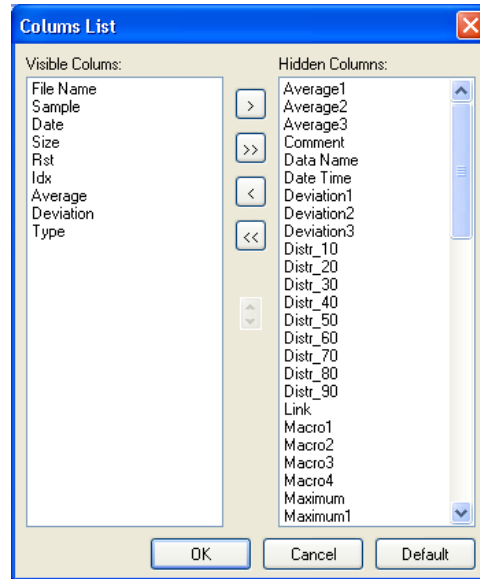


Figure 9-3: Configure Columns dialog window

Figure 9-3 shows the default settings, which you can recall with the **Default** button.

With the  button you can add selected items to *Hidden Columns* (to select items use left mouse button, or for multiple selection the <Ctrl> key and left mouse button). These types of information will not appear in the file information list.

If you use the  button, only the file names are displayed.

To add items to *Visible Columns* use ; or to add all, use the  button.

Show Grid Lines

Show or hide the grid lines between file names and parameters.

Disable File Indexing

The software creates a **filelist.res** list file in every opened directory. When the directory is opened again the file information is loaded much faster from this file. A drawback is that this list is not up-to-date. If **Disable File Indexing** is selected, this feature is not used.

9.2 The Map Window

After selecting the file to open, you will see a window similar to Figure 9-4.

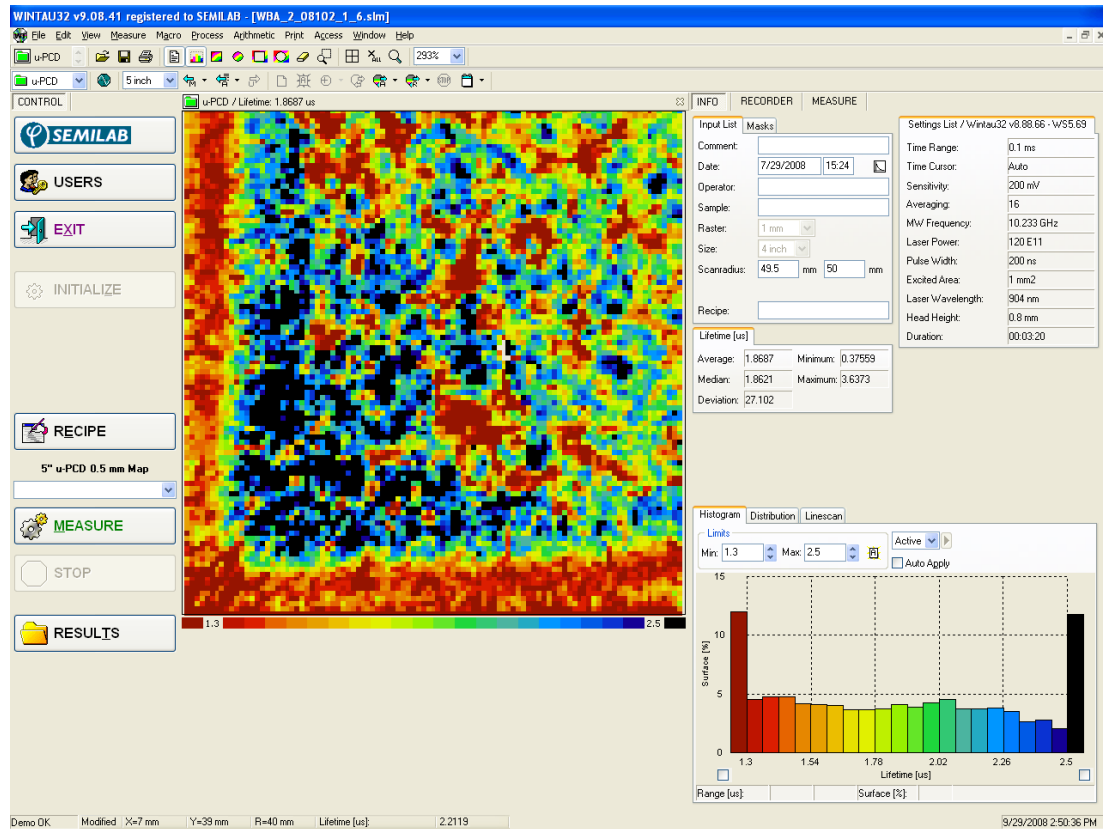


Figure 9-4: Main Window with a map and the Measurement Information window

The map is displayed with its color scale at the bottom in the left window. If you move the mouse over the map the *Status Bar* displays information about the mouse position and the measured value at that position. *X* and *Y* are the coordinates of the point at the mouse position. The center point (0,0) position is located at the center of the map. *R* (the radius) is the distance from the center. The last displayed value depends on the type of the data; it may be lifetime or diffusion length.

9.3 Display Options

Scaling of the active map and the user interface are configurable in the **View** menu.

9.3.1 Zoom

You can change the displayed map size by selecting the **View | Zoom In** or **View | Zoom Out** menu items. There are also some useful buttons on the *Toolbar* to control the display of the map. See section 7.3 for a description of the *Toolbar* buttons.

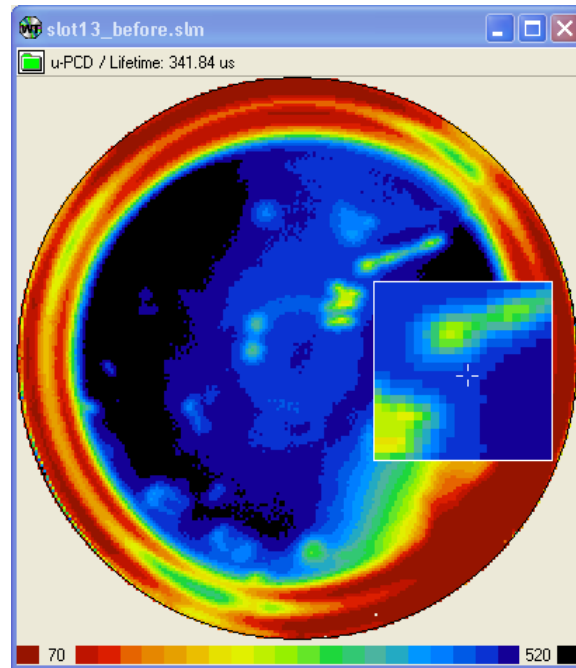


Figure 9-5: Using the Zoom feature

9.3.2 Changing the Displayed Map Colors

The following options are available in the **View** menu. These menu items change the appearance of the displayed map.

- | | |
|--------------------------|---|
| Inverse Colors | When selected, it reverses the sequence of the colors of the Color Scale used to generate the map. |
| Log Scale | When selected, it changes the Color Scale calculation to a logarithmic function, instead of the linear one. |
| Monochrome Colors | When selected, it changes the map scale to monochrome. It may be useful for emphasizing small structures. |

9.3.3 View Options

Selecting the **View | Options** menu item, the *View Options* window appears (see Figure 9-6). This window is used to set the default settings for the map display.

When exiting the software, the options are saved in the **Wintau.ini** file, so at every start-up Wintau restores all the options you previously set. By pressing the **Default** button, the default settings are restored. The **Apply** button activates the settings without closing the window, while the **OK** button activates the settings and also closes the window.

9.3.3.1 General Options

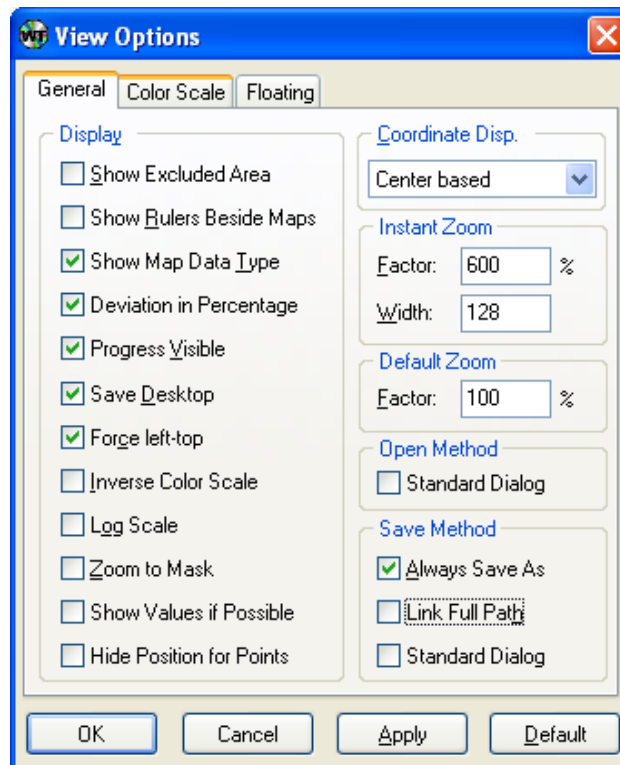


Figure 9-6: General tab of View Options window

Show Excluded Area	The masked out regions are visible but shaded.
Show Rulers Beside Maps	Displays a vertical and a horizontal ruler beside the map for showing the dimensions of the sample.
Show Map Data Type	Displays the Data Types information at the top of maps. You can hide it clicking on the data type area on the map with the right mouse button.
Deviation in Percentage	Deviation values can be displayed in the original unit (μ s, for example) or relative to the average value in percentage.
Progress Visible	Displays the progress bar during time-consuming processes.
Save Desktop	Use this option if you want to keep your opened files for the next session of Wintau.
Force Left-Top	Use this option if you want all newly opened windows to be placed at the left-top position of the <i>Main Window</i> .
Inverse Color Scale	If the View Inverse Colors menu item is selected, it reverses the sequence of the colors of the Color Scale of the map. This sequence will be the default color scale for the map to be measured. This menu item is only available in Measurement Mode.
Log Scale	If the View Log Scale menu item is selected, it changes the Color Scale calculation to a logarithmic function, instead of linear. This sets the default scale type (linear or logarithmic) for the map to be

	measured. This menu item is only available in Measurement Mode.
Zoom to Mask	When the user defines a mask, the map window will show only the selected area.
Show Values if Possible	Values of measured points on maps will be visible with selection of this option.
Hide position for points	Use this option, when you want not see the coordinates of prescan points.
Coordinate Display	This function shows how the position coordinates are calculated on the map. The reference point (0,0) may be at the left-bottom corner or in the center of the map.
Instant Zoom Factor	Using the  <i>Toolbar</i> button this factor determines the magnification in percentage.
Instant Zoom Width	Using the  <i>Toolbar</i> button this factor determines the size of the magnified area.
Default Zoom Factor	This function determines the zoom factor when the  <i>Toolbar</i> button or the View Zoom In or Zoom Out menu items is used.
Open Method/Standard Dialog	If checked, the standard windows <i>Open</i> dialog will be visible instead of the <i>Open Semilab Map File</i> dialog when opening a map.
Save Method/Always Save As	Saving the map always forces the user to give path and filename
Link Full Path	If checked, the link file will contain the path of the linked maps. If the linked data maps are copied to another folder, this makes the link file useless.
Standard Dialog	If checked, the standard windows <i>Save</i> dialog will be visible instead of the <i>Save Semilab Map File</i> dialog when saving a map.

9.3.3.2 Color Scale Options

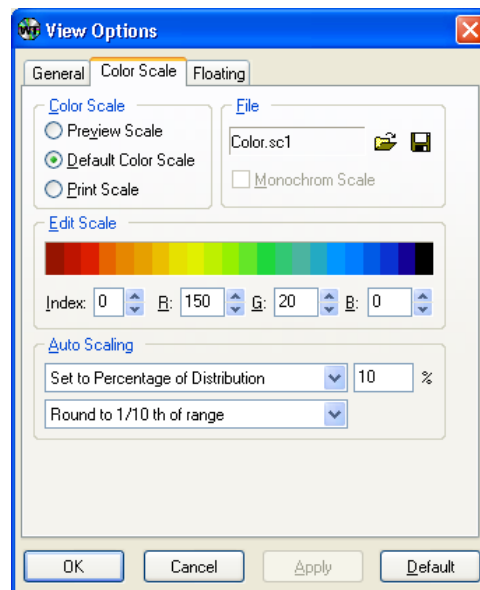


Figure 9-7: Color Scale tab of View Options window

There are three *Color Scales* that Wintau uses, one for the Preview images (**Preview Scale**), one for the map on the screen (**Default Color Scale**) and one for the map on the printout (**Print Scale**). All these types of scales reference a file that tells Wintau the selected color values. When Wintau cannot find these files (at the first start-up) it creates them automatically, and assigns them to the Color Scales. The Preview Scale is different because it has fewer colors defined, but if you have a color printer, then Wintau assigns the same color file to the Default Color Scale and the Print Scale.

Usually you do not need to modify these files, and it is not recommended. However if you are not satisfied with the colors you can define your own Color Scales (by *Edit Scale*), and save them under different filenames (in *File*). This may be the case if your printouts have very different colors from what you see on the screen, but usually today's printer technology provides quite good reproduction of colors.

There is a monochrome flag saved with the file. This is useful for Wintau, because depending on the printer (monochrome or color printer) it changes to its default monochrome or color scale. **Monochrome Scale** checkbox switches to this scale and makes the map appear in grayscale.

In *Edit Scale* field, the colors of the scale can be modified one by one. Index specifies the place of the color in the scale, it can be in the range of [0:21], [0:41] or [0:81], depending on the type of color scale that specifies the number of colors between the upper (*Max*) and lower (*Min*) limit. The 0 index always belongs to the values that are under the *Min* limit, the last index to the values over the *Max* limit. *R*, *G* and *B* specifies the color in the RGB space.

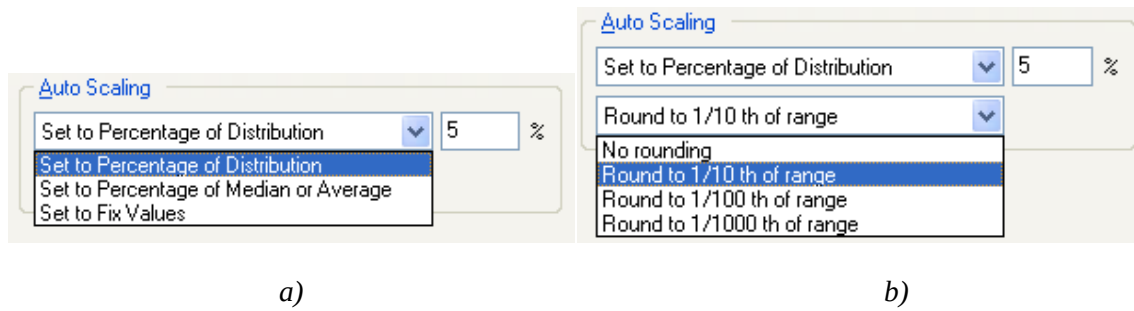


Figure 9-8: Auto Scaling options

There are three options for setting the *Min* and *Max* limits of the color scale by the *Auto Scaling*, see Figure 9-8, a).

Set to Percentage of Distribution sets the limits so that the specified percentage of the data points (5% in Figure 9-8) is outside both limits. The percentage can be set in the field on the right side. In the field on Figure 9-8, b), rounding can be set for the determination of the limits.

Set to Percentage of Median or Average sets the middle of the range to the average of the distribution, and sets the limits to a defined percentage to it. A rounding can be applied here too.

Set to Fix Values sets the limits to predefined values, independent from the distribution. Useful if similar measurements are to be compared.

9.3.3.3 Floating Options

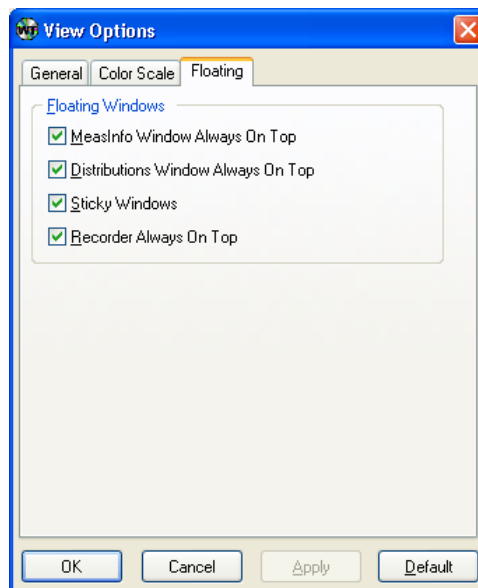


Figure 9-9: Floating tab of View Options window

The floating windows are the *Measurement Information* window and the *Distributions* window. The behavior of these windows can be controlled by the following options.

MeasInfo Window Always On Top

If this is selected, the *Measurement Information* window is always visible on the screen, i.e. it is above the windows of the *Main Window*.

Distributions Window Always On Top

If this is selected, the *Distributions* window is always visible on the screen, i.e. it is above the windows of the *Main Window*.

Sticky Windows

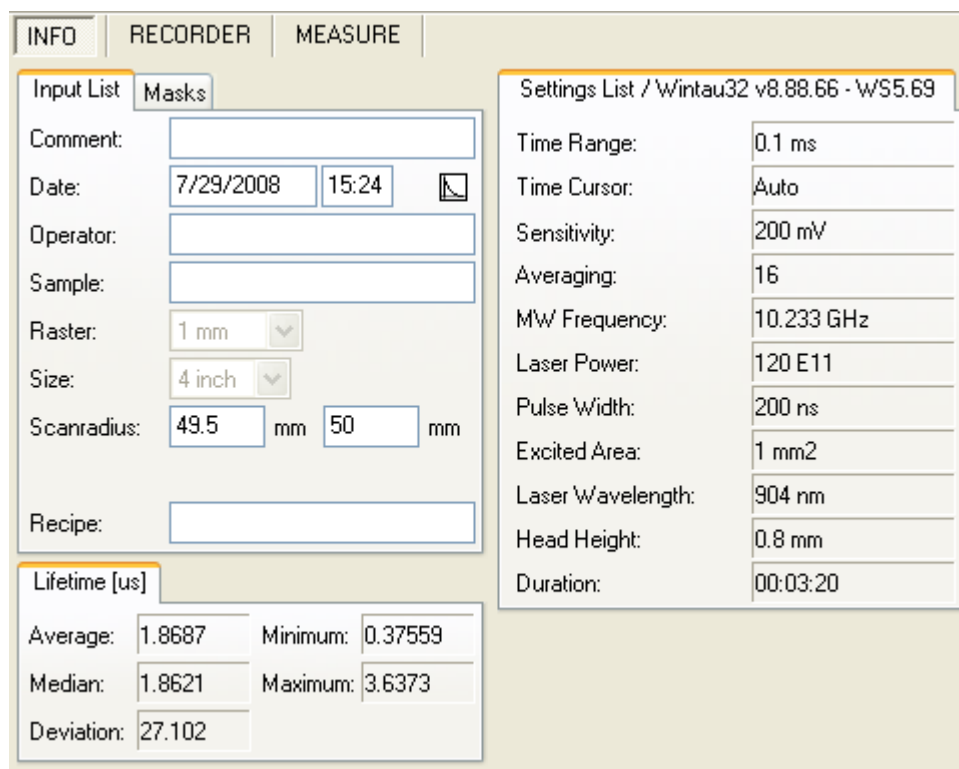
If this is selected, the floating windows will follow the movement of the *Main Window*, but they are placed to touch the *Main Window*'s border.

Recorder Always On Top

If this function is selected, the *Recorder* window is always visible on the screen, i.e. it is above the windows of the *Main Window*. This menu item is only available in Measurement Mode.

9.4 The Measurement Information Window

The *Measurement Information* window appears on the right side of the working area. It can be shown/hidden by selecting the **View | MeasInfo** menu item or the  *Toolbar* button.



Input List		Masks	
Comment:			
Date:	7/29/2008	15:24	
Operator:			
Sample:			
Raster:	1 mm		
Size:	4 inch		
Scanradius:	49.5 mm	50 mm	
Recipe:			

Lifetime [us]	
Average:	1.8687
Median:	1.8621
Deviation:	27.102

Settings List / Wintau32 v8.88.66 - WS5.69	
Time Range:	0.1 ms
Time Cursor:	Auto
Sensitivity:	200 mV
Averaging:	16
MW Frequency:	10.233 GHz
Laser Power:	120 E11
Pulse Width:	200 ns
Excited Area:	1 mm ²
Laser Wavelength:	904 nm
Head Height:	0.8 mm
Duration:	00:03:20

Figure 9-10: Measurement Information window (μ-PCD mode)

The *Measurement Information* window has two tabs: the *Input List* (described below) and the *Masks* tab (described in section 9.5).

The *Input List* in this window displays the comment, date, operator name, sample description, raster size, wafer size and the Scanradius. (See section **Hiba! A hivatkozási forrás nem található..**) Most of the fields can be edited by clicking the mouse on them. Some field labels may be changed by right-clicking on them.

Sometimes it is useful to enter a lengthy comment to describe the sample or the results. To improve readability on printouts, you can enter several lines as a *Comment* using the <Ctrl-Enter> key combination to force a line feed.

The statistics (average, median, deviation, minimum and maximum) of the displayed map data can be seen at the bottom part of this window. Some measurement methods may have several tabs to display measurement statistics.

Right-clicking on a field name in the lower part of this window causes a menu to appear (Figure 9-11) with several options. The options shown in the menu may be different depending on the selected measurement method.

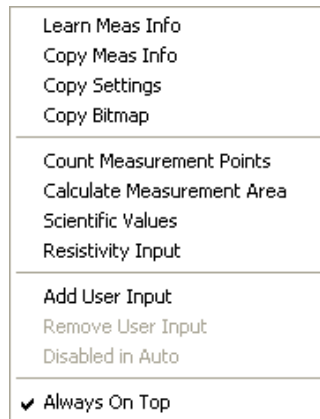


Figure 9-11: The menu that appears after right mouse click on *Measurement Information* window

<i>Learn Meas Info</i>	Learns entry contents and recalls them at every new measurement. If you want to get empty fields again, after wafer loading delete all the entries and then select this option again.
<i>Copy Meas Info</i>	Copies measurement information to the clipboard.
<i>Copy Settings</i>	Copies the measurement settings to clipboard.
<i>Copy Bitmap</i>	Copies all the measurement information to clipboard as a bitmap image.
<i>Count Measurement Points</i>	Counts the number of valid and invalid measured points.
<i>Calculate Measurement Area</i>	Calculates the measurement area in mm ² units and displays the total area and the area of valid points in a separate window.
<i>Scientific Values</i>	Changes the format of the displayed measurement results to scientific.
<i>Resistivity Input</i>	For V-Q measurement method, changes the displayed <i>Doping Concentration</i> value to <i>Resistivity</i> .
<i>Add User Input</i>	User defined input field can be added (see section 7.8.5 for further information about the accessibility of this option).
<i>Remove User Input</i>	Allows removing a previously added user-defined input field.
<i>Disabled in Auto</i>	Disables the user input field when automatic measurement is started.

On the right side of the *Measurement Information* there is the *Measurement Settings* information window. The information in the window depends on the type of measurement (for example, Figure 9-12 shows the window for μ -PCD measurements).

Settings List / Wintau32 v8.88.66 - W\$5.69	
Time Range:	0.1 ms
Time Cursor:	Auto
Sensitivity:	200 mV
Averaging:	16
MW Frequency:	10.233 GHz
Laser Power:	120 E11
Pulse Width:	200 ns
Excited Area:	1 mm ²
Laser Wavelength:	904 nm
Head Height:	0.8 mm
Duration:	00:03:20

Figure 9-12: Measurement Settings information window (μ -PCD)

By right-clicking on the *Measurement Settings* area, you can copy these settings to the clipboard. There is another tab in the *Measurement Information* window marked *Masks*. This window allows manipulations on the current map with masks of different shape. For a further description of this window see section 9.5.

For μ -PCD measurements, pressing the  button displays the *Transient Evaluation* window.

For V-Q measurements, pressing the  button displays the *V-Q Evaluation* window.

9.5 Using Masks and Erasing Data Points

Sometimes it is desirable to measure or display only a part of a whole map. It may also be necessary to exclude areas from the wafer image. For these purposes the so-called scan masks can be used on the wafer image. Masks may be applied before measurement (section 12.6.3) or on a displayed map.

There are rectangular and circular masks both for inclusion and exclusion. If there is a map on the screen, the four mask buttons can be accessed on the *Toolbar*.



Inclusive Rectangle Mask



Inclusive Circle Mask



Exclusive Rectangle Mask



Exclusive Circle Mask

By clicking on these *Toolbar* buttons, the corresponding masks can be activated and can be set by dragging the mouse while pressing the left button. The masks can be deleted by selecting the corresponding mask button and clicking on the map area. One map may include a maximum four masks, one from each type. The relationship between the masks is AND-type for the exclusive masks and OR-type for the inclusive masks.

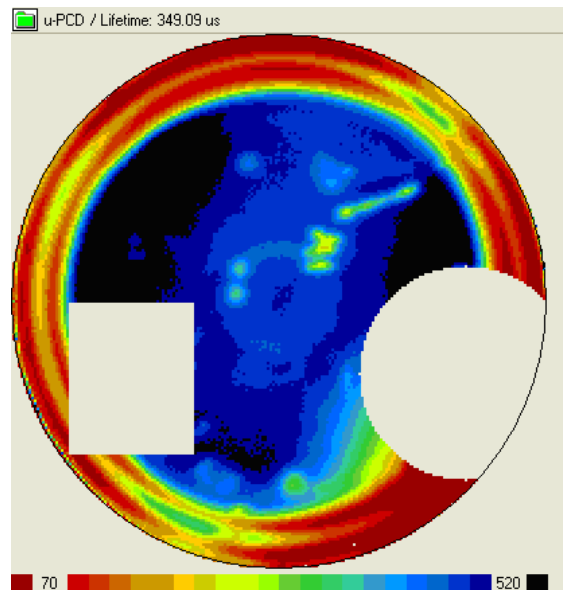


Figure 9-13: Masks on the map

There is another way to operate with masks on a measured map. By activating the *Measurement Information* window  (see section 9.4) and selecting the *Masks* tab, a window appears like Figure 9-14.

INFO		RECORDER		MEASURE	
Input List Masks					
	X1 [mm]	Y1 [mm]	X2 [mm]	Y2 [mm]	
	-9	51.5	12	-51.5	
	X [mm]	Y [mm]	R [mm]		
	-23	9	9		
	X1 [mm]	Y1 [mm]	X2 [mm]	Y2 [mm]	
	-13.5	31.5	17	-35.5	
	X [mm]	Y [mm]	R [mm]		
	0	0	0		
Lifetime [us]					
Average:	1.8117	Minimum:	0.71054		
Median:	1.7579	Maximum:	3.1129		
Deviation:	27.073				

Figure 9-14: Masks tab on the Measurement Information window

By clicking on the icon of a mask type, an entry field of the appropriate mask appears. The user can specify the **top-left** and **bottom-right** corners for rectangle type masks, or the **center coordinates** and the **radius** for circle type masks. All the positions and distances are in millimeter units.

Pressing the Erase  button on the *Toolbar* allows erasing any points from the selected map by marking them with a left mouse button click. The erased points can be put back on the map by a right mouse button click. After finishing the erase procedure, press the  button to cancel the erase operation. This procedure affects all the statistical parameters (average, median, deviation, minimum and maximum) of the map, and only the visible data points are included in the reported statistics.

9.6 Copying Maps and Adding Comments

When the right mouse button is pressed on the map, a pop-up menu appears (Figure 9-15).

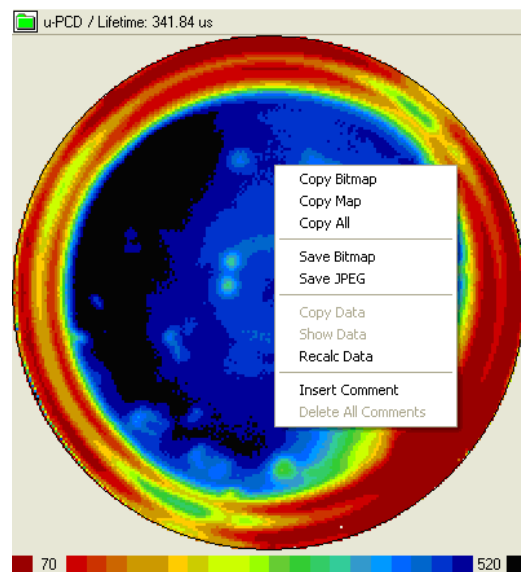


Figure 9-15: Right mouse button pop-up menu on the map

9.6.1 Using the Right Mouse Button on the Map

When the cursor is over the map, click the right mouse button to display a pop-up menu with the following options.

<i>Copy Bitmap</i>	This function copies the map into the Windows clipboard without the black border.
<i>Copy Map</i>	This function copies the map into the Windows clipboard with the black border.
<i>Copy All</i>	This function copies the map together with the color scale into the Windows clipboard.
<i>Copy Data</i>	This function is available only when a rectangular mask is selected. In this case, a table with data values is copied to the clipboard.
<i>Save Bitmap</i>	This function saves a bitmap graphical file to the directory of the map file with the file name of the map and .bmp extension.

Save JPEG This function saves a compressed graphical JPEG file the same way as above with **.jpg** file extension.

Insert Comment This function inserts a *Comment* window to the mouse cursor position on the map. The default comment shows the coordinates and the measured value at the mouse cursor position. The comment text can be edited after double clicking on the comment window. Accept the changes by pressing the <Enter> key.

Delete All Comments This function removes all previously inserted comments from the map.

Copy Map and **Copy All** items can be activated from the **Edit** menu as well.

9.6.2 Editing Comments

Pressing the right mouse button on the *Comment* window activates the *Comment* pop-up menu (see Figure 9-16).

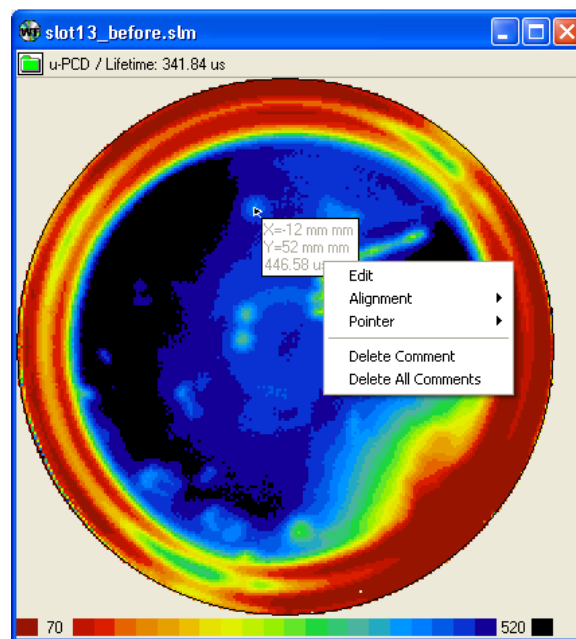


Figure 9-16: Pop-up menu on the *Comment* window

Edit The text in the *Comment* window can be edited. Pressing the <Enter> key terminates the edit process.

Alignment The text in the *Comment* window can be aligned to the left, to the right or center aligned.

Pointer Sets the color (black and white) and the orientation (left-top, left-bottom, right-top and right-bottom) of the pointer (the small arrow at the corner of the *Comment* window).

Delete Comment This function removes the selected *Comment* window from the map.

Delete All Comments This function removes all previously inserted comments from the map.

The *Comment* windows can be dragged above the map area by keeping the left mouse button pressed.

9.7 Data Distributions Windows

Additional representations of the map data are possible. The distribution windows can be accessed by clicking the  *Toolbar* button or by selecting **View | Distributions** in the menu. The X-Y values belonging to the mouse cursor position are displayed in the *Status Bar*.

The *Distributions* window has three tabs as shown in Figure 9-17.

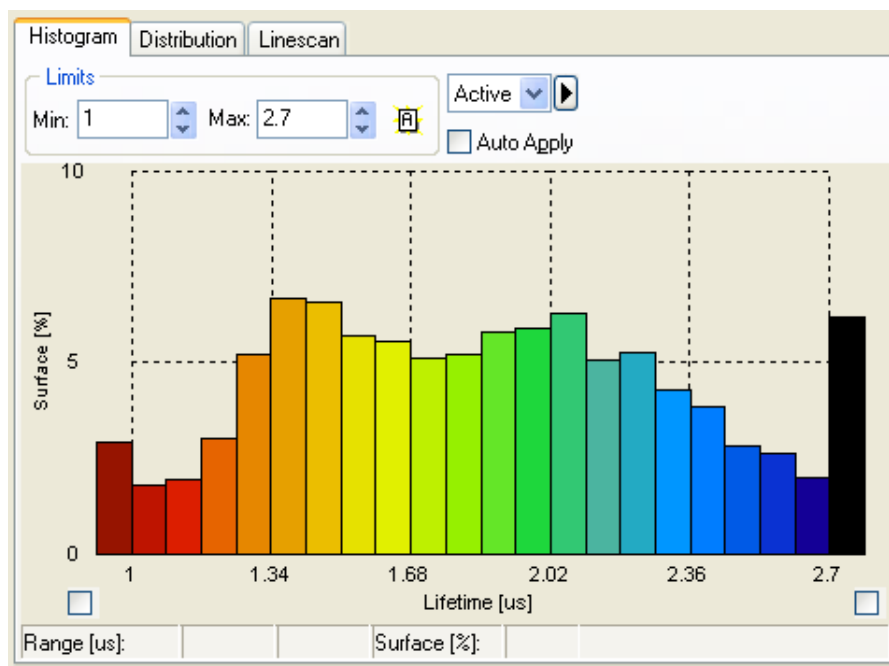


Figure 9-17: *Histogram* tab in *Distribution* window

9.7.1 Histogram

This function gives a way to re-scale the limits of the color gradation by showing the data distribution histogram. The re-scale function affects only the color representation of the map and does not affect the data and the statistics of the map. The AutoScale procedure, which can be accessed by pressing the  button in the *Distribution* window, calculates the upper and lower limits so that about the 80% of the points belong to the range between the limits. If the **Auto Apply** box is checked, the map scale automatically changes when you change the *Min* or the *Max* limits; otherwise an **Apply**  button appears, which you must click to validate the changes. With the two checkboxes below the plot you can exclude the data below the *Min* limit and/or above the *Max* limit. These values are excluded also from the reported statistics. Right-clicking on the histogram area, a menu appears (Figure 9-18).

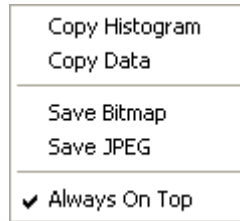


Figure 9-18: Menu displayed by right mouse click on the histogram area

Copy Histogram	Copies the histogram to clipboard as an image.
Copy Data	Copies numeric data of the histogram to clipboard.
Save Bitmap	Saves histogram chart as a bitmap image.
Save JPEG	Saves the chart as a JPEG image.
Always On Top	Causes the <i>Distribution</i> window to appear in front of all other windows.

These functions are also available in other windows.

There is another right click option in the histogram window. You can cut, copy, paste and delete limit setting values, and it is possible to undo the last change.

By double-clicking with the left mouse button on the histogram area, you can change the font type of the chart. This capability is also present in the *Distribution*, *Line Scan* and *SPV* windows.

9.7.2 Cumulative Distribution

This function is the logarithmic representation of the data distribution (percentage above the value on the X-axis). The value of the 50% percentile is the median value displayed in the statistics (see the *Measurement Information* window, Figure 9-10).

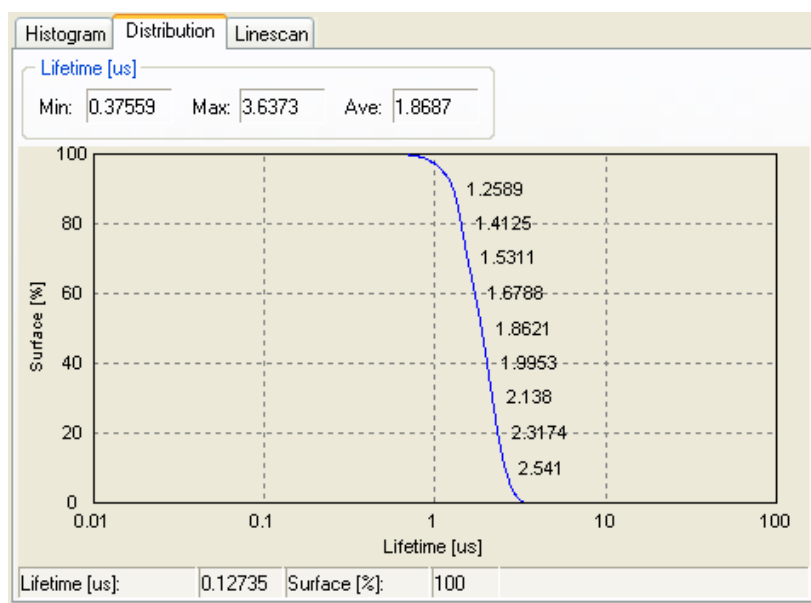


Figure 9-19: Cumulative distribution

9.7.3 Line Scan

Any cross section of the map can be displayed as a 2D plot; this is the so-called Line Scan (Figure 9-20). If you set the *Width* higher than the raster size, the rows inside this area are averaged. In the *Angle* box there are four predefined values (0°, 45°, 90°, 135°). You can use these values, or set the angle to any value between 0° and 180°.

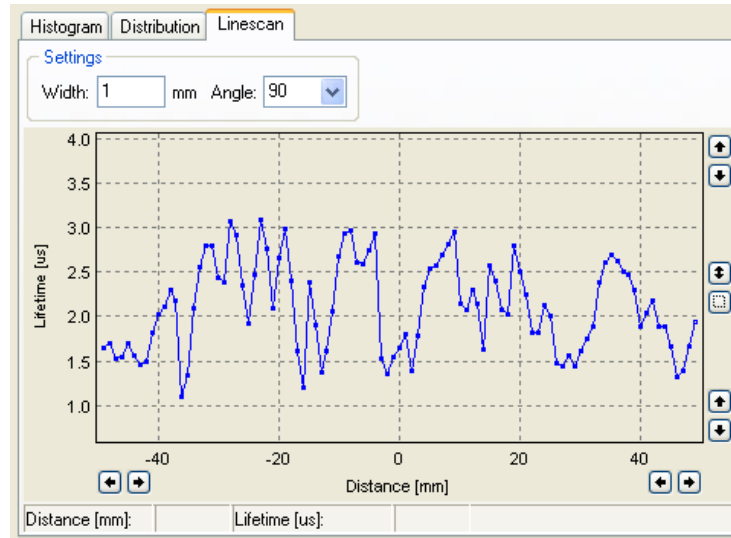


Figure 9-20: Line Scan

9.8 Saving Data

After measuring or modifying any data of the map, it is worth saving the map to keep it for further work. At the bottom of the *Main Window*, in the *Status Bar*, you can check whether your data has been changed or not. If it has been changed, the *Modified* label appears in the left of the *Status Bar*. This label disappears after saving the file. You can save it by clicking on the  button on the *Toolbar*, or by selecting the **File | Save** menu item. You can save all open files selecting **File | Save All** menu item.

If you want to save the map with a different name, choose the **File | Save As** menu item. A similar window appears as the open file window. See section 9.1 for a detailed description of the window options. **File | Save Link As** saves the linked files also.

9.9 Link Files

Sometimes it is useful to keep more than one map together. It is especially true for SPV measurements, where even during the measuring process more than one map is created. Semilab also recommends that the output file (diffusion length map) be kept together with the raw measured SPV amplitude maps. You can link maps of any type and size to have them together for further use.

When you link files, a separate file, the link file (LNK) is created with an **.sll** extension. This file contains the necessary references to the linked files so opening the **sll** file will open all files linked to the active **.sll** file.

9.9.1 Linking Files

By activating the **File | Link** menu item, the *Linked Data Files* window appears (Figure 9-21). The active map is listed in the window. Click the **Add Link** button. The *Link Loaded Map* window appears (Figure 9-22). The other displayed maps are listed. To link a displayed map to the active map, click on the map name to select it, then click **OK**. To link other files stored on the disk, click the **From File** button, and select additional files from the pop-up list. You can select multiple files by pressing the <Ctrl> key while clicking the left mouse button.

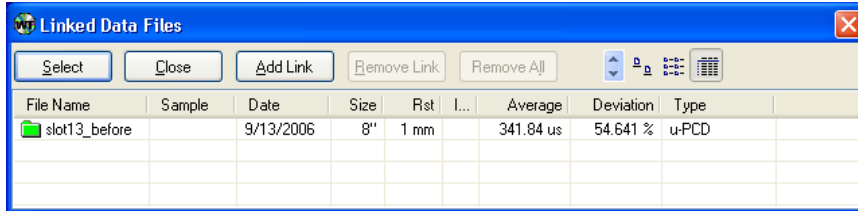


Figure 9-21: The *Linked Data Files* window

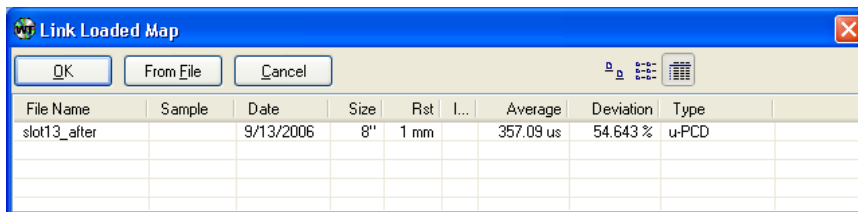


Figure 9-22: The *Link Loaded Map* window

You can remove a file from the *Linked Data Files* list by selecting a file then clicking **Remove Link**. To remove all links between the files, click **Remove All**.

You can change the appearance of the file list. The order of the linked files can be changed by the two small arrows next to the **Remove All** button. The linked files can be displayed by previews, by a simple list of the filenames or by a detailed list (as in Figure 9-21). To change the display, use the three buttons at the top right side of the window. (See section 9.1.1 for information about these three buttons.)

9.9.2 Displaying Linked Files

If a link file is active, there are two small arrows near the file type description  u-PCD on the left side of the *Toolbar*. By clicking on these arrows, you can switch the display to the next or previous linked file. If you click on the file type description, a pop-up list will appear so you can directly select the file to display. The caption of the link file map window always contains the name of the link file, while the captions of the floating windows (*Measurement Information* and *Distributions*) show the name of the displayed file.

Linked files can be printed individually or separately. See section 9.10.1 for details.

9.10 Printing Data

The map files and the corresponding distributions can be printed to any printer installed in Windows. To set up the default printing parameters, choose the **Print | Print Setup** menu item.

9.10.1 Print Setup

Set the default printing conditions using the *Print Setup* window (Figure 9-23 and Figure 9-24). The same *Print Setup* window appears before each map is printed, so print settings can be changed if necessary. If the **Enable Bitmap Acceleration** checkbox is checked, the printed map may be black.

9.10.1.1 Map Size Selection

Maps may be printed in Normal Size or Mini Size. More than one map may be printed per page, depending on the setup, as described below.

For Normal Size printing one of the windows in Figure 9-23 will appear, depending on the state of the **Show Details** checkbox.

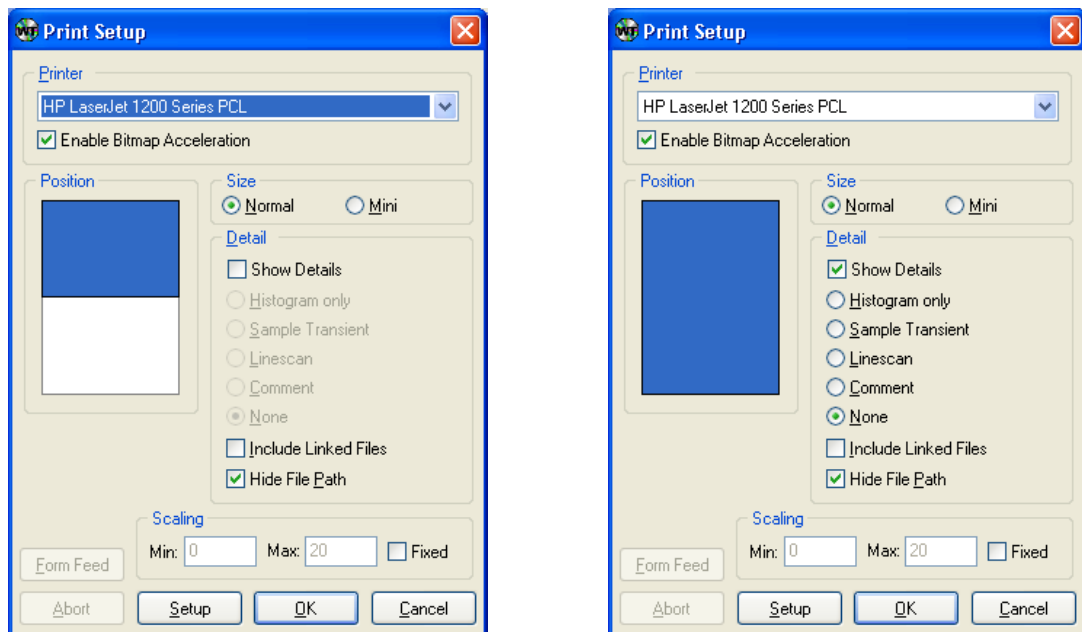


Figure 9-23: Print Setup – Normal Size

Mini Size maps can also be printed. This way several maps can be positioned on a page, 4 or even 8 depending on the **Detail** options (see Figure 9-24).

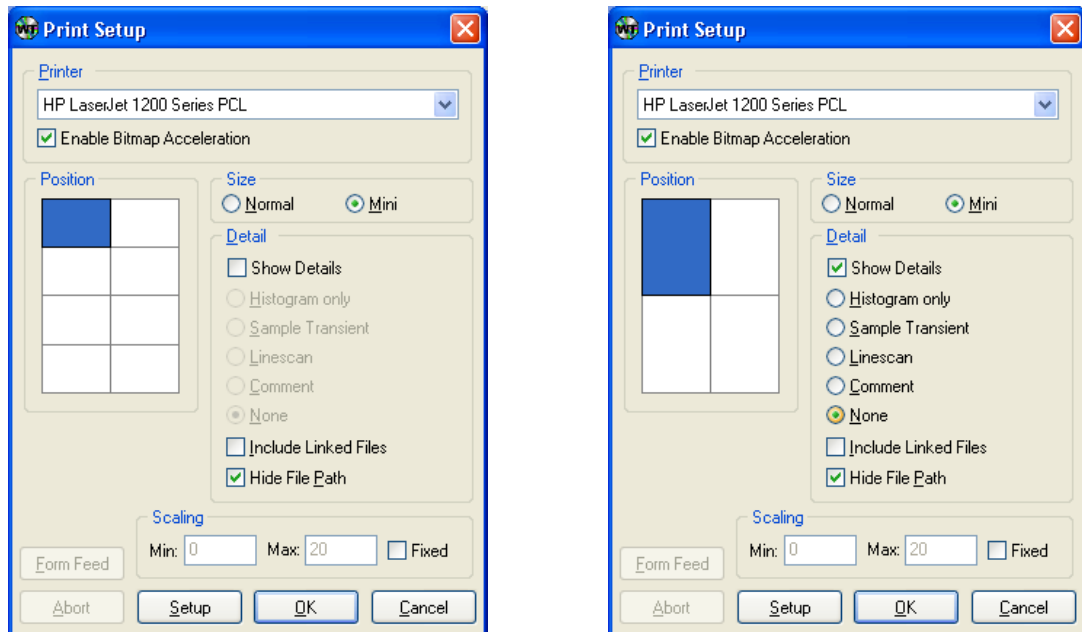


Figure 9-24: Print Setup – Mini Size

9.10.1.2 Print Setup Options

All maps are printed with summaries of the measurement setup and the results. The entries in the *Measurement Information* window (see 9.4) are printed in small font at the top of the page.

If the **Show Details** option is selected, additional details can be shown on the printout. Not only the **Histogram** but also additional information can be printed below the map according to the checked options (**Sample Transient** for lifetime measurements, **Linescan**, **Comment** or **None**). If **Comment** is selected, the text written in the *Comment* field of the *Measurement Information* window appears at the bottom of the page with larger characters.

If **Hide File Path** is checked, the file path and name will not appear on the printout.

If the details are also printed together with the map, one Normal Size map is printed on a page, otherwise each page has two Normal Size maps without details.

The printer can be selected by pressing the **Setup** button; the standard Windows print setup dialog appears.

9.10.2 Printing Displayed Maps

Select **Print | Print** menu item to print the active map. The *Print Setup* window will appear. Verify that the setup is correct, then click **OK** to begin the printing.

If you print Mini Size maps on one page, you can click on the position you want to print on the *Position* page layout display in the *Print Setup* window (Figure 9-24). The used positions are grayed out. When the last position is filled the page will be finished and an automatic form feed is produced. If you do not want to fill all positions select the **Print | Form Feed** to finish the page.

If the displayed map is a link file, you can print all the linked maps together by selecting the **Print | Print Link** menu item.

Selecting the **Print | Print All** menu item, all displayed files are printed and an automatic form feed is produced.

When you press the  *Toolbar* button the selected position on the page (to be printed) will be filled and the next empty position will be selected.

9.10.3 Printing from Files

You can print files without opening them by selecting the **Print | Print Files** menu item. A window similar to the *Open File* window appears (see section 9.1 for a detailed description). You can select the maps one-by-one and the print order of the selection is displayed to the left to the file name. Click again on the file name to remove the file from the print list. The **Select All** button selects all files in the directory.

When you print from files, the linked files are also printed if the **Include Linked Files** box is checked.

10. Measurement Setup Options

Measurement related commands are found on the **Measure** menu (Figure 10-1). A description of these top-level menu items is provided in section 8.4. Some of the **Measure** menu items cause the display of additional menus or windows, which are described in this section.

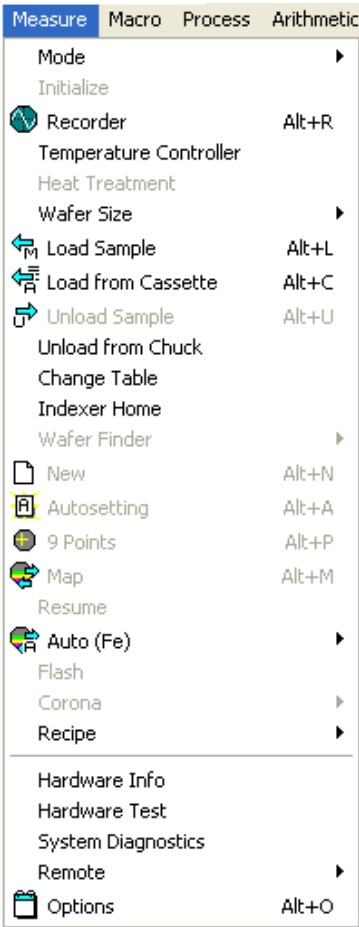


Figure 10-1: Measure menu options

Measurement Menu Item	Section
Selecting the Measure Mode	10.1
Initializing the System	10.2
Setting the Sample Size	10.3
Loading and Unloading a Sample	10.4
Wafer Finder	10.5
Resuming Map Measurements	10.6

Corona	10.7
Recipes	10.8
Hardware Info	10.9
Hardware Test	10.10
<i>Measure</i> <i>Remote Functions for WSOS2 Software</i>	10.11
<i>Measure</i> <i>Options Tabs</i>	10.12
<i>Recorder Windows and Evaluation Windows</i>	10.13

✓ **NOTE:** *Not all features may be available, depending on the options installed in your WT-2000 system.*

10.1 Selecting the Measure Mode

The measurement method can be selected from the **Measure** | **Mode** menu and from the drop-down list of the *Measurement Toolbar*. All measurement methods available for your system will be listed.

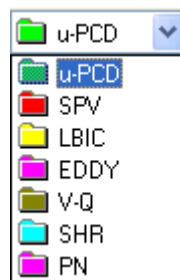


Figure 10-2: Drop-down list of measurement methods

10.2 Initializing the System

The initialization procedure checks the movements of the measuring head and the sample stage (X-Y movements) and searches for the end positions for the head and the stage movements. It sets the indexer to zero position. No sample loading and mapping can be performed until the initialization procedure is complete.

10.3 Setting the Sample Size

Clicking the **Wafer Size** menu item, a list appears (Figure 10-3). The list is divided into parts by separator lines. In each part, select the correct options as described below:

- In the first part of this list, set the size of the sample to be measured.
- In the second part, select whether the wafer is standard shape (round) or square.
- In the third part, indicate if the sample should be oriented by a flat or notch, or not oriented.

- In the last part of the list, select the stage type to be used, Table or Console (or if nothing is installed but a silicon block itself).

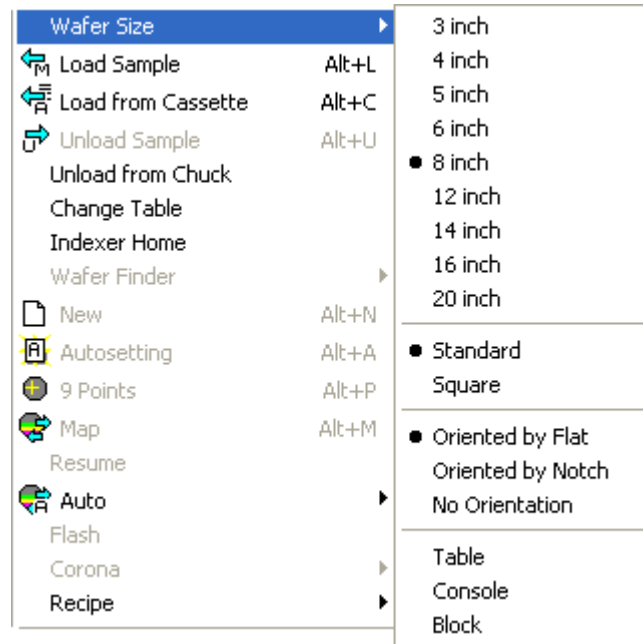


Figure 10-3: Wafer Size options

10.4 Loading and Unloading a Sample

Load Sample

Selecting **Measure | Load Sample** the Table/Console moves to the front, and a window appears requesting the user to put a wafer on the Table/Console. Clicking **OK** the Table/Console moves back. Once the wafer is loaded, further operations depend on the measurement settings (e.g. automatic wafer finder).

Load from Cassette

After selecting **Measure | Load from Cassette**, the **Load Wafer** window appears where you can select the slot from which you want to load the wafer. After loading, the Wafer Finder procedure is automatically performed (if it is selected in the **Measure | Measurement Options Loading** tab). After a successful loading process a new map window is created and the empty map image of the loaded wafer appears.

Unload Sample

The **Measure | Unload Sample** menu item puts the wafer back into the same slot of the cassette from where it was loaded or let you remove the wafer from the stage manually. For manual wafer removal, the “Remove the sample!” instruction appears.

Unload from Chuck

This menu item allows you to remove a sample from an optional hot chuck.

10.5 Wafer Finder

Before map measurements, use the **Wafer Finder** menu item to specify the exact contour of the previously loaded sample.

Clicking the **Wafer Finder** menu item, a list appears (Figure 10-4). The list is grouped into three parts by function.

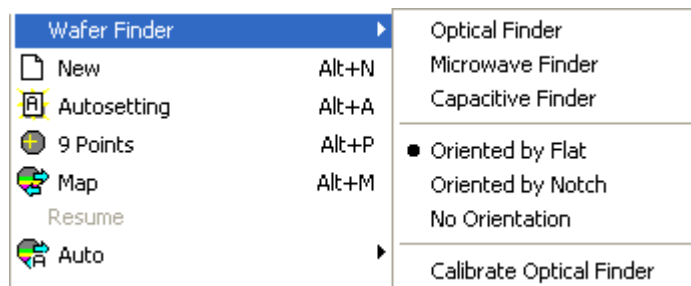


Figure 10-4: Wafer Finder options

The following choices are available for the Wafer Finder procedure:

Optical Finder

The Optical Finder should be used in most cases. The Optical Finder is preferred for polished, shiny samples or samples with antireflection coating on the standard (ESD) measurement stage.

The Optical Finder is not recommended for as-cut, non-polished wafers because the reflection of these samples is in the same range as the measurement stage.

Microwave Finder

Microwave Finder is preferred for as-cut, non-polished samples. It should be used if the sample surface is rough and Optical Finder cannot identify the edges (Optical Finder calibration fails). The Microwave Finder should also be used if the chemical passivation envelope is used. Error of this Finder is the highest among the different Finders.

Capacitive Finder

The Capacitive Finder is preferred for as-cut, non-polished samples. It should be used if both Optical and Microwave Finders fail because of special sample conditions.

Calibrate Optical

If Optical Finder fails (for example, if an unpolished sample is on the stage), use this option to recalibrate the Optical Finder to the changed surface conditions. After the calibration procedure, use the Optical Finder to verify that the sample is correctly detected.

In the Wafer Finder procedure, the contour of the sample is recognized by a reflective object sensor mounted to the measuring head next to the antenna. The stage surface absorbs the infrared light used in the sensor in contrast to the normal sample surfaces. The reflection of the mechanically and chemically polished samples can be distinguished from the background reflection of the stage.

If the incorrect wafer size is set (**Measure | Wafer Size**), a window appears at the end of the finding procedure to warn the user that the wafer size is incorrect. After that the software corrects the wafer size setting automatically.

✓ **NOTE:** In some cases wafer finding may fail if the difference is too high between the set wafer size and the loaded wafer size.

Crystal orientation is usually marked with flat or notch. If either of them is checked the finder will search for a flat or notch. The default orientation is set in the **Options** menu, *Loading* tab (section 10.12.3).

10.6 Resuming Map Measurements

Restart an aborted measurement using the **Measure | Resume** menu option.

Selecting **Resume** continues the current map. A window appears before continuing measurement that asks the user if the invalid points should be measured again. If **Yes** is selected, the head moves back to that points where the measured value was invalid (e.g. because of improper measurement parameters). If **No** is selected, the measurement will go on from the point where it was interrupted.

10.7 Corona

The **Measure | Corona** options (Figure 10-5) allows linear corona charge to be deposited to passivate the wafer surface or for V-Q measurements. The **Corona** menu items are only available for specific measurement methods and only appear after the wafer has been loaded.



Figure 10-5: Corona menu options

Linear Charge Front-side

Selecting this option causes the *Linear Charge Front-side* window to appear (Figure 10-6). Charge polarity and charge density can be set here. Clicking the **Start** button will start the sequence of the charge deposition on the front of the wafer. After the charge procedure is completed, click the **Cancel** button to close the window.

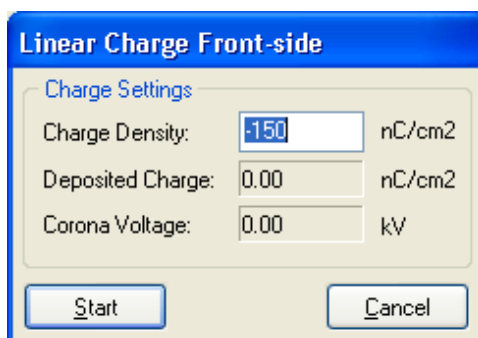


Figure 10-6: Linear Charge Front-side settings

- Linear Charge Back-side** Similar to the **Linear Charge Front-side** option, this option will deposit charge on the back side of the wafer. Back side deposition is only available if the wafer is on the Console stage.
- Linear Charge Both-sides** This option will deposit charge on both sides of the wafer. Deposition on both sides is only available if wafer is on the Console stage.
- Calibrate** It is possible to calibrate the corona charge unit. Use this option if the *Deposited Charge* differs from the set *Charge Density* value by more than 20%. (Corona calibration is described in Chapter 1.)

! CAUTION: A bare wafer must be present on the stage during calibration.

10.8 Recipes

Measurement recipes are used for performing measurements with the same settings. First the measurement conditions and parameters are set. After entering the settings, the recipe is saved to a file with a **.rec** extension. Reloading a recipe file allows the user to perform measurements with the same settings without setting the parameters again. See section 12.9 for more information about creating and using recipes.

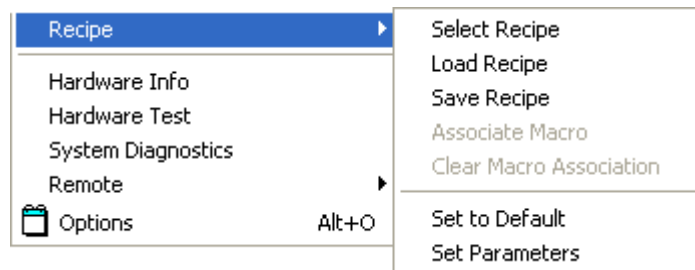


Figure 10-7: Recipe menu options

In the **Measure | Recipe** menu (Figure 10-7) the following options are available:

- Select Recipe** Recipes saved to the default Wintau32 directory can be selected from a list.
- Load Recipe** Recipes can be loaded from any path.
- Save Recipe** The current settings can be saved as a recipe.
- Associate Macro** Besides the current settings a formerly written macro can be saved with the recipe. This macro will be automatically executed when the recipe is loaded.
- Clear Macro Association** Clears the macro association done by **Associate Macro** option.
- Set to Default** After a recipe has been loaded, all settings are predefined according to the recipe. The name of the recipe is displayed at the top of the *Measurement Information* window. To clear the recipe settings, use the **Set to Default** menu item. This menu item is only available

when no wafer is loaded.

Set Parameters

Used to enter settings for creating a recipe.

10.9 Hardware Info

Selecting **Measure | Hardware Info** displays the *Hardware Info* window (Figure 10-8), with status information about the hardware. It characterizes the system and helps in case of service or maintenance. The listed items may be different, depending on the given configuration or measuring head version. By right-clicking in the window, all the *Hardware Info* data can be saved or copied to the Clipboard.

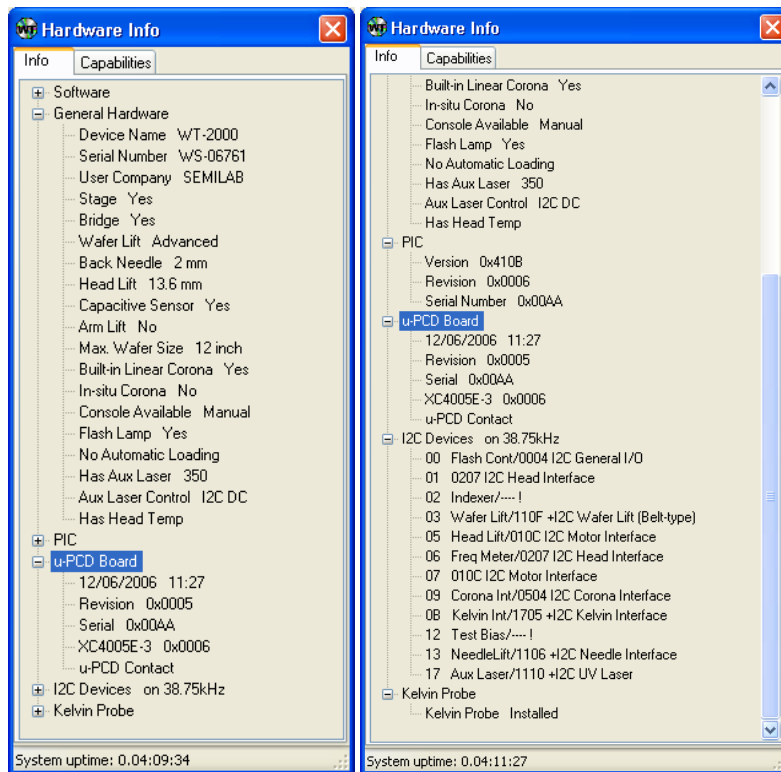


Figure 10-8: The *Hardware Info* window

10.10 Hardware Test

The *Hardware Test* window is used to check the digital input and output communication between the computer and the optical sensors, switches, vacuum valves and moving parts such as the measuring head and the sample stage etc. You do not need to check these signals if the system operates well because it means all the digital functions are fine.

There are five tabs in the *Hardware Test* window: *ATX00*, *Finder*, *Interlock*, *Interlock2*, and *Indexer*.

10.10.1 ATX00 Tab

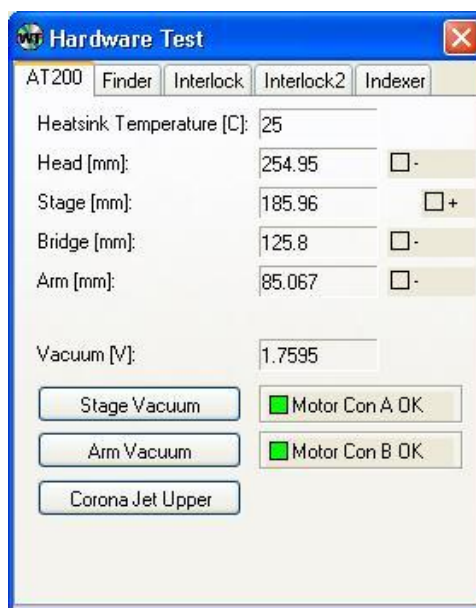


Figure 10-9: ATX00 tab

The ATX00 tab is used to test various aspects of stage operation as described in Diagnostics and Troubleshooting Manual.

<i>Heatsink Temperature °C</i>	Displays the temperature of the heat sink.
<i>Head (mm)</i>	Displays the actual position of the measuring head. If head is in the right end-position the limit indicator ☐ - “lamp” will turn on (green color).
<i>Stage (mm)</i>	Displays the position of the stage. If the stage is in on of its end positions the corresponding limit indicator “lamp” will turn on.
<i>Bridge (mm)</i>	Displays the position of the bridge. If bridge is in its end position the limit indicator “lamp” will turn on.
<i>Arm (mm)</i>	Displays the position of the arm. If arm is in its end position the limit indicator “lamp” will turn on.
<i>Vacuum (V)</i>	Displays the output from the vacuum sensor. By pressing the Stage Vacuum or Arm Vacuum button vacuum can be checked.
<i>Motor Con A(B) OK</i>	The signal “lamps” should be on. This indicates the proper installation of the connector cables between the system and industrial PC.
Stage Vacuum	Turns the stage vacuum on or off.
Arm Vacuum	Turns the arm vacuum on or off.
Corona Jet Upper	Turns the air jet on or off.

10.10.2 Finder Tab

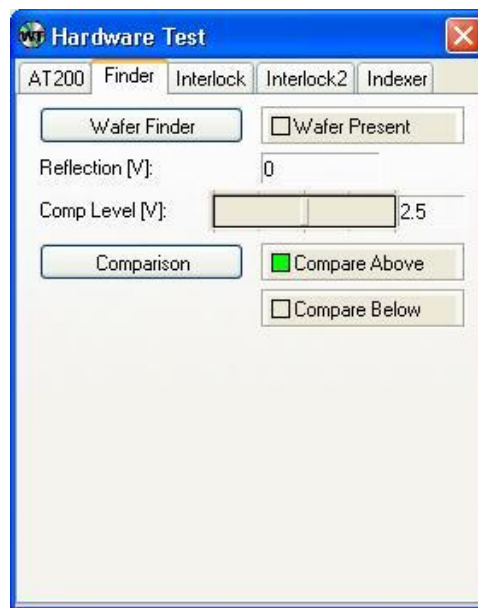


Figure 10-10: *Finder* tab

This tab includes the **Wafer Finder** button and the *Wafer Present* indicator. These buttons are used to check the operation of the Wafer Finder sensor. The **Comparison** button and *Comp Level* slider are used to adjust the sensitivity of the Wafer Finder. These procedures are explained in Diagnostics and Troubleshooting Manual.

10.10.3 Interlock Tab

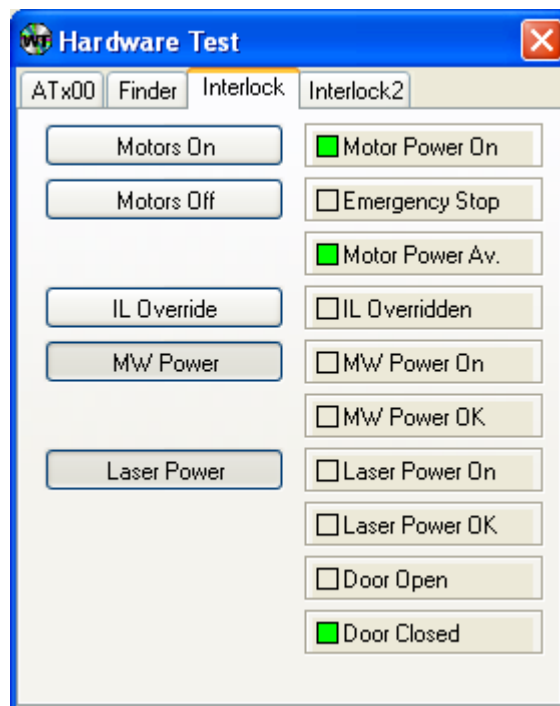


Figure 10-11: *Interlock* tab

The *Interlock* tab is used to test the motors, interlock, microwave power, and laser power. Test procedures are given in Diagnostics and Troubleshooting Manual.

Motors On button	Turns on the motors' power; indicated by turning on the <i>Motor Power On</i> control indicator.
Motors Off button	Turns off the motors' power; indicated by turning off the <i>Motor Power On</i> control indicator.
<i>Motor power Av.</i> control indicator	Indicates that the motor power is available.
IL Override button	Pressing it will override the laser and flash interlocks (works if cover is opened). When the button has been pressed, the condition is shown by the <i>IL Overridden</i> control indicator. Override is on until first close of the cover.
MW Power	Microwave power can be turned on/off by pressing this button. If microwave power is turned on the <i>MW Power On</i> and <i>MW Power OK</i> indicator indicators will turn on.
Laser Power	Laser power can be turned on/off by pressing this button. If Laser power is turned on the <i>Laser Power On</i> and the <i>Laser Power OK</i> indicator indicators will turn on.
<i>Door Open</i> control indicator	This indicator is on when the cover is opened (<i>Door Closed</i> control indicator is off).
<i>Door Closed</i> control indicator	This indicator is on when the cover is closed (<i>Door Open</i> control indicator is off).

10.10.4 Interlock2 Tab

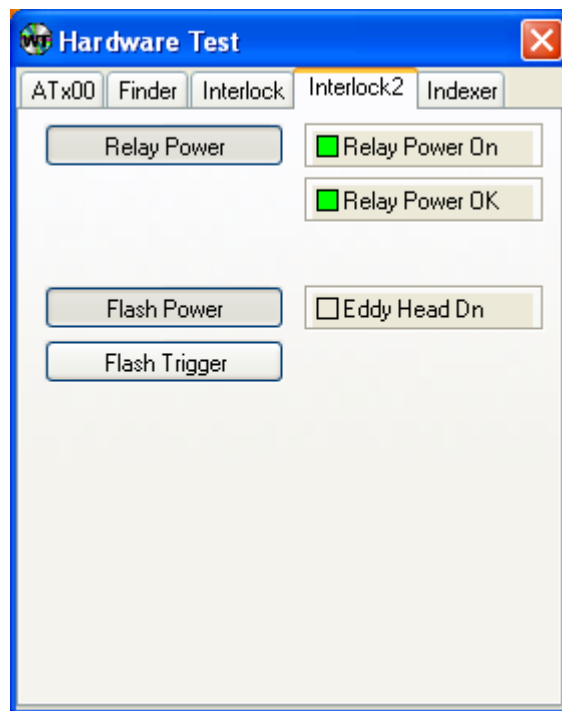


Figure 10-12: Interlock2 tab

The *Interlock2* tab tests the relay unit and indicates the status of the interlock override switch key. The **Flash Power** and **Flash Trigger** buttons can be seen only if flash unit is installed for iron contamination measurement.

Relay Power

Relay power can be turned on/off by pressing this button. If relay power is turned on the *Relay Power On* and the *Relay Power OK* indicators will turn on.

10.10.5 Indexer Tab



Figure 10-13: Indexer tab

Indexer SW A(B,C) indicator lamps indicate if the corresponding switch is on.

10.11 Measure | Remote Functions for WSOS2 Software

The software which directly controls the WT-2000 main unit (WSOS2) runs on the DOS (Measuring) computer. The **Remote** menu items control the WSOS2 software.

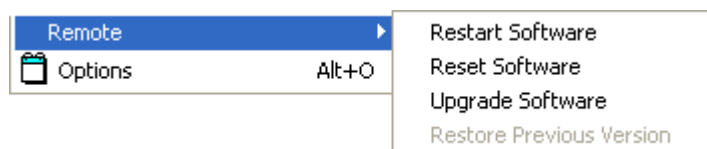


Figure 10-14: Measure | Remote functions

- | | |
|---------------------------------|--|
| Restart Software | This initiates a command that reboots the DOS computer. |
| Reset Software | It does the same as the function above and deletes the configuration file. This means that the wafer size, masks, number of points etc. will be reset to default value. |
| Upgrade Software | If you receive a new, upgraded version of the WSOS2.exe software, you can upgrade it automatically selecting this menu item (see section 15.1.5.2). You can set the path for the upgrade software in the Measure Options Interface tab (10.12.14). |
| Restore Previous Version | Undoes last upgrade changes, copies back older file versions. |

10.12 Measure | Options Tabs

To select measurement options, select the **Measure | Options** menu item, or click the **Options** button . Either action displays the *Measurement Options* window (Figure 10-15).

The *Measurement Options* window has several tabs.

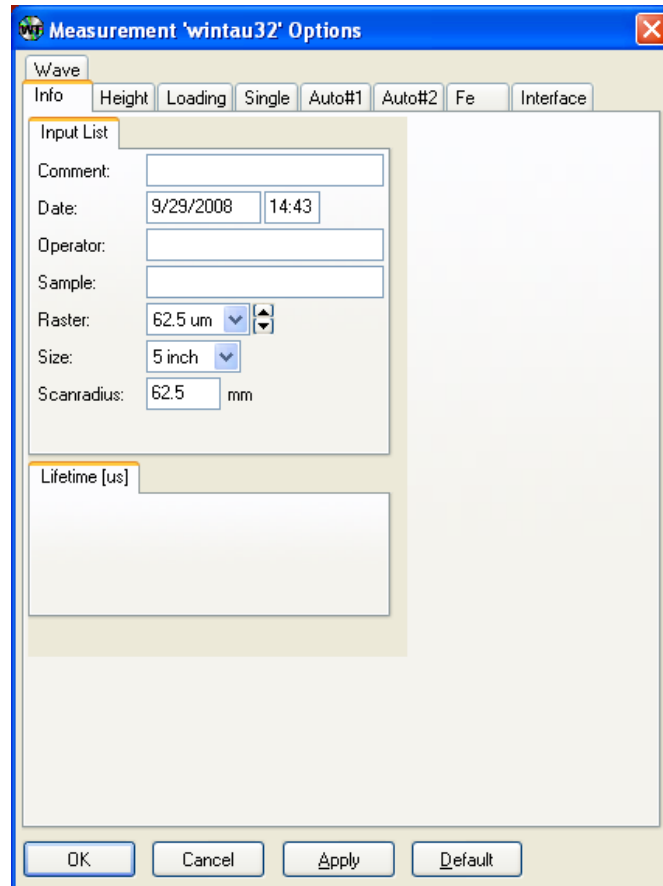


Figure 10-15: The Options window has several tabs

The measurement options can be modified only on the Engineer access level. (See section 7.8 for a description of access levels.)

Each tab has the same button choices at the bottom.

- | | |
|----------------|---|
| OK | Applies all changes to the tabs, and then closes the <i>Measurement Options</i> window. |
| Cancel | Closes the <i>Measurement Options</i> window without making any changes. |
| Apply | Applies the changes without closing the <i>Measurement Options</i> window. |
| Default | The options of the active tab are reset automatically to the default values. |

The available options on the tabs depend on the configuration of the system.

10.12.1 Info Options

The screenshot shows a software window with a tabbed interface. The 'Info' tab is selected, and within it, the 'Input List' sub-tab is active. The 'Masks' sub-tab is also visible. The 'Input List' section contains several input fields: 'Comment' (a text box), 'Date' (a date field showing '9/10/2008' and a time field showing '10:28'), 'Operator' (a text box), 'Sample' (a text box), 'Raster' (a dropdown menu showing '1 mm' and a vertical scroll bar), 'Size' (a dropdown menu showing '8 inch'), and 'Scanradius' (a text box showing '100' followed by 'mm'). Below the 'Input List' section is a section labeled 'Lifetime CPD [us]' with a large empty text box.

Figure 10-16: The Info tab with the Input List active

It is possible to set default values in the *Input List*. These default values will be applied to all measurements. See section 9.4 for more information about the *Input List*. See section 9.5 for more information about the *Masks*.

10.12.2 Height Options

The *Height* tab (Figure 10-17) contains all the head height related settings for all measurement methods installed in your WT-2000 system. After finalizing settings (by clicking **OK**), the heads will move to the set heights.

Info	Height	Loading	Single	Auto#1	Auto#2	Fe
Head Height Options						
μ -PCD Head Height:		2.5		mm		
SPV Head Height:		2.5		mm		
LBIC Head Height:		4		mm		
Eddy Head Height:		3		mm		
Eddy Jump Height:		2		mm		
Capacitive Finder Height:		3		mm		
SHR Height:		2.5		mm		
Pass. Board Used:		☒ No		☐ Yes		
Passivation Board Offset:		6		mm		
Use Capacitive Sensor:		☒ No		☐ Yes		
Always Remind Me:		☒ No		☐ Yes		

Figure 10-17: The Height tab

Typical values for the various head heights are shown in Table 10-1. However, head height settings are fine-tuned for each WT-2000 system. Your head height settings are likely to be different from the values shown in the table. It is recommended to record your head height values so that they are easily available if they need to be checked at a later date.

For certain applications, it may be necessary to use different head heights depending on the type of sample being measured. Contact Semilab for help in determining the correct head height for your application.

! CAUTION: *It is not recommended to change the height settings without instruction from Semilab. An improper head height setting can either cause a damage of the sample or the head (too low position) or decreased measuring sensitivity (too high position).*

Table 10-1
Typical Head Heights

Measuring Head	Typical Height
μ -PCD	The typical head height is approximately 2.5 mm for μ-PCD head for the standard wafer measurement without chemical passivation. In case of μ-PCD measurement of low resistivity (below 0.5-1 Ω-cm) material, 1.5 mm head height may be

Measuring Head	Typical Height
	recommended because of the relatively weak signal. For μ -PCD measurements on non-oxidized, bare semiconductor wafers, chemical passivation is sometimes performed. When using a chemically passivated sample with the protective stage, set the <i>Passivation Board Offset</i> (the default is 6 mm), and select Yes in the <i>Pass. Board Used</i> field.
SPV	Typically ~2 to 2.5 mm
LBIC	Typically ~3.5 to 4 mm
Eddy	Typically 0.5 to 1.5 mm
V-Q	Typically 2.5 mm
SHR	Typically 1 mm

The *Height* tab contains additional height parameters as described below.

Eddy Head Height

Two head height parameters are associated with the eddy current measurement method. Eddy Head Height is the head height while measuring the eddy signal.

The sliding eddy sensor touches the sample surface at this position.

The non-contact eddy sensor does not touch the sample at this distance.

Eddy Compensation Height

Applicable only for the non-contact eddy head. The head height where the eddy compensation is performed before each measurement. It is advised to be the same value as the *Eddy Head Height*.

Eddy Jump Height

Applicable only for the sliding eddy head. The *Eddy Jump Height* parameter sets the distance between the sensor and the sample during the loading procedure, wafer finding procedure, etc. At this position the sensor does not touch the sample surface.

Capacitive Finder Height

The capacitive finder also has the possibility to set its own head height by this entry field.

Pass. Board Used

If the passivation board is installed on the stage, there is a possibility to give an offset (*Passivation Board Offset [mm]*) for the head height settings. It makes the re-setting of head heights in different measuring modes unnecessary.

Use Capacitive Sensor

This feature allows setting the head heights on the basis of the capacitive sensor signal. (In the current software version, the use of this feature is not recommended; it is under testing.)

Always Remind Me

If **Yes** radio button is checked, then after any measurement method change, the *Measurement Options* window will appear with the

Height tab being active.

! CAUTION: *If the passivation board is on the table, but Pass. Board Used radio button is set to No the head may crash the table or the sample! The same situation may occur in case of incorrect head height settings!*

If the system is equipped with block scanning option, it includes a proximity sensor, and the head heights are relative to the actual sample surface. In this case, after loading a sample the head moves to the center of the sample stage (or to the center of the last measured sample), and starts lowering from the topmost position. The proximity sensor is measuring the distance from the sample surface continuously, and stops lowering at a preset point. Afterwards this vertical position of the head will be regarded as 0 head height, and head heights are counted from this point, until the next sample is loaded.

10.12.3 Loading Options

The *Loading* tab contains settings related to manual sample loading. It is used in recipe setup.

The screenshot shows the 'Loading' tab selected in a software interface. The 'Sample Loading Options' section contains the following settings:

- Wafer Shape: Standard (dropdown)
- Oriented by: ☒ Flat, ☐ Notch, ☐ None
- Optical Finder: Not Used (dropdown)
- Microwave Finder: ☐ Used, ☒ Not Used
- Capacitive Finder: ☐ Used, ☒ Not Used
- Orientation (Opto): ☒ As Found, ☐ Force 0
- Orientation (MW, Cap): ☒ As Found, ☐ Force 0
- Scanradius: ☐ Keep, ☒ Force 100 %
- Wafer Holder: Table (dropdown)
- Follow Sample Slope: ☐ Yes, ☒ No
- Detect Table Surface: ☐ Used, ☒ Not Used
- Trace Wafers: ☐ By Vacuum, ☒ Not Used
- Threshold: 3 V 3 V

Figure 10-18: Loading options

Wafer Shape

The shape of the wafer to be loaded can be selected from the drop-down list. The following shapes are available:

- standard (circle)
- square
- quarter circle

- square with radius
- half circle
- line

Orientation

Crystal orientation is usually marked with flat or notch. If either of them is checked, the finder will search for a flat or notch.

Optical Finder

If **Force Center** is selected, the finder procedure is not done, the user can enter the position of the wafer instead, by clicking the  button.

If **Used** is selected, the Optical Finder starts to locate the sample automatically after loading. If **Not Used** is selected, wafer finding may be activated from the **Measure** menu command. If **Used (Auto Calibration)** is selected, the finder calibrates itself according to the reflection of the loaded sample and then starts to locate that.

Clicking the  button when the finder is used, a pop-up window appears (Figure 10-19) to allow the user to set *Offset* and *Scaling Factor* for square samples.

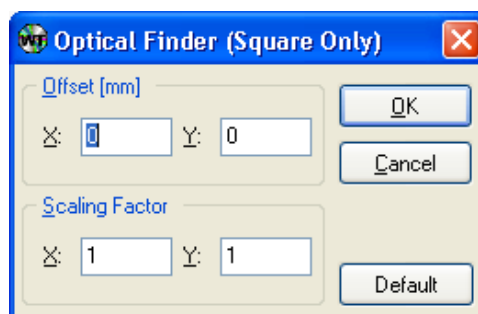


Figure 10-19: Settings for Optical Finder

Microwave Finder

If **Used** is selected, the Microwave Finder starts to locate the sample automatically after loading. If **Not Used** is selected, wafer finding may be activated from the **Measure** menu command.

Clicking the  button, a pop-up window appears to allow the user to set *Offset*.

Capacitive Finder

If **Used** is selected, the Capacitive Finder starts to locate the sample automatically after loading. If **Not Used** is selected, wafer finding may be activated from the **Measure** menu command.

✓ **NOTE:** *If the Optical Finder and another Finder are both selected, the WT-2000 will use only the Optical Finder. If the Optical Finder is not used, but the other two are selected, the Microwave Finder will be used.*

Orientation

If **As Found** is checked then the flat will be used in its real position as the wafer finder found it. In other case the contour will be found

properly, but the flat position is supposed to be in the specified angle (*Force[Degrees]*). The optical finder (*Opto*) can be set independently from the microwave and capacitive finders (*MW*, *Cap*).

Scanradius

If **Keep** is checked, the last used scan radius will be used when a new sample is loaded, or when clearing the measurement. If **Force** is checked, the scan radius will be the given percent (*Force[%]*) of the found sample diameter. If **Exclude** is selected, an edge exclusion can be set in millimeters: the outer area of the sample will not be measured

Wafer Holder

Wafer holder can be a **Table** with vacuum sample fixing and a **Console** (allows the possibility of charge passivation on both sides), or **Block**. Be sure that the correct radio button is checked.

Detect Table Surface

If **Used** is checked, then the wafer finder detector measures the reflection of the table. By this method, it decides if the metal measuring stage (needed for LBIC measurements) is installed or not, and informs the user if the metal measuring stage should be installed or removed.

V-Q Back-contact Force

Only available in V-Q mode. Sets the force of the pin used to contact the back of the sample. The unit is % of the maximal force, **25 %** is the default.

Trace Wafers:

The wafers can be traced by the help of the vacuum system: when the wafer is on the table or on the loading arm, the vacuum is blocked by the wafer.

10.12.4 Single Options

These options determine several important parameters of the measurement. There are some fields only available for a specified measurement method. Figure 10-20 shows the *Single* options.

The screenshot displays the 'Single' tab of a software interface for measurement setup. The tabs at the top are 'Info', 'Height', 'Loading', 'Single' (selected), 'Auto#1', 'Auto#2', and 'Interface'. Below the tabs, the 'Single Measurement Options' section contains the following controls:

- Number of Points:** A dropdown menu showing '9 points' and a small '...' button to the right.
- Movement Control:** A dropdown menu showing 'Measure on the fly'.
- Warn if not Saved:** Two radio buttons, 'No' (selected) and 'Yes'.
- Scan Every:** A numeric input field showing '1' and a unit dropdown showing 'line(s)'.
- Measurement Count:** Two input fields, 'Points: 0' and 'Maps: 0'.

Figure 10-20: Single tab options (shown in all modes)

Number of Points

The number of measured points can be set for x-point measurement (all modes). The user can select the number of measured points from the drop-down list. The measurement points are placed to the last saved or default positions. Going back to the measurement window (clicking **OK**) these points can be dragged on the sample area with left-mouse click.

Clicking on  button, the *Prescan Points Setup* window appears (see Figure 10-21). The *Prescan Points Setup* window gives the possibility to specify the exact coordinates of the measuring points. With selecting the **Polar** checkbox, the coordinates are converted to polar (radius, angle).

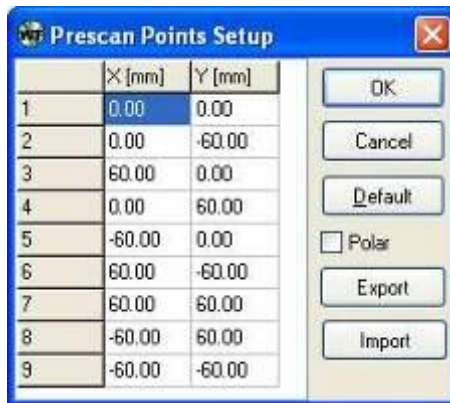


Figure 10-21: Prescan Points Setup window

Export Exports coordinate data to a *.ppe file, a text file listing each point and its coordinates. It can be useful in case of large number of points. In such a case user may edit a file in many other software programs (e.g. Notepad, Excel) which can be easier to use in this manner.

Import Imports coordinates from a *.ppe file.

Movement Control

If **Measure on the Fly** is checked, the head moves with a constant speed. This type of measurement is recommended. If **Stop at every point** is checked, the head makes discrete movement from a point to another. **Quasi-smooth** is a transition between the two other modes.

Warn if not Saved

If the **Yes** radio button is checked a warning message appears if the user closes an unsaved map and allows the user to save it. It can be easily decided if the map is saved or not taking a look at the name of it at the top of the map window. If **New** is displayed, the map is possibly unsaved. In other case the name of the map is displayed in that field.

Scan Every n line(s)

This option creates line scan measurements on the sample. Every *n* lines are scanned during mapping, where *n* is the set value in the entry field. The number of line scans depends on the *Raster* setting (section 9.4) and the size of the sample. This option makes the

measurement faster, and the resolution will be decreased only in the vertical direction.

Measurement Count This counter makes possible to track the number of measurements, done by the equipment, points measurements, and maps, implicitly.

10.12.4.1 Single Tab Options for μ -PCD Measurements

The *Single* tab has several options available only in the μ -PCD measurement method (Figure 10-22).

The screenshot shows a software interface with a tabbed menu at the top: Info, Height, Loading, Single (selected), Auto, μ -PCD, Fe, Cu, Charge. Below the tabs is a section titled 'Single Measurement Options'. It contains several input fields and checkboxes:

- 'Number of Points:' with a dropdown menu showing '9 points' and a small '...' button.
- 'Slowing Near Edge:' with two radio buttons, 'No' (selected) and 'Yes'.
- 'Movement Control:' with a dropdown menu showing 'Measure on the fly'.
- 'Warn if not Saved:' with two radio buttons, 'No' (selected) and 'Yes'.
- 'Scan Every:' with a numeric input field showing '1' and a unit label 'line(s)'.
- 'Bias Light:' with a numeric input field showing '0' and a unit label 'uW/cm2'.
- 'Used hours:' with a numeric input field showing '1.64'.
- 'Measurement Count:' with sub-fields for 'Points:' (showing '1') and 'Maps:' (showing '0').

Figure 10-22: Single tab options for μ -PCD with Bias Light

Number of Points In addition to the functions listed in section 10.12.4, the *Number of Points* value is used to define the number of PreScan measurements performed during the Autosetting procedure.

Slowing Near Edge This function helps avoid the signal distortion especially in μ -PCD mode at the edge of the wafer, because the DC level of the reflection changes rapidly there. If it is used, the measurement takes a little bit longer since the measurement speed is decreased near the edges.

Bias Light (Only if bias light is installed for μ -PCD or SPV) The bias light intensity, in the first box, will be set when a wafer is loaded. The usage of the bias lamp can be tracked in *Used hours*.

10.12.4.2 Single Tab Options for SPV Measurements

The *Single* tab has several options available only in the SPV measurement method.

Bias Light The intensity of the SPV bias light can be set here in the actual units. Units can be μ W/cm² or mSun. Extended bias light control is possible in SPV Recorder. For SPV measurement, it is

recommended to use $10\mu\text{W}/\text{cm}^2$ bias intensity.

10.12.5 Auto Options

The *Auto* tab is used to set measurement parameters for automatic measurement (section 12.9).

The screenshot shows the 'Auto' tab selected in a software interface. The tab bar at the top includes 'Info', 'Height', 'Loading', 'Single', 'Auto', 'u-PCD', 'Fe', 'Cu', and 'Charge'. The 'Auto' tab is active, displaying 'Automatic Measurement Options'. The options are as follows:

- Measurement Type:** Radio buttons for 'Points' and 'Map'. 'Map' is selected.
- Autosetting:** Radio buttons for 'Yes' and 'No'. 'Yes' is selected. There is a button with three dots to the right.
- Do Unrecognized:** Radio buttons for 'Yes' and 'No'. 'No' is selected.
- Linear Charge:** Checkboxes for 'Used' and 'Remove after meas.'. 'Used' is checked.
- Front:** Text box containing '-150' followed by 'nC/cm2'. **Back:** Text box containing '-150' followed by 'nC/cm2'.
- Temp. Cont. Startup:** Radio buttons for 'Yes' and 'No'. 'No' is selected. A text box with '30' and 'C' is to the right.
- Check for Vacuum:** Radio buttons for 'Yes' and 'No'. 'No' is selected. There is a button with three dots to the right.
- Filter:** Radio buttons for 'Yes' and 'No'. 'No' is selected.
- Align:** Radio buttons for 'Yes' and 'No'. 'No' is selected. A text box with '180' is to the right.
- Print:** A dropdown menu showing 'None'.
- Save:** Radio buttons for 'Yes' and 'No'. 'Yes' is selected. There is a button with three dots to the right.
- Export:** Radio buttons for 'Yes' and 'No'. 'No' is selected.

Figure 10-23: Auto tab

Measurement Type

The measurement type (map or x-point) can be selected with the radio buttons. By **Points**, several points are measured on the wafer, the number of points is defined in the *Single* tab. By **Map**, an x-y scanning is done, and all points of the sample are measured. By **Parallel PTC (No Link)**,

Autosetting

If **Yes** is selected, then the Autosetting procedure is activated for each wafer before measurement. If it is necessary to use the same parameter settings for the whole cassette, set the correct parameters and check the **No** radio button. In this case, Autosetting is not performed for any wafers, and the stored parameters are used for all wafers.

Do Unrecognized

If **Yes** radio button is checked, then mapping measurement will start even if a wafer loaded from cassette cannot be recognized by the Optical Finder. In such a case, the software supposes that the loaded wafer is in the same position as the previous one and starts mapping.

Linear Charge

Linear charging of the wafers before measurement can be automatically performed if the checkbox is checked. The amount of charge to be deposited can be set in the *Front* and *Back* entry fields

	in nC/cm ² units. (The <i>Back</i> entry field is only available if measurement is done on Console. See Measure Options Loading menu.) If it is necessary to reset original surface charge conditions, check Remove after measure checkbox. In this case the wafer will be charged with the same amount but opposite polarity after measurement.
<i>Temp. Cont. Startup</i>	If the system includes a hot chuck, this field allows starting heating of the hot chuck right after starting the Wintau32. The chuck is then held on the temperature given here while the software runs.
<i>Check for Vacuum</i>	The system checks if vacuum support is present, and gives error message if not.
<i>SHR Leakage</i>	Only available in SHR and JPV modes. Leakage is measured together with sheet resistance if checked.
<i>Filter</i>	If the Yes radio button is checked, the Filter procedure is applied on the map automatically right after measurement.
<i>Align</i>	If the Yes radio button is checked, the map will be aligned to the specified angle right after measurement.
<i>Print</i>	After finishing a map, there is a possibility to print it automatically by selecting Normal or Mini from the selection field. In Normal mode the printout will be a normal size map (1 or 2 maps per page), while in Mini mode the printout may contain 4 or 8 maps.
<i>Save</i>	The measured maps will be saved to a specified folder with a specified name plus the slot number. Path and name specification can be done after pressing  button, or they can be specified at the time of measurement.
<i>Save Raw Maps</i>	(Not available for μ-PCD but present if Fe concentration map is selected.) If selected, saves the raw files related to the measurement. In case of Fe measurement, not only the calculated Fe concentration map but also the lifetime (μ-PCD method) or the diffusion length (SPV method) maps measured before and after the Fe activation are saved. When using a method, that uses more than one signal channels, such as SPV, LBIC or SHR, the raw measured maps (i.e. the SPV amplitude maps) are saved if <i>Yes</i> or <i>All</i> option is selected. The results of the choices <i>Yes</i> and <i>All</i> is different only if Fe concentration is measured by SPV. If <i>Yes</i> is checked, only the diffusion length maps, before and after flash (and the Fe map) are saved. If <i>All</i> is checked, the SPV signal maps are saved for all wavelengths too.
<i>Export</i>	If Yes is checked, the measurement result is exported to an ascii file that contains the measurement settings, the coordinates of the measurement points and the corresponding measurement results, after measurement.

10.12.6 *u*-PCD Options

The options of this tab are only available in μ -PCD mode.

The screenshot shows the 'u-PCD' tab selected in the software interface. Below the tab are two sections of options:

- uPCD Options:**
 - Average: ☐ Preset ☒ Auto
 - Laser Power: ☐ Preset ☒ Auto
 - Amplitude Range: ☐ Preset ☒ Auto
 - Low Time Ranges: ☒ Used ☐ Excluded
 - Time Window: ☐ Preset ☒ Auto -> max [dropdown menu]
 - Amplitude Correction: ☐ Used ☒ Not Used
 - Slope Correction: ☐ Yes ☒ No
- Multicrystalline Block Options:**
 - Bidirectional Scanning: ☐ Yes ☒ No

Figure 10-24: *u*-PCD tab

<i>Average</i>	The Averaging used during measurement, can be set manually (Preset checked), or can be set by Autosetting procedure (Auto checked).
<i>Laser Power</i>	The Laser Power used during measurement, can be set manually (Preset checked) or can be set by Autosetting procedure (Auto checked).
<i>Amplitude Range</i>	The Amplitude Range used during measurement, can be set manually (Preset checked) or can be set by Autosetting procedure (Auto checked).
<i>Low Time Ranges</i>	If Used is checked, enables using time ranges below 0.1ms for Autosetting. It is recommended to use in case of low lifetime samples (lower than 1 μ s).
<i>Time Window</i>	Time Window used during measurement, can be set manually (Preset checked), or can be set by Autosetting procedure for each wafer (Auto checked). In case of Auto is selected, a maximum range can be defined by the <i>Max</i> selection field next to the Auto button. If Autosetting sets a higher time window, it would be decreased to the set maximum value.
<i>Amplitude Correction</i>	Corrects the microwave non-linearity of the system by measuring a MW-DC curve and modifying the transient amplitude.
<i>Slope Correction</i>	The microwave DC level sometimes changes even during the measurement and it results in a distorted transient shape. The Slope Correction procedure takes the slope of the microwave DC level into

account when evaluating the transient.

- Bidirectional Scanning

This option is only available if the *Block* is set as wafer holder. If **Yes** is chosen, the movement during measuring the horizontal lines of the scan, is bidirectional: odd lines are measured from the left to the right, even lines are measured from the right to the left.

This option makes the measurement faster, but it produces a slight zigzag artifact in the map.

10.12.7 QSS u-PCD Options

This tab can be accessed in u-PCD mode, if quasy steady-state u-PCD (QSS u-PCD) is installed in the system.

InfoHeightLoadingSingleAutou-PCDQss u-PCDCassette

QSS-uPCD Options

Qss-uPCD:

Yes

No

Wafer Thickness

200

um

Doping Concentration:

1.00E+15

1/cm3

Laser Power:

120

E11

Number of Steps for Scan:

20

Qss Light for J0 map:

1000

mSun

1500

mSun

Figure 10-25: QSS u-PCD options

- Qss-uPCD

The QSS measurement can be switched on and off here.
- Wafer thickness

The thickness of the wafer that is needed for J_0 calculation.
- Doping Concentration

The doping concentration of the substrate is to be entered here.
- Laser Power

The laser power of the u-PCD excitation can be given here. The default is 120E11 photons/pulse.
- Number of Steps for Scan

The number of steps during the injection level scan can be entered here. This is the number of points that are measured in the lifetime-injection level function. This scan is done in the prescan points.
- Qss Light for J0 map

If J_0 is mapped, the lifetime is meaasured in every points with two Qss light values. These values can be given here.

10.12.8 SPV Options

The SPV tab has several options available only in the SPV measurement method (Figure 10-26).

Figure 10-26: SPV tab options

Refractive Index

The refractive index of the dielectric layer (SiO_2 or SiN) on the sample surface. This parameter is used when the thickness of the oxide or nitride is given, and its reflectance is calculated and the injection level is compensated with it.

Point Meas. Average

Averaging used in case of x-point measurement.

*Thickness Corr.,
Correlation*

This function provides thickness correction and applies a correlation function for SPV measurements. If the **Used** radio button is selected, the measured values will automatically be corrected with the selected functions. Clicking on  button a window appears (Figure 10-27) with the following choices:

- **Thickness Correction:** The *Wafer Thickness* should be entered in μm units and the *Backside SRV* (Surface Recombination Velocity) in cm/s units. If the SRV is not known, use a default value for calculation (**10000 cm/s**). This function corrects the measured SPV diffusion length by a factor that depends on the wafer thickness and backside SRV.
- **Diff. Length Correlation:** Using the default value converts the measured values to correlate well with the other commonly used SPV method.

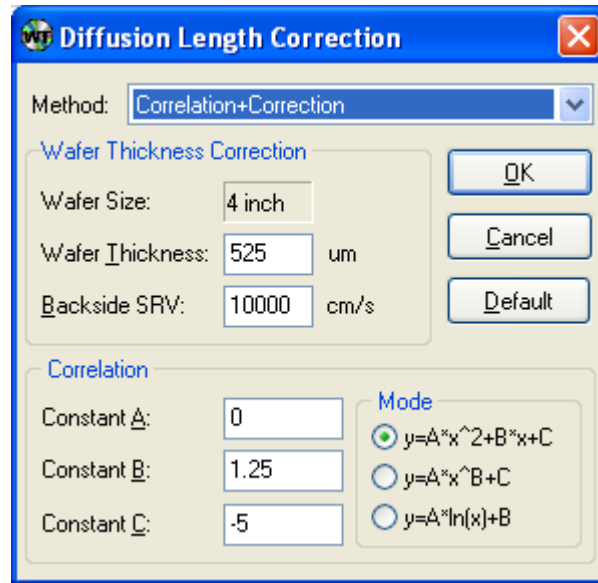


Figure 10-27: Diffusion Length Correction window

SPV Autosetting

Three Autosetting modes can be selected from the drop down list.

400 Hz sets averaging to 16, measuring frequency to 404 Hz and switches the filter on and internal amplification to 10x. This function takes the SPV Amplitude versus Light Flux curve and selects the amplitude, where the error function has a minimum.

1.6 kHz is the same as above except that the averaging is 8 and the frequency is 1616 Hz.

Keep frq is the same as above except that all the parameters can be changed freely.

400 Hz mode is recommended for standard measurements. However 1.6 KHz mode gives better signal to noise ratio, in the case of long diffusion lengths (above 100 μm), but gives lower values. **Keep frq** mode is for experimental purposes.

Autosetting Position

This option specifies whether the Autosetting procedure is performed before the first point (*One Point*) or before every measurement point (*All Points*).

10.12.9 Fe Options

The *Fe* tab (Figure 10-28) is available when using μ -PCD or SPV. In other modes, the *Fe* tab does not appear.

μ-PCD
SPV

Figure 10-28: Fe tab options for μ-PCD and SPV

- Fe Conc. Measurement** If **Yes** is selected, the Fe concentration measurement is activated.
- Fe Concentration Constant** Depending on the selected measurement method, the Fe concentration is calculated in slightly different way with different constants. It is recommended to leave the constant values unchanged. The defaults are: μ-PCD: 3400; SPV: 1000.
- Number of Flashes** The number of flashes that are already done by the current lamp can be tracked in this window. When a flash lamp is replaced, this field should be reset by double clicking on it.
- Detect Flash** If the system includes a calibrator diode, it can be used to ensure that the flash works properly.
- Flash Count Setup** The Fe concentration measurement is performed via two consecutive measurements of the same wafer, one before and one after the flash lamp illumination. Different numbers of flashes can be defined here for different lifetime intervals. The default value is **10 for μ-PCD** and **30 for SPV**. The assumed dissociation factor of Fe-B pairs with the specified flash count can be given in the *Diss.Fact.* field.
- Flash Init Delay** When turning on the flash unit it needs some time to get ready. This is the *Init Delay*. The default delay is **15** seconds.
- Flash Delay** After flashing the lamp, the unit needs time to be ready to flash again. This is why a delay is needed before the next flash. The default delay is **6** seconds.
- Flash Cooling** After flashing the lamp, the system needs time to cool down. This cool down time can be set here. The recommended minimum value is **180 s for μ-PCD** and **120 s for SPV**.

<i>Flash Recovery</i>	After a number of flashes the sample surface needs time to stabilize. This time can be set here in second units.
<i>Autosetting After Flash</i>	If Yes is checked, then the Autosetting procedure automatically starts after flashing. The Yes setting is recommended.
<i>Calculate Other Conc.</i>	Any type of contamination (other than Fe) can be measured by this feature, provided that the recombination effect of the contamination changes by flashing the sample. The parameter that is used in the calculation of the concentration can be set in <i>Other Conc. Constant</i> field.
<i>Fe Conc. Limit</i>	If a smaller Fe concentration is measured and calculated, then the displayed Fe concentration will be this number.
<i>Pre Anneal</i>	If the system includes a hot chuck, it can be used to do a heat treatment (pre anneal) before the measurement. By pre anneal, it can be ensured that all Fe atoms in the sample get paired by B atoms, and therefore they will be possible to be measured. This feature can be used for Fe repeatability measurements too.

10.12.10 Charge Options

The *Charge* tab (Figure 10-29) is available in μ -PCD mode when the system has the C-PCD option. In other modes, the *Charge* tab does not appear.

Figure 10-29: Charge tab options

If *Use Charge PCD* is set to **Yes**, measurements will be performed in C-PCD mode. The settings below determine in situ charge parameters:

<i>Use Charge PCD</i>	If Yes is selected, the C-PCD option is used in μ -PCD mode. In situ charging of the wafer is done during map measurement.
<i>Pin Voltage</i>	The voltage applied on the charger pin can be specified in the appropriate entry field. The same possibility is available for the line charger also. The minimum and maximum values are shown in brackets next to the entry field (e.g., -7...3 kV).

<i>Autosetting Delay</i>	If C-PCD mode is used for mapping, the Autosetting is followed by further parameter setting which is for setting the time and voltage ranges for the C-PCD measurement. This parameter setting starts with a pre-charge process whose duration can be set in this entry field. If the Auto checkbox is checked, the duration is determined automatically. After pre-charge a lifetime-time function is measured. The time and amplitude ranges are continuously set for the measured transients until this function saturates.
<i>Autoset.Point.Meas. Delay</i>	This option is the same as <i>Autosetting Delay</i> but is used for x-point measurements.
<i>Point Meas. Delay</i>	A delay before measuring a single point can be set in case of x-point mode.
<i>Pin Offset</i>	The relative position of the measuring head and the upper pin charger can be set precisely. Usually the default 0 value is adequate, but it should be set in some special cases e.g. if two measuring heads are built in.
<i>LTT Interval</i>	It is possible to limit the duration of the lifetime-time function measurement during parameter setting. In this case the measurement stops if the curve is saturated or the set value is reached.

10.12.11 Eddy Options

The screenshot shows a software interface with several tabs: Info, Height, Loading, Single, Auto, Eddy (highlighted), Cassette, and Interface. Below the tabs is a section titled 'Eddy Options'. It contains three rows of settings:

- Check Distance Range:** Radio buttons for 'Yes' and 'No'. 'No' is selected.
- Find Optimal Height:** Radio buttons for 'Yes' and 'No'. 'Yes' is selected.
- Single Sample Method:** Radio buttons for 'Yes' and 'No'. 'No' is selected. To the right, there is a text input field labeled 'Distance +/-:' containing the value '0.5' followed by 'mm'.

Figure 10-30: Eddy tab options

The *Eddy* tab (Figure 10-30) is available only in Eddy measurement mode, in case of non-contact eddy head. The following settings can be made:

<i>Check Distance Range:</i>	The non-contact eddy sensor gives accurate results when the sample - probe distance is in the range of 1 - 2 mm. If <i>Yes</i> is checked, the values measured outside this range are regarded invalid. If <i>No</i> is checked, these values are measured too. In this case caution has to be taken, as the values measured in such points can be less accurate.
<i>Find Optimal Height:</i>	If <i>Yes</i> is checked, the head height is adjusted to optimal before measurement. This overrides the head height set in <i>Height</i> tab, section 10.12.2. <i>Yes</i> is the advised setting.

Single Sample Method: The distance sensor compensates most of the height dependence of the resistivity measurement, but there remains a weak dependence that changes from sample to sample. This can be eliminated by measuring the resistivity-head height function before each measurement. This feature is advised to be switched on by checking Yes when huge variation in the sample - probe distance is expected during mapping. E.g. uneven sample or block with not parallel sides.

The *Distance* +/- range is advised to be set so that the maximal and minimal sample to probe distance fit into the range.

10.12.12 V-Q Options

The V-Q tab (Figure 10-31) is available only when the V-Q measurement method is selected. For other methods, the V-Q tab does not appear.

V-Q Options

Measurement Method: **V-Q (Tox,Vfb,Dit,Qtotal)**

Aver. Point Meas.: **Preset** Map: **Preset**

Si Type: ☒ n-type ☐ p-type

Doping Concentration: **1.00E+15** 1/cm3

Resistivity: **4.54** Ohmcm

Point Meas. Delay: **2** s

Automatic Settings: ☒ Yes ☐ No

Precharge Correction: ☐ Used ☒ Not Used

Minimal Q Step: **5** nC/cm2 **Learn**

Q Step Factor: **100** %

End Q (Map): **-200** nC/cm2

Measure Bright: ☒ Yes ☐ No

Dielectric Constant: **3.82**

Q Limit: **1000** nC/cm2

Max. Measurement Time: **0** min

Figure 10-31: V-Q tab options (V-Q measurement method shown)

The settings of the currently chosen measurement method are displayed in this window. To view the settings of a different measurement method, click the drop-down *Measurement Method* list. Note that this action also changes the Sub Measurement Mode in the *Measurement Toolbar*.

The options displayed in the V-Q tab vary depending on the selected measurement method. Possible options are described below.



**WARNING: Do not touch the bottom of the measuring head.
There may be high voltage present!**

<i>Preset</i>	It is possible to set the averaging of x-point measurements (section 12.6.2) and mapping independently. If a field is set to Preset , then the averaging set in the <i>Recorder</i> window will be used during measurement.
<i>Oxide Thickness</i>	The oxide thickness should be entered here. This may also change to <i>dox</i> in tunnel electric field and PDM modes.
<i>Si Type</i>	Select the conductivity type of the sample.
<i>Doping Concentration</i>	The doping concentration of the sample should be entered in cm^{-3} units in the <i>Doping Concentration</i> entry field. When <i>Doping Concentration</i> is entered, the <i>Resistivity</i> value is automatically calculated. The value entered in the V-Q tab becomes the default value in the <i>Measurement Information</i> window's <i>Input List</i> (see section Hiba! A hivatkozási forrás nem található.). The <i>Input List</i> can be modified to accept <i>Resistivity</i> input instead of <i>Doping Concentration</i>. See section 9.4.)
<i>Resistivity</i>	The resistivity should be entered into the <i>Resistivity</i> field in $\Omega\text{-cm}$ units. When <i>Resistivity</i> is entered, the <i>Doping Concentration</i> value is automatically calculated. The resistivity entered in the V-Q tab determines the default doping concentration in the <i>Measurement Information</i> window's <i>Input List</i> (See section Hiba! A hivatkozási forrás nem található.).



NOTE: The conductivity type must be correct before the doping concentration or resistivity is entered.

<i>Point Meas. Delay</i>	The <i>Point Meas.Delay</i> entry field gives the time elapsed between the head movement to the measuring site and the refreshing of the surface potential voltage. This delay is necessary because some time is required for the stabilization of the surface potential. The default delay is 0.5s .
<i>Precharge Correction</i>	As charging to negative and then to positive has a hysteresis, the Q_{tot} calculation has to be corrected. This correction can be done by using the <i>Precharge Correction</i> . To enable the correction, select the Used button.
<i>Automatic Settings</i>	If <i>Automatic Settings</i> Yes is selected, special fields appear as shown in Figure 10-32.

Automatic Settings:	☒ Yes	☐ No
Precharge Correction:	☐ Used	☒ Not Used
Minimal Q Step:	5	nC/cm ² Learn
Q Step Factor:	100	%
End Q (Map):	-200	nC/cm ²

Figure 10-32: Automatic Settings

Minimal Q Step Sets the minimum step of the charging process. Pressing the [Learn](#) button, the physically possible minimum step with the maximum corona wire voltage is calculated automatically in order to ensure homogeneous charging of sample.

Q Step Factor The Wintau32 software sets a charge step automatically. The given percent of this step is used during the measurement. The software ensures that the minimum charge step will not be smaller than the minimum possible attainable with the maximum corona voltage.

End Q (Map) Gives the final surface charge condition used to calculate an oxide thickness map. (A more detailed description is provided below.)

If **Automatic Settings No** is selected, the following measurement parameters (Figure 10-33) must be set manually. (If **Yes** is selected, they are set automatically.)

Automatic Settings:	☐ Yes	☒ No
Precharge Correction:	☒ Used	☐ Not Used
Start Q:	-300	nC/cm ²
End Q (Points):	200	nC/cm ²
End Q (Map):	-200	nC/cm ²
Q Step:	20	nC/cm ²

Figure 10-33: Manual settings (typical values for p-type wafers are shown)

StartQ The initial surface charge condition (first point of the V-Q curve)

EndQ (Points) The final surface charge condition (last point of the V-Q curve)

EndQ (Map) In the oxide thickness map measurement, two maps are performed. The initial map is performed at the surface charge condition specified in the **StartQ** field. The final map is performed at the surface charge condition specified by the **EndQ (Map)** field. The oxide thickness map is then

calculated from these two maps.

Q Step Gives the charge steps of the V-Q curve.

Measure Bright Makes a “bright” measurement (with the light on). The result is typically compared to a “dark” measurement. If the two results are the same, the surface is in accumulation. If the results are different, the surface is in depletion or inversion.

Dielectric Constant Enter the dielectric constant of the film on the sample. For example, the dielectric constant of silicon dioxide is 3.9.

Q Limit The *Q Limit* specifies the maximum amount of charge that will be applied. If the V-Q curve does not reach saturation before this limit, no additional charge will be applied and the measurement will stop.

Max. Measurement Time The *Max. Measurement Time* specifies the longest time for the measurement. If the measurement is not completed by this time limit, it will stop.

The E+, E- and E+/E- measurements have identical parameter setting possibilities. Note that E_{tunnel} measurements are x-point measurements.

The selections for the Q_m measurement method have different options, as shown in Figure 10-34 and described below.

The screenshot shows the 'V-Q Options' dialog box with the 'Measurement Method' set to 'Qm (Mobile Charge)'. The parameters are as follows:

Parameter	Value	Unit
Measurement Method	Qm (Mobile Charge)	
Aver. Point Meas.	Preset	
Map	Preset	
Temperature	180	C
Heat Treatment Time	240	10 s
E+	7E5	V/cm
E-	-4E5	V/cm
Bias Laser	☒ Used ☐ Not Used	
Qm Limit	1E9	1/cm2
Q Step	20	nC/cm2
Oxide Thickness	0	nm

Figure 10-34: Options for Q_m measurement method

Temperature The heat treatment temperature can be set here. A recommended value is **180°C**.

✓ **NOTE:** It is possible to preheat the built-in hot chuck of the WT-2000 wafer tester using the Windows Scheduler. *Hotch_on.vbs* and *Hotch_off.vbs* files can be opened with Windows Scheduler, and the switch on and switch off times can be predefined by them.

<i>Heat Treatment Time</i>	The length of the heat treatment can be set here in seconds. A recommended value is 240 seconds (4 minutes).
<i>E+</i>	Defines the electric field used to set the initial condition. The built-in corona charger deposits charge until the specified electric field condition is reached. A typical value of <i>E+</i> is 7E5.
<i>E-</i>	Defines the electric field used to move mobile ions from the initial condition. The built-in corona charger deposits charge till the specified electric field condition is reached. A typical value of <i>E-</i> is -4E5.
<i>Bias Laser</i>	The default value is Used . Using the Bias Laser means a “bright” measurement condition, which results in more precise mobile charge measurements.
<i>Qm Limit</i>	Defines the detection limit. Values below this limit are not shown on maps. The default value is 1E9 ; please do not decrease it! Values below this limit may give false results.
<i>Q Step</i>	When corona charging device charges the sample to E-/E+ it automatically charges, in one step, to a lower E than the set value. After that, it continues charging using small steps until it reaches the specified electric field condition precisely. This charge step can be modified here. The recommended value is 20 nC/cm ² (the default value).
<i>Oxide Thickness</i>	Oxide thickness of the measured sample. To calculate mobile charge, it is necessary to enter the oxide thickness.

10.12.13 Cassette Options

The *Cassette* options tab (Figure 10-35) is available for systems with automatic wafer loading capability.

Cassette Parameters [mm]

Wafer Size: 5 inch

Offset: 1 Depends on type

Pitch Size: 4.75 Standard 4.75

Partial Step: 2.5 Advised 2.5

No of Slots: 25 25 - 51

☐ Inverted Wafer Indexes ☐ Inverse Auto Process

Enabled Wafer Sizes

- ☒ 2 inch
- ☒ 3 inch
- ☒ 4 inch
- ☒ 5 inch
- ☒ 6 inch
- ☒ 8 inch
- ☒ 12 inch
- ☒ 14 inch
- ☒ 16 inch
- ☒ 20 inch

Figure 10-35: Cassette tab

The *Measurement Options* window *Cassette* tab has two parts.

In the first part, the cassette parameters and cassette test possibilities can be found. The variable cassette parameters are *Offset*, *Pitch Size*, *Partial Step*, *No of Slots*.

! CAUTION: Please contact Semilab before changing these parameters. The sample and/or the whole cassette can be damaged due to incorrect parameter settings.

Offset

The values of cassette offset can be set manually or with pressing **Check Cassette Offset** button. After pressing this button the *Check Cassette Offset* window will appear (Figure 10-36).

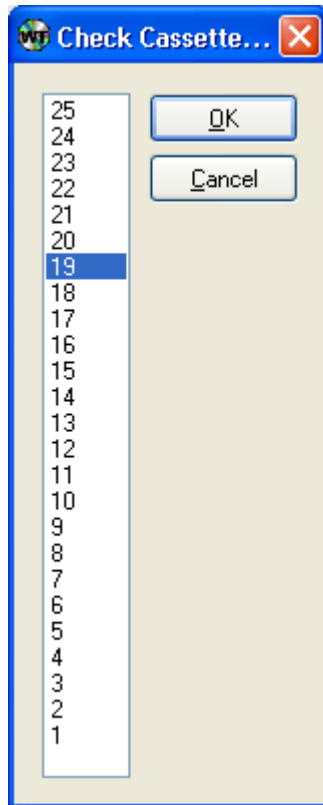


Figure 10-36: Check Cassette Offset window

In this window, a slot must be selected where the offset will be set. After selecting a slot and pressing the **OK** button, the *Cassette Offset Setting* window will appear (Figure 10-37) and the arm of the indexer will be moved near to the cassette.

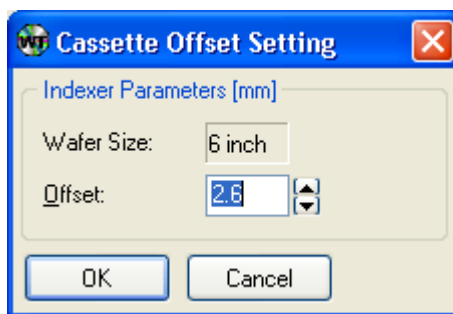


Figure 10-37: Cassette Offset Setting window

Push the arm carefully between the wafers and set the correct offset by clicking the up and down arrows . If the desired values are set, click **OK** button to end the procedure. After pressing the **OK** button, the arm will be moved to the home position. With pressing the **Cancel** button, the changes are not stored.

! CAUTION: After changing the cassette offset settings, please check cassette offset at least in three position (e. g. first, last and one from the middle region).

<i>Pitch Size</i>	The distance between two slots. The default standard values are predominantly good.
<i>Partial Step</i>	The value of arm lifting by loading a wafer.
<i>No of Slots</i>	The number of slots in the cassette.
Check Loading	With the help of the Check Loading button, a test loading can be made from a selected slot
Inverted Wafer Indexes	There are two type of cassette indexing. Normal indexing begins from the bottom of the cassette; the inverted indexing begins from the top of cassette. Choose between these two possibilities with clicking the Inverted Wafer Indexes selection box.
Inverse Auto Process	If the Inverse Auto Process option is selected, then the measurement will begin with the last wafer.

In the second part of the *Cassette* tab, the enabled wafer sizes can be seen.

10.12.14 Interface Options

The screenshot shows the 'Interface' tab of a software application. It has four tabs at the top: 'SRV', 'Cassette', 'Interface' (which is active), and 'Wave'. The 'Interface' tab is divided into three main sections. The first section, 'Communication Setup', contains several input fields: 'Communication Protocol' is a dropdown menu set to 'UDP'; 'Remote IP Address' is a text box with '192.168.2.1'; 'Remote Command Port' is a text box with '50000'; 'Local Command Port' is a text box with '50000'; 'Local Data Port' is a text box with '50002'; 'Min. Timeout/Reestablish' has two text boxes, the first with '0' and the second with '15000'; 'Retry/Reestablish Count' has two spinners, both set to '2'. The second section, 'Upgrade Software', contains two file selection buttons (represented by '...' icons) and a text box for 'Remote Directory' containing 'c:\'. The third section, 'Alarm Logging', contains a checked checkbox labeled 'Log Alarm' and a 'Clear Log' button.

Figure 10-38: Interface tab

Communication between the User PC and the Remote (Measuring) PC can be set in this tab.

Communication Protocol The protocol for remote communication can be selected here. The possibilities: UDP, TCP, IPX.

Remote IP Address This field should contain the IP address of the remote computer.

<i>Remote Command Port</i>	Command port address of the remote computer can be specified.
<i>Local Command Port</i>	Command port address of the user computer.
<i>Local Data Port</i>	Data port address of the user computer.
<i>Min. Timeout/ Reestablish</i>	If connection is lost the software waits for a specified time before starting the reestablish process (first field). The second field gives the maximum duration of the connection process. If it is exceeded, a timeout-error message appears.
<i>Retry/Reestablish Count</i>	The connection will be tried to reestablish n times if connection is lost. The n can be specified in the first entry field of this line. The second field gives the number of reconnections. If this number is exceeded an error message appears and the software stops trying reestablishing remote connection.
<i>Upgrade Software</i>	Two paths for the software upgrade can be defined by the two entry fields. The Windows-based system will search for the upgrade at these locations when Remote Upgrade Software is activated in the Measure menu.
<i>Remote Directory</i>	The destination directory of software upgrade on the remote computer.

10.12.15 Wave Options

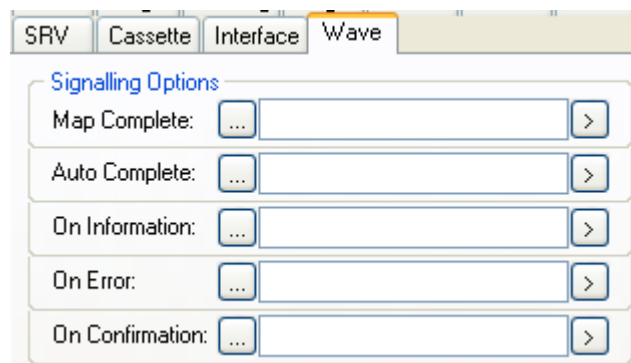


Figure 10-39: Wave tab

There is the possibility to add sound signals to different actions. To specify a signal for an action, press the appropriate  button and select a file with ***.wav** extension from your hard disk.

10.13 Recorder Windows and Evaluation Windows

Clicking the **Measure | Recorder** menu item, the *Recorder* window of the currently set measurement method appears. This window is the basic control panel of the measurements. All the measurement related parameters can be modified here.

! CAUTION: For all measurement methods it is recommended to use Autosetting for setting measurement parameters.

Different *Recorder* windows are explained in this section

10.13.1 Transient Recorder (μ -PCD and C-PCD Modes)

For the μ -PCD and C-PCD methods, the parameters and the signal parameter setting procedure itself are displayed in the *Transient Recorder* window. There are the measuring parameters and the *Scope* window, which displays the transient in the *Transient Recorder* window.

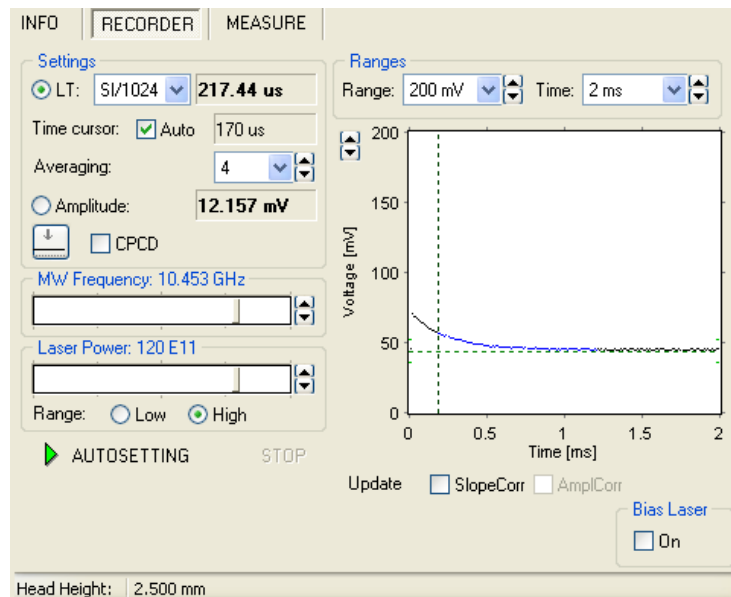


Figure 10-40: Transient Recorder window

The Microwave Frequency, the Averaging, the signal Range (Y-axis of the *Scope* window) and the Time Window (X-axis of the *Scope* window) are automatically set by the Autosetting procedure.

LT (transient evaluation method)

The *LT* selection defines the method for transient evaluation. There are three different methods, selectable with a drop-down menu: the so-called Semilab method, which was developed by Semilab; the $1/e$; and the $1/e^2$ methods.

Several selections are available for the Semilab method. It can be used with different number of points taken from the transient from the *Time Cursor*. The *Time Cursor* is the vertical line in the *Scope* window, which determines the point from where the recorded data are evaluated to avoid the initial, distorted part of the transient. The evaluated points are displayed with blue color and their number may be 64, 128, 256, 512 or 1024 (the entire recorded transient consists of 2048 data points). The number of the selected method indicates the number of points used. The recommended method is **SI1024**.

During the lifetime measurement the decay of the microwave reflectance and thus the decay of the resistivity is monitored in time.

If we assume that the recombination is governed only by the Shockley-Reed-Hall process and there is only deep level nivo in the gap then the recombination will be pure exponential. Unfortunately this is a very simplified view of the process.

If you select 1/e evaluation that means that the software will fit a real exponential to the measured curve. As it can be checked in the transient evaluation window, this value will give something near to the SI/1024 evaluation. Also it is worth to plot the measured values and the fitted exponential, you will see they are not in perfect correlation. There is also a main drawback of fitting exponential. The lifetime value will strongly depend on the offset of the curve. To overcome all these issues we use auto-correlation functions. This evaluation gives the exponential value on mathematically correct exponential decays, and will give reasonable and reliable values when this curve is distorted. Moreover this correlation function method is much faster.

The number after SI/... refers to the number of points that are used during the calculation. The more points you use the better the reliability of the evaluation. All customers are using the SI/1024. Only in very special cases we advice to use SI/512 or SI/256 (when the lifetime is extremely short <100ns) We have the option of 1/e evaluation just for "research" purposes.

The fitted lifetime is displayed near the *LT* drop-down box in μ s units.

Time cursor

The *Time Cursor* position is a point in the time scale from which the signal is used during evaluation. (Refer to the description for *LT* for more information.) Any measured point before the *Time Cursor* is neglected in the exponential fitting procedure. By the proper setting of the cursor the non exponential initial part of the decay (e.g. due to redistribution of carriers) is avoided and only the carrier recombination transient is analyzed.

The *Time Cursor* is set to automatic (**Auto**) by default i.e. the software automatically determines its position after recording a transient according to a built-in algorithm.

If the checkbox is **not ticked**, the *Time Cursor* can be set manually by dragging the cursor line with the mouse in the *Scope* window.

If the checkbox is **ticked**, two methods can be used to determine its position. The method can be changed by right-clicking in the Settings area, and calling **Autosetting Options | Auto Time Cursor**.

- **Semilab** method is the default, it is advised to use this in most cases. This ensures the best repeatability possible.
- **Maximum Amplitude** method sets the time cursor to the very beginning of the transient, where the amplitude of the signal is maximal. This option is useful when the signal is

very small.

It is recommended to use automatic mode to get reliable and reproducible results.

The position of the *Time Cursor* is displayed near the *Time cursor* checkbox in μs units.

Averaging

Averaging is the number of transients taken and averaged at a given position. The possible values are 1, 4, 16, 64, 256, and 1024. More averaging means an improved signal-to-noise ratio, at the expense of longer time necessary for the measurement.

Amplitude

The signal amplitude in the transient at the *Time Cursor* position is displayed also in the *Settings* panel in mV units.

The signal amplitude can be mapped also. The displayed amplitude corresponds to the measured signal among time cursor (the offset usually is not 0 and is extracted).

CPCD

There is a button  and a checkbox for Charge-PCD control. By pressing the button, the corona charge is activated immediately and works continuously until the button is pressed again. It can be useful when measuring, e.g. Lifetime vs. Time curve at a single point. To use the C-PCD measurement method, simply check the related checkbox. In this case the current C-PCD settings appear next to the checkbox. See **Options** menu *Charge* tab, section **10.12.10**.

Microwave Frequency (for tuning)

The reflected microwave power depends on the microwave frequency. As a consequence, the measured transient amplitude also varies. Tuning consists of finding the microwave frequency where the signal has its maximum amplitude. The microwave frequency can be modified within a 500 MHz range by dragging the slider or using the small arrows near the slider. The basic frequency is **10 GHz**.

Laser Power

The excitation level (number of excess carriers generated) is proportional to the laser power. By the application of different laser powers, transients at high and low excitation levels can be measured and compared leading to valuable information on the contamination species. The intensity of the laser is measured continuously, the laser power is indicated accordingly.

In the default case, high power (**120E11**) is used in order to obtain a good signal-to-noise ratio.

Range (for Laser Power)

The Laser Power can be changed easily, within an order of magnitude (low and high range).

Autosetting

This function activates the Autosetting procedure.

Stop

Stops the measurement or autosetting procedure. Only active during measurement.

<i>Bias Light</i>	This function is only visible if bias light option is built in the WT-2000 device. The bias light intensity can be set by the slider in mSun units.
<i>Range (in Scope window)</i>	The Y-axis of the <i>Scope</i> window is the signal amplitude: the output voltage of the microwave detector. Depending on the signal level, different sensitivities can be selected such as 1, 2, 5, 10, 20, 50, 100, 200, 500, 1000 and 2000 mV. In the case of highly conductive material (0.1-1 Ω -cm) the relative change in conductivity due to the laser pulse is small. As a consequence high sensitivity should be selected. On the other hand, measurements on wafers with high resistivity are normally performed with low sensitivity (large amplitude range), due to their high signal amplitude. The proper selection of the sensitivity range is when the signal amplitude is between the 50% and 70% of the whole range.
<i>Time (in Scope window)</i>	The X-axis is the time elapsed after starting of the transient recording (the zero point is before excitation). The time ranges available are 0.025, 0.05, 0.1, 0.2, 0.4, 1, 2, 4, 10, 20 and 40 ms. When set properly, the fitted decay time should be between 1/5 and 1/10 of the whole range.
<i>Zero offset</i>	The zero offset is the constant part of the signal, the DC level due to microwave reflection that is not influenced by laser excitation. This useless part is subtracted from the signal before amplification and evaluation. In case of a proper zero offset setting the whole transient is inside the <i>Scope</i> window. The horizontal zero line is automatically set by the Autosetting procedure but can be dragged manually as well.
<i>Signal amplitude limit</i>	In some cases the signal amplitude can vary dramatically across the wafer. For example in patterned wafers, the signal level drops where the surface is covered by metal. To avoid misleading analysis of small, decreased signals a signal amplitude limit can be set. A transient below that limit is ignored during evaluation and no lifetime value is recorded. The corresponding position in the map remains gray (invalid point), like unmeasured places. The signal amplitude limit range is shown by two points on the Y-axis below and above the zero line. Pressing the <Shift> key, the range can be modified by dragging one of these points up or down along the Y-axis.
<i>Update</i>	It is possible to perform measurements continuously after each other by checking the Update checkbox. Pressing the right mouse button on the area of the <i>Scope</i> window also makes the system perform a prompt measurement.
<i>Arrows Up/Down</i>	There are two arrows pointing upwards and downwards situated in the bottom right of the <i>Scope</i> window. These arrows are for moving the measuring head up or down. Another way to move the measuring head is to use a sliding scale, which becomes visible when you move the mouse pointer above the area where the <i>Voltage [mV]</i> scale label is displayed. The current head position is displayed in the bottom of the <i>Transient Recorder</i> window.

! CAUTION: When modifying head height be sure that the measuring head will not touch the sample. If the head height is set too low, it can break the sample and even damage the head.

SlopeCorr

If checked, the slope of the “transient’s tail” is compensated, thus resulting in more precise lifetime values.

Laser Source

It is possible to use optional laser sources for μ -PCD (or μ -PCD based) measurements. **IR (904)** is the standard excitation laser. Aux (904) is used for an optional laser.

Bias Laser

The Bias Laser is optional. If checked, the continuous mode Bias Laser will be switched on during the measurement.

10.13.1.1 Right-Click Features

There are some additional right-click features.

Right-clicking in the *Recorder Window* on the area outlined in red (Figure 10-41), the following options are available:

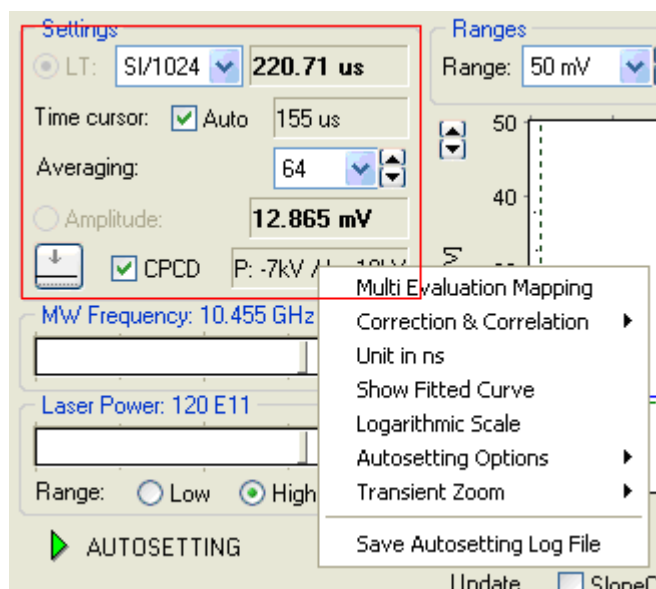


Figure 10-41: Right-click in the area shown by the red box to display additional options

Multi Evaluation Mapping

Three maps are generated at one measurement: one with Semilab method, one with $1/e$ and one with $1/e^2$.

Correction & Correlation

Different type of correlation functions can be applied to the measured lifetime. If correlation is applied, it is signed in the type of the map.

Units in ns

If selected, the lifetime is displayed in ns units. If not selected, the unit is μ s.

Show Fitted Curve	Shows the fitted curve with purple color.
Logarithmic Scale	Changes the voltage scale to logarithmic.
Transient Zoom	Zooms the transient window. Possible values are 1, 2, 5, 10 (in case of short lifetime, to see more detail in the transient, use transient zoom to 5 or 10).
Save Autosetting Log File	Saves a log file for the Autosetting procedure to the selected destination.
Autosetting Options	Opens a list of settings for the Autosetting procedure (see Figure 10-42). • Correlations should be set to y=x. • Auto Time Cursor should be set to Semilab. • Tuning Polarity should be set to Auto.

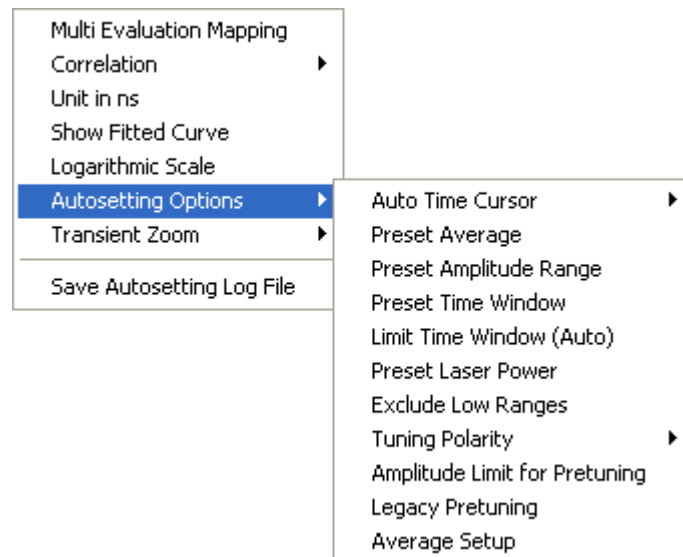


Figure 10-42: Autosetting Options

10.13.1.2 Measurement with Bias Light

The control slider of the *Bias Light* is placed on the μ -PCD Recorder window. By default, the light intensity is displayed in mSun units.

Several options can be activated by the right mouse button on the *Bias Light* control slider.

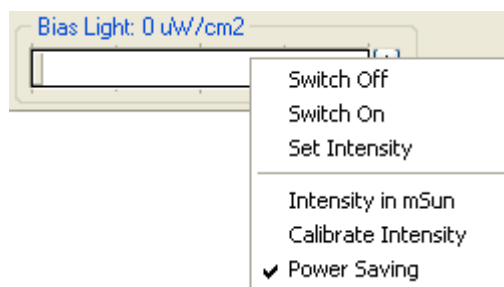


Figure 10-43: Bias Light options

Switch Off	Switch off the bias lamp.
Switch On	Switch on the bias lamp.
Set Intensity	Sets the value of bias light intensity. After you type the required value, the software searches for the nearest valid value and sets the intensity to this value. (Calibration is needed for this function. See Chapter 1.)
Intensity in mSun	Switch the intensity unit between mSun and $\mu\text{W}/\text{cm}^2$.
Calibrate Intensity	Activates the calibration process of the bias light intensity. See Chapter 1.
Power Saving	When this option is active, the ☒ symbol can be seen in front of this button. In this case the software will turn off the bias light after a period of time (selected when the option was activated).

10.13.1.3 Menu of Transient Recorder Window

In the transient recorder, a menu with several functions can be accessed by right-clicking in the rest of the recorder window.

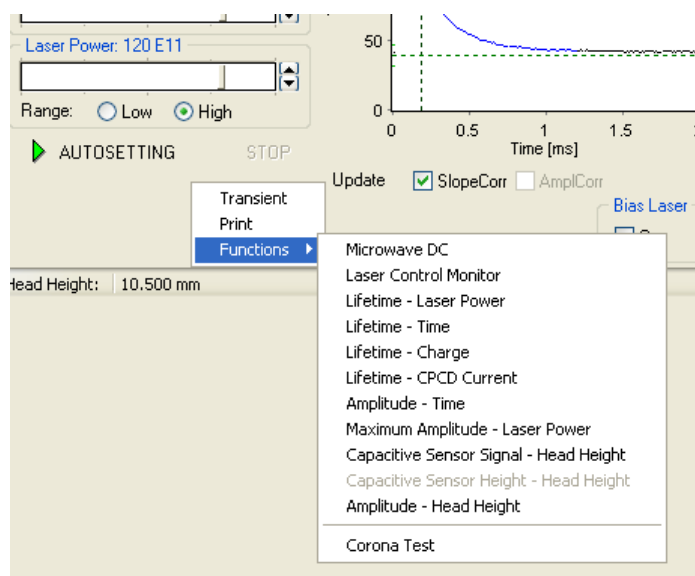


Figure 10-44. Menu in the Transient recorder

Transient	This function opens the <i>Transient Evaluation</i> window (see next section.).
Print	This function prints out the transient. In the appearing <i>Print</i> window you can set the position of the transient on the paper, and change the printer properties and settings.
Functions	This function contains the menu items shown in Figure 10-44. Menu in the Transient recorder. Many of the functions open a window and record a curve on a plot. Others are used to set the parameters. The commonly-used functions are described in the following sections.

Some commonly-used curves are described in the following sections. If there is a problem with the signals or the measurement, these curves may help to find the reason.

To refresh a curve, use the **New** command in the curve window. The curve can be loaded, saved or printed, by using the menu displayed by right clicking on it. The curve can also be copied to the Clipboard as a graph or as data.

10.13.1.3.1 Microwave DC

The *Microwave DC* window shows the Microwave DC signal versus Microwave Frequency curve. The curve is recorded continuously if the checkbox at the top left corner of the window is checked. The Microwave DC curve should be a non-monotonic (typically it has a minimum) curve between the 0V and the 5V limits measured on samples from a wide resistivity range (from insulator to metal).

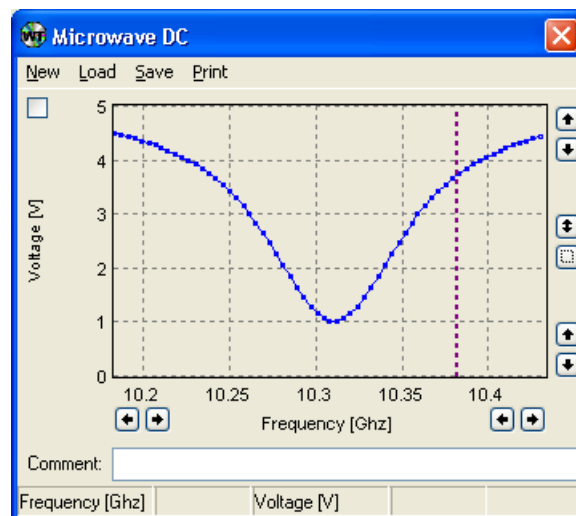


Figure 10-45: Microwave DC curve

10.13.1.3.2 Laser Control Monitor

The Control Voltage versus Laser Power curve is measured in this window. It serves for checking the stabilization of the laser. The curves should increase monotonically and should remain below the *Control Limit*.

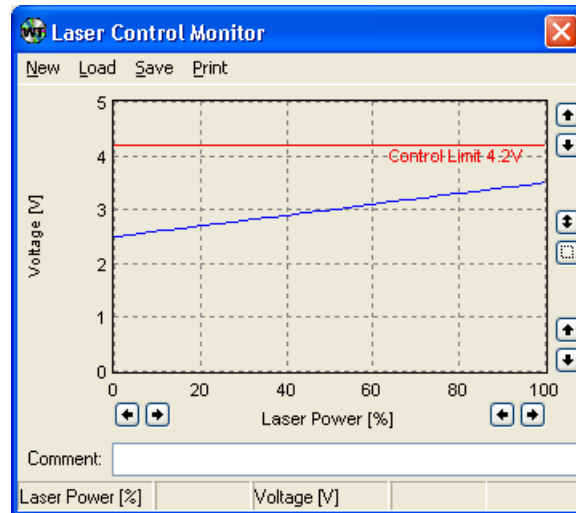


Figure 10-46: Laser Control Monitor

10.13.1.3.3 Lifetime – Laser Power

The Lifetime versus Laser Power curve is measured in this window. The resulting curve (logarithm of Lifetime versus Laser Power) is typically increasing and it can be fitted with a straight line. The slope of the line may be characteristic of the dominant contaminant in the wafer.

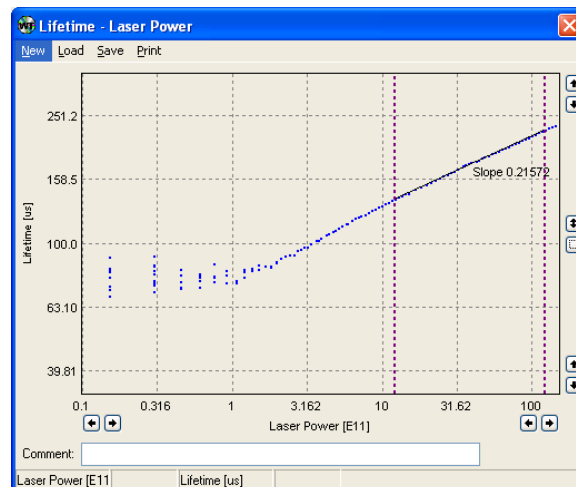
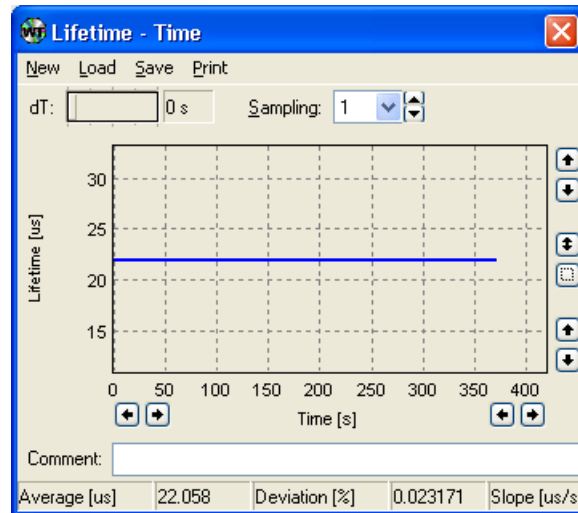


Figure 10-47: Lifetime – Laser Power

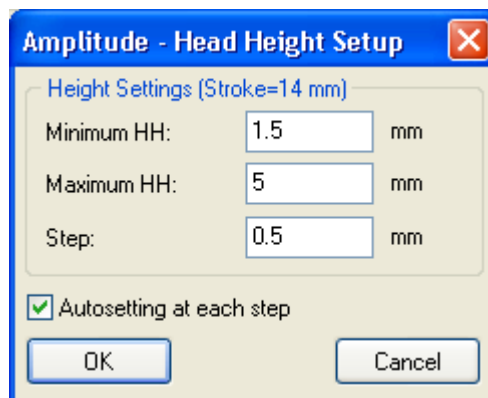
10.13.1.3.4 Lifetime – Time

The curve of Lifetime as a function of Time is recorded in this window. The measuring interval can be set between 0-250 seconds by dragging the dT slider. When dT is set to 0, the maximum available measuring speed is selected. The *Sampling* determines how many measured values are averaged before recording a point on the plot.

Figure 10-48: *Lifetime – Time*

10.13.1.3.5 Amplitude-Head Height function

The **Amplitude – Head Height** function can be started from the *Recorder Window Functions* menu. When this menu item is selected, the *Amplitude – Head Height Setup* window appears.

Figure 10-49: *Amplitude – Head Height Setup* window

This function is used to define the head height for non-contact and near surface measurement modes.

10.13.1.4 Transient Evaluation Window

When displaying a μ -PCD map, an additional button  appears in the *Measurement Information* window *Input List*. By pressing this button the *Transient Evaluation* window appears with a sample transient measured on the map. The **View | Transient** menu item also opens this window.

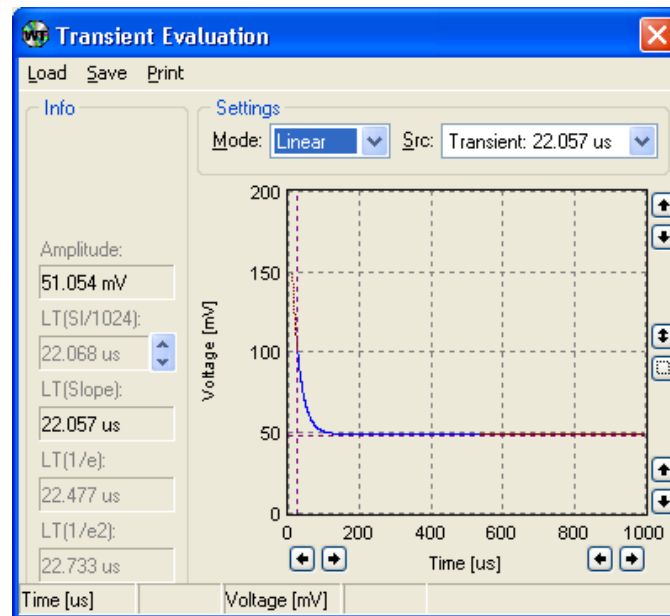


Figure 10-50: Transient Evaluation window

The *Info* area displays information about the displayed transient curve: the position where the transient was recorded, the amplitude of the transient and the lifetime evaluated with different methods.

To select a measurement to display, select the desired transient from the *Src* drop-down list.

There are four methods to illustrate the transient data. They can be changed in the *Mode* drop-down list. The transient can be plotted on linear (**Linear**) or logarithmic (**Log**) scale. It can also be illustrated as the output signal of a frequency scan (**Scan**) performed on the transient or as a function of the time cursor position (**Cursor-LT**).

The axes of the plot area can be re-scaled by the small arrows at the bottom and the right side of the plot. By pushing the  button at the middle of the right side the original scaling can be set again.

The  button allows you to select a rectangular area in the plot. This handling is true for all plots appearing in the software.

If the mouse cursor is on the plot area, its coordinates appear in the *Status Bar* of this window.

There are several right-click menu options available in this window.

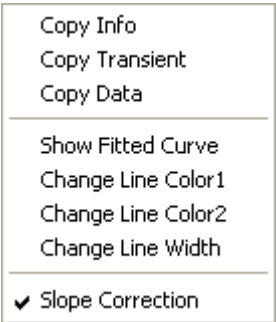


Figure 10-51: Right-click menu in *Transient Evaluation* window

<i>Copy Info</i>	Copies the <i>Info</i> panel.
<i>Copy Transient</i>	Copies the transient itself as a bitmap.
<i>Copy Data</i>	Copies transient data to the Clipboard.
<i>Show Fitted Curve</i>	If checked, shows a fitted curve of the transient.
<i>Slope Correction</i>	If checked, the slope of the “transient’s tail” is compensated. Reduces edge effect of high lifetime samples.

The transient can be loaded or printed, or the current transient can be saved to disk. These functions can be activated using the menu of the *Transient Evaluation* window.

10.13.1.5 QSS u-PCD recorder

If Qss is installed in the system, and it is switched on, the u-PCD recorder is changed to Qss u-PCD recorder:

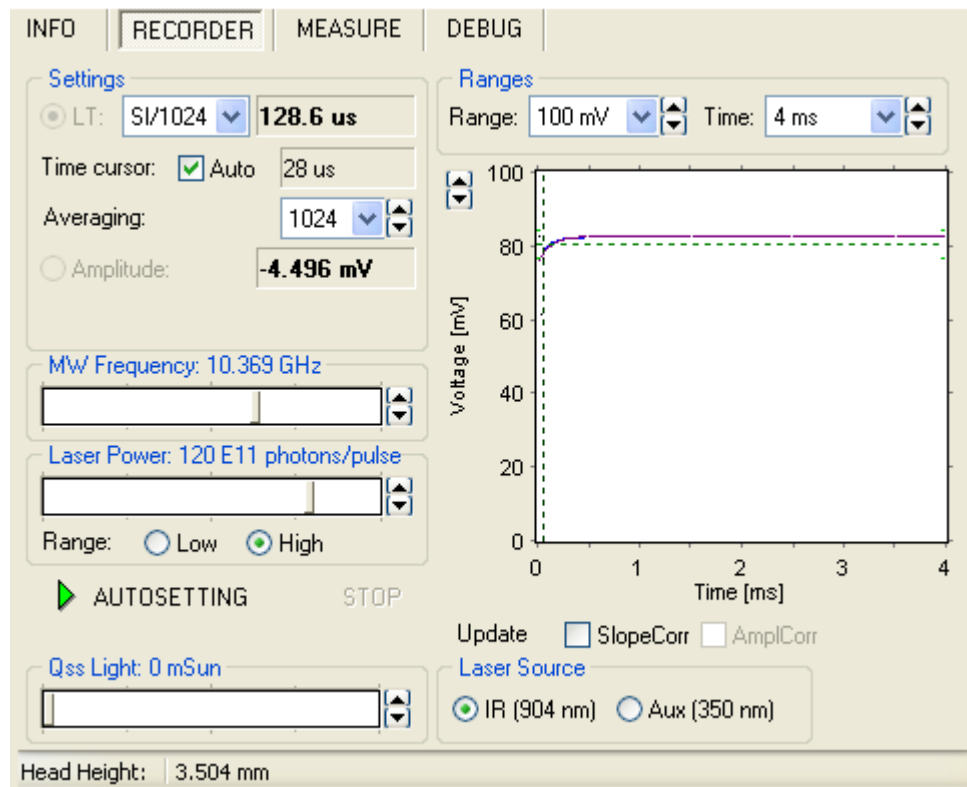


Figure 10-52: Qss u-PCD recorder

The option that Qss adds to the u-PCD recorder is the *Qss Light* slide bar. The intensity of the Qss light can be set by this.

10.13.1.6 Qss evaluation window

When displaying a Qss u-PCD measurement (map or points), an additional  button appears in the info window:

Figure 10-53: Info window of a Qss map

By clicking on this button, the Qss-uPCD evaluation window appears:

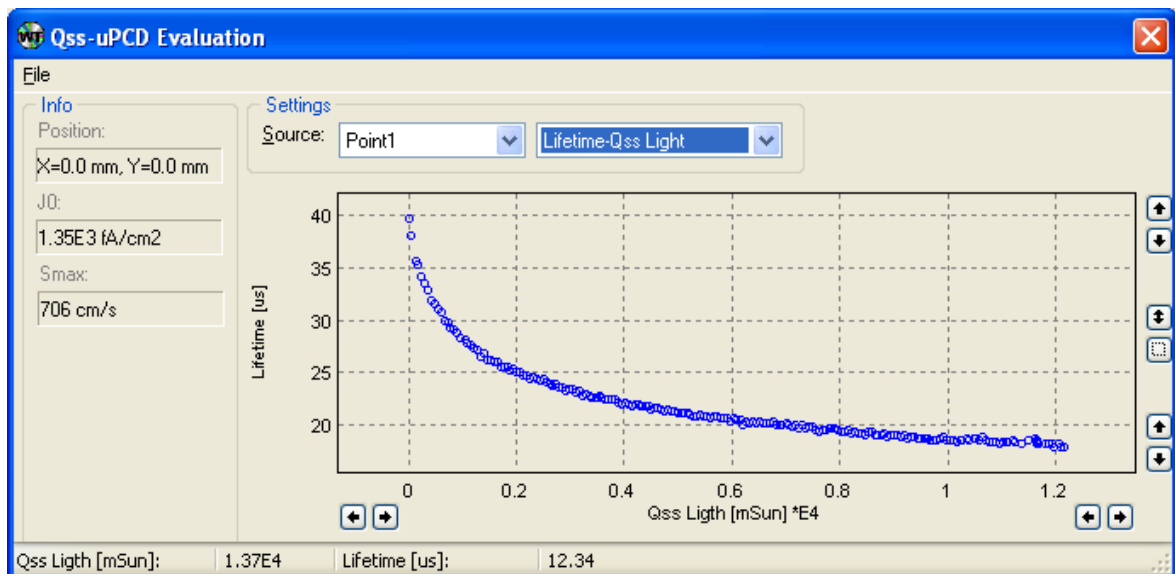


Figure 10-54: Qss-uPCD evaluation

This window allows evaluating the qss measurement result. In settings, there are two scroll-down menus: in the left the measurement point can be chosen, in the right the function can be chosen. The following functions are available:

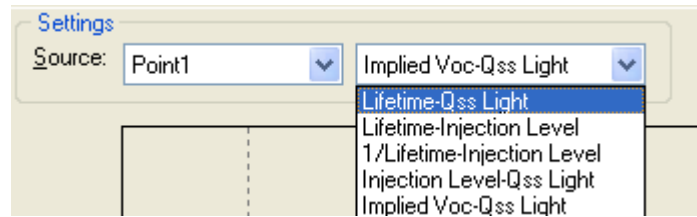


Figure 10-55: Functions of Qss evaluation

J_0 is calculated from the 1/lifetime – injection level function:

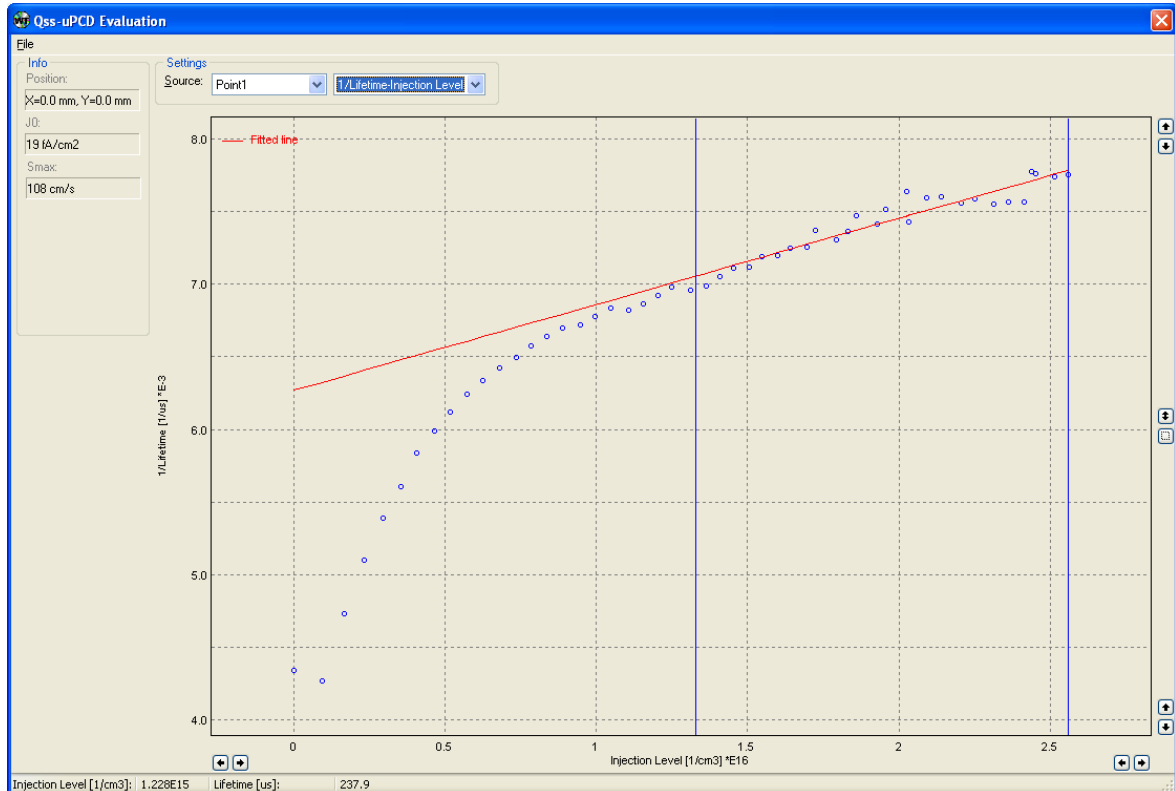


Figure 10-56: The reciprocal of the lifetime in the function of the injection level. J_0 is calculated from the fitted line.

10.13.2 SPV Recorder (SPV Mode)

In SPV mode the parameters and the signal parameter setting procedure itself are displayed in the *SPV Recorder* window. There are the measuring parameters and the *SPV Plot*, which displays the measured SPV signals versus light penetration depth with the fitted line that is used to calculate the diffusion length.

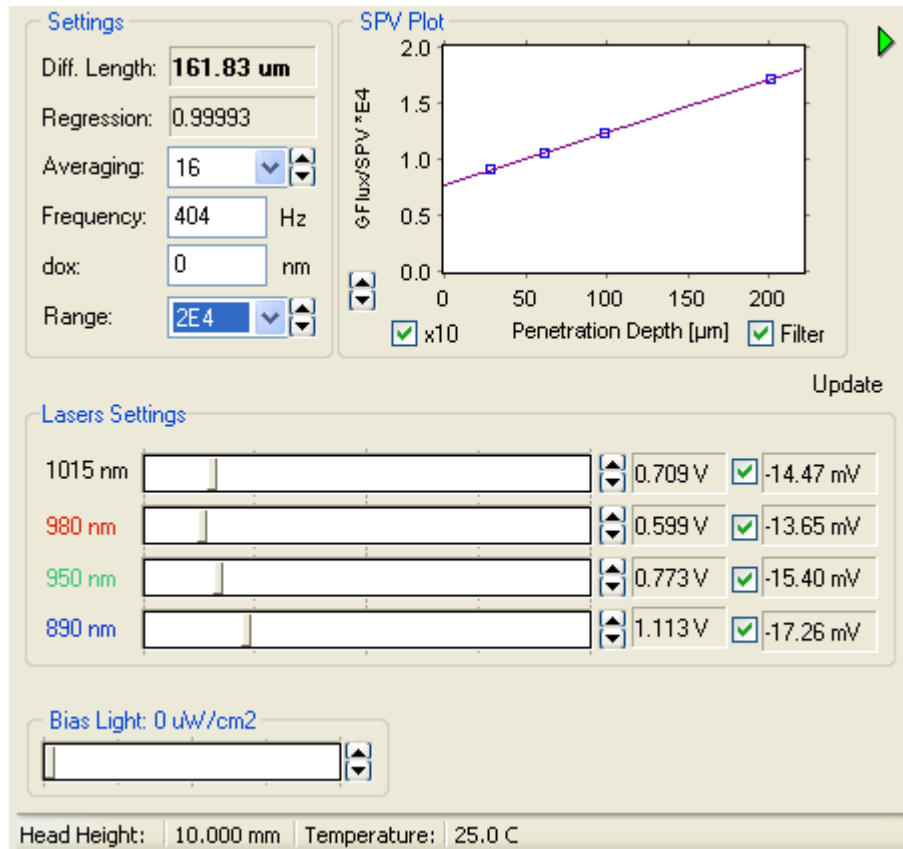


Figure 10-57: SPV Recorder

The measuring *Frequency*, the *Averaging* and the laser control voltages, according to the set SPV signal amplitudes, are automatically set by the Autosetting.

- Diffusion Length* The fitted diffusion length is displayed in μm units.
- Regression* The regression factor shows if the measured Gflux/SPV amplitude points are on a straight line or not. The closer the regression factor is to 1, the better is the fitted straight line. If you use only two lasers for the SPV measurement, the regression factor is always 1.
- Averaging* Averaging is the number of SPV measurements taken and averaged at a given position. The possible values are 4, 8, 16, 32, 64 and 128. More averaging means an increased signal-to-noise ratio at the expense of longer time necessary for the measurement.
- Frequency* Although the higher the measuring frequency is the smaller the noise of the measurement is, if you measure long diffusion lengths (close to or over the wafer thickness), lower frequencies should be set. The frequency can be set in the 100Hz to 2kHz range.
- dox* The oxide thickness (*dox*) is an important parameter in the SPV measurement since the wavelength dependent light reflection from the sample surface should be taken into account in the evaluation. If the wafer is non-oxidized (bare), **0** can be written as oxide thickness, because at the thickness of the native oxide, this parameter does not

influence the evaluation.

Range (GFlux/SPV)

The *Gflux/SPV* vs. *Penetration* depth is plotted in the *SPV Recorder* window. The range can be set according to the expected maximum value to have the whole fitted line in the window. This parameter does not influence the measurement. The possible values are **200, 500, 1000, 2000, 5000, 1E4, 2E4, 5E4** and **1E5**.



(Autosetting)

The green triangle in the right-top corner of the *SPV Recorder* calls the autosetting procedure. There are three Autosetting types which can be set in the **Options** menu at the *Single* tab.

x10

In the *SPV Plot*, a ten-fold amplifying can be activated by this checkbox. It is recommended to measure without additional amplifying.

Filter

A filter can be activated by this checkbox. It is recommended to measure without filtering.

Laser Settings

There are four or eight different wavelength lasers built in the SPV head (depending on the SPV head configuration installed in your system). For diffusion length measurements the use of at least two lasers is necessary to get a fitted straight line on the *SPV Plot* to calculate the diffusion length. The wavelength, the control voltage and the SPV signal amplitude measured at each laser are displayed in the *Laser Settings* panel. The light intensity can be set by changing the control voltage. The control voltages and thus the SPV signals can be modified by dragging the slider with the mouse or using the small up and down arrows at the right side of the slider fields.

The Autosetting procedure sets the control voltages of the lasers automatically. Each laser can be operated in the range of 0 to .5V. A laser operates if its checkbox, right to the value indicating the laser power, is checked. If the mouse cursor is above the measured signal value of the laser wavelengths, the signal of the selected laser is measured continuously. It is possible to record the SPV signal amplitude versus time curve at each laser just like the diffusion length versus time curve (see **Functions** menu) in the *SPV Recorder* window.

Update

It is possible to perform measurements continuously by checking the **Update** checkbox. Pressing the right mouse button on the area of the *SPV Plot* window also makes the system perform a prompt measurement.

Bias Light slider

The *Bias Light* slider can be found below the *Laser Settings* if bias light hardware is installed. This slider sets the bias light intensity measured in $\mu\text{W}/\text{cm}^2$ or mSun units.

Right clicking the *Bias Light* control area a popup menu appears (Figure 10-58) with the following choices:

- **Switch Off:** Switches off the bias light.

- **Switch On:** Switches on the bias light, using the intensity last used.
- **Set Intensity:** A window appears where the exact bias intensity can be entered in the actually set units ($10\mu\text{W}/\text{cm}^2$ is recommended for SPV).
- **Intensity in (units):** Changes the unit to mSun or $\mu\text{W}/\text{cm}^2$.
- **Calibrate Intensity:** Performs intensity calibration.
- **Power Saving:** If no measurement is running the bias light is switched off automatically after the set time elapsed.

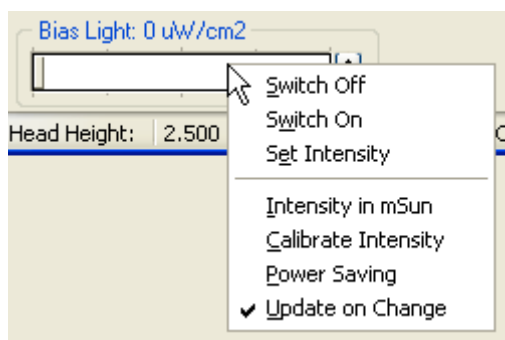


Figure 10-58: Pop-up menu for *Bias Light* options

Head Height

In the bottom of the *SPV Recorder* window the actual SPV head height is displayed in mm units.

10.13.2.1 Menu of SPV Recorder Window

A menu can be accessed by right-clicking in the SPV recorder window.

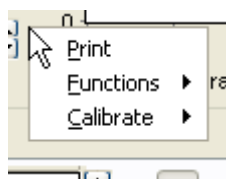


Figure 10-59: Right-click menu in *SPV Recorder*

Print

Prints the SPV plot with the fitted curve for diffusion length calculation.

Calibrate

The SPV head should be calibrated after installation; because the photon flux of each laser at the given control voltage has to be known to evaluate the signal. Only do calibration after the warm-up time of the system! Calibration is described in Chapter 1.

Functions

This function contains the menu items listed below. All of them open a window and record a curve on a plot. A new curve is recorded if the **New** command is activated. The curves can be loaded, saved and printed; or by right clicking on them, they can be copied to the

Clipboard as a graph or data. If there is a problem with the signals or the measurement, these curves may help to find the reason.

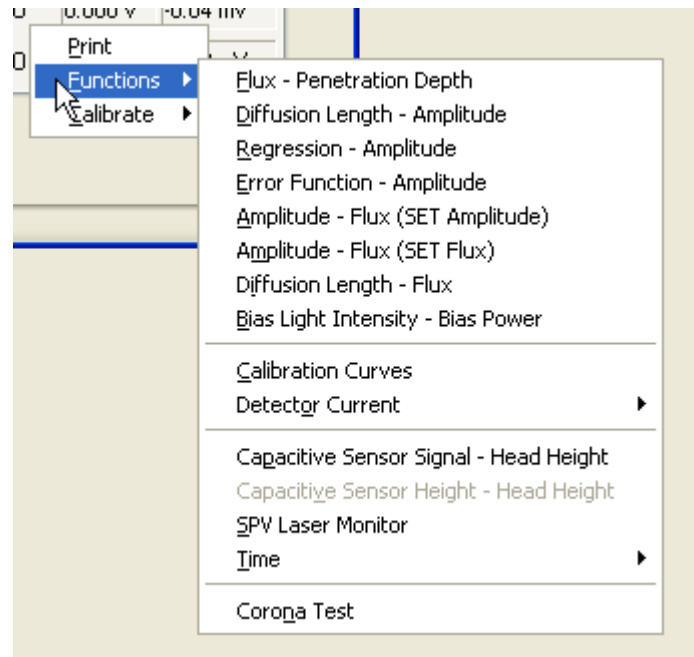


Figure 10-60: The *Functions* menu in the *SPV Recorder*

10.13.2.1.1 Amplitude – Flux

The SPV signal amplitude (*Voltage*) versus light flux (*GFlux*) curves are measured at every wavelength in this window. The resulting curves are typically linear at low light fluxes. Then they may reach the saturation values at very high flux. Positive amplitudes are measured on n type semiconductors, while negative signals are measured on p type. There is a horizontal cursor line that can be picked up and dragged by the mouse (keep any of the mouse buttons pressed). The actual cursor position determines the desired SPV signal level and so the necessary light flux.

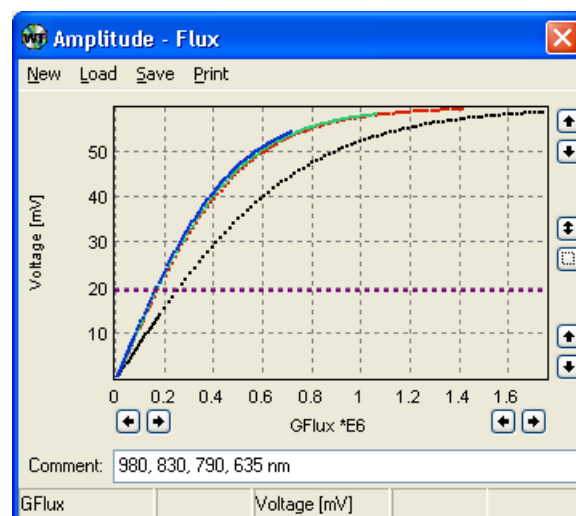


Figure 10-61: Amplitude – Flux

10.13.2.1.2 Diffusion Length – Amplitude

The fitted diffusion length versus the SPV signal amplitude curve is calculated in this window.

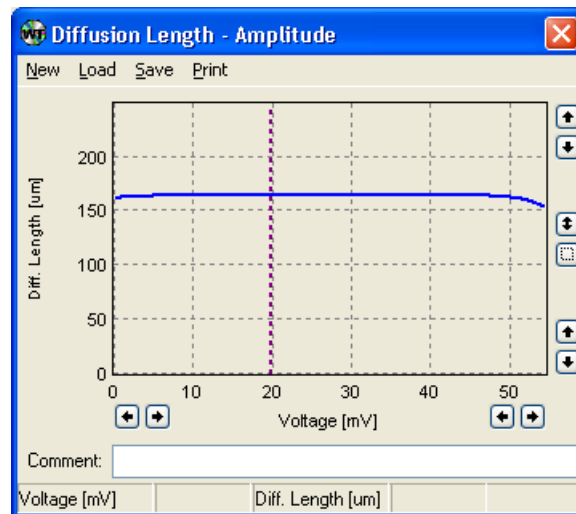


Figure 10-62: Diffusion Length – Amplitude

10.13.2.1.3 Regression – Amplitude

The fitted regression factor versus the SPV signal amplitude curve is calculated in this window.

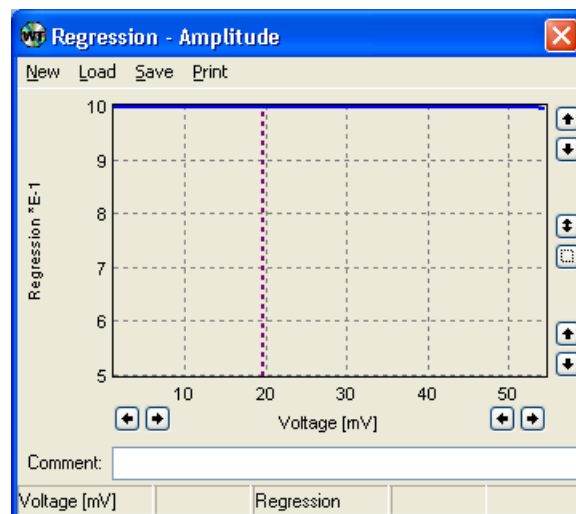


Figure 10-63: Regression – Amplitude

10.13.2.1.4 Error Function – Amplitude

The error function versus the SPV signal amplitude curve is calculated in this window. This error function is used in the Autosetting procedure to determine the laser amplitudes for the SPV measurements.

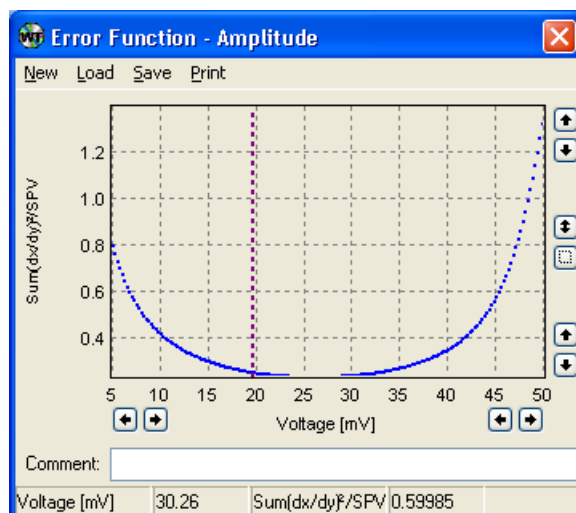


Figure 10-64: Error Function – Amplitude

10.13.2.1.5 Calibration Curves

The Calibration Curves (i.e. light flux versus the control voltage of the lasers curves) are displayed in this window. These curves are used to calculate the necessary control voltages if the light fluxes are already determined according to the set amplitudes.

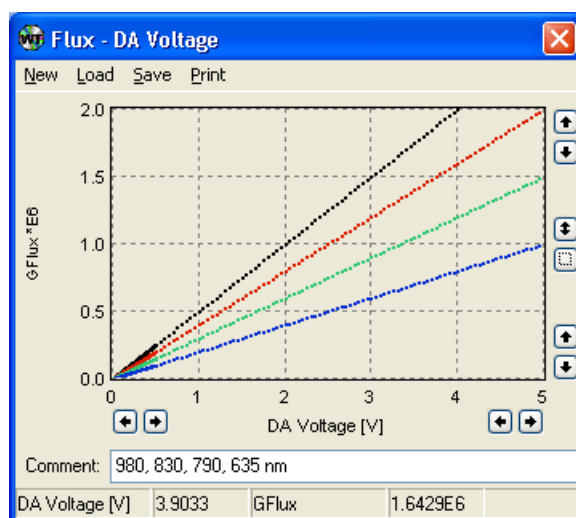
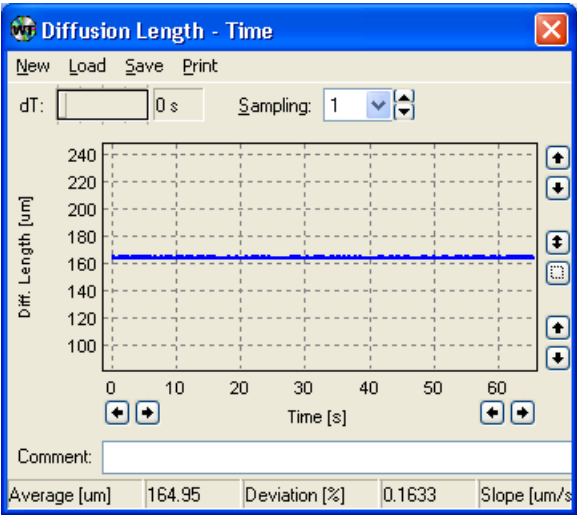


Figure 10-65: Calibration Curves

10.13.2.1.6 Diffusion Length – Time

The Time curves are accessed from the **Functions | Time** menu item in the *SPV Recorder* window.

The Diffusion Length – Time curve is recorded in the window shown below. The measuring interval can be set between 0-250 seconds by dragging the *dT* slider. When *dT* is set to **0**, the maximum available measuring speed is selected. The *Sampling* determines how many measured values are averaged before recording a point on the plot.



10.13.3 V-Q Recorder (V-Q Mode)

The V-Q Recorder (Figure 10-66) appears after the wafer has been loaded. It can also be displayed by clicking the **Measure | Recorder** menu item.

The V-Q Recorder window has the same appearance for all V-Q measurement methods.

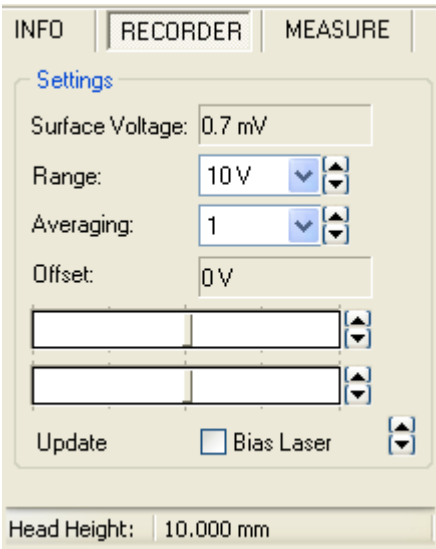


Figure 10-66: V-Q Recorder window

- Surface Voltage

The *Surface Voltage* shows the actual measured Surface Potential in millivolt units.
- Range

The *Range* should be set to **100V** for all V-Q measurement methods.
- Averaging

The *Averaging* sets the number of measurements at one point of the wafer during map measurement. It can be 1, 2, 4, 8, 16, 32, 64, 128, or 256. For a map, **8** is typically used.
- Offset

The probe head should be calibrated after installation or maintenance. When calibrating, the head moves above a graphite plate and adjusts the offset until the measured voltage equals -0.2V.

This *Offset* value is displayed in the *V-Q Recorder* window. Calibration is described in Chapter 1.

Update

The **Update** button is used for activating continuous refreshing of measured voltage.

Bias Laser

When a Bias Laser is built into the probe head, there is a checkbox in this window. Checking the **Bias Laser** checkbox the Bias Laser switches on, providing continuous illumination for “bright” measurements.

Head Height

The *Head Height* field displays the head height for the V-Q mode. The head height can be changed using the up/down arrows .

Several functions can be activated from the **Functions** menu (Figure 10-67) of the V-Q Recorder window.

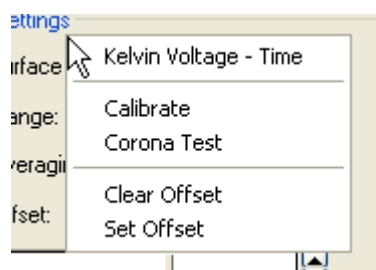


Figure 10-67: Functions in the V-Q Recorder window

Kelvin Voltage – Time

Displays a graph of Kelvin Voltage vs. Time.

Calibrate

Begins the V-Q calibration procedure (calibration is described in Chapter 1).

Corona Test

Displays a pop-up window (Figure 10-68) with control over the charge settings. It is possible to set the voltage and to see the current deposited on the stage. The slider allows adjustment of the upper corona wire.

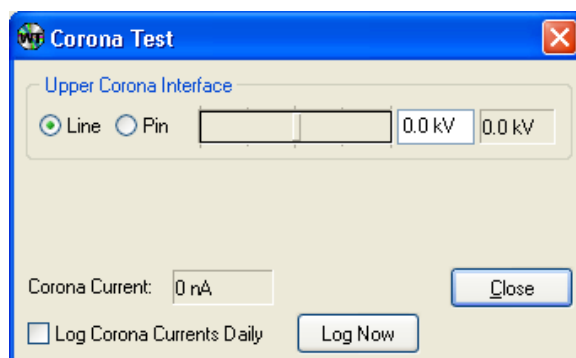


Figure 10-68: Corona Test window

10.13.3.1 V-Q Evaluation

When starting a V-Q measurement, the *V-Q Evaluation* window (Figure 10-69) appears and is displayed during the measurement. The window can be displayed at a later time by clicking the  button in the *Info List* of the *Measurement Information* window.

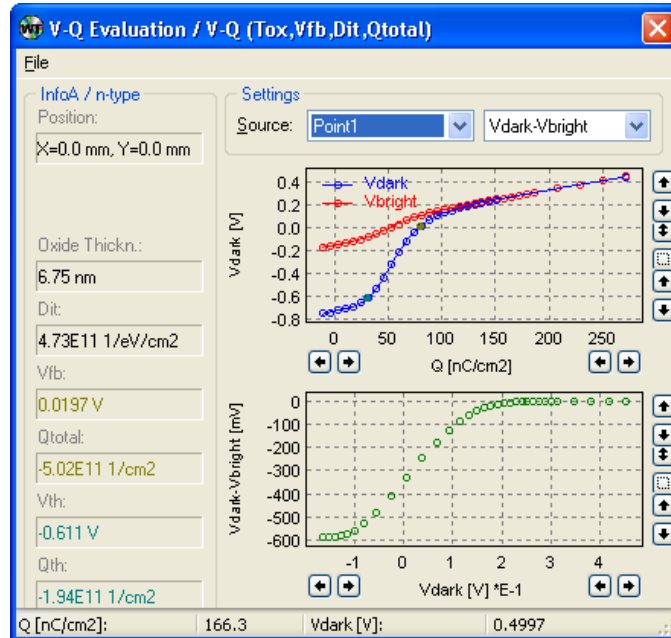


Figure 10-69: V-Q Evaluation window (V-Q measurement shown)

Use the *Source* drop-down list to select the point to display.

The upper section of the window displays the “bright” and “dark” V-Q curves.

The curve displayed in the lower section of this window is selected by the drop-down list at the top right of the window.

Vdark - Vbright When a “bright” V-Q curve has been measured, displays a curve created by subtracting the “bright” curve from the “dark” curve.

C-V Displays the calculated C-V curve.

Band bending The ideal and fitted curves are displayed in the same graph. The curves are marked to identify the energy band regions.

The *Info* area on the left displays information about the measured point, depending on the measurement method. The measurement position and the measurement results will be shown. If the measurement results were calculated from the curve, the results will be color-coded to match the point on the curve used for the calculation.

10.13.4 JPV Recorder

The *JPV Recorder* window (Figure 10-70) can be used for the JPV measurement mode as well as p/n type determination with the JPV measurement head.

Figure 10-70: JPV Recorder window

<i>JPV</i>	Shows the actual measured sheet resistance in ohm/square units.
<i>Type</i>	Shows the type of the bulk of the sample.
<i>Rin-, Rin+</i>	Signal components of the inner ring (the square root average of the two components is displayed next to these fields.)
<i>Rout-, Rout+</i>	Signal components of the outer ring (the square root average of the two components is displayed next to these fields.)
<i>Frequency</i>	Chopping frequency of the excitation light can be set here between 15.3Hz and 500kHz. The recommended default value is 7.8 kHz .
<i>Averaging</i>	Sets the number of measurements in one point of the wafer during prescan and mapping. It can be 1, 2, 4, 8, 16, 32, 64, 128 or 256. Averaging of at least 8 is recommended.
<i>Update</i>	Activates continuous refreshing of measured voltage.
<i>dox</i>	Thickness of the oxide layer on the sample is to be entered here.
<i>Laser/LED</i>	Laser or LED power can be set by the slider in the bottom part of the window. It is recommended to use the maximum power for measurements.

Several functions can be activated by right-clicking in the recorder window, and selecting the **Functions** menu (Figure 10-71).

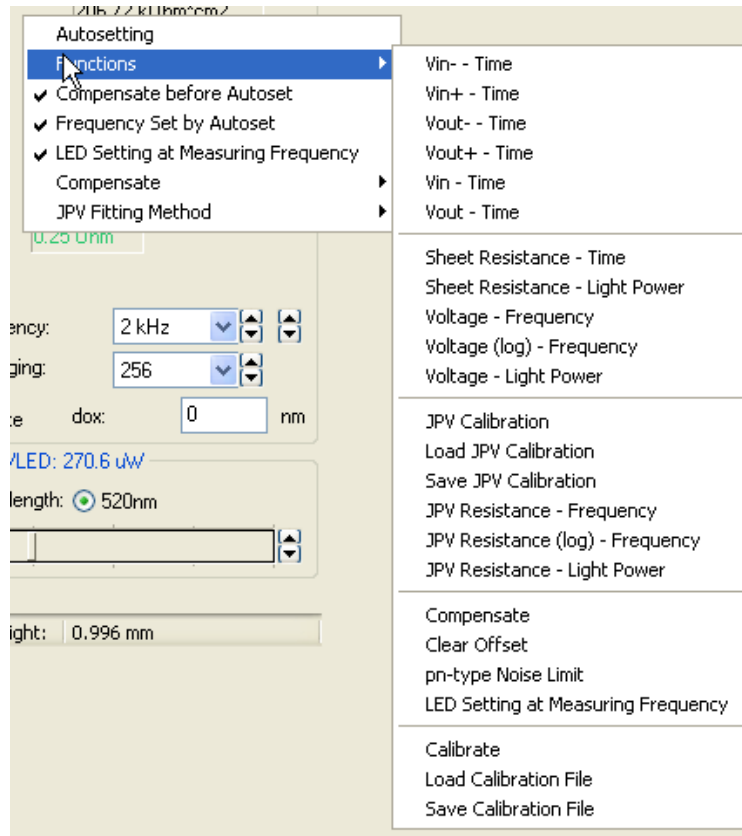


Figure 10-71: Functions in the JPV Recorder window

The first functions in the menu are measured plots according to their names. These are mainly used at device setup and for experimental purposes.

JPV Calibration Calibration of the JPV head. See details in Chapter 1.

Compensate As the head has a small voltage offset it has to be taken into consideration during evaluation of measured signal. Thus the probe head should be compensated after installation (or maintenance) by this option. During compensation to zero, a grounded sample must be placed under the head.

Clear Offset Clears the offset value stored during the Compensation procedure.

pn-type Noise Limit It is possible to set the p/n type noise limit. This value is a limit that is used by the system to distinguish between the noise and the signal. If the measured signal is above the limit, the system will evaluate it, and the measurement point will be a valid point on the map. After this menu item is selected, the *SHR Setup* window will appear (Figure 10-72) to allow entry of the desired limit value.

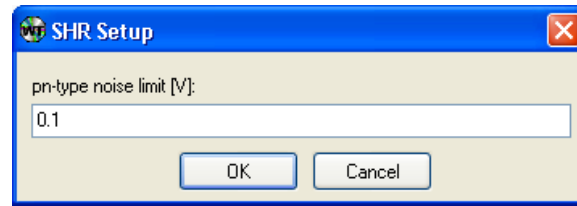


Figure 10-72: *SHR Setup* window for *pn-type noise limit*

- Calibrate*** Activates calibration of the JPV head. See details in Chapter 1.
- Load Calib.File*** Loads a previously saved calibration file.
- Save Calib. File*** After calibration, the calibration data can be saved to a file.

11. Macros

Macro language has been implemented in Wintau32 to give users a wider range of control over software capabilities. It is a special program language, which starts as a sequence of ASCII characters representing the source code, created by keystrokes using the macro window of the Wintau32 software. The basic program unit is the file, which has the extension **.mac**.

This chapter gives a brief overview of the macro capabilities in Wintau32. The organizational structure of this chapter is shown in Table 11-1. (Not all subsections are shown in this table.)

Users who will run Semilab-supplied macros should read section 11.1. Users who are interested in understanding the functionality of the Semilab-supplied macros may wish to review the remaining sections in the chapter.

! CAUTION: Improper use of macros can cause damage to your system. Users who wish to modify or create macros should contact Semilab for assistance.

Table 11-1
Organization of Macro Topics

Section name	Section
Programming Environment	11.1
The Macro Menu	11.1.1
The <i>Macro Editor</i> Window	11.1.2
Lexical Grammar of Macro Programs	11.2
Comments	11.2.1
Tokens	11.2.2
Declarations	11.2.3
Statements	11.3
Labeled Statements	11.3.1
Goto Statement	11.3.2
If Statements	11.3.3
While Statement	11.3.4
For Statement	11.3.5
Break Statement	11.3.6
Continue Statement	11.3.7
Writing Macro Programs	11.4

Call Function	11.4.1
Components	11.4.2

11.1 Programming Environment

11.1.1 The Macro Menu

To handle the macros you have the following selections from the **Macro** menu (Figure 11-1) on the *Main Window*:

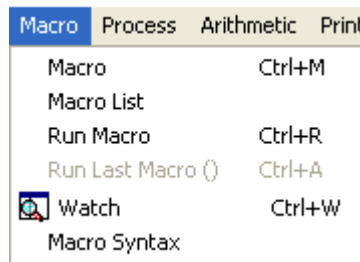


Figure 11-1: The Macro menu

Macro	Displays the available macros in the <i>Macro Editor</i> window (Figure 11-2).
Macro List	Displays the available macros.
Run Macro	Runs a macro from the open dialog box.
Run Last Macro	Runs the last executed macro.
Watch	Displays the <i>Macro Watch</i> window (Figure 11-5).
Macro Syntax	Displays the <i>Macro Syntax</i> window (Figure 11-4).

11.1.2 The Macro Editor Window

On the left-hand side of the *Macro Editor* window (Figure 11-2) there is a list which shows the macros. If one of them is selected the source code of the selected macro can be seen on the right. The editing of the macro is possible there.

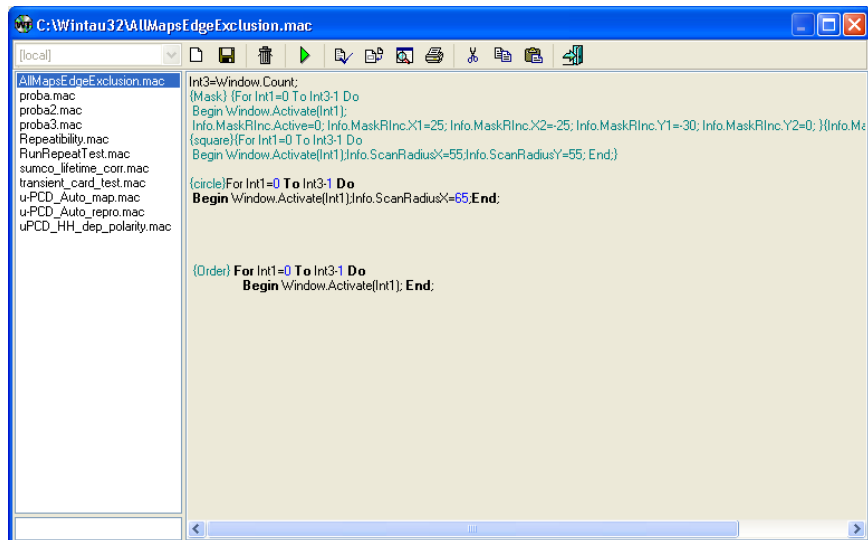


Figure 11-2: The *Macro Editor* window

You can also give menu access to a macro by typing a name in the left-bottom edit-box of the *Macro Editor* window. The new menu item appears at the bottom of the **Macro** menu in the *Main Window*.

11.1.3 Macro Editor Toolbar

The following list explains the meaning of toolbar icons in the *Macro Editor* window.



Figure 11-3: Toolbar in *Macro Editor* window

- | | |
|---|--|
|  | Create new macro. |
|  | Saves the macro. |
|  | Deletes the selected macro. |
|  | Runs the selected macro. Function key <F9> can activate the Run process also. |
|  | Checks for syntax errors in the selected macro. Function key <F10> can activate the check process. |
|  | Displays the <i>Macro Syntax</i> window. |
|  | Opens the <i>Watch Window</i> for macro debugging. |
|  | Prints the selected macro. |
|  | Cuts the selected text from the selected macro. |



Copies the selected text from the selected macro to the Clipboard.



Pastes text from the Clipboard to the selected macro.



Closes the *Macro Editor* window.

11.1.4 The Macro Syntax Window

The *Macro Syntax* window (Figure 11-4) lists all available macro variables, functions and procedures.

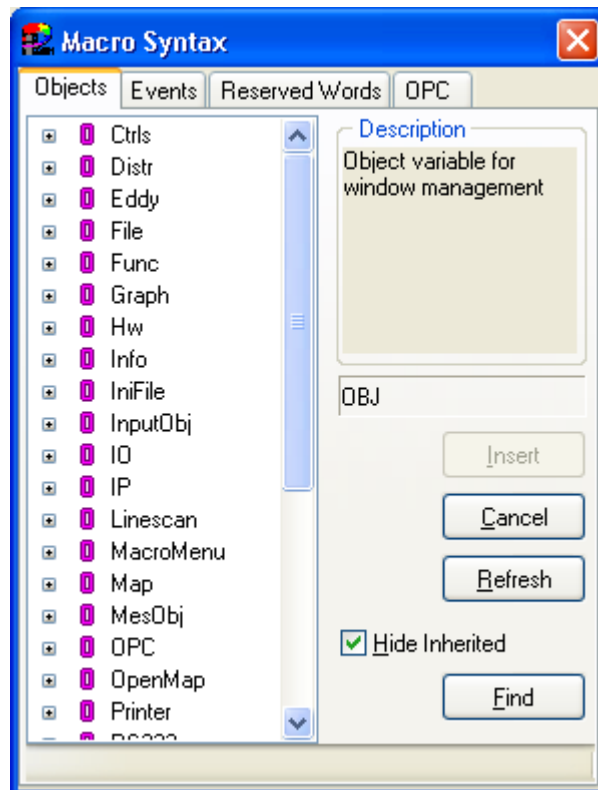
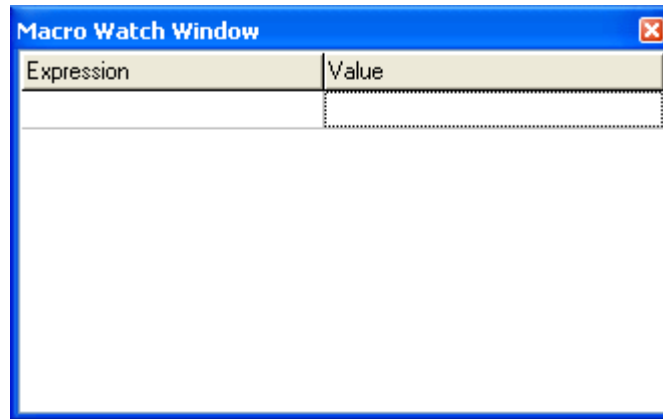
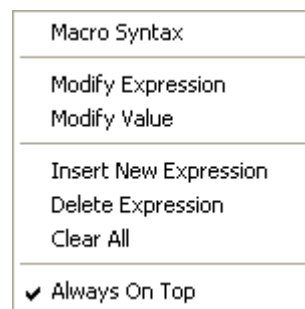


Figure 11-4: The *Macro Syntax* window

11.1.5 The Macro Watch Window

There is a debugger for the macro language built into Wintau32. You can add watches by the **Macro | Watch** menu item. The *Macro Watch* window is shown in Figure 11-5. Use the right mouse button to choose from the possibilities (Figure 11-6).

Figure 11-5: The *Macro Watch* windowFigure 11-6: Right-click menu in *Macro Watch* window

Macro Syntax	Displays the <i>Macro Syntax</i> window..
Modify Expression	Modify a watch added previously by Insert New Expression option.
Delete Expression	Delete an added expression.
Clear All	Remove all added watches.
Always On Top	If this option is checked, the <i>Macro Watch</i> window will not be hidden by other windows.

11.2 Lexical Grammar of Macro Programs

Lexical grammar is concerned with the different categories of word-like units, known as tokens. When the macro is compiled the compiler parsed the source code file into tokens and white space. White space is the collective name given to spaces (blanks), horizontal tabs, new line characters and comments. White space can serve to indicate where tokens start and end.

The macro language is not case sensitive.

11.2.1 Comments

Comments are pieces of text used to annotate a program. Comments are for the programmer's use only; they are stripped from the source text before parsing.

Wintau32 comments are any sequence of characters placed between the symbol pair {} with maximum length of 255 character.

Examples:

Message('Hello world!'); {Display a message with Hello world!}

or

Message('Hello world!'); {*****Display a message with Hello world!*****}

11.2.2 Tokens

The Wintau32 macro compiler recognizes the following class of tokens:

- Keyword
- Identifier
- String literal
- Operator
- Punctuator

11.2.2.1 *Keywords*

Keywords are words reserved for special purposes and must not be used as normal identifier names. The reserved words are listed in the *Reserved Words* tab of the *Macro Syntax* window (Figure 11-7).

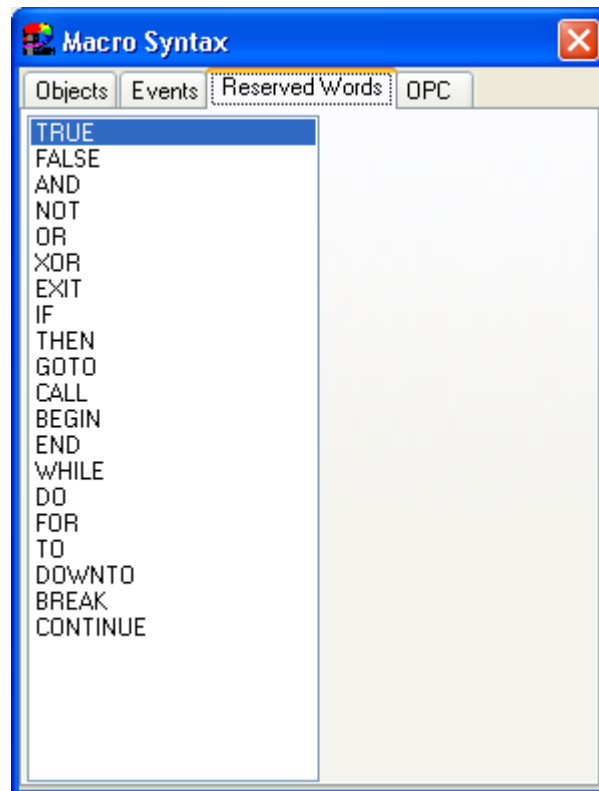


Figure 11-7: *Reserved Words* tab of *Macro Syntax* window

11.2.2.2 Identifiers

Identifiers are arbitrary names of max. 255-character length given to variables and user defined variables. The list of built-in Wintau32 macro language identifiers can be seen in the *Objects* tab of the *Macro Syntax* window if the **Hide Inherited** checkbox is turn off.

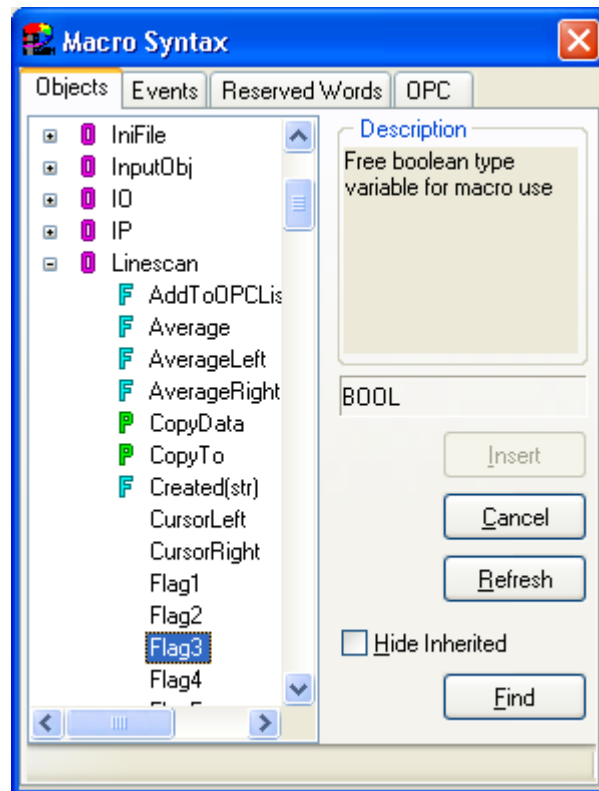


Figure 11-8: Built-in Wintau32 macro language identifiers in *Objects* tab

Examples:

```
int1 = 1;
str1 = 'Hello world!';
uPCD.Int1 = 154;
uPCD.Flag1 = FALSE;
Map.Float5 = 1.365;
```

11.2.2.3 String Literal

String literals are used to handle fixed sequences of characters.

Example:

'Hello world!'

11.2.2.4 Operator

Operators are tokens that trigger some operations when applied between two variables and/or objects in an expression.

An operator is one of the following:

+ - * / . < > >= <= =

11.2.2.5 Punctuator

The punctuators are separators of lines of macro.

Example:

```
str1 = 'Hello world! This is an example of using punctuators when the string constant is longer
than the editor window';
Message(str1);
```

11.2.3 Declarations

All variables of the Wintau32 macro language are objects. An object is an identifiable region of memory that can hold fixed or variable value (or set of values). Each value has an associated name and type. The name is used to access the object and this name can be simple identifier or it can be a complex expression.

The Wintau32 macro language has many types of built in objects. A list of built in objects can be seen in the *Objects* tab of the *Macro Syntax* window.

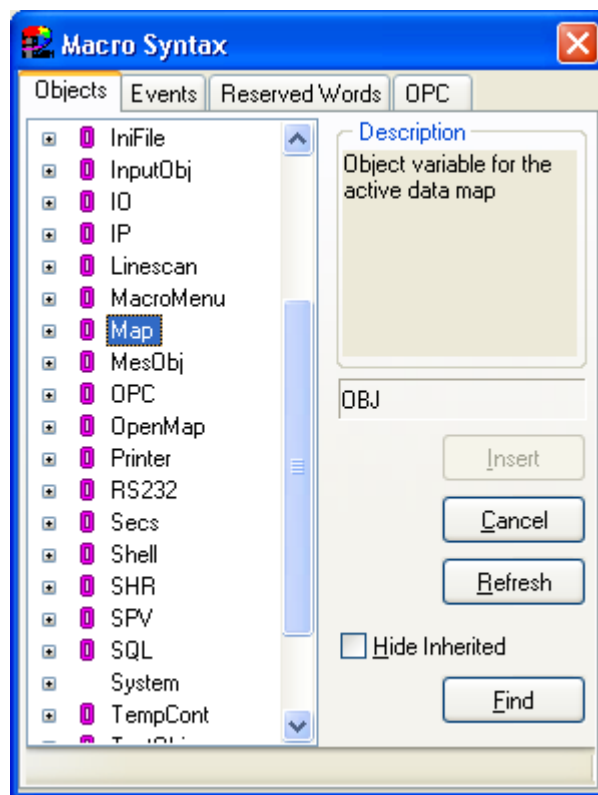


Figure 11-9: Built in objects in *Objects* tab

The built-in objects are connected to the hardware of the WT-2000 system, the windows of the Wintau32 operational software, or the interfaces of the system.

It is possible to create new objects.

Example:

```
newint('created_integer');
newfloat('created_float');
newstr('created_string');
newobject('created_object');
created_object.newfloat('object_float');
created_object.newstr('object_str');
```

```
created_integer = 32459;  
created_float = 32459.8;  
created_string = 'Wintau32';  
created_object.object_float = 4.26;  
created_object.object_str = 'WT-2000';
```

```
created_object.float1 = 3.15; { Every created object has the simple built in variable set}  
created_object.str1 = 'Lifetime'; {and the function connected to the created variable set}
```

If you check the syntax by <F10> button you can see the created object with all available variables and functions of the object in the *Macro Syntax* window.

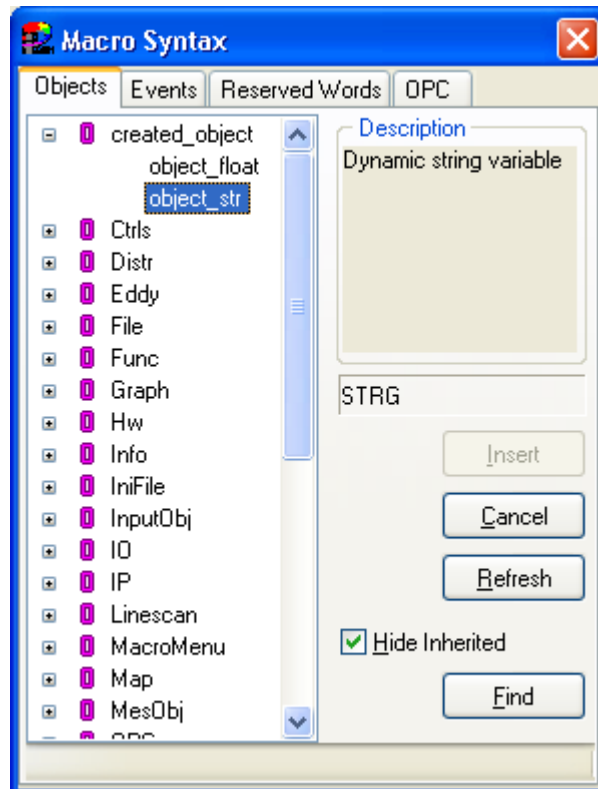


Figure 11-10: Created object shown in *Object* tab of *Macro Syntax* window

Figure 11-11 shows the created object if the **Hide Inherited** checkbox is turned off.

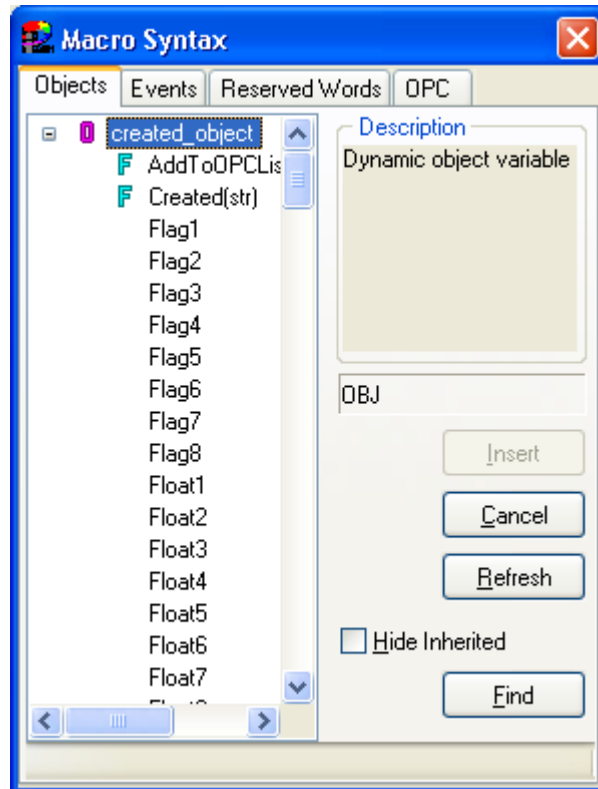


Figure 11-11: Created object shown when Hide Inherited is not selected

To destroy these objects you should type:

```
free('created_integer');
free('created_float');
free('created_string');
free('created_object');
```

If the code of the macro has the lines which destroy the objects before the first syntax check (<F10>), the *Macro Syntax* window does not display the created object.

So the complete code:

```
newfloat('created_float');
newstr('created_string');
newobject('created_object');
created_object.newfloat('object_float');
created_object.newstr('object_str');
```

```
created_integer = 32459;
created_float = 32459.8;
created_string = 'Wintau32';
created_object.object_float = 4.26;
created_object.object_str = 'WT-2000';
```

```
created_object.float1 = 3.15; { Every created object has the simple built in variable set}
created_object.str1 = 'Lifetime'; {and the function connected to the created variable set}
```

```
free('created_integer');
free('created_float');
```

```
free('created_string');
free('created_object');
```

11.3 Statements

Statements specify the flow of control as a program executes. If there are no jumps and selection statements, the lines of the source code will be executed in the order of appearance.

11.3.1 Labeled Statements

The identifier of the label serves as a target for the goto statement.

11.3.2 Goto Statement

The goto statement transfers the control to the labeled statement.

11.3.3 If Statements

Selection or flow control statement by testing certain values.

Example:

```
int1 = 1;
int2 = 0;
go_there: {***** Label *****}

if int1 < 25 then begin {***** if statement *****}
int1 = int1 + 1;
end;

int2 = int2 + int1;

if int1 < 25 goto go_there; {***** if with goto statement *****}

message(str(int2,0));
```

11.3.4 While Statement

The while statement lets you loop a set of statements. The loop statement will be executed repeatedly until the conditional expression compare equal to false.

Example:

```
int1 = 1;
int2 = 0;
while int1 < 25 do begin {***** int1 < 25 is the conditional expression *****}
int1 = int1 + 1;
int2 = int1 + int2;
end;
message(str(int2,0));
```

11.3.5 For Statement

The for statement lets you loop a set of statements. The loop statement will be executed repeatedly until the counter is lower than the target value.

Example:

```
int1 = 1;
int2 = 12;
int3 = 1;
for int1 = 1 to int2 do begin                                {***** int1 is the counter, int2 is the target *****}
int3 = int3* int1;
end;
message(str(int3,0));
```

11.3.6 Break Statement

The break statement can be used only inside a loop. It terminates the loop.

11.3.7 Continue Statement

The continue statement can be used only inside a loop statement. It transfers control to the beginning of the next loop.

Example:

```
int2 = 25;
int3 = 0;
for int1 = 1 to int2 do begin
if int1 < 6 then continue;
if int1 = 8 then break;
int3 = int3 + int1;
end;
message(str(int3,0));
```

11.4 Writing Macro Programs

11.4.1 Call Function

It is possible to insert a source code that is in another macro file.

Example:

```
int3 = 0;
for int1 = 1 to 25 do begin
call square_calculator;
int3 = int2 + int3;
end;
message(str(int3,0));
```

The square_calculator.mac file contains the following code:

```
int2 = int1 * int1;
```

11.4.2 Components

This section tries to give an introduction to the usage of macro software components.

As mentioned previously all variable are objects in the Wintau32 macro program language. Information about any object, or object variable/function can be found in the *Macro Syntax* window. The *Description* section on the right-up corner of the *Macro Syntax* window displays a short information about the selected object/variable/function/procedure. Below this section the type of item can be seen.

11.4.2.1 Ctrls – Object variable for visible controls

Using these objects it is possible to create and handle forms, buttons, text forms, radio buttons etc.

The following example will show the creating of form and buttons:

```
Ctrls.NewForm(0,'Wsize_form');           {*****Creating new empty form ****}
Ctrls.Wsize_form.Width = 210;
Ctrls.Wsize_form.Height = 300;
Ctrls.Wsize_form.Caption = 'Please select wafer size!';
```

```
Ctrls.NewButton(Ctrls.Wsize_form.Handle,'Fiveinch_but');  {**Creating new button **}
Ctrls.Wsize_form.Fiveinch_but.Caption = '5 inch';
Ctrls.Wsize_form.Fiveinch_but.Top = 50;
Ctrls.Wsize_form.Fiveinch_but.Left = 60;
Ctrls.Wsize_form.Fiveinch_but.Modalresult = 1;
```

```
Ctrls.NewButton(Ctrls.Wsize_form.Handle,'Sixinch_but');
Ctrls.Wsize_form.Sixinch_but.Caption = '6 inch';
Ctrls.Wsize_form.Sixinch_but.Top = 100;
Ctrls.Wsize_form.Sixinch_but.Left = 60;
Ctrls.Wsize_form.Sixinch_but.Modalresult = 2;
```

```
Ctrls.NewButton(Ctrls.Wsize_form.Handle,'Eightinch_but');
Ctrls.Wsize_form.Eightinch_but.Caption = '8 inch';
Ctrls.Wsize_form.Eightinch_but.Top = 150;
Ctrls.Wsize_form.Eightinch_but.Left = 60;
Ctrls.Wsize_form.Eightinch_but.Modalresult = 3;
```

```
Ctrls.NewButton(Ctrls.Wsize_form.Handle,'Cancel_but');
Ctrls.Wsize_form.Cancel_but.Caption = 'Cancel';
Ctrls.Wsize_form.Cancel_but.Top = 220;
Ctrls.Wsize_form.Cancel_but.Left = 60;
Ctrls.Wsize_form.Cancel_but.Modalresult = 4;
```

```
if Hw.WaferLoaded then begin
message('Please Unload the sample before wafer size change!');
exit;
end;
```

```
if Hw.WaferRemained then begin
message('Please Unload the sample before wafer size change!');
exit;
end;
```

```
int1 = Ctrls.Wsize_form.ShowModal;
if int1 = 4 then exit;
if int1 = 1 then begin
```



```

Hw.WaferSize = 2;
end;

if int1 = 2 then begin
Hw.WaferSize = 3;
end;

if int1 = 3 then begin
Hw.WaferSize = 4;
end;

```

Using this form it is possible to change the wafer size from 5 inch to 8 inch.

11.4.2.2 *Distr – Object Variable for the Active Distributions*

Using this object it is possible to adjust the properties of distributions, color scale of maps, etc.

Example:

```
Distr.InvColors = TRUE;
```

This macro changes the color scale to inverse: high values become red, low values become blue. An example is shown in Figure 11-12.

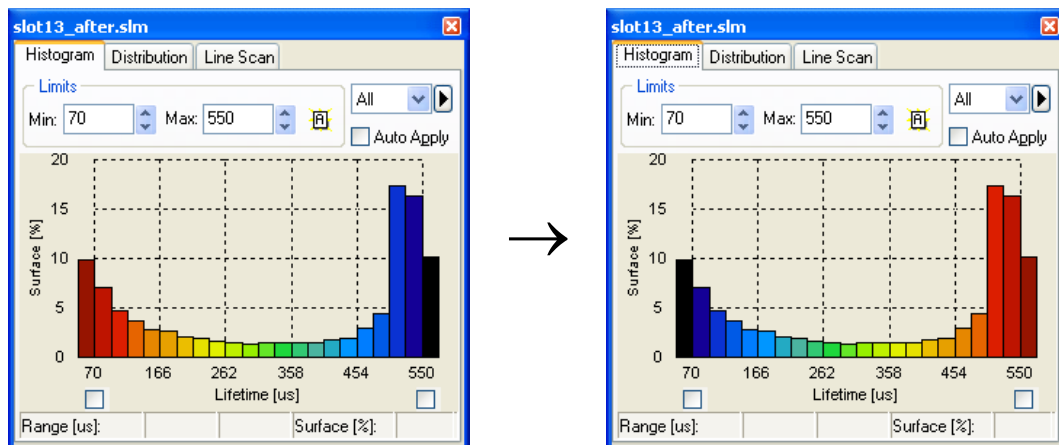


Figure 11-12: Example of inverted color scale

11.4.2.3 *Eddy – Object Variable for the Eddy Recorder Window*

Using this object it is possible to adjust the average of the *Eddy Recorder* in resistivity measurement mode. It is possible also load and save calibration files etc. This works only in eddy current mode.

Example:

```
Eddy.Average = 5;
```

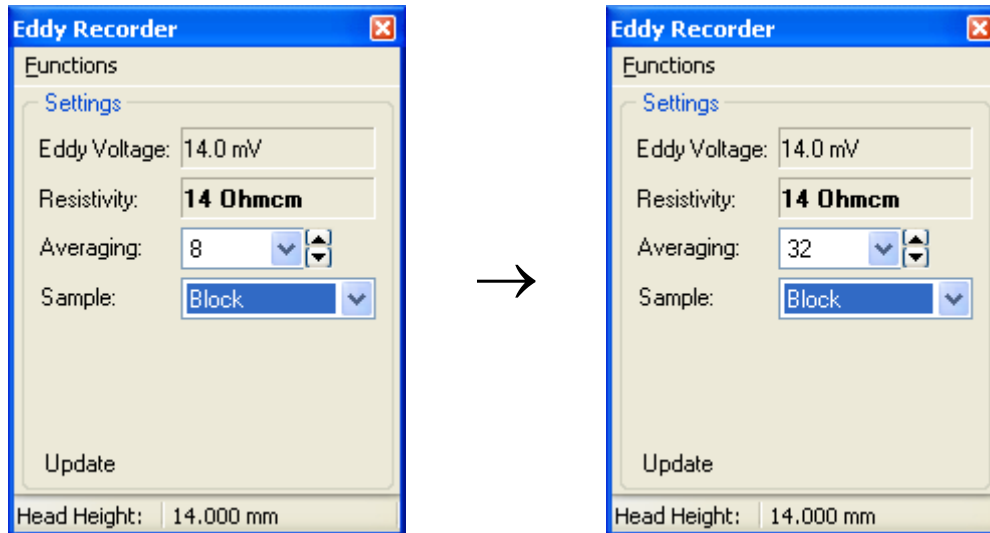


Figure 11-13: Example of change in Eddy averaging mode

This macro changes the average in eddy/resistivity measurement mode.

11.4.2.4 File – Object Variable for File Management

Using this object it is possible to perform file manipulations.

Example:

```
str1 = 'c:\temp\text.txt';
```

```
File.Rewrite(str1);
```

```
File.WriteLineStr('This procedure will write this string to the str1 file.');
```

```
File.Close;
```

This macro will create or overwrite the **c:\temp\text.txt** file and put a line “This procedure will write this string to the str1 file.” into the file.

11.4.2.5 Func – Object Variable for the Various Functions Available from the Recorder Windows

Using this object it is possible to manipulate the various functions of the recorder windows, e.g. *Microwave DC* curve, *Lifetime-Time*, etc. in μ -PCD measurement mode.

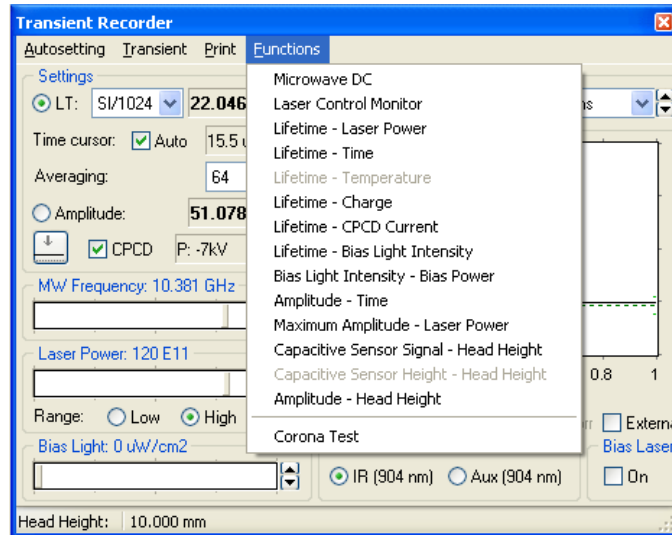


Figure 11-14: Functions in a Recorder window (μ -PCD shown)

Example:

```

if NOT(Func.Visible) then begin
Func.Show(0);                                     {*****Display the MWDC curve *****}
end;
delay(20000);                                     {**** Wait 20 sec*****}
if Func.Ready then begin
Func.SaveToFile('c:\temp\mwdc.txt');               {*****Save the curve*****}
end;
Func.Hide;                                       {*****Hide the MWDC window*****}

```

If the system was initialized and the μ -PCD recorder is present, this macro will save the *Microwave DC* curve.

11.4.2.6 Graph – Object Variable for Graphicon

Using this object it is possible to draw graphicons.

Example:

```

Graph.Caption = 'MWDC Curve';
Graph.Show;
Graph.LoadFromFile('c:\temp\mwdc.txt');
Graph.Xhi = 10.6;                                {*****Set the display limit*****}
Graph.Xlo = 10;
Graph.Yhi = 5;
Graph.Ylo = 0;

```

This macro displays in graph the previously saved curve.

The result is shown in Figure 11-15.

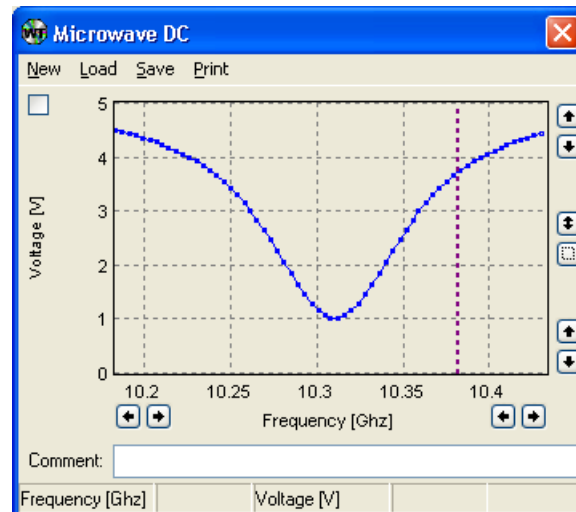


Figure 11-15: Example of previously saved curve

11.4.2.7 Hw – Object Variable for the Measurement

Using this object it is possible to control the hardware of the system.

Example:

if Hw.WaferRemained **then begin**

message('There is sample on the stage. Please unload the sample!');
end;

File.Dialog.Title = 'Please give the file name of the map!';
File.Dialog.Filter = 'Semilab files (*.slm) | *.slm';
File.Dialog.ShowSave;
str1 = File.Dialog.Path + File.Dialog.FileName;

If Not Hw.Initialized **then begin**

Hw.Initialize;
end;

Hw.LoadWafer(0);
Hw.SetMode(0);
Hw.FindWafer(0);
Hw.AutoSetting;
Info.Raster = 2;
Hw.Map;
Map.SaveFileAs(str1);
Hw.UnloadWafer;

This macro loads a sample, asks the filename, performs the autosetting and the map procedure, and unloads the sample.

Hw.SetMode(x); command sets the measuring mode to x. The codes for this command are detailed in the following table.

Value	Function	Value	Function
0	u-PCD	5	

1	EPI	6	V-Q
2	SPV	7	SHR
3	LBIC	8	PN
4	EDDY		

The current wafer size is stored in Hw.WaferSize variable. See next table.

Value	Function	Value	Function
0	3in	3	6in
1	4in	4	8in
2	5in	5	12in

! ***CAUTION: Changing the value of the variable SHR.Mode can cause an improper measurements or even damage to the system. Only use this object if it is expressly advised by Semilab.***

11.4.2.8 Info – Object Variable for the Active Measurement Information

Using this object it is possible to control the info panel, put on masks and get information about the active data file.

Example:

Info.Raster = 2; {The number typed here is the resolution of mapping in mm. Valid values are 2^n where $-4 \leq n \leq 6$ is whole number.}

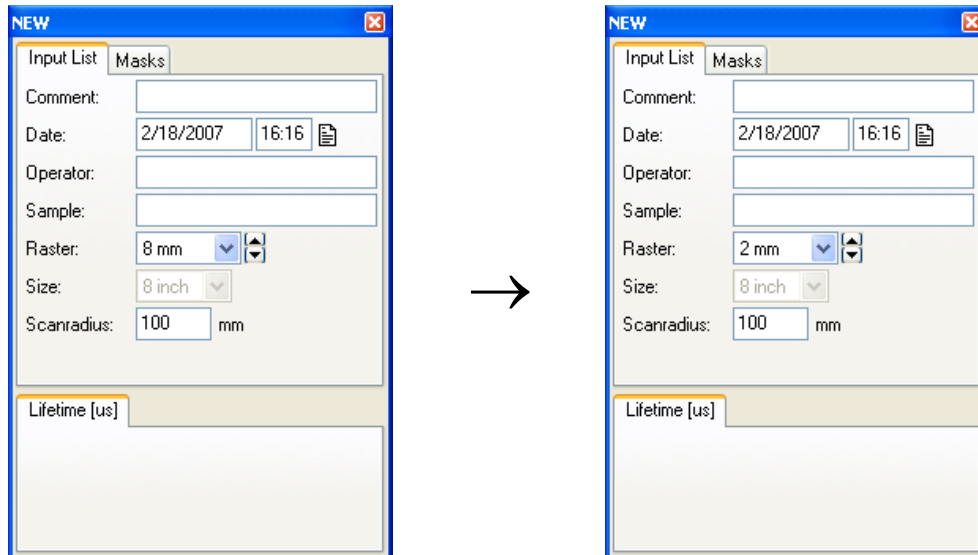


Figure 11-16: Example of change in raster

11.4.2.9 Infile – Object Variable for Windows Ini File Manipulation

Using this object it is possible to manipulate the Windows **ini** file.

11.4.2.10 Inputobj – Object Variable for Nonmodal Input Window

Using this object it is possible to create and control the simple input window.

Example:

```
InputObj.ShowOKCancel("Type a number");
InputObj.Wait;
message(str(InputObj.TextValue*InputObj.TextValue,3)); {*****displays the square of the
previously typed number *****}
```

11.4.2.11 Object Variables for Communication and Database Access

Using this type of object, it is possible to control communication via different interfaces and different protocols.

It is possible to control the different interfaces of the system with the following objects:

- IO object – Optional optical interface
- RS232 object – RS232 serial communication interface

The following objects are connected to different protocols

- IP object - TCP/IP communication protocol
- OPC object – OPC communication protocol
- SECS object – SECS communication protocol

The SQL object can be used for ODBC database access via SQL statement.

11.4.2.12 Linescan – Object Variables for the Linescan Window

Using this object it is possible to get information from the linescan window.

11.4.2.13 Macromenu – Object Variable for Macro Menu

Using this object it is possible to manipulate the permissions and check status in the macro menu.

If one macro has a macro menu name (filled the bottom edit box below the file list) the macro will appear in the macro menu.

11.4.2.14 Map – Object Variable for the Active Data Map

Using this object it is possible to manipulate the active data map.

Example:

```
str1 = File.ExtractFileName(Map.FileName);  
Map.Smooth;  
Map.SaveFileAs(str1 + '_smooth.slm');
```

This macro makes a smooth procedure on the active data map and save it on different name.

11.4.2.15 Mesob – Object Variable for Nonmodal Message Window

Using this object it is possible to create and control the simple message window.

11.4.2.16 OpenMap – Object Variable for the Open Window

Using this object it is possible to display and get information from the open map window.

Example:

```
OpenMap.Show;  
str1 = OpenMap.Directory;  
message('Number of Semilab files in the ' + str1 + ' folder: ' + str(OpenMap.ItemCount,0));
```

This macro displays the number of Semilab files in the folder where the open map window was closed.

11.4.2.17 Printer – Object variable for printer

Using this object it is possible to configure the printer.

11.4.2.18 Shell – Object variable for executing applications

Using this object it is possible to execute and control another window's applications.

Example:

```
Shell.Show = 1;  
Shell.File = 'c:\windows\notepad.exe';  
Shell.Parameters = 'c:\temp\text.txt';
```

```
File.Rewrite(Shell.Parameters);  
File.WriteLine('Some text to the text file.');
```

```
File.Close;  
  
Shell.Execute;  
Shell.Wait(0);  
message('Some text was written in the file.');
```

11.4.2.19 *SHR – Object Variable for SHR Recorder Window*

Using this object it is possible to get raw SHR voltage values, set the average, set the frequency etc. on the *SHR Recorder* window in JPV measurement mode.

11.4.2.20 *SPV – Object Variable for SPV Recorder Window*

Using this object it is possible to get raw SPV voltage values, set the average, set the frequency etc. on the *SPV Recorder* window in SPV measurement mode.

11.4.2.21 *System Declarations*

System is not a real object; it collects the remaining declarations. These are mathematical functions, variable conversation functions etc.

11.4.2.22 *Tempcont – Object Variable for the Temperature Controller*

Using this object it is possible to control the temperature controller in systems which have a hot-chuck.

11.4.2.23 *Texobj – Object Variable for Nonmodal Text Window*

Using this object it is possible to create and control the simple edit window.

11.4.2.24 *Upcd – Object Variable for the Transient Recorder Window*

Using this object it is possible to set the time range, amplitude range, average etc on the μ -PCD Recorder window in μ -PCD measurement mode.

11.4.2.25 *VQ – Object Variable for the V-Q Recorder Window*

Using this object it is possible to control the *VQ Recorder* window in V-Q measurement mode.

11.4.2.26 *Window - Object Variable for Window Management*

Using this object it is possible to configure the layout of (map) windows, change the active map to another map etc.

12. System Operation

This chapter provides detailed procedures for performing measurements. To perform a measurement, the user must first set up the parameters then begin the measurement process. Both tasks may be performed manually or automatically:

- Parameters may be set individually before each sample measurement, or they may be set automatically using a “recipe” with frequently-used setup conditions.
- The measurement may be performed “manually,” with the user responsible for starting the measurement, saving the data, printing the results, and performing other tasks sequentially. Alternately, a measurement may be performed automatically. In this case, the user pre-specifies all the tasks that must be performed; and once started, all tasks proceed without operator intervention.

This chapter describes how to perform fully manual setup and measurement in sections 12.3 through 12.6. Recipes are described in section 12.9, and automatic measurements are explained in sections 12.10 and 12.11.

Engineer-level users should first learn to perform fully manual setup and measurement. However, frequent users, and users in a production environment, will certainly want to take advantage of the convenient operation provided by recipes and automatic measurements.

Information in this chapter is organized as shown in Table 12-1.

Table 12-1
Organization of System Operation Procedures

✓ Measurement Procedure	✓ Section
✓ Preparation for Use	✓ 12.1
✓ Loading the Sample	✓ 12.2
✓ Wafer Finder	✓ 12.3
✓ Setting the Signal Parameters	✓ 12.4
✓ Setting the Measurement Parameters	✓ 12.5
✓ Making a Measurement	✓ 12.6
✓ Saving Data	✓ 12.7
✓ Making Multiple Measurements	✓ 12.8
✓ Recipes	✓ 12.9
✓ Automatic Measurements	✓ 12.10
✓ Automatic Measurements Using Recipes	✓ 12.11

✓ **NOTE:** An overview of the fully manual measurement process is given in the Quick Start Guide, section 6.1.

12.1 Preparation for Use

Before loading a sample, verify that the hardware and software are ready for measurement.

- [1.] If the WT-2000 was shut down, start it again following the procedures in section 6.4.
- [2.] Check to see if the system needs to be initialized. In the **Measure** menu, the **Initialize** menu item will be black if initialization is necessary, and it will be gray if the initialization has already been performed.
 - [2.1.] To initialize the system, select the **Initialize** menu item.
 - [2.2.] Depending on the setup of your system, the *Height* tab of the *Measurement Options* window may appear (Figure 12-1). Click **OK** to close the window.

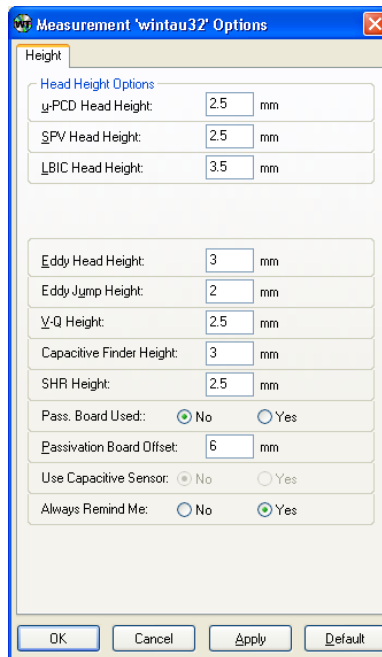


Figure 12-1: The *Height* tab of the *Measurement Options* window

- [2.3.] During the initialization procedure the stage, the bridge and the measuring head make a series of movements. This procedure initializes and checks the motors and mechanics.
- [3.] Select the measurement method from the **Measure | Mode** menu, or choose it from the drop-down menu (Figure 12-2) in the left top of the *Main Window*.



Figure 12-2: Selecting the measurement method

✓ **NOTE:** *The software screens may look different from the figures, depending on the options installed in your WT-2000 system.*

- [4.] Many methods allow different types of measurements (Sub Measurement Modes) within the same measurement method. Select the desired measurement type by clicking on the sub measurement in the *Buttons area* (Figure 12-3).

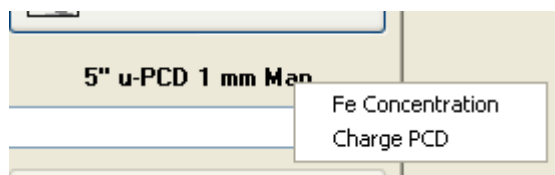


Figure 12-3: Sub Measurement Mode drop-down list in the *Buttons area* (μ -PCD method shown)

After the initial hardware and software setup procedure is complete, load the sample into the system as described in section 12.2.

JPV measurements require additional setup before the sample is loaded.

12.2 Loading the Sample

Samples can be loaded onto the stage either manually or automatically. Each method is described separately. Manual wafer loading is described in section 12.2.1. Automatic wafer loading using an indexer is described in section 12.2.2.

12.2.1 Manual Loading

To load a sample manually, follow the procedure described below.

- [1.] Verify that the *Wafer Shape* is set properly in **Measure | Options, Loading** tab (see section 10.12.3).
- [2.] Check or set the sample size in **Measure | Wafer Size** or in the appropriate drop-down list (Figure 12-4).

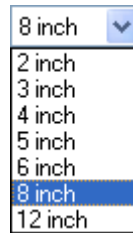


Figure 12-4: Selecting the sample size

- [3.] Select the **Measure | Load Sample** menu item or the **Load Sample** button  on the *Toolbar*.
- [3.1.] The  button is available even if there was no initialization yet. Clicking on this button, the Initialize procedure is performed automatically if necessary.
- [3.2.] If there is a wafer on the stage already, the following message appears in order to avoid the damage caused by the previously loaded wafer and the one to be loaded: “*There is a wafer remaining on the stage!*”
- [4.] The *Load Sample* window appears (Figure 12-5). Place the sample on the stage, centered above the vacuum grooves.

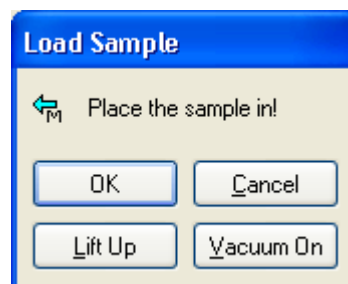


Figure 12-5: Load Sample window for manual loading

! ***CAUTION: Please, place the wafer in the middle of the stage as accurately as possible! If a hot-chuck is installed in the WT-2000, placing the wafer improperly can make the wafer stuck in the hot-chuck. Semilab refuses responsibility for damage of the wafer or the WT-2000 that comes from inaccurate positioning of the wafer during manual loading***

- [4.1.] When using the chemical passivation envelope, it is necessary to install the protective plexi passivation board before installing the sample. If the board is not already installed, place it on the stage as described in Chemical Passivation Guide.
- [4.2.] Put the envelope on the plexi passivation board and fix it with adhesive tape.
- [5.] When the sample is in position, click the **OK** button. This action turns on the stage vacuum.
- [5.1.] If the sample moves, use the **Vacuum On** button to turn on the vacuum, then check the wafer position.

- [5.2.] If you need to reposition the sample, click the **Vacuum Off** button. (If the stage has lift pins, click **Lift Up** and **Lift Down** to move the wafer as required.) Click **Vacuum On** to restart the vacuum.
- [5.3.] When the wafer position is acceptable, click **OK**.

✓ **NOTE:** To remove the wafer after the measurement, click the **Unload Sample** button  on the **Measurement Toolbar**, or select the **Measure | Unload Sample** menu item. The stage will move to the unload position and the vacuum will be turned off. Open the lid and remove the sample.

- [6.] Close the lid of the WT-2000.

After the sample is loaded, continue to the Wafer Finder procedure (section 12.3).

12.2.2 Automatic Wafer Loading

Standard wafer cassettes can be used for automatic loading. To load a wafer automatically from the cassette, follow the procedure described below. Be sure that the lid is closed before loading a wafer.

- [1.] Check or set the wafer size in **Measure | Wafer Size** or in the appropriate scroll-down list (Figure 12-6).

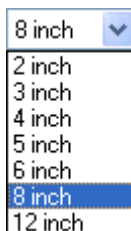


Figure 12-6: Selecting the wafer size

- [2.] The cassette should be placed to the middle of the indexer platform. In the proper position, the middle of the cassette “H-bar” keeps platform Switch-1 pressed (see Figure 12-7). For 12” (300 mm) cassettes, Switch-2 and Switch-3 detect the proper position.

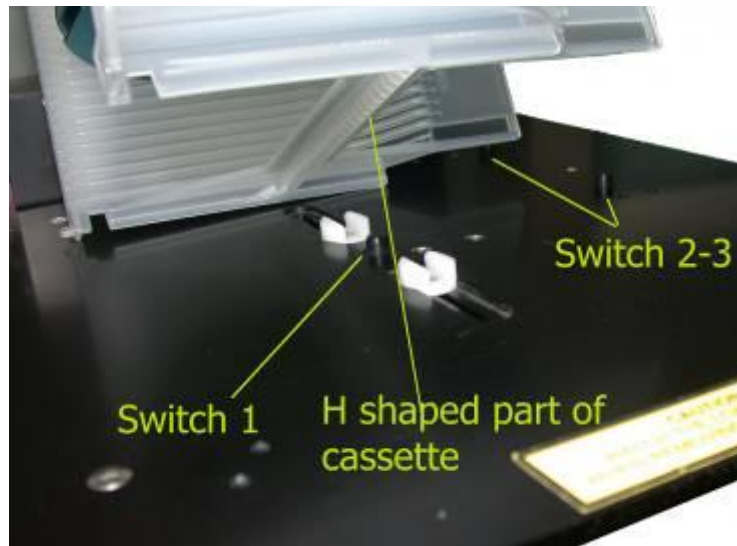


Figure 12-7: Mounting the cassette

- [2.1.] When the system is first installed, or when changing the cassette type (size), it is necessary to check the parameter setting on the **Measure | Options** menu, **Indexer** tab. See section 10.12.13.
- [3.] Select the **Measure | Load from Cassette** menu item or the **Load from Cassette** button  on the **Toolbar**.
- [3.1.] The  button is available even if there was no initialization yet. Clicking on this button, the Initialize procedure is performed automatically if necessary.
- [3.2.] If there is a wafer on the stage already, the following message appears in order to avoid the damage caused by the previously loaded wafer and the one to be loaded: *“There is a wafer remaining on the stage!”*
- [4.] The **Load Sample** window appears (Figure 12-8). Select the wafer to load, then click the **OK** button.

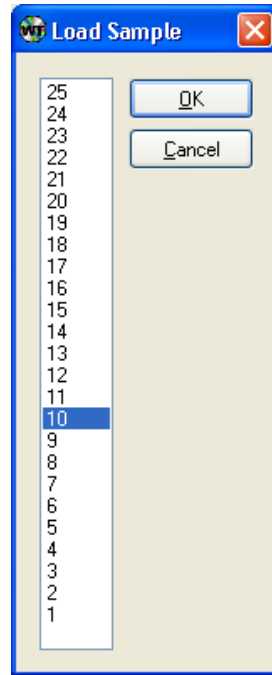


Figure 12-8: Select the wafer for automatic loading.

- [5.] The wafer will be loaded onto the stage automatically.
- [5.1.] The wafer loading starts with the movement of the indexer: The cassette is lifted to a position where the selected wafer could be reached by the tongue. The stage moves toward the cassette and the end effector is inserted into the cassette below the selected wafer.
- [5.2.] In case of a proper parameter setting (**Measure** | **Options**, **Indexer** tab) the end effector should not touch the selected wafer and the one below it when moving. The indexer makes a partial step down placing the wafer on the tongue. The end effector vacuum is automatically switched on to hold the wafer firmly during its transport. By the relative movement of the stage and the end effector, the sample is placed to the middle of the stage.
- [5.3.] The wafer is held in this position by the stage vacuum until the end of the measurement sequence.

✓ **NOTE: The wafer will be unloaded automatically after the measurement is complete.**

After the sample is loaded, continue to the Wafer Finder procedure (section 12.3).

12.2.3 Loading a sample on the console

In case of C-PCD measurement, the sample holder has to be changed to console. This wafer holder allows applying corona charge on the backside of the wafer. See section 4 for mounting the console wafer holder.

Loading a wafer on the console is possible either manually or automatically.

12.3 Wafer Finder

After loading the wafer, complete the Wafer Finder procedure so that the system detects the placement of the wafer on the stage.

! CAUTION: After loading the wafer, the WT-2000 top cover should be kept closed during operation. If the cover is not closed, the laser cannot operate and no measurement can be carried out. In this case the “Laser is interlocked! Close the cover” warning appears on the screen if a measurement has been started.

Several Wafer Finder methods are available. Generally, the Optical Finder is used. However, when using the chemical passivation envelope for μ -PCD measurement, the Microwave Finder should be used. (See section 10.5 for more information.)

All Wafer Finder methods can be started from the *Measure | Wafer Finder* menu item. If the Optical Finder is used, the procedure may be activated automatically (if selected from the *Measure | Options Loading* tab; see section 10.12.3).

In the first step of the Wafer Finder procedure, the dimensions of the sample are measured (Diameter, ScanRadius or ScanLimitX and ScanLimitY). If the sample is circular, the second step is the exact determination of the flat (or notch) orientation (depending on the setting of the *Measurement Options Loading* window, see section 10.12.3). In case of more flats only the largest one is identified.

At the end of the Wafer Finder procedure, the measuring head goes to the center of the sample by default, and a window with the sample contour image appears showing the flat (or notch) orientation as well. The appropriate *Recorder* window also appears. Figure 12-9 shows an example of the *Main Window* appearance at the end of the Wafer Finder procedure.

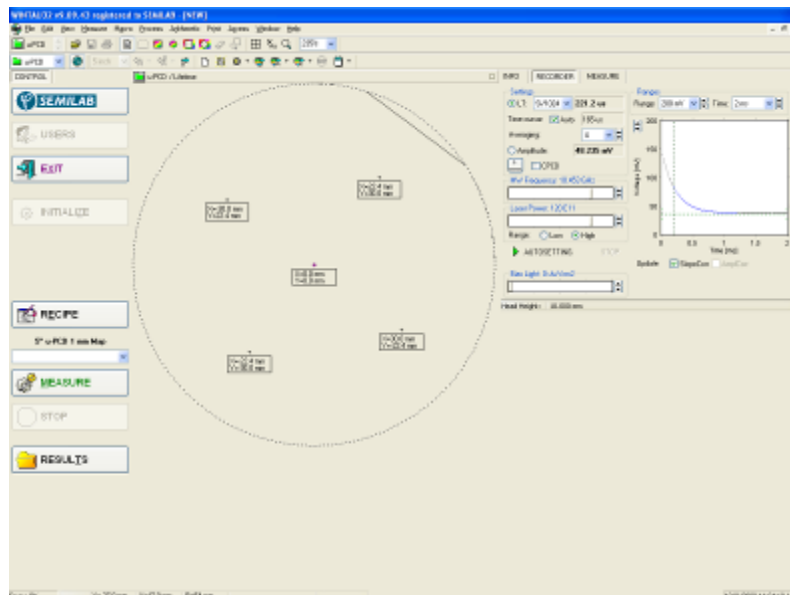


Figure 12-9: Typical appearance of the *Main Window* after the Wafer Finder procedure (μ -PCD mode shown)

✓ **NOTE:** *The direction upwards on the screen corresponds in the WT-2000 system to the direction opposite from the operator.*

After the Wafer Finder procedure is complete, set the signal parameters as described in section 12.4 before proceeding further.

12.4 Setting the Signal Parameters

Before measurement, both the signal parameters and the measurement parameters must be set. Signal parameters optimize the signal. Measurement parameters define the measurement pattern or map. This section describes the procedure for setting the signal parameters. (Signal parameters can be saved as part of a recipe, as described in section 12.9.1.)

The signal parameters are shown in the appropriate *Recorder* window for the measurement method. This window is the basic control panel of the measurements. All the measurement related parameters can be viewed and modified here. *Recorder* windows are automatically displayed after the Wafer Finder procedure is complete. *Recorder* windows can be activated at any other time by the **Measure | Recorder** menu item or by clicking the  button.

12.4.1 Autosetting

The intent of the signal parameter setting is to optimize all parameters that determine the measurement: for example, photo conductive decay signal (for μ -PCD) or light flux settings (for SPV). With the help of the Autosetting function in the **Measure** menu, the optimum signal parameter settings are realized automatically; no user intervention is necessary.

Although signal parameters in the *Recorder* windows can be manually adjusted, it is strongly recommended to use the automatic parameter setting procedure in all measurement modes. It optimizes the measuring parameters according to built-in algorithms. Using the Autosetting procedure makes the measurement faster, more reproducible and operator independent. After any modification of the measuring parameters a prompt measurement is performed to refresh the *Recorder* display.

! **CAUTION:** *For all measurement methods requiring parameter settings, it is recommended to use Autosetting for setting measurement parameters.*

After wafer loading, begin the Autosetting procedure using one of the following methods:

- select the **Measure | Autosetting** menu item in the *Main Window*,
- select the **Autosetting** menu item in the *Recorder* window,
- or click the  button in the *Measurement Toolbar*.

When the Autosetting procedure is complete, set the measurement parameters as described in section 12.5.

The following sections contain notes about the Autosetting procedure for specific measurement modes.

12.4.2 Signal Parameters for μ -PCD and C-PCD Measurements

For the μ -PCD and C-PCD methods, the signal parameters are displayed in the *Transient Recorder* window (Figure 12-10).

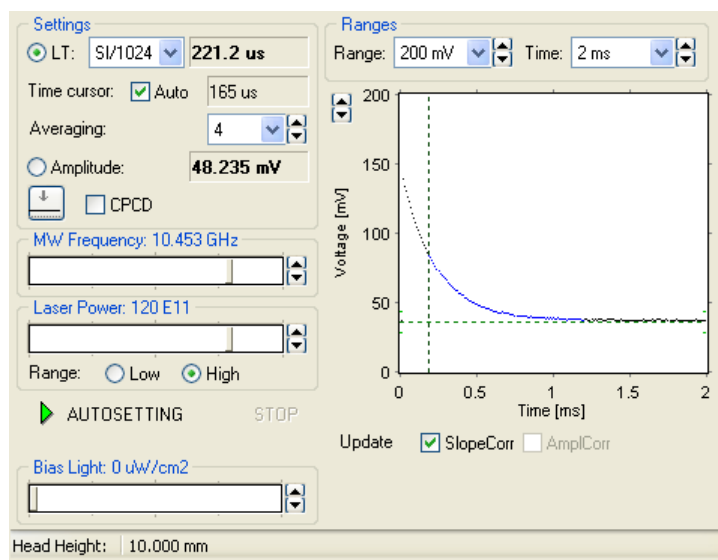


Figure 12-10: Transient Recorder window

The MW (Microwave) Frequency, the Averaging, the signal Range (Y-axis of the Scope window) and the Time window (X-axis of the Scope window) are automatically set by the Autosetting procedure. See section 5.1.6 for a description of the Autosetting procedure.

As a result of this procedure the signal (decay) should appear in the Scope of the *Transient Recorder* window.

✓ **NOTE:** For a complete description of the values in the Transient Recorder window, and their recommended settings, see section 10.13.

When there is proper operation of the signal excitation and detection units, a smooth decay curve should be positioned in the Scope window as shown in Figure 12-10. If the decay signal looks correct, continue to set the measurement parameters as described in section 12.5.

! **CAUTION:** The measurements during the signal parameter setting are carried out at the same position (the default position is the middle of the wafer). In the case of a sample extremely sensitive to light (e.g. oxide charging effect of oxides with pure quality) the decay can be modified by subsequent light excitation. As a result the map shows a different lifetime value at that point.

If there is no signal (flat line) or the noise is larger than half of the whole range, review Diagnostics and Troubleshooting Manual to determine if a hardware problem exists.

12.4.3 Signal Parameters for SPV Measurements

For SPV measurements, the signal parameters are displayed in the *SPV Recorder* window (Figure 12-11).

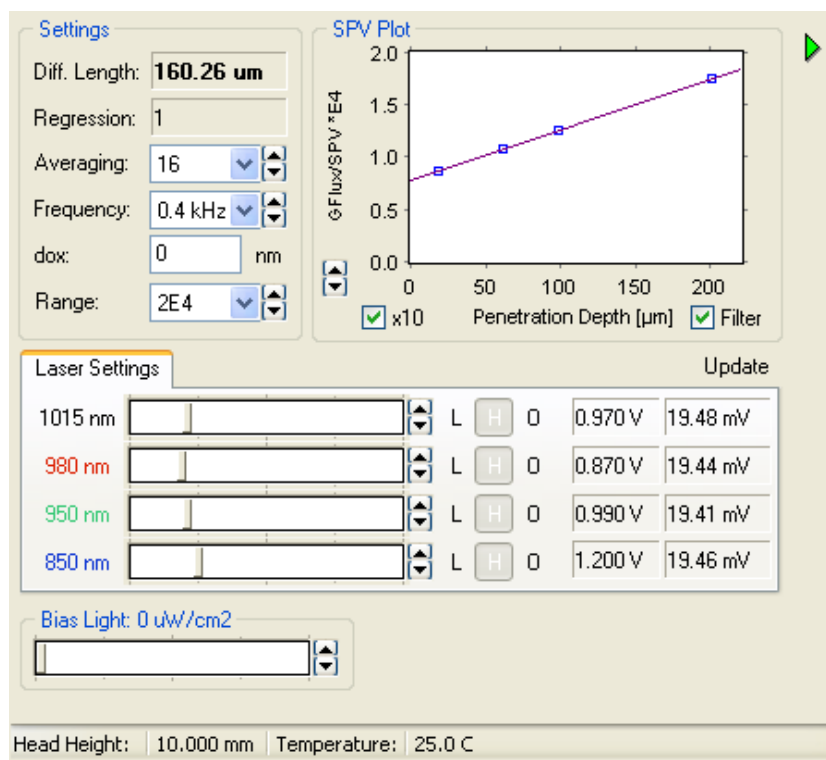


Figure 12-11: SPV Recorder window

Before performing the Autosetting procedure, set the thickness value in the *Input List* tab (section 12.5).

During the Autosetting procedure, the SPV signal versus absorbed flux curves are measured (positive signal for the n-type wafer and negative for the p-type). After recording the curves, the software sets the flux values for all wavelengths in accordance with the Autosetting routine. See section 5.4.1 for a description of the Autosetting procedure.

The *Measuring Frequency*, the *Averaging* and the Laser control voltages, according to the set SPV signal amplitudes, are automatically set by the Autosetting.

As a result of the Autosetting procedure, a line appears in the *SPV Plot*.

✓ **NOTE:** For a complete description of the values in the *SPV Recorder* window, and their recommended settings, see section 10.13.1.5.

A straight line, fitted on the points, is displayed as shown in Figure 12-11. If the plot looks correct, continue to set the measurement parameters as described in section 12.5.

12.4.4 Signal Parameters for V-Q Measurements

V-Q measurements do not require an Autosetting procedure. The current signal parameters are displayed in the *V-Q Recorder* window.

✓ **NOTE:** For a complete description of the values in the V-Q Recorder window, and their recommended settings, see section 10.

12.4.5 Signal Parameters for JPV Measurements

Autosetting is required for JPV measurements. However, the correct calibration file must be loaded.

In the *JPV Recorder* window, use the **Functions | Load Calib. File** menu item to load the correct calibration file.

✓ **NOTE:** For a complete description of the values in the JPV Recorder window, and their recommended settings, see section 10.13.4.

12.5 Setting the Measurement Parameters

Measurement parameters describe the sample being measured, including the date of the measurement, operator name, size of the sample, and similar identifying information. All parameters of the measurement are saved and printed together with the measured data.

Measurement parameters are set in the *Input List* tab (Figure 12-12) of the *Measurement Information* window (**View | MeasInfo**).

Figure 12-12: Input List tab for μ -PCD measurement

The *Input List* serves for storing the measurement parameter settings. Before beginning the measurement, update the fields in this list as described below. Most of them can be edited by clicking the mouse on them. Note that the *Input List* fields can be modified by someone with

Engineer-level access, so additional fields may be present that are not included in the list below (see section 7.8.5 and section 9.4).

✓ **NOTE: It is possible to enter default values for the Input List that will be applied to every measurement. See section 9.4 for information.**

<i>Date</i>	This field is automatically refreshed.
<i>Comment</i>	Enter a comment. This field can be edited before or after the measurement or even later after loading the saved measurement.
<i>Operator</i>	Enter the operator identification. This field can be edited before or after the measurement or even later after loading the saved measurement.
<i>Sample</i>	Enter the sample name. This field can be edited before or after the measurement or even later after loading the saved measurement.
<i>Raster</i>	The <i>Raster</i> defines the step size between the X and Y scans across the sample. If you plan to make a map, set the <i>Raster</i> value before starting the measurements. The recommended <i>Raster</i> settings depend on the resolution of the measurement method. For example, a common <i>Raster</i> for μ-PCD is 1 or 2 mm. For SPV, a common <i>Raster</i> is 4 to 8 mm.
<i>Wafer Size</i>	The <i>Wafer Size</i> was specified before loading the wafer (section 12.2), and it cannot be modified if the wafer is already loaded.
<i>Scanradius</i>	The <i>Scanradius</i> is half the distance across the sample. In the case of a round sample, it is a radius. In the case of a square sample, it is half the width of the side. <i>Scanradius</i> can be set from 0 up to the size of the sample. It can be modified also on the map display by dragging the outline of the displayed area while pressing the left mouse button. The <i>Scanradius</i> can be set before or after the measurement. Engineer-level users can change the name of the field to <i>Scandiameter</i> by right-clicking on the label.
<i>Doping Concentration</i>	In V-Q mode, enter the doping concentration of the wafer. The entered value should be as accurate as possible, to ensure greatest accuracy of D_{it} calculations. (This field can be changed to display <i>Resistivity</i> . See section 9.4.)
<i>dox</i>	In V-Q mode, the oxide thickness should be entered in the <i>dox</i> field.

The time required to obtain a complete measurement is highly dependent on the selected measurement method and the measuring parameters (for example, *Raster*, *Scanradius*).

12.6 Making a Measurement

After the signal parameters and measurement parameters have been set, the sample can be measured. Several different pattern designs are available as described in the following sections:

- A single point measurement (section 12.6.1)
- A number of measurements in a pattern, called an “x-point” measurement (section 12.6.2)
- A map of the sample (section 12.6.3)

Additionally, special measurement options are available, depending on measurement mode:

- Fe concentration can be determined in μ -PCD or SPV mode (section 12.6.4)
- Mobile charge (Q_m) can be determined in V-Q mode (section 12.6.5)

! CAUTION: *It is important that the lid of WT-2000 should be kept closed during operation. If the cover is not closed, the laser cannot operate (interlocked) and no measurement can be carried out. In this case the “Laser is interlocked! Close the cover!” warning message appears on the screen.*

12.6.1 Single Point Measurements

A single point measurement can be performed (even without initialization) if you put the wafer below the measuring head and open the *Recorder* window (**Measure | Recorder** or  button). In this case the position on the wafer is not known.

If the wafer is loaded in the regular way, the position is known and single point measurement is possible anywhere on the wafer. Position the cursor on the wafer image in the map window where you want to measure, then press the right mouse button. The measuring head will go to the selected position (identified by a purple star) and perform a measurement there. The coordinates are displayed in the *Status Bar*, and the result of the measurement can be seen in the *Recorder* window.

For continuous measurement at the same measurement point, check the **Update** box in the *Recorder* window.

12.6.2 X-Point Measurements

After the parameter setting procedure described in section 12.4.5, fast x-point measurements can be performed at a predefined number of measuring sites.

The number of measuring sites can be selected two different ways:

- Click the drop-down list next to the **X-Point** button  on the *Measurement Toolbar* (Figure 12-13). Select from the number of points (1, 5, 9, 17, 25, or 50).

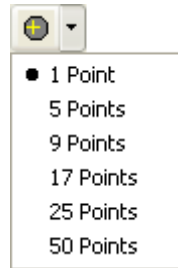


Figure 12-13: Selecting the number of points to measure

- Or, select the number of points from the *Measurement Options* window *Single* tab (section 10.12.4). This window also allows you to select a **Custom** number of points.

After selecting the number of points to measure, the name of the **X-Point** button  will change to the selected number of points, and the **Measure | X-Point** menu item will be renamed to the current number of points.

The positions of the points are shown on the wafer image (see Figure 12-14). The position of a point can be modified by dragging the box below the mark (cross) with the mouse. Press the left mouse button, drag the box and move it wherever you wish inside the wafer area. The new location will be shown inside the box. (The measurement locations can also be modified using the *Prescan Points Setup* window; see section 10.12.4.)

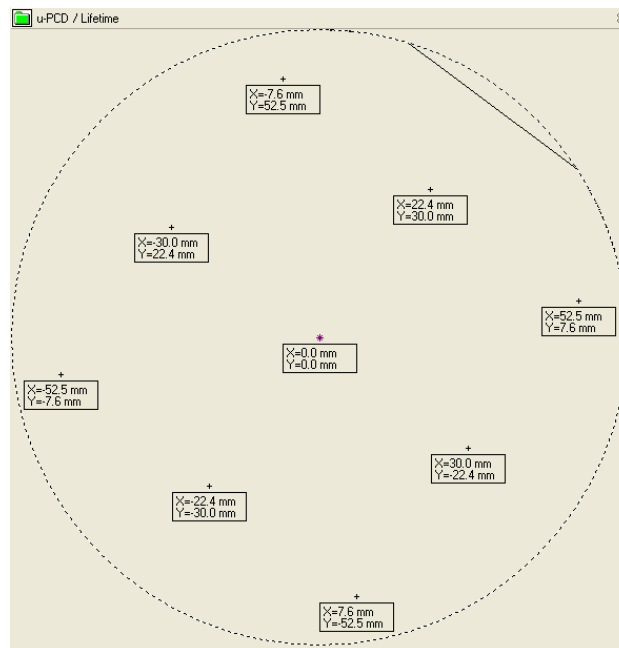


Figure 12-14: A 9-point measurement pattern

When you are ready to measure, click the **X-Point** button, or select the **Measure | X-Point** menu item. The measurements will be completed automatically at the predefined locations.

During the measurement, the measured values at each point are displayed in the *Recorder* window. After the measurement, the measured values are displayed on the wafer image below the coordinates for each point (Figure 12-15). The results can be saved for later review if desired (see section 9.8).

In V-Q mode, the *V-Q Evaluation* window disappears after the measurement but can be displayed by pressing the  button in the *Input List*.

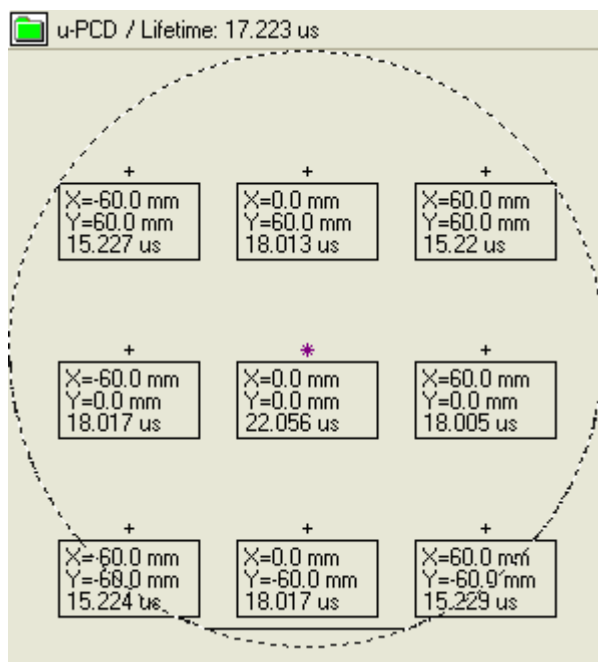


Figure 12-15: After x-point measurement, the measured values are displayed for each point.

The positions are saved in the configuration file of the Wintau software, so the next time you start the software the measuring positions will be the same. One of the small crosses at the points is a different color. This is the default position of the measuring head after the wafer loading procedure (the center of the wafer by default), so the Autosetting is performed there even if you move the head to somewhere else by pressing the right mouse button on the wafer image.

12.6.3 Map Measurements

If there is a map already in the map window that belongs to the loaded wafer, save the map before the next measurement if you need the data, otherwise you will lose it.

Before starting the mapping, set the necessary *Scanradius* and *Raster* size in the *Measurement Information* window *Input* list (Figure 12-12).

If you wish to map only a defined region on the sample, you can use masks to define the area to be measured. If you select a mask button on the *Toolbar*, you can use the mouse button to define the masked area. If you use the mask buttons on the *Mask* tab in the *Measurement Information* window, you can enter the mask coordinates by typing the values in the boxes. See section 9.5 for more detailed information about masks.

Note that certain measurement methods may only be available in x-point patterns. If maps are not available, the map buttons on the *Measurement Toolbar* will not be active. For the V-Q measurement method, Table 12-2 lists the measurement methods for which mapping is possible.

Table 12-2
Mapping Capability for V-Q Measurements

	x-Point	Mapping
Surface Potential	X	X
PDM	X	X
Q_{eff}	X	X
V-Q (V_{fb} , D_{it} , Q_{total})	X	–
V-Q (T_{ox})	X	X
E+, E-	X	–
Q_{m}	X	X

Start the map measurement from the **Measure | Map** menu or by pressing the **Map/Resume** button  on the *Measurement Toolbar*.

During the map measurement, the displayed results are constantly updated:

- During μ -PCD measurement, the measured transient and the actual evaluated lifetime value can be seen in the *Transient Recorder* window.
- During SPV measurement, the measured SPV signals and the evaluated diffusion length can be seen in the *SPV Recorder* window.
- The statistics of the already measured values (average, minimum and maximum) are continuously refreshed in the *Measurement Information* window.

The ratio of the already measured area over the whole area in percentage, as well as the extrapolated time until the end of the measurement, is displayed and continuously refreshed in the top caption of the map window (Figure 12-16).

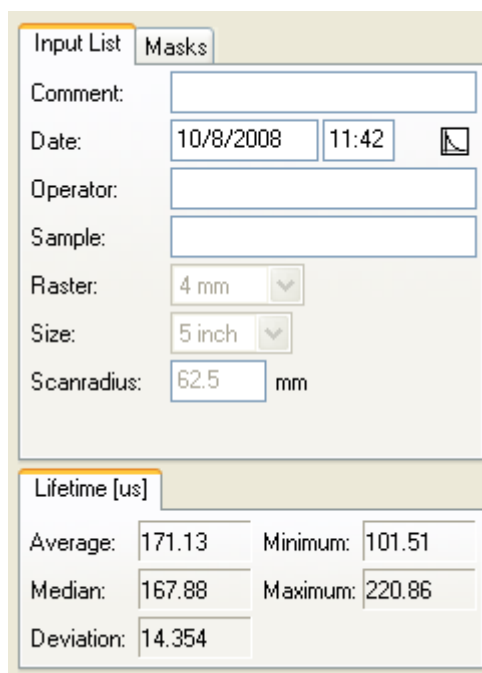


Figure 12-16: Measurement timer

To view a histogram of the measurements, click the  *Toolbar* button or select the **View | Distributions** menu item. To refresh the histogram, click the  button in the *Distribution* window. (See section 9.7.1 for more information.)

The map measurement can be stopped at any time, even before it has been completed, by pressing the <Esc> key or by clicking the **Stop**  button in the *Measurement Toolbar*. Measurement can be resumed by selecting the **Measure | Resume** menu item or by pressing the **Map/Resume**  button in the *Measurement Toolbar*.

After the measurement, the map is re-scaled according to the new color scale calculated from the map data collected up to this point; and the statistics of the map are calculated and displayed in the *Measurement Information* window *Input List* (Figure 12-17).



Input List	
Comment:	
Date:	10/8/2008 11:42
Operator:	
Sample:	
Raster:	4 mm
Size:	5 inch
Scanradius:	62.5 mm
Lifetime [us]	
Average:	171.13
Minimum:	101.51
Median:	167.88
Maximum:	220.86
Deviation:	14.354

Figure 12-17: Map statistics in the *Input List*

12.6.4 Fe Concentration Measurement

Depending on the configuration of your WT-2000 system there may be the capability to measure Fe concentration and produce maps. The Fe concentration measurement can be performed in both μ -PCD and SPV measurement modes.

To enable the Fe concentration measurement, open the *Measurement Options* window, and select the *Fe* tab. Select **Yes** in the *Fe. Conc. Measurement* section. (See section 10.12.9 for other *Fe* setup values.) An alternate method is to select the **Fe** sub-measurement mode from the drop-down list **Fe Concentration Map** in the *Buttons area*.

Start the measurement by selecting the **Measure | Fe Conc.** menu item or clicking the **Auto Measure Fe Conc.** button  on the *Toolbar*. (The drop-down arrow to the right of the button allows you to select x-point measurements or a map.)

After the Autosetting routine, the lifetime or diffusion map will be measured then the flash unit will turn on and the pre-selected number of flashes will be performed. After this, the map will be measured again and a Fe concentration map will be calculated. By selecting **File Save**, the map can be saved, together with its raw files.

12.6.5 Making C-PCD Measurements

In most cases it is worth to make a standard μ -PCD measurement before the C-PCD on the same wafer in order to compare the results.

The corona charge applied for surface passivation is advised to be the same type as the excess carrier: if the sample is P type, apply negative charge, if the sample is N type apply positive charge.

If the sample has a native oxide, in-situ corona charge has to be applied continuously during measurement in order achieve good surface passivation.

12.6.5.1 C-PCD on oxidized wafers

If the sample has a thick and good thermal or chemical oxide on the surface, it is enough to apply a homogeneous corona charge on both sides of the wafer once before mapping. The advised charging voltage is (+ or -) 10 kV.

12.6.5.2 C-PCD on wafers with native oxide

If the sample has a native oxide, in-situ corona charge has to be applied continuously during measurement in order achieve good surface passivation.

There are three ways to switch on continuous corona charging:

- Click on **CPCD** checkbox in the *Transient Recorder*. See section 10.13.1.
- Change Sub Measurement Mode to *Charge PCD* in the *Sub Measurement mode* field of the *Buttons* area. See section 7.5.
- Go to *Measurement | Options | Charge* and select **Yes** checkbox. See section 10.12.9.

12.6.6 Using Linear Corona Charge

If your system has a built in linear corona charger, it can decrease the surface recombination velocity, effectively passivating the surface for the μ -PCD measurement and increasing the barrier for the SPV measurement of n-type wafers. (See section 5.3.1 for a discussion of charging techniques for μ -PCD measurements.) If the corona charge is desired, it must be deposited before the sample is measured.

To put corona charge on the top of the sample, select **Corona | Linear Charge Front-side** from the **Measure** menu. The *Linear Charge Front-side* window appears (Figure 12-18). Select the desired charge sign and value (see section 10.7), then click the **Start** button. After the charge procedure has finished, the display shows 100% and the **Start** button appears again. At this time, close the window or click **Cancel**, then you can make the μ -PCD or SPV measurement.

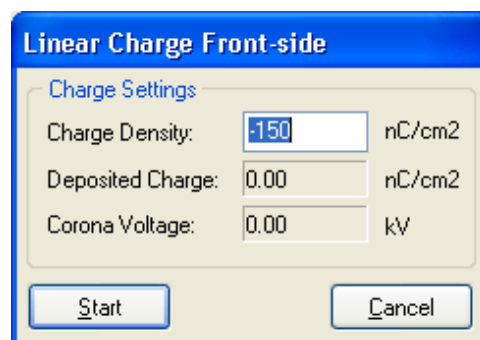


Figure 12-18: Linear charge control window

12.6.7 Mobile Charge Measurement

The steps of the mobile charge measurement cycle proceed automatically as follows:

- Negative corona charge deposition until E- is reached (see section 5.7.4).
- Heat treatment at a specified temperature, for a specified time.
- A point measurement (using the number of points in the **Measurement Options Single** tab).
- Positive charge deposition until E+ is reached.
- Map measurement of the initial condition.
- Heat treatment at a specified temperature, for a specified time (see section 10.12.6).
- Map measurement of the treated condition.
- Mobile charge map calculation.

12.7 Saving Data

To save the displayed data, use the **File** menu items to save the current data files as described in sections 8.1 and 9.8. Information is saved in a proprietary format that allows Wintau software to save the measurement parameters together with the measured data.

Maps may be saved separately as image files, either bitmap or JPEG. See section 9.6.1 for information on saving image files.

12.7.1 Saving SPV Maps as Linked Files

During the SPV measurement the diffusion length map is plotted on the screen. This map is calculated from the SPV maps measured with different wavelengths and collected in the memory. Raw maps as well as the diffusion length map are linked and can be saved by selecting the menu option **File | Save Link As**. In this case, a link file together with raw files and diffusion length map will be saved. (See section 9.9 for more information about link files.) If the **File | Save** command is used, then only the diffusion length map will be saved.

12.8 Making Multiple Measurements

You can make multiple measurements on a sample without removing it. You can also change to a different measurement method.

✓ **NOTE:** *Be sure to save the displayed data before making the next measurement, otherwise the data will be lost.*

To make another measurement, click the **New** button  in the **Measurement Toolbar**. This action refreshes the screen by removing the results from the previous measurement. You will not be prompted to save the existing data first. After clicking **New**, you may begin the next measurement with the same measurement method, or you may change the measurement method.

After you have changed the measurement method, perform the Autosetting procedure before starting the new measurement.

12.9 Recipes

A recipe is a previously-saved set of conditions for performing a measurement. Using a recipe saves time and reduces the chance of error during measurement setup. Recipes can be used for systems with manual loading as well as automatic wafer loading.

12.9.1 Creating a Recipe

To create a recipe, follow the procedures below to:

- Set the measurement parameters
- Specify the wafers for measurement
- Specify *Recorder* settings
- Save the recipe

The procedure to create a recipe has several steps.

- [1.] Begin by selecting the **Measure | Recipe | Set Parameters** menu item. (This menu item is only available if no wafer is loaded.) The *Measurement Options* window will be displayed (Figure 12-19).

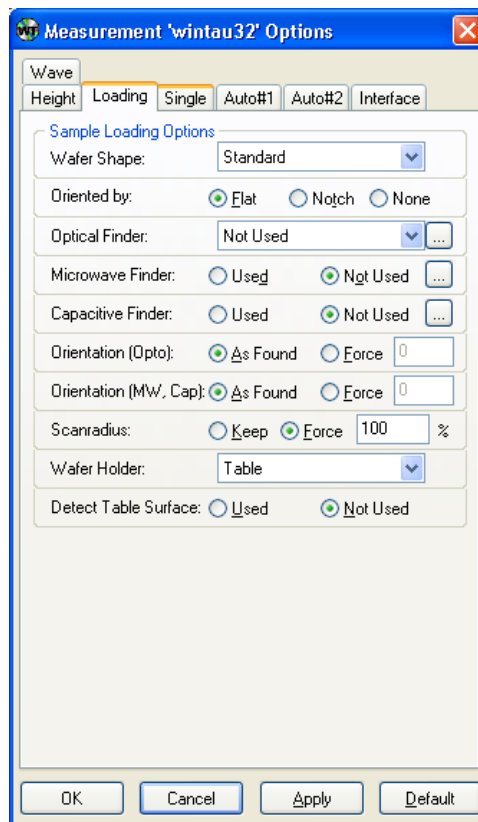


Figure 12-19: The Loading tab of the Options window

- [2.] Check all the tabs of the *Measurement Options* window except for *Auto*, and set the parameters for the measurement. (See section 10.12 for more information about the *Options* tabs.) The saved recipe contains all the settings available in all the

Measurement Options tabs. Do not forget to check settings such as head height, sample holder settings and Wafer Finder.

- [2.1.] The *Auto* tab is used for automatic measurements. If the recipe will be used for an automatic measurement, set the parameters as described in section 12.10.
- [3.] The *Measurement Information* window, with the *Input List* tab, will also be displayed. Set the resolution (*Raster*), size and scan diameter parameters, and fill the information fields if needed. See section 12.5 for more information about the *Input List*.
- [4.] All this data will be saved to the recipe. When you have finished checking the settings, click **OK** in the *Measurement Options* window to close both windows.
- [5.] On systems with automatic wafer loading, the *Select Samples* window appears next (Figure 12-20). If you have a typical wafer distribution in your cassettes, or if you always need to load the same slots, define the selected slots here. *File Names* also can be specified. If these parameters are not necessary for your recipe, leave these fields empty.
 - [5.1.] Select each wafer for measurement by clicking once on the checkbox next to its slot number (clicking again deselects the wafer). To select all wafers, click **Select All**; to clear all selections, click **Deselect All**.
 - [5.2.] You can assign file names and sample names to the wafers by simply clicking on the *File Name* or *Sample* field of the slot and entering a name.
 - [5.3.] To set the filename to the slot number, click the **Set Names** button. A pop-up window appears. Enter the prefix for the filename, then click **OK**. The slot number will be added to the end of the *File Name* for each slot.
 - [5.4.] To set the sample name to the slot number, click the **Set Samples** button. A pop-up window appears. Enter the prefix for the sample name, then click **OK**. The slot number will be added to the end of the *Sample Name* for each slot.
 - [5.5.] Clear these entries by using the **Clear Names** and **Clear Samples** buttons.
 - [5.6.] Click **OK** to close the *Select Samples* window.

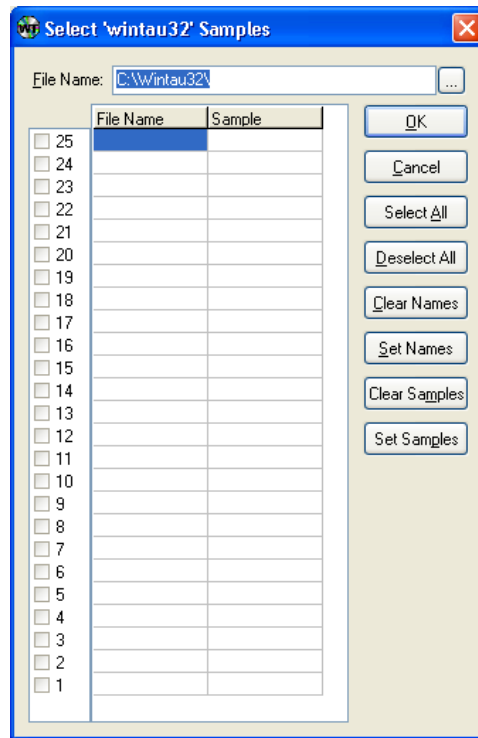


Figure 12-20: The *Select Samples* window

- [6.] Next, the *Recorder* window of the active measurement method is displayed. Specify all settings (see section 12.4 for more information). These settings will also be saved. Leave the *Recorder* window open.
- [7.] After setting all the parameters needed, you must save the recipe. Click the **Measure | Recipe Save Recipe** menu item. The *Save Recipe As* window will appear (Figure 12-21). The default location for saving a recipe is the Wintau32 directory, but you can navigate to a different location if you prefer. Enter the name of the recipe and click **Save**. The **.rec** file extension will automatically be added to the name of the recipe.

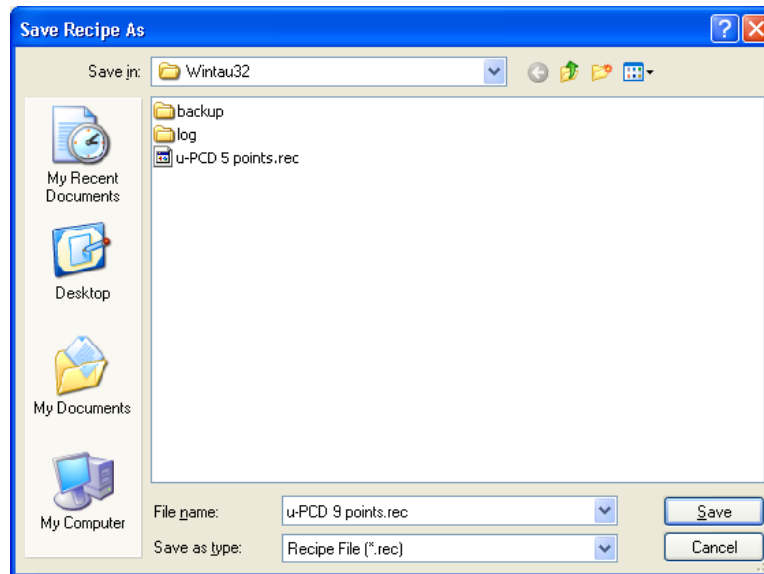


Figure 12-21: Saving a recipe

12.9.2 Loading a Recipe

When a recipe was created, it saved a set of conditions for performing a measurement. By loading the recipe, you restore the saved measurement conditions, saving time during measurement setup.

There are different methods for loading a previously-saved recipe:

- Select the **Measure Recipe | Select Recipe** menu item to choose a recipe stored in the Wintau32 default directory, or use **Measure | Recipe | Load Recipe** menu item to choose a recipe stored in a different location.
- Choose a recipe from the **Recipe List** drop-down menu u-PCD 9 points on the *Buttons area*.

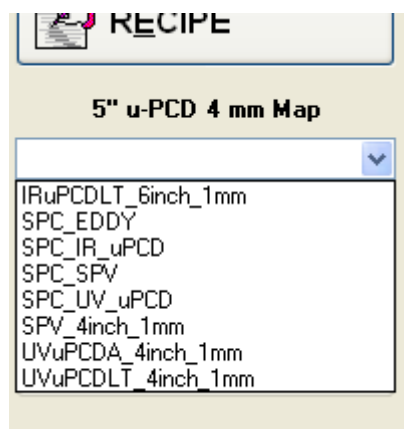


Figure 12-22: Reloading previously saved recipes

Once the recipe is loaded, you can review the recipe. While the recipe is loaded, you may also load a sample and start the x-point or map measurement using a normal method. The measurement will be performed using the conditions specified by the recipe. You may also

perform fully automatic measurements using the recipe as described in section 12.11. (If the automatic measurement option is not used, the settings in the *Auto* tab are ignored.)

12.10 Automatic Measurements

Automatic measurements automate many tasks which are normally performed by the user, such as saving and printing results. Automatic measurements may be performed with manual parameter setting or with recipes that use a previously saved set of parameters.

Before performing the automatic measurement, first perform the operational checks and select the measurement mode as described in section 12.1.

Automatic measurements will be performed with the currently loaded measurement conditions, so you must set the signal parameters and measurement parameters, or load a recipe, before starting the automatic measurement.

The WT-2000 system has a **Measure Recipes** feature (described in section 12.11) which automatically begins the measurement after specifying the recipe. However, sometimes it is necessary to modify the signal parameters before beginning the measurement. In this case, the automatic measurement is a two step process: set the parameters, then begin the automatic measurement. This procedure is described in the following sections.

✓ **NOTE:** *The Quick Start Guide provides an overview of the procedure for performing automatic measurements. See section 6.2.*

12.10.1 Systems with Manual Wafer Loading

- [1.] To perform an automatic measurement with manual wafer loading, first set the measurement conditions.
 - [1.1.] Select **Measure | Options** and then set the measurement conditions in the *Auto* tab of the *Measurement Options* window (see section 10.12.5). If making an x-point measurement, set the number of points on the *Single* tab (see section 10.12.4). The *Input List* also appears. Fill in all required information as described in section 12.5. Click **OK** in the *Measurement Options* window to close both windows.
 - [1.2.] Alternately, load a previously saved recipe as described in section 12.9.2.
- [2.] If it is necessary to change a parameter that is not included on one of the *Options* tabs, make the change before proceeding to the next step.
 - [2.1.] For μ -PCD measurements with bias light, activate the *Transient Recorder* window by using the **Measure | Recorder** menu option or clicking on the  button. Set the bias light intensity using the *Bias Light* slider.

! **CAUTION:** *If use of the bias light is needed in automatic measurement mode, the proper bias light intensity must be set before selecting the automatic measurement option.*

- [3.] Start the automatic measurement by clicking the **Auto Measure**  button or by using the **Measure | Auto** menu selection.
- [3.1.] Clicking on the arrow next to the button  displays a drop-down menu (Figure 12-23) that identifies the type of measurement to be performed.

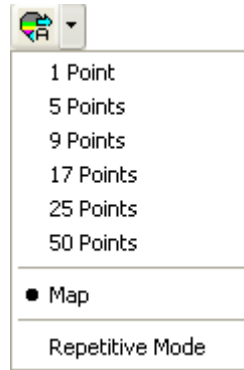


Figure 12-23: Drop-down menu next to Auto Measure button

- [3.2.] The drop-down menu also allows selection of a different measurement method if desired. If **Repetitive Mode** is selected, the sample will be measured repeatedly until cancelled.
- [4.] If *Save* is selected on the *Auto* tab, you will be required to provide a filename for saving. The results will be saved automatically after the measurement is complete.
- [5.] When prompted, place the sample on the stage, then close the cover.
- [6.] If *Autosetting* is selected in the *Auto* tab, Autosetting will be performed.
- [7.] If *Linear Charge* is selected in the *Auto* tab, it will be deposited before the measurement.
- [8.] The measurement will proceed without operator intervention.
- [9.] If *Print* is selected in the *Auto* tab, the measurement results will be printed automatically.
- [10.] When the measurement is complete, remove the sample.

12.10.2 Systems with Automatic Wafer Loading

- [1.] To perform an automatic measurement, first set the measurement conditions.
- [1.1.] Select **Measure | Options** and then set the measurement conditions in the *Auto* tab of the *Measurement Options* window (see section 10.12.5). If making an x-point measurement, set the number of points on the *Single* tab (see section 10.12.4). The *Input List* also appears. Fill in all required information as described in section 12.5. Click **OK** in the *Measurement Options* window to close both windows.
- [1.2.] Alternately, load a previously saved recipe as described in section 12.9.2.

- [2.] If it is necessary to change a parameter that is not included on one of the *Options* tabs, make the change before proceeding to the next step.

- [2.1.] For μ -PCD measurements with bias light, activate the *Transient Recorder* window by using the **Measure | Recorder** menu option or clicking on the  button. Set the bias light intensity using the *Bias Light* slider.

! CAUTION: *If use of the bias light is needed in automatic measurement mode, the proper bias light intensity must be set before selecting the automatic measurement option.*

- [3.] Start the automatic measurement by clicking the **Auto Measure**  button. If using the menu, select **Auto | X-Point from Cassette** or **Auto | Map from Cassette**, depending on the type of measurement desired.

- [3.1.] Clicking on the arrow  next to the button displays a drop-down menu (Figure 12-24) that identifies the type of measurement to be performed.

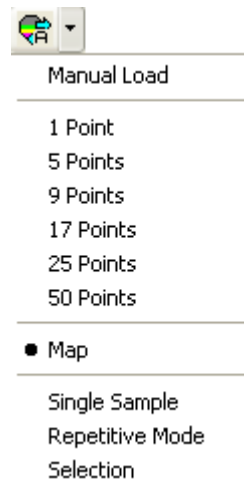


Figure 12-24: Drop-down menu next to Auto Measure button for systems with automatic wafer loading capability

- [3.2.] The drop-down menu also allows selection of a different measurement method if desired. If **Repetitive Mode** is selected, the sample will be measured repeatedly until cancelled. See section 7.4.
- [3.3.] The **Selection** option causes the *Select Samples* window to appear immediately. If **Single Sample** is selected, only one wafer can be selected from the *Select Samples* window.
- [3.4.] To measure a wafer without using the indexer, select the **Manual Load** option. Place the wafer on the stage when prompted by the software.
- [8.] After selecting automatic measurement, the *Select Samples* window appears (Figure 12-25).

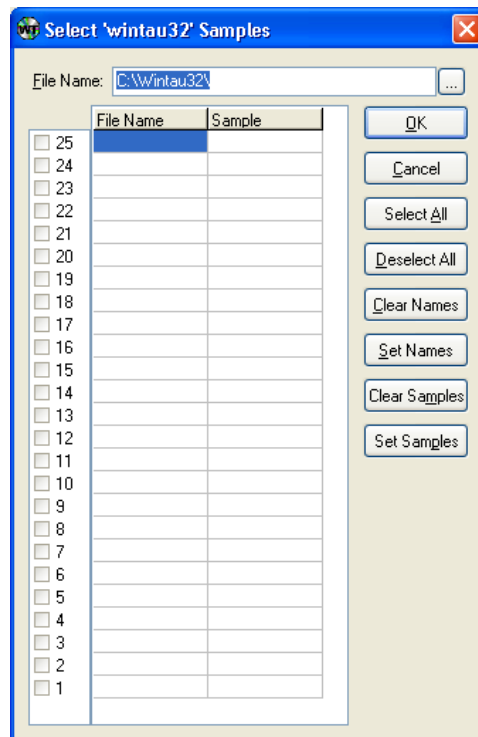


Figure 12-25: Select Samples window

- [9.] Select each wafer for measurement by clicking once on the checkbox next to its slot number (clicking again deselects the wafer). To select all wafers, click **Select All**; to clear all selections, click **Deselect All**.
- [10.] You can assign file names and sample names to the wafers by simply clicking on the *File Name* or *Sample* field of the slot and entering a name.
 - [10.1.] To set the filename to the slot number, click the **Set Names** button. A pop-up window appears. Enter the prefix for the filename, then click **OK**. The slot number will be added to the end of the *File Name* for each slot.
 - [10.2.] To set the sample name to the slot number, click the **Set Samples** button. A pop-up window appears. Enter the prefix for the sample name, then click **OK**. The slot number will be added to the end of the *Sample Name* for each slot.
 - [10.3.] Clear these entries by using the **Clear Names** and **Clear Samples** buttons.
- [11.] Specify the path for saving the data by entering it in the *File Name* edit box at the top of the window, or by pressing the  button to browse to the desired directory. The same path will be used for all wafers.
 - [11.1.] If you also enter the file name after the path, the same file name will be used for all the wafers. You can use the **Set Names** button to fill all the *File Name* fields. In this case, the last two characters of the file names will contain the slot number automatically.
 - [11.2.] In case the *File Name* written in the list is different from the one in the *File Name* edit box, the *File Name* in the list has higher priority. However, the path is always taken from the *File Name* edit box.

- [12.] After the wafers for measured are selected, and the path and file names/sample names are properly entered, click the **OK** button.
- [13.] At this time, the *Measurement Options* window will appear. If the *Auto* tab is not active, click on the tab to select it (Figure 12-26). Here the parameters of the auto-measurement can be set. For detailed description of these fields please refer to section 10.12.5. The parameters set here will apply for the whole measurement series.

✓ **NOTE:** Depending on the measurement mode and selection of the Fe concentration measurement, the appearance of the Auto tabs may be different from the example in the figure.

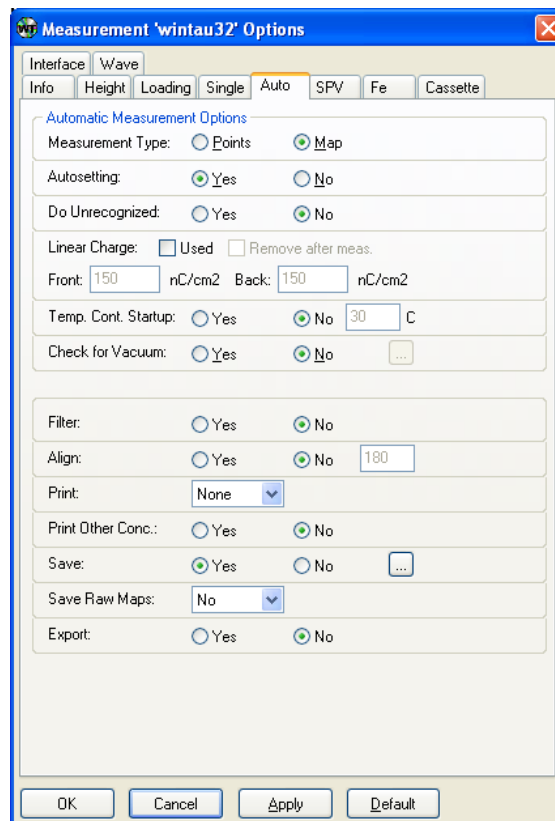


Figure 12-26: Auto tab

- [14.] After the *Auto* parameters are properly set, press the **OK** button to start the automatic measurement.
- [15.] The selected wafers will be automatically loaded, measured, and unloaded. If specified on the *Auto* tab, Autosetting will be performed and linear corona charge will be deposited. The results will be printed or saved as per the settings on the *Auto* tab.
- [16.] When the automatic measurement is complete, a message appears to remind the user to remove the cassette.

12.11 Automatic Measurements Using Recipes

The WT-2000 system has a **Measure Recipes** feature which automatically begins the measurement after specifying the recipe. This feature also allows multiple recipes to be used on each sample. For systems with automatic wafer loading, it is possible to specify different recipes for different wafers in the cassette. Note that the **Measure Recipes** feature does not allow modification of the signal parameters before beginning the measurement. If modification is required (for example, setting the bias light intensity), use the **Auto Measure** feature described in section 12.10.

✓ **NOTE:** *The Quick Start Guide provides an overview of the procedure for performing automatic measurements using recipes. See section 6.3.*

12.11.1 Systems with Manual Wafer Loading

[1.] To perform an automatic measurement using a recipe, click the **Measure Recipe(s)** button  on the *Measurement Toolbar*.

[1.1.] Clicking on the arrow next to the button  displays a drop-down menu (Figure 12-27) that identifies the type of measurement to be performed.

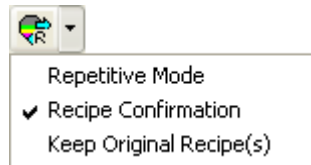


Figure 12-27: Drop-down menu next to Measure Recipe(s) button

- [1.2.] If **Repetitive Mode** is selected, the sample will be measured repeatedly until cancelled.
 - [1.3.] If **Recipe Confirmation** is selected, the *Measurement Options* window will appear before starting the measurement, to allow the user to review (and possibly change) the recipe settings.
 - [1.4.] If **Keep Original Recipe(s)** is selected, the recipe will be run with any changes made during Recipe Confirmation. However, the original recipe will not be changed.
- [2.] First the *Select Recipes* window appears (Figure 12-28). Select the recipe to be used for measurement, then click **OK**.
- [2.1.] To measure a sample using different recipes, select all the recipes to be used by pressing the <Ctrl> key while clicking with the left mouse button on the recipe names.

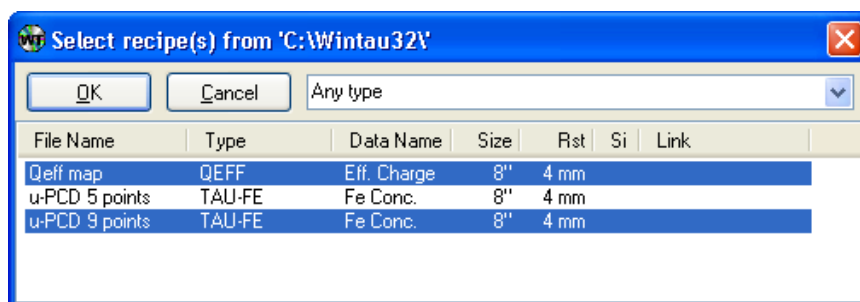


Figure 12-28: Recipe selection window with two recipes selected using <Ctrl> and left mouse button

- [3.] If *Save* is selected on the *Auto* tab, you will be required to provide a filename for saving. The results will be saved automatically after the measurement is complete.
- [4.] When prompted, place the sample on the stage, then close the cover.
- [5.] If *Autosetting* is selected in the *Auto* tab, Autosetting will be performed.
- [11.] If *Linear Charge* is selected in the *Auto* tab, it will be deposited before the measurement.
- [6.] The measurement will proceed without operator intervention.
- [7.] If *Print* is selected in the *Auto* tab, the measurement results will be printed automatically.
- [8.] When the measurement is complete, remove the sample.

12.11.2 Systems with Automatic Wafer Loading

On systems with automatic wafer loaders, multiple wafers can be measured in a cassette. Different recipes can be used for the different wafers. It is not necessary to use the same recipe for all wafers.

- [1.] To perform an automatic measurement using a recipe, click the **Measure Recipe(s)** button  on the *Measurement Toolbar*.
- [1.1.] Clicking on the arrow next to the button  displays a drop-down menu (Figure 12-29) that identifies the type of measurement to be performed.

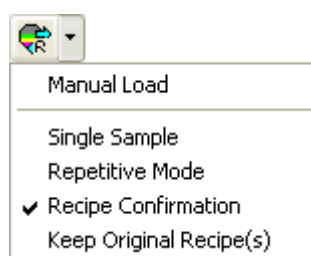


Figure 12-29: Drop-down menu next to Measure Recipe(s) button for systems with automatic wafer loading capability

- [1.2.] If **Single Sample** is selected, only one wafer can be selected from the *Select Samples* window.
- [1.3.] If **Repetitive Mode** is selected, the sample will be measured repeatedly until cancelled.
- [1.4.] If **Recipe Confirmation** is selected, the *Measurement Options* window will appear before starting the measurement, to allow the user to review (and possibly change) the recipe settings.
- [1.5.] If **Keep Original Recipe(s)** is selected, the recipe will be run with any changes made during Recipe Confirmation. However, the original recipe will not be changed.
- [1.6.] To measure a wafer without using the indexer, select the **Manual Load** option. Place the wafer on the stage when prompted by the software.
- [2.] First the *Select Recipes* window appears (Figure 12-30). Select the recipe to be used for measurement, then click **OK**.
 - [2.1.] If you need different recipes for different slots (for example, different measuring methods needed), select all the recipes to be used by pressing the <Ctrl> key while clicking with the left mouse button on the recipe names. You will assign the recipes to the different slots at a later step.

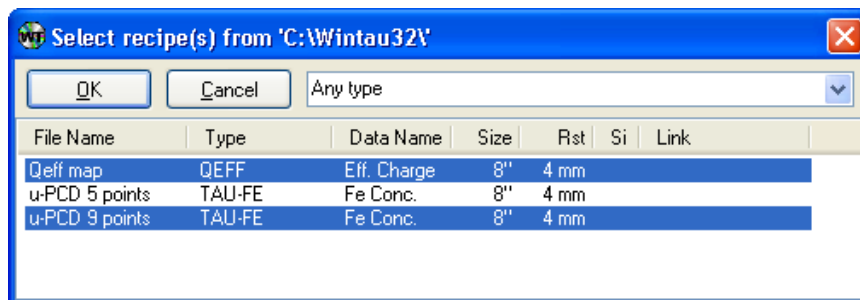


Figure 12-30: Recipe selection window with two recipes selected using <Ctrl> and left mouse button

- [3.] Next, the *Select Samples* window appears (Figure 12-31). If you selected more than one recipe, the window appears for each recipe, one after another. In the header of the window, the recipe name is displayed. Select the checkbox for the slots to be measured with the active recipe, and set the *File Names*. The same window will appear for the next selected recipe.

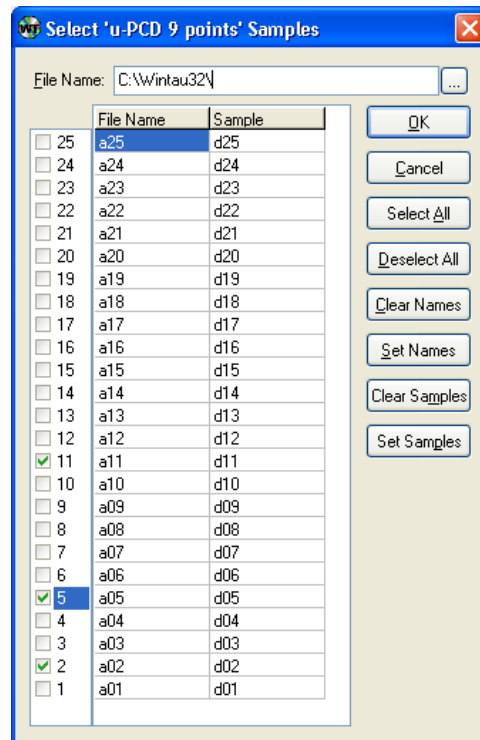


Figure 12-31: Selecting samples and setting file names for a recipe

✓ **Note:** If multiple recipes are selected for a sample, and you have specified the same File Name for different recipes, the later overwrites the earlier. For this reason, you should specify different File Names for each recipe.

- [4.] Next to the *Select Samples* window, the *Input List* window also appears (see section 12.5). Enter comments, operator name, and sample name. You can also change the *Raster* or the *Scanradius* (these fields display the values specified by the recipe).
- [5.] Click **OK** in the *Select Samples* window when all entries are complete.
- [6.] After clicking **OK**, the measurement starts with the set parameters.
- [7.] The selected wafers will be automatically loaded, measured, and unloaded. If specified on the *Auto* tab, Autosetting will be performed and linear corona charge will be deposited. The results will be printed or saved as per the settings on the *Auto* tab.
- [8.] If a wafer is associated with more than one recipe, it will be measured according to all of the associated recipes.
- [9.] The results will be printed or saved as per the settings.
- [10.] When the automatic measurement is complete, a message appears to remind the user to remove the cassette.

13. Calibration

This chapter includes calibration procedures for those measurement modes that require calibration:

✓ Calibration	✓ Section
✓ Calibration of the μ -PCD Bias Light Option	✓ 13.1
✓ SPV Calibration	✓ 13.2
✓ Corona Calibration	✓ 13.3
✓ V-Q Calibration	✓ 13.4
✓ JPV Head Calibration	✓ 13.5

13.1 Calibration of the μ -PCD Bias Light Option

The Bias Light option has a built in photo sensor to measure the light intensity. The sensor should be calibrated once at the time of installation, then once in a month. To begin the calibration, open the *Transient Recorder* window. In the *Bias Light* section, right-click on the slider (see section 10.13.1). In the pop-up menu, select **Calibrate Intensity**. The *Bias Light Intensity – Bias Power* window will appear, and the calibration will begin. The software determines the Bias Light Intensity – Bias Power curve during the calibration process. After the calibration is complete, click **Save** in the *Bias Light Intensity – Bias Power* window to save the file. To display the curve at a later time, use the *Transient Recorder Functions* menu and select **Bias Light Intensity – Bias Power**.

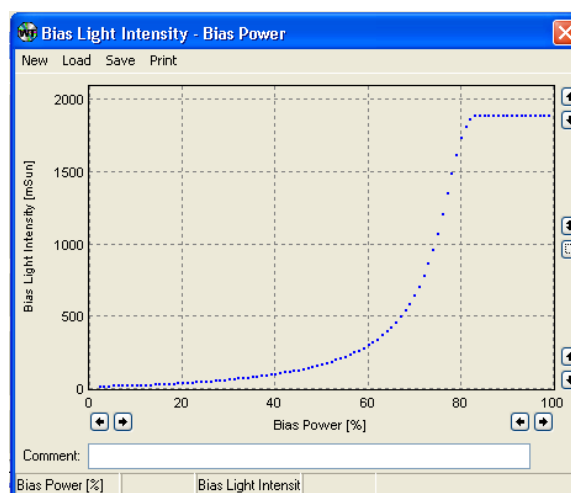


Figure 13-1: Typical calibration curve for Bias Light Intensity vs. Bias Power

13.2 SPV Calibration

The SPV head should be calibrated once in six month, because the photon flux of each laser at the given control voltage has to be known to evaluate the signal. The calibration procedure creates a file (**SPVcal.dat**), which contains the calibration data. The wavelengths of the lasers are read from the SPV head. The **detector.dat** file is also needed for the calibration. It contains the sensitivity data of the detector.

To perform the calibration, open the *SPV Recorder* window (see section 10.13.1.5). After selecting the **Calibrate** menu item, the calibration procedure is performed automatically. The calibration curves can be displayed by using the **Functions | Calibration Curves** menu item in the *SPV Recorder* window.

13.3 Corona Calibration

The corona charger is used for C-PCD and V-Q measurements. It may also be used by itself. The corona charger should be calibrated after maintenance, or when the *Deposited Charge* differs from the set *Charge Density* value by more than 20%.

Before calibrating the corona charger, load a bare wafer onto the sample stage. Select the **Measure | Corona | Calibrate** menu item (Figure 13-2).



Figure 13-2: Corona calibration menu item

The *Current-Voltage* window will appear during the calibration.

Once the calibration is complete, a confirmation message appears to indicate that the calibration completed successfully (Figure 13-3).

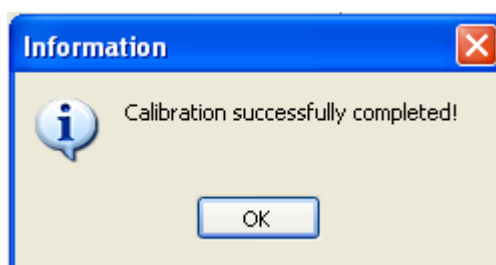


Figure 13-3: Successful calibration

Click OK. The calibration curve is indicated, if the calibration was good, it is like Figure 13-4.

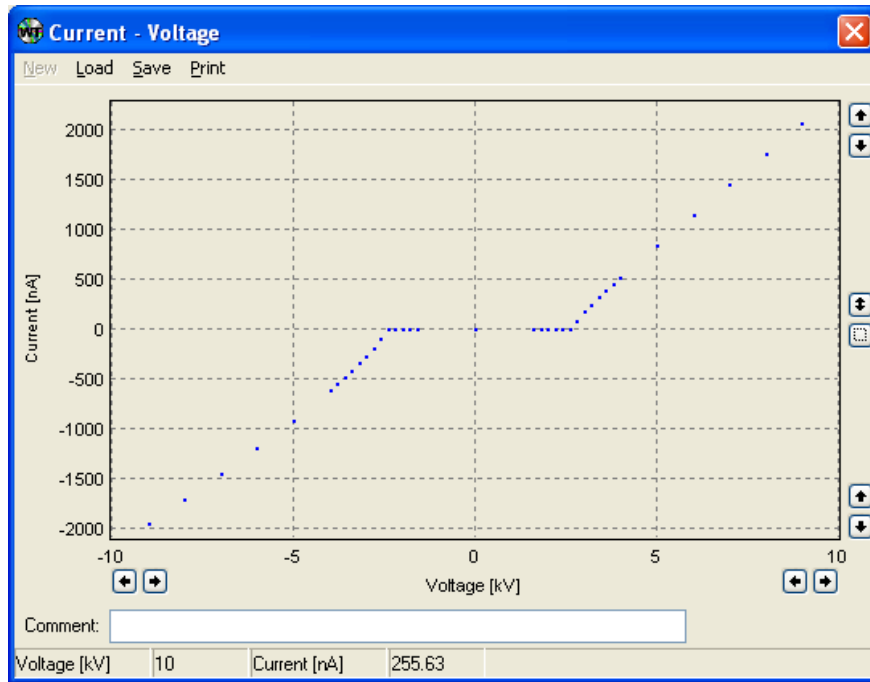


Figure 13-4: Calibration curve of the corona charger

13.4 V-Q Calibration

The probe head should be calibrated after installation or maintenance. No sample is required for the calibration procedure.

Select the **Functions | Calibrate** menu option in the *V-Q Recorder* window (Figure 13-5). The calibration procedure will begin.

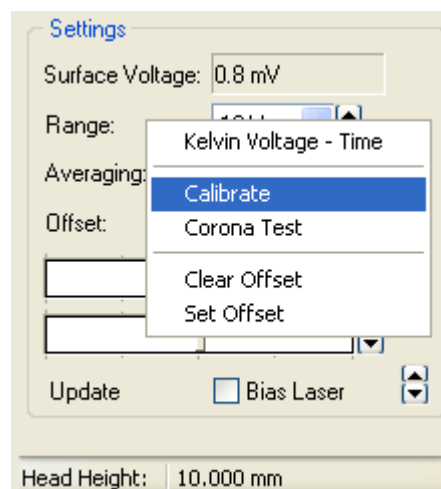


Figure 13-5: The *Calibrate* option in the *V-Q Recorder* window

When calibrating, the head moves above a graphite plate and adjusts the offset until the measured voltage equals -0.2V. This offset value is displayed in the *V-Q Recorder* window.

Semilab recommends using this automatically determined offset. However, the offset can be set manually by the two sliders below. The upper slider is the rough adjustment, while the lower one adjusts the fine tuning of the offset. Both sliders can also be adjusted using the up/down arrows . Right-clicking in the *Settings* area of the *V-Q Recorder* window displays a pop-up menu allowing you to clear or set the *Offset* values.

13.5 JPV Head Calibration

The JPV head should be calibrated after installation or maintenance, or after changing the measurement frequency. The calibration consists of three steps: the LED intensity calibration, the JPV voltage calibration and the EQE calibration.

13.5.1 Calibrating the LED intensity

- Switch on the WT-2000 and perform initialization.
- Switch to JPV mode and call the recorder window.
- Click on the *LED intensity* slider bar with the right button.
- To find out whether intensity calibration is needed or not, select *Check Intensity*. The head moves above the detector and LED intensity is measured. If the difference of the intensity in the popup window and the intensity indicated in the *Recorder* window, *LED intensity area*, is bigger than 5%, it is necessary to recalibrate the intensity.

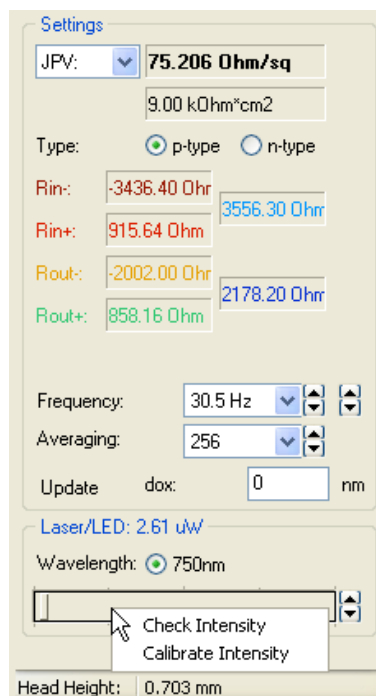


Figure 13-6: Right click on the intensity slider bar to check on calibrate intensity

- Again, right click on the intensity bar and choose *Calibrate Intensity*.
- A popup window informs when calibration is finished.

13.5.2 Calibrating the JPV voltage by the metal reference plate

For the calibration of the JPV head, there are three reference voltage plugs on the JPV probe. To come at these plugs, remove the front cover of the measurement head. The upper and the lower ones of these plugs are on ground potential, the potential of the middle one is 25 V.

Setting the point measurement averaging setup to 256

Before calibration, the point averaging has to be set to 256 at every frequency. Please, do the following steps:

- Call the JPV Recorder window
- Right-click in the Averaging field
- Choose *Average Setup* (Figure 13-7)

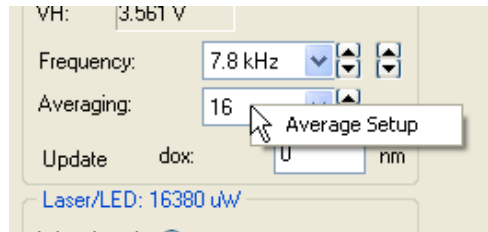


Figure 13-7. Right click on the intensity slider bar to set Average Setup

- A popup window comes up with the preset averages for the measurements. The first column shows the averaging for one-point measurements, the second column shows the averaging for mapping.

Frequency	Point	Map
500 kHz	256	16
250 kHz	256	16
125 kHz	256	16
62.5 kHz	256	16
31.3 kHz	256	16
15.6 kHz	256	16
7.8 kHz	256	16
3.9 kHz	256	16
2 kHz	256	16
977 Hz	256	16
488 Hz	256	16
244 Hz	256	16
122 Hz	256	16
61 Hz	256	32
30.5 Hz	256	64
15.3 Hz	256	128

Figure 13-8. JPV Averaging Setup table. Point measurement averaging in the first column.

- If the point averaging is less than 256, please, change every values in the first column to 256, as on Figure 13-8.

Compensation before calibration

- Please, load the reference metal plate, as if it was a sample. (Don't run wafer finder.) Align it, so that the connection wire points towards the operator.
- Make sure that the sensor of the JPV probe is over the middle of the metal plate.

- Please plug the wire of the metal plate into one of the ground plugs of the JPV probe. (Upper or lower one.)
- Run compensation from the *SHR Recorder* window, **Functions | Compensate** menu item. Please, save the compensation curves.
- Please, check the result of the compensation. For this, set the light intensity 0 by the slider bar, and call **Functions | Voltage – Frequency**. If the compensation was effective, all the values are not more than 10 mV. Please, save the curves.

JPV Calibration

- Please, plug the wire of the reference plate into the 25 V plug on the JPV probe (middle one).
- Please, set the light (Laser or LED) intensity to 100%. Leave the mouse cursor over the **Update** button for one minute to warm up the head.
- Call Right-click | **Functions | JPV Calibration** in the *Recorder* window.
- If the sensor is over the middle of the metal plate, click OK in the pop up window.
- Enter 25 mV as reference voltage on the metal plate in the appearing window and Click OK.
- Enter 1E11 as Reference Frequency on the metal plate.
- Please, save the curves.
- Please, check the result of the calibration by calling Right-click | **Functions | Voltage – Frequency**. All the curves have to be either constant 25 mV or 0 mV.

Checking the noise level

- Please, plug the metal plate into one of the ground plugs of the JPV probe. (Upper or lower one.) Leave all other settings the same.
- Call Right-click | **Functions | Voltage – Frequency**. The noise should be less than 0.1 mV.

Please, unplug and unload the reference metal plate. Reset the light intensity to its initial value. Fit back the front cover of the head.

13.5.3 Calibrating to the external quantum efficiency of the samples

The calibration has to be adjusted to the typical external quantum efficiency of the measured samples. If the measured samples have highly different EQE from the reference sample (>5%), a new calibration must be done for the different type of sample. Change in reflectance changes the EQE.

- Please, load a sample that has a known sheet resistance. Make sure that the JPV head is over the middle of the sample. Please, close the lid of the WT-2000!
- Please, make sure that Right-click | **Compensate Before Autose** is not checked.
- Please, make sure that the distance of the head from the sample is the same as it was between the head and the metal plate during calibration.
- Call Right-click | **Functions | Calibration**. The window on Figure 13-9. comes up.

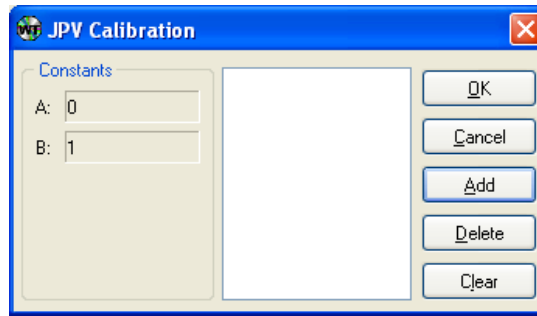


Figure 13-9. Calibrating the JPV to EQE

- Click **Clear**, then **Add**. Wait until the measurement is done.
- A window calls to “Enter non-compensated value [Ohm]”. Don’t change the number, and press **OK**!
- A window calls to “Enter target value [Ohm]”. Please, enter the nominal sheet resistance of the sample!

If there are more than one JPV heads in the WT-2000, run this procedure for all!

The JPV calibration is now finished.

Regular calibration of the equipment is required to maintain the accuracy of the measurements. Semilab does not bear the responsibility for inaccurate measurements which occurred due to the lack of calibration or to the neglecting of the precautions.

14. Maintenance

The WT-2000 system has been designed for almost maintenance-free operation. It requires very little periodic maintenance.

There are only two optional parts of the WT-2000 system that have limited lifetime: the flash unit for Fe concentration measurement, and the bias light for μ -PCD.

Table 14-1 lists the periodic maintenance procedures that are included in this chapter.

Table 14-1
Periodic Maintenance

Procedure	Frequency	Section
Cleaning the System	Every 6 months in a cleanroom Every month in a laboratory Every week in a factory	14.1
Lubrication	Once a year	14.2
Replacement of the flash bulb	After 50 000 flashes	14.3
Replacing the μ -PCD bias lamp	After 200 hours	14.4
Replacing the pins of four point probe	After 40 000 touch-downs	14.5

The high reliability of the WT-2000 system results from Semilab's philosophy of addressing recurring failures as a design issue needing correction. The system design has been improved to the point where recurring failures are absent; thus no installation instructions are necessary for this manual, except replacing the flash lamp and the bias lamp.

In the event that a part fails, a replacement will be provided with complete installation instructions.

If a software upgrade is required, see section 15.1 for instructions.



WARNING: Always disconnect the computer unit of the WT-2000 from all sources of electrical power before performing any maintenance procedures.

14.1 Cleaning the System

When using the WT-2000 in a cleanroom, cleaning may be required about every 6 months. When using the WT-2000 in common laboratory conditions, the cleaning procedure should be done every month. When using the WT-2000 in a dusty ambient, such as a factory, the cleaning is required once a week.

Cleaning (decontamination) procedures should be done with alcohol using a cleanroom-compatible lint-free cloth.

Clean the rails directing the movement of the stage and the rail of the head.

Dust can be removed from the not easily accessible parts of the WT-2000 by compressed air flow. If the side covers of the WT-2000 are removed, all parts can be cleaned by compressed air. The side covers can be removed by a 3 mm hex key. After cleaning, please make sure to reconnect grounding of all side covers.

! CAUTION: Before removing a side cover always make sure that the equipment is switched off
--

The stage should be cleaned at the same time with a dry cloth.

! CAUTION: Do not clean the ESD table with alcohol or any other solvent!

14.2 Lubrication

This chapter describes the lubrication procedures allowed to perform on the WT-2000 system.

Lubrication material: Original THK grease. AFB-LF is the standard grease, while AFF can be used in clean room.

Amount of material: approximately 3g/maintenance procedure

It is advised to lubricate the rail of measurement head, measurement stage, bridge, loading arm and the bearing of indexer (this can be reached when the indexer platform is at lower position) once a year, or if the movement produces noise. Loading arm and indexer are only present in equipments with automatic loading option.

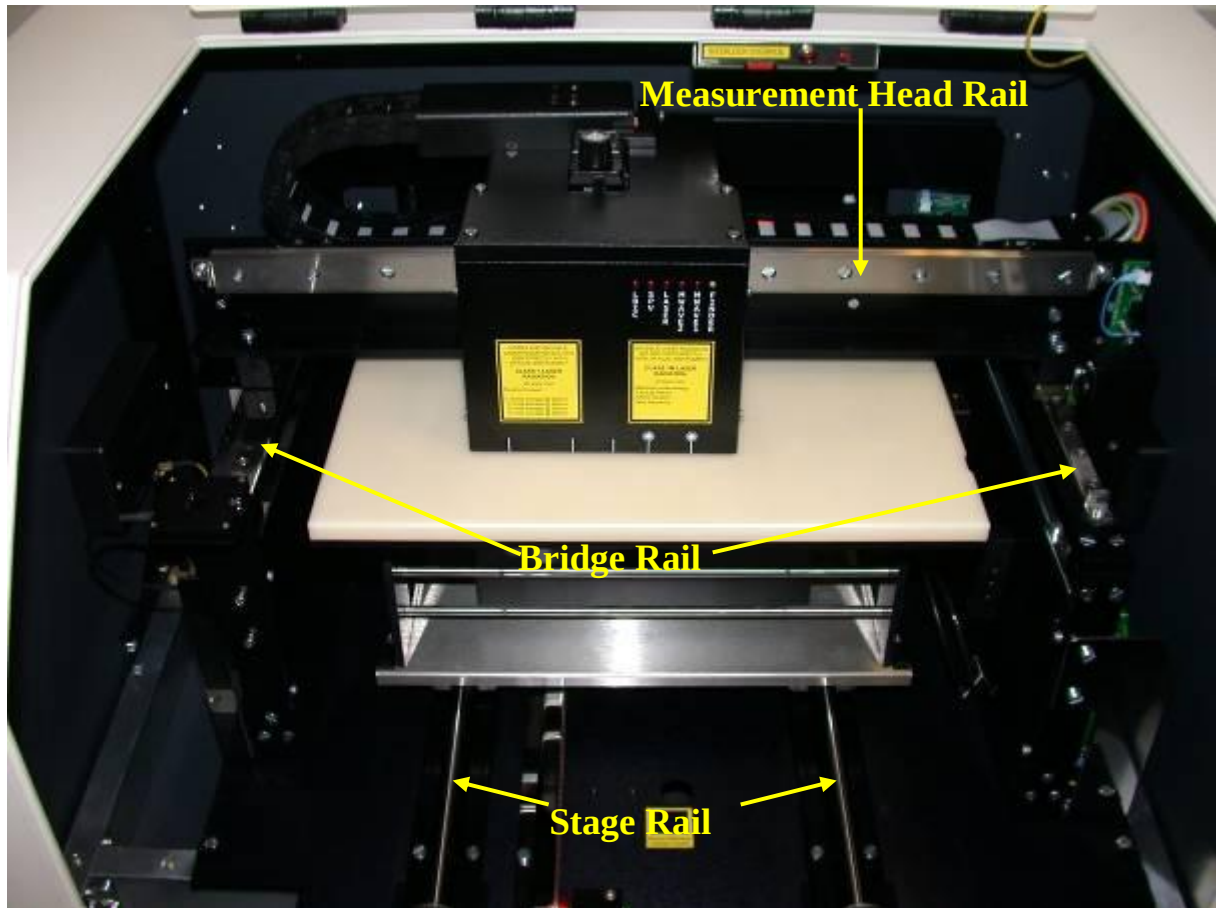


Figure 14-1: Parts of the WT-2000 that need lubrication.

14.3 Replacement of the flash bulb

Table 14-2 Guaranteed usage of flash bulbs

Type of flash lamp	Number of flashes guaranteed
Hensel EHT 3000	50 000
Comet CB-25H	50 000

The warranty of the flash bulbs, applied to WT-2000 systems can be seen in Table 14-2. After flashes are performed more times than guaranteed, it is recommended to change the flash bulb.

The number of flashes, done by the lamp, can be tracked in *Measure | Options | Fe* tab, *Number of Flashes* counter. (See also sec. 10.12.9.) If this value reaches the warranted number, it is recommended to change the bulb of the flash lamp.

After the bulb is replaced, please reset the flash counter by double left click in the counter area and click **Yes** in the popup window to confirm resetting the flash counter.

14.3.1 Steps of replacing the flash bulb

- [1.] Turn off the switch on the flash lamp. Please turn off the lamp generator and remove the power cable from it.
- [2.] Remove the lamp from the reflector on the system door. The lamp is fixed by two screws that can be driven by hand. First loosen up these screws, then gently turn the lamp right in the reflector, then pull it upwards.

! CAUTION: In the next steps you have to wear plastic gloves. DO NOT TOUCH THE BULB BY BARE HAND!

- [3.] Remove the bulb from the lamp. Please take off the bulb by hand and pull it out from its outlet.
- [4.] Push the new bulb into its outlet.
- [5.] Attach the lamp body to the reflector. Align the lamp so that the holder rings are backwards, compared to the WT-2000. Slip the lamp into the reflector, so that it fits in the three slots. Turn it left. Fix the screws.
- [6.] Connect the power cable and then turn on the generator and the lamp.
- [7.] Perform some test flashes. It is possible from the Wintau32 operational software. Please check that there is no any sample in the WT-2000 system, and close the system door. Please turn on the system and activate the *Measure | Hardware Test* menu item. Please select the *Interlock2* tab and click on the **Relay Power** and the **Flash Power** button. (See also sec. 10.10.4.) The flash lamp generator is turned on. After the “Test” led is lit on the generator you can check the performance of the new flash bulb. If you click on the **Flash Trigger** button, the lamp has to make a flash.

14.4 Replacing the μ -PCD bias lamp

The warranty of the bias lamp for μ -PCD is 200 hours usage. The net time of using the bias lamp can be tracked *Measure | Options | Single* tab, Bias Light area, Used hours field:

Bias Light:	-100	$\mu\text{W}/\text{cm}^2$	Used hours:	2.90
-------------	-----------------------------------	---------------------------	-------------	-----------------------------------

If the indicated number reaches 200 hours, it is recommended to change the bulb of the bias lamp.

After the bulb is replaced, please reset the bias counter by double left click in the counter area. Click **Yes** in the popup window to confirm resetting the counter.

14.4.1 Steps of replacing the bias halogen bulb

- [1.] Turn off the system
- [2.] Take off the back cover of the WT-2000. Eight bolts fix it.
- [3.] Remove the cover of the bias lamp unit. Four bolts fix it.
- [4.] Remove the bulb from the bulb holder. Four bolts fix it.
- [5.] Pull both cables from the bulb and connect them to the new bulb.
- [6.] Fix the new bulb to the holder, put on the cover and test the bulb. Please, always perform bias lamp calibration procedure after changing the bulb. (Right click on the *Bias Lamp* slider in the *Transient Recorder* window, **Calibrate Intensity**. See section 10.13.1.2)

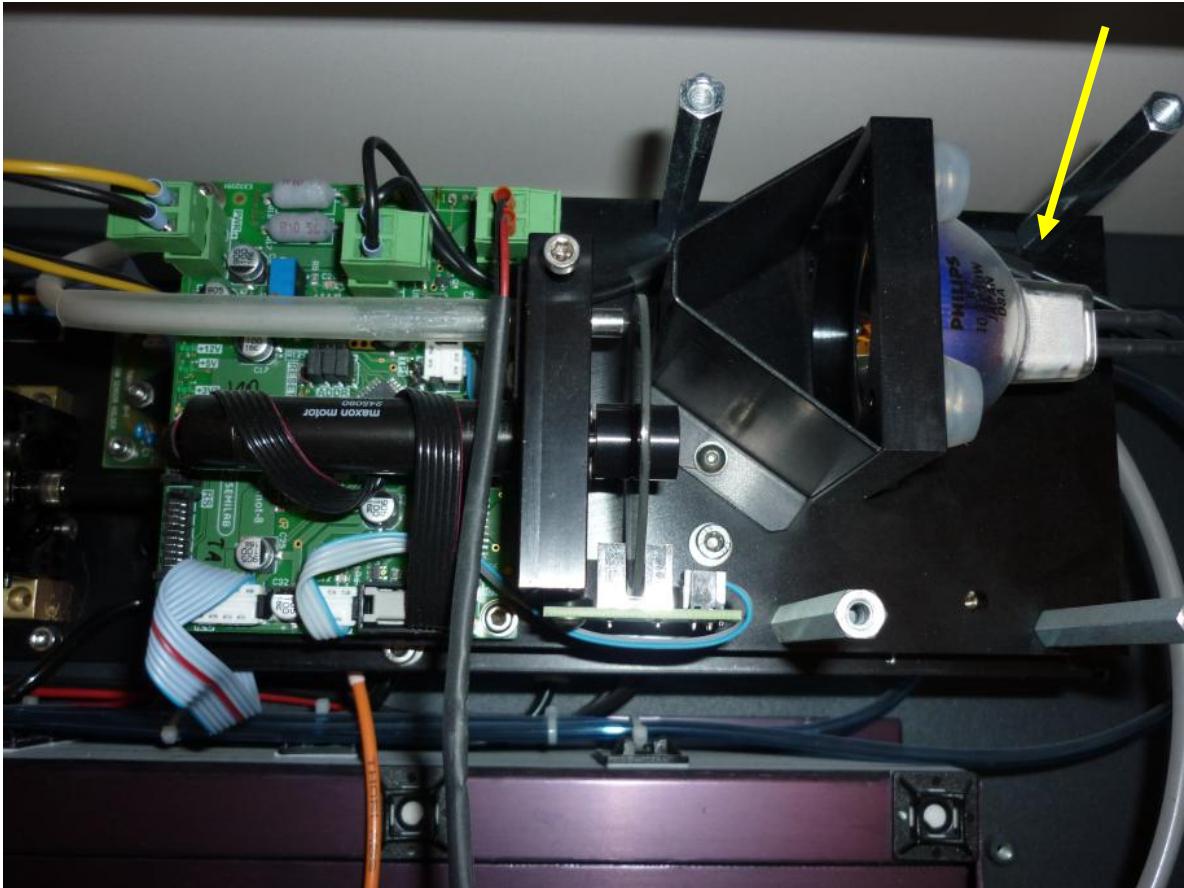


Figure 14-2: Replacing the bias halogen bulb

14.5 Replacing the pins of 4pp

If a four point probe is installed in the system, it needs regular maintenance as follows.

The pins of four point probe gets worn at an average of 50 000 touch-downs, so it is recommended to replace the pins after 40 000 touch-downs in order to get reliable results. The number of touch-downs can be checked in the Recorder window of 4pp.

14.5.1 Steps of replacing the 4 point probe

- [1.] Turn off the system
- [2.] Remove the front cover of the measuring head. To do this, unscrew the four screws at the sides of the front cover. See Figure 14-3.



Figure 14-3: Measuring head of WT-2000 with a 4pp. Remove front cover by the screws at the arrows.

[3.] Remove the plug of the 4 point probe. See Figure 14-4, A.

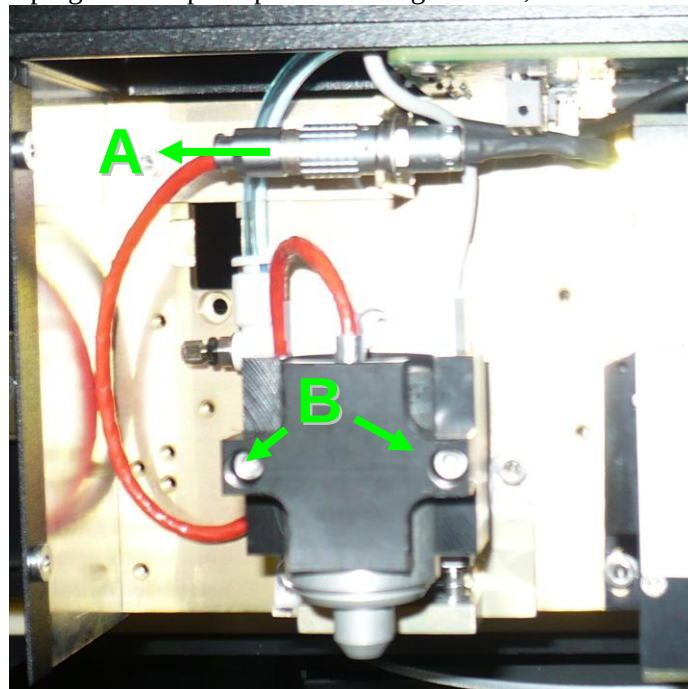


Figure 14-4: Remove the plug of the 4 point probe (A) and unscrew the two screws of the probe holder (B)

- [4.] Unfix the probe by the two screws as seen on Figure 14-4, B.
- [5.] Replace the probe by the new one.
- [6.] Fix it by the screws of Figure 14-4, B.
- [7.] Plug the cable of the new probe as in Figure 14-4, A.
- [8.] Fix the front cover by its four screws as on Figure 14-3.
- [9.] Reset the counter of the measurement by double-clicking on the counter in the Recorder of 4pp.

15. Appendix

15.1 Installing Wintau32 Software

The WT-2000 computer is pre-loaded with software. In the event that you must reinstall the software, this section provides information about the installation procedure. Contact Semilab if you have any questions about the software installation.

This section also includes instructions for installing software upgrades.

15.1.1 Requirements

For proper operation the requirements in Table 15-1 should be fulfilled.

Table 15-1
Computer Requirements for Wintau32 Software

Operating system requirements:	Microsoft Windows 98, NT, 2000 or XP
Hardware requirements:	Pentium-compatible computer running the Microsoft Windows operating system Dedicated 10/100 Ethernet card for measuring capabilities Optional Windows-compatible color printer

15.1.2 Installing the Software from the CD-ROM

✓ **NOTE: If the WT-2000 system is purchased with standard configuration, and the WT-2000 is shipped with two industrial PC-s, the software is already installed on the PC-s. It is neither needed, nor advised to reinstall the software, because it removes the calibration and configuration files of the system, and it needs to be calibrated again.**

The software is distributed on a CD-ROM and can be installed by running the **setup.exe** program. If you accept the defaults in the appearing dialogs, the software will be installed to the C:\wintau32 directory, and a shortcut will be created on the Desktop and in the Windows Start menu.

15.1.3 Windows Networking Configuration

✓ **NOTE:** *Connecting the measuring computer to a network indicates potential harm from viruses, spywares or crackers. Semilab doesn't bare responsibility for any harm, data loss or malfunction of the WT-2000 that is caused by these.*

The Wintau32 software communicates with the WT-2000 Main Unit via NIC (Network Interface Card). There can be two NICs in the PC; in this case one NIC can be used for LAN access and the other one for connecting the WT-2000. There is a cross-wired CAT5 UTP cable connecting the Windows computer to the tool; if it is unplugged, a notification should display in the *Status Bar* (see Figure 15-1).

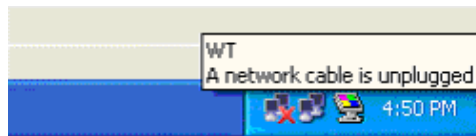


Figure 15-1: UTP connection notification

Click on the icon with the right mouse button, and select **Open Network Connections**. The *Network Connections* window will open (Figure 15-2).

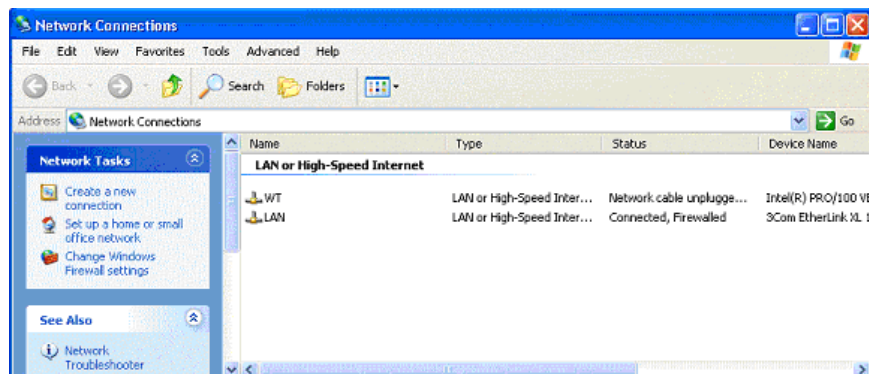


Figure 15-2: Network Connections window

Right-click on the connection for WT-2000 and select **Properties** (Figure 15-3).

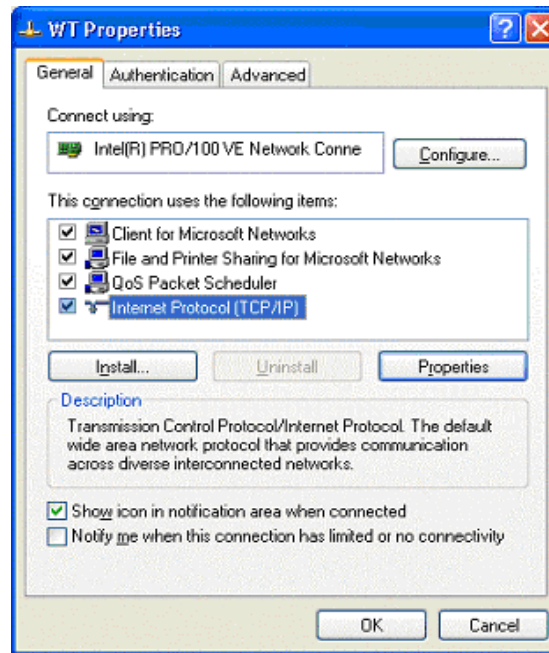


Figure 15-3: WT Properties window

Select **Internet Protocol (TCP/IP)** and click **Properties**. In the appearing window set the parameters as shown in Figure 15-4:

IP Address: 192.168.2.3

Subnet mask: 255.255.255.0

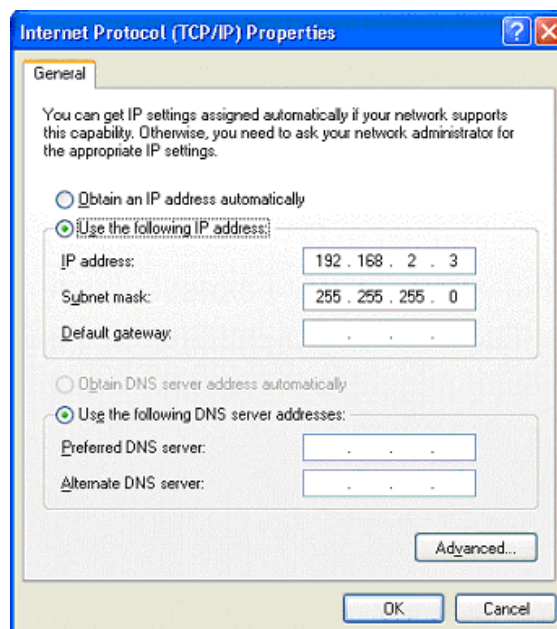


Figure 15-4: Internet Protocol Properties window

When done, the cross-wired cable should be connected again.

✓ **NOTE:** The network firewall for the WT-2000 communication should be turned off for the network interface. If it is activated it blocks communication between the user interface and the Main Unit.

15.1.4 Interface Settings in Wintau32

Start Wintau32 and select **Measure | Options** and the **Interface** tab (Figure 15-5).

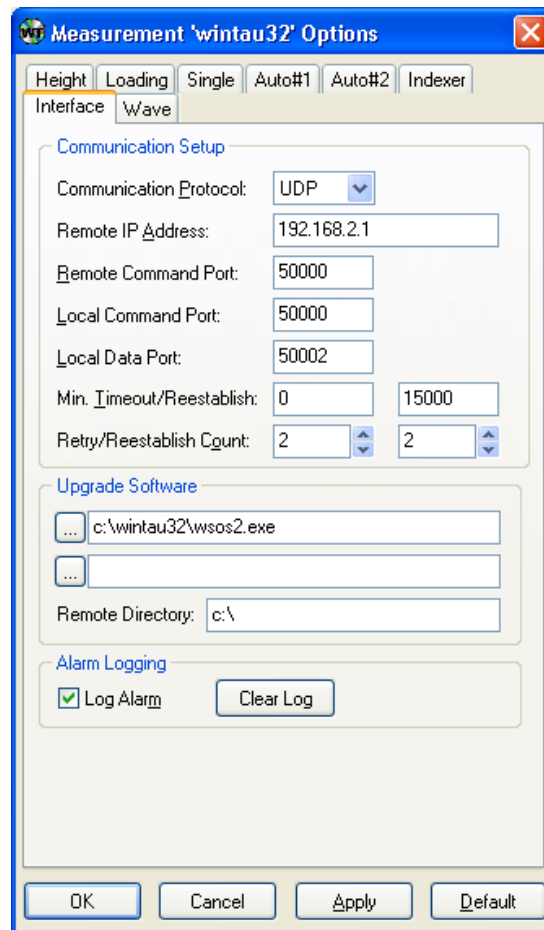


Figure 15-5: Interface tab on Measurement Options window

Check the fields and correct if necessary, then click **OK**. If the tool starts by switching on both computers (DOS, Windows), a message **WSx.xx OK** should appear in the Wintau32 *Status Bar* at the bottom left.

15.1.5 Software Upgrades

15.1.5.1 Wintau32 Upgrade

If you receive an upgrade for Wintau32, it is an executable file named **wintau32.exe**. Exit Wintau32 and copy the new file over the old one (overwriting) in the C:\wintau32 folder. The older versions can be found in the C:\wintau32\backup folder after upgrade.

15.1.5.2 WSOS2 Upgrade

If you are upgrading the DOS (lower level) software, copy the received **wsos2.exe** file to the C:\wintau32 folder.

Both computers should be switched on. Next, select the **Measure | Remote | Upgrade Software** menu item.

Select **OK** in any windows that appear.

15.1.6 Registration

After installation, Wintau32 needs to be registered for full functionality. If you do not register you cannot save data, and the *unregistered version* text appears on every printout. Measuring capabilities are also unavailable without registration.

You need a *User* name and a *Registry ID* for registration. Semilab recommends using your company name as the *User* name. It is displayed in the title bar of Wintau and on every printout. To validate the *User* you need the *Registry ID* specific to the *User*, which can be requested from the vendor (Semilab). There are different registries IDs for the Data Evaluation Mode and the Measurement Mode of the Wintau software. (See section 7.7 for an explanation of these modes.)

To perform the registration, select the **Help | Register** menu item. Type the *User* name and your *Registry ID* in the window that appears (Figure 15-6).

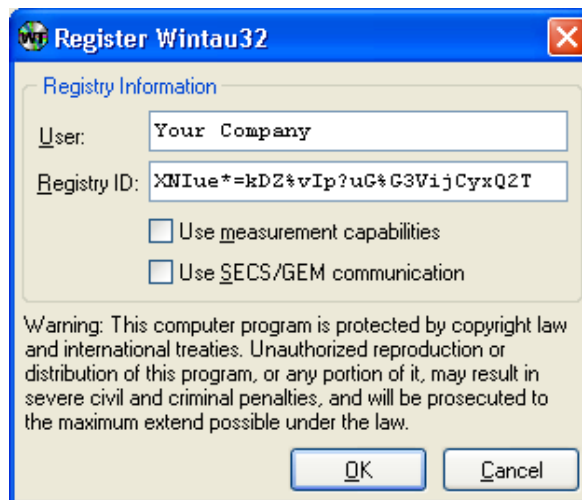


Figure 15-6: Registration window

- ✓ **NOTE:** Check the *Use measurement capabilities* if available box if you want to measure with the software, and if you have the appropriate hardware and the appropriate Registry ID. If you do not have measuring capabilities, or if you want to use the software for data manipulating only, you do not need to check the above box.
- ✓ **NOTE:** The *User* name and the *Registry ID* are both case-sensitive.

Alternatively, the file **id.txt** can be obtained from the root folder of the CD-ROM or it can be sent by the vendor. Copy this file into the C:\wintau32 folder.

16. Index

A

ASCII files 8-2
 Autologin 7-11
 Autosetting..... 10-27, 12-9, 12-20
 location on sample..... 12-16
 SPV 5-24, 10-55
 u-PCD..... 5-7, 10-40

B

Bias light
 theory..... 5-8

C

Calibration
 frequency 13-1, 13-2
 frequency 13-3
 frequency 13-4
 Child windows 8-15
 Cleaning frequency..... 14-1
 Color scales
 default setting..... 9-8
 displayed map 9-7, 9-18
 Comments
 Input List 9-12
 Input List 12-13
 map..... 9-17
 Communication between computers.10-38, 15-2
 Computers..... 2-1, 2-2
 Console stage 4-16
 Conventions 1-3
 Copying data 8-4
 Copying maps 9-16
 Corona wire..... 4-14
 C-V curve 5-29, 10-63

D

Default settings
 display 9-7
 Input List 9-13
 recipe..... 10-6
 zoom..... 9-9
 Desktop
 saving 8-2, 9-8
 dox..... 12-13

E

Exporting files
 ASCII 8-2
 bitmaps 8-3, 9-16, 9-19

HTML 8-3
 JPEG 9-17, 9-19

F

Fe concentration units 8-13
 Fe-B pairs 5-24
 File list
 configuring 9-4
 copying..... 9-4
 creating..... 9-4
 disabling..... 9-5
 navigating..... 9-2
 refreshing 9-2
 Filelist.res 9-2
 Flash interlock 3-2, 10-10
 Floating windows..... 9-11
 Form feed 9-23

H

Head height..... 10-14
 Help
 how to get..... 1-4
 manual..... 7-13
 Histograms..... 9-18
 Hot chuck 8-7, 10-3

I

Image processing 8-9
 Initialize..... 4-24, 6-9, 8-6, 10-2, 12-2
 Input List 9-12, 12-12
 Iron detection..... 5-24

K

Kelvin probe 5-28

L

Laser
 μ-PCD 3-6
 interlock 3-2, 10-10
 penetration depth 5-1
 SPV 3-8
 VQ 3-9
 Linear corona charge..... 5-19, 10-5, 12-19
 automatic measurements 10-22
 calibrating..... 10-6
 calibrating..... 13-2

M

Macros..... 11-1
 running..... 11-2, 11-3

Masks	8-4	Screens	
buttons	9-14	printing.....	1-5
zoom	8-5	<i>Select Samples</i> window	12-22, 12-27
MeasInfo.....	8-5	Semilab	
Measurement heads	2-1	contact information	1-4
Measurement mode	4-23	Single point measurements	12-14
Measurement Mode		sll files	9-20
setting	7-8	Software	
Mini size prints.....	9-22	default directory.....	15-1
Mobile charge	5-30	default path.....	4-22
Modes		installation	15-1
Data Evaluation.....	7-8	registration.....	15-5
Measurement.....	7-8, 15-5	upgrade.....	15-4
P		Sounds.....	10-39
Pin charger	4-14, 10-29	SPV	
Point coordinates	10-20, 12-15	comparison to u-PCD.....	5-20
Power		maps.....	5-23
shutdown procedure	6-9	SPV plot	
starting the system.....	6-8	normal appearance	12-11
<i>Prescan Points Setup</i> window.....	10-20	Stage types.....	10-3
Printing		changing.....	8-7
details	9-23	Stage, console	4-16
form feed	9-23	Sub Measurement Modes	7-7, 12-3
size selection.....	9-22	Surface recombination velocity	5-6, 8-12
R		T	
Raster.....	12-13	Transient curve	
Raw files	10-23, 12-20	normal appearance	12-10
Reducing file size.....	8-10	V	
Resolution.....	8-10	Vacuum connections	4-10
Rulers	9-8	V-Q curve.....	5-29
S		W	
Safety symbols	3-1	Wafer Finder.....	10-4
Scanradius.....	12-13	Wintau32	
Screenprints	1-5	registration.....	15-5